

Configuration Guide | PUBLIC
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Tuning Guide

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1 Welcome

Giving you the best user experience is our priority. Therefore, we do our best to provide you with the clearest and the most complete documentation possible.

For this purpose, feel free to leave additional feedback by using the dedicated feature on the Help Portal. Please write some details about your suggestion. Based on SAP user feedbacks, we improve your documentation and deliver Frequently Asked Questions (FAQs) about all aspects of working with SAP Convergent Charging efficiently. We complete these solutions with our best time-saving and task-completion tips and tricks.

You can also visit our page Community Questions and Answers for SAP Convergent Charging.

2 Before You Start

About This Document [page 6]

Target Audience [page 7]

What's New in This Document [page 8]

Additional Information [page 8]

Document Abbreviations [page 9]

Document Definitions [page 9]

2.1 About This Document

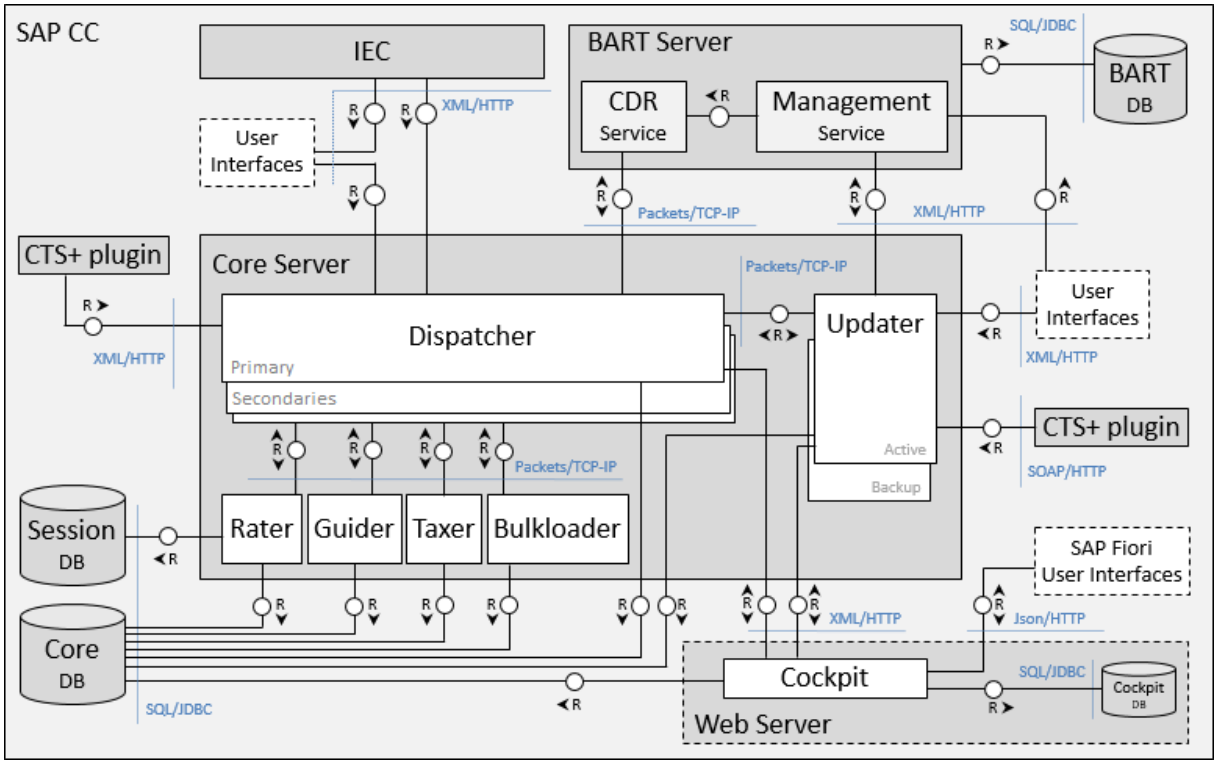
> → Remember
>
> The commands used in this guide have been tested in the different scenarios with the dedicated tools. However, using these tools is not mandatory: other tools are available according to your preferences. In this case, the commands may differ.

SAP Convergent Charging is made up with a set of Java-based software components that you can install within your landscape to perform your business operations. For further information about the different software components, refer to the Architecture section of the SAP CC 2021 Application Help documentation.

These software components are grouped upon the following categories:

- Server systems (Core Server, BART Server and Import/Export Connector)
- User interfaces (Cockpit, Core Tool, BART Tool, CAT Tool, Admin+, BART+, Setup Tool, BART Setup Tool, Config Tool, HTTP Client and Message Client)
- Databases (Core Database, BART Database, Session Database and IEC Database)

The following schema provides you with an overview of the interactions between the different software components within an SAP CC landscape:



This Administration Guide is an SAP document that contains:

- Recommendations about the **sizing** of the different aspects of your landscape, such as:
  - Deployment strategy for the SAP CC instances
  - Network, disks and operating system specifications and requirements
  - JVM[1] settings
- Procedures for **configuring** some technical and business aspects of the different software components
- Procedures for **tuning** some technical mechanisms used by the different software components
- Orientations for **troubleshooting** problems using the mechanisms provided by SAP Convergent Charging

2.2 Target Audience

This guide is intended for the following audiences:

- Application and System Administrators
- Solution and Technology Consultants

[1] Java Virtual Machine

2.3 What's New in This Document

What's New in SAP Convergent Charging 2021 FPS 2?

As of FPS 2, SAP Convergent Charging 2021 provides you with:

- Modification and enhancement of the Configuration Settings for Cockpit [page 20] section to consider the single sign-on (SSO) SAML 2.0 protocol for authentication of Cockpit users

What's New in SAP Convergent Charging 2021?

As of major release 2021, SAP Convergent Charging provides you with:

- Modification of the Configuration Settings for Cockpit [page 20] section with the addition of a new setting related to the activation of the web assistant

2.4 Additional Information

For more information about specific topics, see the quick links as shown in the table below:

Content	Quick Link
SAP CC documentation	help.sap.com/cc2021 **Note** SAP CC is part of the SAP solution *SAP Billing* based onSAP Business Suite or SAP S/4HANA. For further information, consult the central solution information at the following address: https://wiki.hybris.com/display/ybilling/Library
Related SAP notes	support.sap.com/notes
Related release notes	https://support.sap.com/en/my-support/knowledge-base.html
Released Platforms	Supported Database Platforms
SAP Solution Manager	http://support.sap.com/solutionmanager

2.5 Document Abbreviations

Abbreviation	Meaning
CEA	Capabilities-Exchanging-Answer
CER	Capabilities-Exchanging-Request
CIF	Charging output Integration Framework
CPU	Central Processing Unit
CRM	Customer Relationship Management
DWA	Device-Watchdog-Answer
DWR	Device-Watchdog-Request
MSCC	Multiple-Services Credit Control
NUMA	Non-Uniform Memory Access
RDBMS	Relational Database Management System
RIF	Rerating Integration Framework
SAPS	SAP Application Performance Standard
SAP MC	SAP Management Console
SAP MMC	SAP Microsoft Management Console
SSD	Solid-State Drive
TIF	Transaction Integration Framework
UDP	User Datagram Protocol

2.6 Document Definitions

Availability [page 10]

Bandwidth [page 10]

Benchmarking [page 10]

Business Scenario [page 11]

Component [page 11]

Configuring [page 11]

Throughput [page 11]

Latency [page 11]

Response Time [page 12]

Scalability [page 12]

Sizing [page 12]

Tuning [page 12]

SAP Application [page 13]

SAPS [page 13]

2.6.1 Availability

The availability of a system represents the probability that this system will work as expected during a processing time, that thus characterizes its reliability. This probability is expressed using “nines” terms:

- **One-nine**, which means that the system is available 90% of the time, and can be unavailable 36,5 days/year
- **Two-nines**, which means that the system is available 99% of the time, and can be unavailable 3,65 days/year
- **Three-nines**, which means that the system is available 99,9% of the time, and can be unavailable 8,76 hours/year
- **Four-nines**, which means that the system is available 99,99% of the time, and can be unavailable 52,5 mins/year
- **Five-nines**, which means that the system is available 99,999% of the time, and can be unavailable 31,5 sec/year

2.6.2 Bandwidth

Bandwidth represents a measurement of bit-rate related to available or consumed digital resources, expressed in bits (or multiple of bits) per second (bit/s, kbit/s, Mbit/s, Gbit/s, and so on).

2.6.3 Benchmarking

Sizing information can be determined using SAP Standard Application Benchmarks. Released for technology partners, benchmarks provide basic sizing recommendations to customers by injecting a substantial load upon a system during the testing of new hardware, system software components or RDBMS[2]. All performance data relevant to the system, user, and business applications are monitored during a benchmark run and can be used to compare the benchmarked platforms.

For more information, refer to the following location: http://www.sap.com/benchmark.

[2] Relational Database Management System

2.6.4 Business Scenario

From a microeconomic perspective, a business scenario is a cycle which consists of several different interconnected logical processes in time. Typically, a business scenario includes several company departments and involves other business partners.

From a technical point of view, a business scenario needs at least one SAP application (SAP ERP, SAP SCM, or others) for each cycle and perhaps other third party systems. A business scenario is a unit which can be implemented separately and reflects the customer's prospective course of business.

2.6.5 Component

A component is the smallest individual unit considered within the Solution Development Lifecycle. Components are separately produced, delivered, installed, and maintained.

2.6.6 Configuring

Configuring a system consists in determining the relevant behavior for each element that is part of this system, in order to provide technical or business functionalities and thus fit specific requirements.

2.6.7 Throughput

Taking part of a system productivity evaluation, the throughput represents the maximum rate of digital resources that a system can process in a given time, that can be measured via benchmarking operations.

2.6.8 Latency

Taking part of a system productivity evaluation, the latency represents the time interval experienced by a system between the incoming request and the provided response, that:

- Characterizes the velocity of this system, and by extension its usability
- Limits the throughput

2.6.9 Response Time

Taking part of a system productivity evaluation, the response time represents the total amount of time that is taken to respond to an incoming request. This total amount is the sum of the following elements:

- **Service time**, that represents the time taken by the system to work on the request, whose value varies according to the workload
- **Wait time**, that represents the time the incoming request had to wait in a queue before being taken into account by the system for working purpose
- **Transmission time**, that represents the travelling time used to transfer the incoming request between the different concerned elements

2.6.10 Scalability

Scalability represents the ability of an element (system, network, process, and so on) to handle an increasing workload or the ability of this element to be enlarged in order to handle that increasing workload. 2 differents categories of scalability exist, that can also be combined together:

- **Horizontal** scalability, that consists in adding new systems to distribute the workload
- **Vertical** scalability, that consists in modifying the resources (CPU[3], memory, storage capacities, networking capacities, and so on) of the available systems

2.6.11 Sizing

Sizing a system consists in determining the hardware requirements for this system, such as:

- Network bandwidth
- Physical memory
- CPU[4] processing power
- I/O capacities

The size of both hardware and database is influenced by business and technological aspects, which means that the number of users and the related workload they generate must be taken into account.

2.6.12 Tuning

Tuning an application consists in optimizing the configuration of this application to ensure its scalability and performance when responding to an increasing workload.

[3] Central Processing Unit
[4] Central Processing Unit

2.6.13 SAP Application

An SAP application is an SAP software solution that serves a specific business area like SAP ERP, SAP CRM, SAP PLM, SAP SRM, SAP SCM, and SAP CC.

2.6.14 SAPS

The SAP Application Performance Standard (SAPS) is a hardware-independent unit that describes the performance of a system configuration in an SAP environment. It is derived from the Sales and Distribution (SD) Benchmark, where 100 SAPS is defined as the computing power to handle 2,000 fully business processed order line items per hour.

For more information, refer to the following location: http://www.sap.com/benchmark → Measuring in SAPS.

3 Configuring SAP Convergent Charging

Configuring the SAP CC User Interfaces [page 14]
Configuring the Core Server System [page 45]
Configuring the BART Server System [page 262]
Configuring the Diameter Server System [page 292]
Configuring the Import/Export Connector System [page 307]

3.1 Configuring the SAP CC User Interfaces

Configuring the Desktop Applications [page 14]
Configuring the Web Applications [page 18]
Configuring the Command-Line Applications [page 44]

3.1.1 Configuring the Desktop Applications

Keywords [page 14]
Preliminary Notes [page 15]
Description [page 15]
Configuring Core Tool [page 15]
Configuring BART Tool [page 17]
Configuring CAT Tool [page 18]

3.1.1.1 Keywords

GUI[5], graphical user interface, Core Tool, BART Tool, CAT Tool

[5] Graphical User Interface

3.1.1.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- Installing and Launching Core Tool section of the SAP CC 2021 Installation and Maintenance Guide documentation
- Installing and Launching BART Tool section of the SAP CC 2021 Installation and Maintenance Guide documentation
- Installing and Launching CAT Tool section of the SAP CC 2021 Installation and Maintenance Guide documentation

3.1.1.3 Description

All desktop applications of SAP Convergent Charging have a default configuration that relies on a configuration file that is automatically created (at first launch time) if this file does not exist. You can use the following procedures to modify the configuration file of the different graphical user interfaces of your SAP CC landscape and thus fit your specific needs:

- Configuring Core Tool [page 15]
- Configuring BART Tool [page 17]
- Configuring CAT Tool [page 18]

> **Note**
>
> - Configuration files are available in the `config` folder of each user interface
> - The settings defined in the configuration files do not relate to an SAP user, but are applied to all SAP users

3.1.1.4 Configuring Core Tool

The configuration of the Core Tool user interface can be modified to fit specific needs. The following table contains the list of available parameters that you can change:

Parameter	Type	Description	Default Value	Possible Values
`language`	string	Language used within the GUI[6]	EN	An ISO-639 code
`country`	string	The country specific language used within the GUI	US	An ISO-3166 code

[6] Graphical User Interface

Parameter	Type	Description	Default Value	Possible Values
`catalog.cache.retention_period`	integer	Maximum period (in seconds) of retention in a cache of some objects referenced in decision trees. The value 0 deactivates the cache. The recommended value for better user experience is 60.	0	Any positive or null integer value
`frame_width`	integer	The width of the window of the GUI	900	1 to 65536
`frame_height`	integer	The height of the window of the GUI	700	1 to 65536

`help.contents.url`
`help.components.url`
`help.portal.url`
	string	Address of the online helps accessible from the *Help* menu of the Core Tool, that can point to local files or to web address (such as SAP Help Portal)	The URL[7] of the SAP CC Core Tool user interface documentation on SAP Help Portal	A valid HTTP[8] or file URL pointing to online helps

`xml.verbose`	boolean	Verbose mode, used to trace communications between Core Tool and the dispatcher or updater instance of the Core Server system	false	true / false

To modify the configuration of the Core Tool user interface, execute the following procedure:

1. Ensure the concerned Core Tool is completely stopped
2. Edit the `config/core_tool.config` configuration file. If this file does not exist, create a copy from the `config/core_tool.config.sk` template file
3. Modify the value of the relevant parameters
4. Save your modifications
5. Restart the Core Tool

> ⚠ Caution
>
> - If a property is defined more than once in the file, only the last value is considered
> - If the imported file contains several definitions for a same property, only the last one is kept

[7] Uniform Resource Locator
[8] HyperText Transfer Protocol

3.1.1.5 Configuring BART Tool

The configuration of the BART Tool user interface can be modified to fit specific needs. The following table contains the list of available parameters that you can change:

Parameter	Type	Description	Default Value	Possible Values
`language`	string	Language used within the GUI[9]	EN	An ISO-639 code
`country`	string	The country specific language used within the GUI	US	An ISO-3166 code
`frame_width`	integer	The width of the window of the GUI	900	1 to 65536
`frame_height`	integer	The height of the window of the GUI	700	1 to 65536
`log/level`	integer	The level of the generated log messages	2	0 (debug) 1 (info) 2 (warning) 3 (error)
`BartHelpContentsURL` `BartHelpPortalURL`	boolean	Address of the online helps accessible from the *Help* menu of the BART Tooll, that can point to local files or to web address (such as SAP Help Portal)	The URL[10] of the SAP CC BART Tool user interface documentation on SAP Help Portal	A valid HTTP[11] or file URL pointing to online helps

To modify the configuration of the BART Tool user interface, execute the following procedure:

1. Ensure the concerned BART Tool is completely stopped
2. Edit the `config/bart_tool.config` `XML` configuration file. If this file does not exist, create a copy from the `config/bart_tool.config.sk` template file
3. Modify the value of the relevant parameters
4. Save your modifications
5. Restart the BART Tool

> ⚠ Caution
>
> - If a property is defined more than once in the file, only the last value is considered
> - If the imported file contains several definitions for a same property, only the last one is kept

[9] Graphical User Interface
[10] Uniform Resource Locator
[11] HyperText Transfer Protocol

3.1.1.6 Configuring CAT Tool

The configuration of the CAT Tool user interface can be modified to fit specific needs. The following table contains the list of available parameters that you can change:

Parameter	Type	Description	Default Value	Possible Values
`language`	string	Language used within the GUI[12]	EN	An ISO-639 code
`country`	string	The country specific language used within the GUI	US	An ISO-3166 code
`log/level`	string	The level of the generated log messages	inform	debug inform warning error
`cat.help.contents.url` `help.portal.url`	string	Address of the online helps accessible from the *Help* menu of the CAT Tooll, that can point to local files or to web address (such as SAP Help Portal)	The URL[13] of the SAP CC CAT Tool user interface documentation on SAP Help Portal	A valid HTTP[14] or file URL pointing to online helps

To modify the configuration of the CAT Tool user interface, execute the following procedure:

1. Ensure the concerned CAT Tool is completely stopped
2. Edit the `config/cat_tool.config` configuration file. If this file does not exist, create a copy from the `config/cat_tool.config.sk` template file
3. Modify the value of the relevant parameters
4. Save your modifications
5. Restart the CAT Tool

> ⚠ Caution
>
> - If a property is defined more than once in the file, only the last value is considered
> - If the imported file contains several definitions for a same property, only the last one is kept

3.1.2 Configuring the Web Applications

Keywords [page 19]

[12] Graphical User Interface
[13] Uniform Resource Locator
[14] HyperText Transfer Protocol

Preliminary Notes [page 19]

Description [page 19]

Configuring Cockpit [page 19]

3.1.2.1 Keywords

GUI[15], graphical user interface, user experience (UX), web application, Tomcat, Cockpit

3.1.2.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- Installing and Launching Cockpit section of the SAP CC 2021 Installation and Maintenance Guide documentation
- Enhanced Logging and Tracing Framework section of the SAP CC 2021 Application Help documentation

3.1.2.3 Description

SAP Convergent Charging provides a web application named Cockpit. With the **Convergent Charging Cockpit** user interface, SAP CC users quickly and efficiently perform your administrative or business configuration tasks. Depending on the user settings, the Cockpit launchpad contains several apps.

A Cockpit web application is connected to one Core Server system.

You can use the following procedures to modify the configuration file of the Cockpit web application and thus fit your specific needs.

3.1.2.4 Configuring Cockpit

The configuration of the Cockpit application relies on a text file named `sap_cc_cockpit_<NAME>.properties`, where `<NAME>` corresponds to the name of the deployed Cockpit web application. This configuration files includes settings for both technical and business features.

> **Note**
>
> - A template file named `sap_cc_cockpit_<WEB_APP_NAME>.properties.sk` is available in the delivered `ZIP` file of the SAP CC Cockpit software unit. You can use this template to create various configuration files (but SAP SE recommends that you do not modify the template itself).

[15] Graphical User Interface

- The Cockpit configuration file uses the ISO 8859-1 character encoding format. As a consequence, characters that cannot be directly represented in this encoding scheme must be written using unicode escapes. For further information, refer to the Java™ Language Specification.

Configuration Settings for Cockpit [page 20]

Modifying the Configuration File of the Cockpit Web Application [page 38]

As an SAP CC system administrator, you want to configure some features or mechanisms. In a production environment, you previously back up the configuration file of the Convergent Charging Cockpit web application (*.PROPERTIES file).

Configuring the App Process a Chargeable Item [page 38]

Java Virtual Machine (JVM) [page 41]

3.1.2.4.1 Configuration Settings for Cockpit

The following table contains the list of settings that you can change [page 38], grouped upon the following categories:

- Core Server communication settings [page 20]
- Core Database communication settings [page 21]
- Cockpit database communication settings [page 25]
- Cache management settings [page 27]
- Data files management settings [page 27]
- Session management and security settings [page 28]
- SAML 2.0 Settings (for identity providers) [page 29]
- OAuth2 settings (for third-party authorization servers) [page 31]
- User Management settings [page 32]
- Tracing settings [page 33]
- App Process a Chargeable Item settings [page 36]
- Web Assistant settings [page 37]

Setting	Values	Type	
Core Server Communication Settings			
`cc.sid` SID[16] of the Core Server system you want to administer. If you modify this SID while sessions are running, the SAP CC power users can continue their sessions and save their modifications using the previously defined Core Server system.	By default, no SID is set. The Web application can thus not start. → Remember This setting is hot modifiable when Cockpit is executed.		■

[16] SAP System Identifier

Setting	Values	Type	
`hci.api.delegator.login` Login used to be authenticated on the HCI API (login for the delegator user). **Note** This setting is necessary in case of SAML 2.0 authentication protocol.	The default value is set to: empty (user delegation disabled). You must set up the login of a valid technical user with the `AUTHORIZATION_DELEGATOR` role. This role allows a technical user to indirectly grant the usage of the Core Server HCI APIs from Cockpit to an individual Cockpit user. **Note** Refer to the Standard Users section of the SAP CC 2021 Security Guide documentation for more information.		☐
`hci.api.delegator.password` Password used to be authenticated on the HCI API (password for the delegator user).	You must set up a password that respects the SAP password policy applicable to SAP CC technical users. **Note** The password is encrypted by Cockpit using the existing passphrase mechanism.		☐
`hci.http.timeout` Maximum period (expressed in milliseconds) allocated to the execution of an HCI operation sent to Core Server	The default value is set to: 300000 You can set up any positive value. Setting this parameter to 0 disables the timeout mechanism.		■
`hci.http.url` CSV[17] list of URLs[18] relating to deployed dispatchers. In a secure environment, use URLs based on HTTPS[19].	By default, no URL is set. The Web application can thus not start. You must set up URLs which correspond to instances that belong to the targeted Core Server system declared in the `cc.sid` parameter. **Example** `http://localhost:9100/`		■
Core Database Communication Settings			

[17] Comma Separated Value
[18] Uniform Resource Locator
[19] HyperText Transfer Protocol Secured

Setting	Values	Type

`sqlhelper.jdbc.uri`
JDBC[20] URI of the reference database instance used by SAP CC Cockpit to start
	You must set up one or two valid URIs[21] which correspond to database instances.
	For SAP HANA, the URI format is:
	`jdbc:sap://<primary_server>:<port>[;<secondary_server>:<port>][/?databaseName=<tenant_name>]&currentSchema=<schema>`
	For Sybase ASE, the URI format is:
	`jdbc:sybase:Tds:<server>:<port>/<database_name>?<prop1>=<value1>&<prop2>=<value2>`
	For Oracle with "thin" driver, the URI format is:
	`- jdbc:oracle:thin:@//<server>:<port>/<ServiceName>`
	`- jdbc:oracle:thin:@<server>:<port>:<SID>`
	For MS SQL Server (IPv4), the URI format is:
	`jdbc:sqlserver://<server>:<port>;DatabaseName=<database_name>;sendStringParametersAsUnicode=true[;IntegratedSecurity=true]`
	For MS SQL Server (IPv4 or IPv6), the URI format is:
	`jdbc:sqlserver://;serverName=<server>;port=<port>;DatabaseName=<database_name>;sendStringParametersAsUni`
		■

[20] Java Database Connectivity
[21] Uniform Resource Identifier

Setting	Values	Type	

	`code=true[;IntegratedSecurity=true]`
	For IBM DB2, the URI format is:
	`jdbc:db2://<server>:<port>/<database_name>:currentSchema=<schema_name>;`		

`sqlhelper.login` Login used for authenticating on the Core Database	You must set up a value that corresponds to an already defined user. **Note** For security reasons, SAP SE recommends that you create a service user dedicated to connections between SAP CC Cockpit and the Core Database. This user shall be granted a read access to database views pointing on the `CERT_STORE`, `FILE_METADATA`, `FILE_METADATA_PATH`, `METRIC_RECORD`, `OBJECT_CHANGE_LOG`, and `USER_OPERATION` tables.		■
`sqlhelper.password` Password of the previously specified login, used for authenticating on the Core Database	You must set up a password that respects the SAP password policy.		■
`sqlhelper.connection.count` Number of connections to be opened with the Core Database	The default value is set to: **20** You can set up any positive value.		□
`sqlhelper.statement.cache.size` Size of the statement cache	The default value is set to: **0** You can set up any positive value. **Caution** SAP SE recommends that you do not modify this parameter except under supervision of the SAP Support team.		□

Setting	Values	Type	
`sqlhelper.id.interval` Size of the unique ID batch reservation	The default value is set to: **`10000`** You can set up any positive value greater than **`1000`**. ⚠ Caution SAP SE recommends that you do not modify this parameter except under supervision of the SAP Support team.		☐
`sqlhelper.max.redo` Number of times an instance will re-execute a request in the Core Database	The default value is set to: **4** You can set up any positive value.		☐
`sqlhelper.reconnect.interval` Duration (in seconds) between two reconnection attempts	The default value is set to: **10** You can set up any positive value.		☐
`sqlhelper.reconnect.attempts` Number of attempts when connecting to the Core Database	The default value is set to: **4** You can set up any positive value.		☐
`sqlhelper.db.instance.count` Number of SAP CC Core Database instances that the application can connect to	The default value is set to: **0** You can set up any positive value.		☐
`sqlhelper.db.timeout` Frequency (in seconds) of "ping" operations that are sent to the SAP CC Core Database	The default value is set to: **0** You can set up any positive value greater than 30. Setting this parameter to 0 disables the ping operations.		☐
`sqlhelper.connection.heartbeat` Frequency (in seconds) of "heartbeat" operations that are sent to the Core Database	The default value is set to: **0** You can set up any positive value greater than 30. Setting this parameter to 0 disables the ping operations.		☐

Setting	Values	Type	
`sqlhelper.jdbc.properties` Additional properties for the SQL[22] connections	You can set up a **CSV** formatted value as specified in RFC 4180, which means that: • Fields are separated by a comma delimiter and optionally enclosed by a text double-quote • Each field is the concatenation of a property name and a property value, using the following format: `property_name=<property_value>` **Note** The user login and password cannot be set in this parameter because there are defined by the `sqlhelper.login` and `sqlhelper.password` parameters.		☐
`query.limit.fetchSize` Maximum number of results per query	The default value is set to: **`10000`** You can set up any positive value greater than 10000		☐
Cockpit Database Communication Settings			

[22] Structured Query Language

Setting	Values	Type

`datasource.jdbc.uri`
The JDBC URI of the H2 embedded database instance used by SAP CC Cockpit to start
	You must set up a valid URI respecting the following format:
	For a file-based data storage, the URI format is:
	`jdbc:h2:file:<absolute_path_of_database_file>`
	For an in-memory data storage, the URI format is:
	`jdbc:h2:mem:<database_name>`
	Example
	• For a file-based data storage: `datasource.jdbc.uri=jdbc:h2:file:C:/Users/Admin/Documents/cockpit_database`
	• For an in-memory data storage: `datasource.jdbc.uri=jdbc:h2:mem:cockpit_database`
	Caution
	If this parameter is not set, the `jdbc:h2:mem:cockpit` in-memory data storage is used by default. Please note that all data stored within an in-memory data storage is lost in case of a SAP CC Cockpit restart.
		■

`datasource.login` The login used for authenticating on the Cockpit Database	You must set up a value that corresponds to an already defined user.	■
`datasource.password` The password used for authenticating on the Cockpit Database	You must set up a password that respects the SAP password policy.	■

Setting	Values	Type	
`datasource.pool.size` The size of the pool dedicated to the connections to the Cockpit Database	The default value is set to: **10** You can set up any positive value.		☐
`datasource.timeout` Maximum period (expressed in milliseconds) allocated to the execution of a query performed on the Cockpit Database	The default value is set to: **`30000`** You can set up any positive value. Setting this parameter to 0 disables the timeout mechanism. **Note** SAP SE recommends that you always set a value for this timeout to avoid any contention problem.		☐
`datasource.cache.size` The size of the cached structure (expressed in Kilobytes) used by the H2 embedded RDBMS to store the frequently used data	The default value is set to: **`131072`** You can set up any positive value. **Note** For further information about this cached structure, refer to the H2 database documentation.		☐
Cache Management			
`cache.type` Type of the cached structure containing queries	The default value is set to: **`simple`** You can set up one of the following values: **`none`**, **`simple`**		☐
`cache.simple.evict.scheduler` Duration (in seconds) between two reset operations performed on the cache of queries	The default value is set to: **`60`** You can set up any value in the following range: **]0,300[**		☐
Data Files Management			

Setting	Values	Type
`http.multipart.max-file-size` Maximum size (expressed in kilobytes) of the file to upload into the Cockpit application	The default value is set to: **`10240`** (about 10 megabytes) You can set up any positive value. Consult the SAP CC users (administrators, pricing specialists) to determine if they need to upload larger files into Cockpit, for example a large mapping table to import. If you increase this maximum size, consider increasing the `hci.http.timeout` setting because of longer upload times when importing larger files into Cockpit. **⚠ Caution** Modifying this value requires a modification of the memory resources (-Xmx option) allocated to the Java web server running a deployed Cockpit application. See Installing and Launching Cockpit. SAP SE thus recommends that you only change this value in case you encounter errors when importing files, for example a large mapping table.	■
Session Management and Security Settings		
`http.session.expiration.scheduler` Duration (in seconds) between two verifications performed on HTTP sessions in order to detect expired ones. When an HTTP session is expired, this session is deleted.	The default value is set to: **`60`** You can set up any positive value.	□

Setting	Values	Type	
`browser.sessionTimeout` Maximum period (expressed in minutes) of inactivity until which the session of an SAP CC power user expires. When this timeout is reached, the unsaved modifications are lost.	The default value is set to: **`120`** You can set up any positive value that corresponds to a period that must be longer than the lock timeout defined by the `browser.sessionLockTimeout` parameter. **→ Remember** This setting is hot modifiable when Cockpit is executed.		☐
`browser.sessionLockTimeout` Maximum period (expressed in minutes) during which a session can be left inactive (meaning that there is no interaction between the user and the web application). When this timeout is reached, the application locks the user session automatically and the user must log on again.	The default value is set to: **`10`** You can set up any period that must be shorter than the timeout defined by the `browser.sessionTimeout` parameter. Setting this parameter to 0 disables the locking mechanism. **→ Remember** This setting is hot modifiable when Cockpit is executed.		☐
SAML 2.0 Settings (for Identity Providers)			
`saml2.idp_entity_id` Identifier of the identity provider. **⚠ Caution** When OAuth and SAML 2.0 are both enabled then Cockpit does not start.	The default value is set to: empty (SAML 2.0 SSO disabled) You can define any valid identifier corresponding to the identity provider. **ⓘ Note** To enable the SAML 2.0 SSO authentication, user delegation [page 21] mechanism must be activated. **→ Remember** This setting is hot modifiable when Cockpit is executed.		☐

Setting	Values	Type
`saml2.idp_sso_service_url` URL to redirect the SAML request to (identity provider location).	The default value is set to: empty You can define any valid URL corresponding to the identity provider. **→ Remember** • This setting is hot modifiable when Cockpit is executed. • This setting is relevant only if `saml2.idp_entity_id` is set.	☐
`saml2.idp_certificate` File name of the file containing the X.509 public key certificate of the identity provider to validate SAML 2.0 assertions.	The default value is set to: empty The certificate must be **DER**-encoded and may be supplied in binary or Base64 encoding. **ⓘ Note** If the certificate is provided in Base64 encoding, it must start with `-----BEGIN CERTIFICATE-----` and must end with `-----END CERTIFICATE-----`. **→ Remember** • This setting is hot modifiable when Cockpit is executed. • This setting is relevant only if `saml2.idp_entity_id` is set.	☐

Setting	Values	Type	
`saml2.sessionMaxTimeout` Period (in minutes) starting after an SAML authentication at the end of which the user's session expires.	The default value is set to: **`720`** (12h) You can set up any positive value. **Note** This timeout is considered only with the SAML 2.0 SSO authentication and when no expiration date is set in the SAML assertion by the identity provider. **→ Remember** • This setting is hot modifiable when Cockpit is executed. • This setting is relevant only if `saml2.idp_entity_id` is set.		☐
OAuth2 Settings (for Third-Party Authorization Servers)			
`oauth2.authorization_server.url` URL of the authorization server. **Caution** When OAuth and SAML are both enabled then Cockpit does not start.	The default value is set to: empty (OAuth2 disabled) You can define any valid URL corresponding to the authorization server.		☐
`oauth2.authorization_server.client_id` Client identifier of the application provided by the authorization server.	You must define a valid client identifier known by the authorization server.		☐
`oauth2.authorization_server.client_secret` Client secret of the application defined in the authorization server.	You must define a valid client secret known by the authorization server.		☐

Setting	Values	Type	
`oauth2.redirect_uri.base_url` Reachable based URL used by the authorization server to redirect the authorized user. **Note** The `redirect_uri` must be the same than the `redirect_uri` registered in the authorization server (see OAuth 2.0 Authorization Framework).	The default value is set to: empty (computed by Cockpit) The URL format must be: `<baseUrl>` where `<baseUrl>` follows the syntax: `https://<IP_OR_DNS_ADDRESS>:<PORT_NB>/<WEB_APP_NAME>` **Example** `https://localhost:8088/cccockpit` **→ Tip** This setting must be performed when Cockpit cannot compute a reachable URI.		☐
`oauth2.authorization_server.revocation_url` URL used to notify the authorization server that a previously granted token is no longer needed. **Note** This URL is used when the discovery process fails.	The default value is set to: empty (no revocation) You can define any valid URL of token revocation endpoint from the authorization provider.		☐
User Management Settings			
`msg.api.login` Login used to be authenticated by the `Message Charging Client` API	You must set up a value that corresponds to an already defined user with the **Message Charging Client** role and the `Security Profile` set to **Service User**. **→ Remember** This setting is hot modifiable when Cockpit is executed.		☐

Setting	Values	Type	
`msg.api.password` Password used to be authenticated by the `Message Charging Client` API	You must set up a password that respects the SAP password policy. **→ Remember** This setting is hot modifiable when Cockpit is executed.		☐
`msg.api.timeout` Maximum period (expressed in milliseconds) allocated to the execution of a query performed by the `Message Charging Client` API.	The default value is set to: **`20000`** You can set up any positive value. Setting this parameter to 0 disables the timeout mechanism. **ⓘ Note** SAP SE recommends that you always set a value for this timeout to avoid any contention problem. **→ Remember** This setting is hot modifiable when Cockpit is executed.		☐
Tracing Settings			
`ls.app.severity` Severity threshold for the generation of log messages related to the business processing (application level)	The default value is set to: **INFO** You can set up one of the following values: **DEBUG, PATH, INFO, WARNING, ERROR** or **FATAL**		☐
`ls.app.destination` CSV list of output destinations for the log messages related to the application level	The default value is set to: **LDC,LDF1** Possible values are: **LDC, LDF1, LDF2, LDF3, LDF4** and **LDF5**		☐
`ls.sys.severity` Severity threshold for the generation of log messages related to the system processing (system level)	The default value is set to: **INFO** You can set up one of the following values: **DEBUG, PATH, INFO, WARNING, ERROR** or **FATAL**		☐
`ls.sys.destination` CSV list of output destinations for the log messages related to the system processing (system level)	The default value is set to: **LDC,LDF1** Possible values are: **LDC, LDF1, LDF2, LDF3, LDF4** and **LDF5**		☐

Setting	Values	Type	
`ls.trc.severity` Severity threshold for the generation of trace messages	The default value is set to: **ERROR** You can set up one of the following values: **DEBUG, PATH, INFO, WARNING, ERROR** or **FATAL** → Remember This setting is hot modifiable when Cockpit is executed.		☐
`ls.trc.domain` CSV list of trace domains that the generated messages relate to	The default value is set to: **SQL,NET,MSG,HB,AK,BLK,CACHE,LOCK,NOTIF,ADMIN,HCI,WS,WSBUS,CIF,RIF,SLDDS,JCO,PPI,PERF,DATA** → Remember This setting is hot modifiable when Cockpit is executed.		☐
`ls.trc.destination` CSV list of output destinations for the generated messages	Possible values are: **LDC, LDF1, LDF2, LDF3, LDF4** and **LDF5** → Remember This setting is hot modifiable when Cockpit is executed.		☐
`ld.console.enabled` Enables/disables the use of the console	The default value is set to: **false** Possible values are: **true** or **false** → Remember This setting is hot modifiable when Cockpit is executed.		☐
`ld.console.formatter` Output format of traces generated within the console	The default value is set to: **TRACE** Possible values are: **TRACE, LIST** or **XML** → Remember This setting is hot modifiable when Cockpit is executed.		☐

Setting	Values	Type	
`ld.fileX.enabled` Enables/disables the use of fileset destination #X (X from 1 to 5)	The default value is set to: **true** Possible values are: **true** or **false** → Remember This setting is hot modifiable when Cockpit is executed.		☐
`ld.fileX.name` Path, name and extension of trace files for fileset destination #X (X from 1 to 5)	The default value is set to: **@CATALINA_HOME@/logs/@WEBAPP_NAME@_LDF2.trc** You can set up any string, using the following variables: • **@CATALINA_HOME@**, that is replaced at runtime by the home folder of Catalina (Tomcat) • **@WEP_APP_NAME@**, that is replaced at runtime by the context path of the deployed application (in which possible URL separators are replaced by '_') → Remember This setting is hot modifiable when Cockpit is executed.		☐
`ld.fileX.size` Maximum size before rotation of the trace files for fileset destination #X (X from 1 to 5)	The default value is set to: **2M** You can set up any value respecting the `<value><unit>` format, where: • `<value>` is a mandatory positive integer • `<unit>` is an optional string representing the unit, that can be: **K** (kilo), **M** (mega) or **G** (giga) → Remember This setting is hot modifiable when Cockpit is executed.		☐

Setting	Values	Type	
`ld.fileX.setsize` Number of trace files before rotation for fileset destination #X (X from 1 to 5)	The default value is set to: **4** You can set up any positive value → Remember This setting is hot modifiable when Cockpit is executed.		☐
`ld.fileX.formatter` Output format of traces when using file-set destination #X (X from 1 to 5)	The default value is set to: **LIST** Possible values are: **TRACE, LIST** or **XML** → Remember This setting is hot modifiable when Cockpit is executed.		☐
App Process a Chargeable Item Settings			
`charging.catalog.sync.enabled` Enables or disables the app to force the pricing catalog synchronization in the rater and bulkloader instances to refresh the catalogs before the charging or simulation operations	The default value is set to: **true** (synchronization is forced) Possible values are: **true** or **false** → Tip If you change this value as a system administrator, your SAP CC users just have to press F5 or reopen the app in their Cockpit user interfaces. → Remember This setting is hot modifiable when Cockpit is executed.		☐

Setting	Values	Type
`charging.output.item.immediate.loading.enabled` Enables or disables the immediate loading of output items (billable/consumption items) generated by the charging process execution into SAP Convergent Invoicing for immediate billing. This only applies to the real execution mode of the charging process but not in estimated mode for simulations. **→ Remember** This setting is hot modifiable when Cockpit is executed.	The default value is set to: **false** (feature disabled) Possible values are: **true** or **false** **→ Tip** If you change this value as a system administrator, your SAP CC users just have to press F5 or reopen the app in their Cockpit user interfaces.	☐
Web Assistant Settings		
`webassistant.url` Enables or disables the web assistant feature.	The default value is set to: `https://<SAP_CLOUD_REGION>/web_assistant/framework/` Possible regions for `<SAP_CLOUD_REGION>` are defined by SAP Enable Now. **ⓘ Note** You can find a list of framework URLs by checking the documentation of SAP Enable Now. The regions and URLs are listed in the document named *Web Assistant Integration*, chapter *Regional Web Assistant Framework URLs*, under task *Integrate*. You can disable the web assistant feature by removing this setting or leaving it empty. **→ Remember** This setting is hot modifiable when Cockpit is executed.	☐

■ Mandatory ☐ Optional

3.1.2.4.2 Modifying the Configuration File of the Cockpit Web Application

As an SAP CC system administrator, you want to configure some features or mechanisms. In a production environment, you previously back up the configuration file of the Convergent Charging Cockpit web application (*.PROPERTIES file).

Modify the Configuration File

As a system administrator or an application administrator, to modify the configuration of the Cockpit web application, complete the following procedure:

1. On the Java web server (Tomcat for example), locate the configuration file named `sap_cc_cockpit_<WAR_FILE_NAME>.properties`, where `<WAR_FILE_NAME>` corresponds to the name of the deployed Cockpit web application.
   This file belongs to the deployed Cockpit web application that you want to reconfigure.
2. In a production SAP system landscape, back up this configuration file.
3. Edit this text file in a text editor.
4. Modify the value of the relevant settings [page 20].
   Some changes can require a restart of the Cockpit back-end application in its Java web server.
5. Save your modifications.
6. In a production SAP system landscape, back up the modified configuration file.

> ⚠ Caution
>
> - If a property is defined more than once in the file, only the last value is considered.
> - If the imported file contains several definitions for a same property, only the last one is kept.

3.1.2.4.3 Configuring the App Process a Chargeable Item

To configure the app Process a Chargeable Item, modify [page 38] the configuration file of the deployed Cockpit web application.

Enabling and Displaying the App Process a Chargeable Item [page 39]
: Before pricing specialists can use the their app Process a Chargeable Item in the Cockpit user interface, a user administrator must create a dedicated and specialized SAP CC user as a service user.

Enabling the Billable Items Immediate Loading Feature in the App Process a Chargeable Item [page 40]

Disabling the Automatic Synchronization of Pricing Catalogs in Rater and Bulkloader Instances [page 41]

Related Information

Process a Chargeable Item

3.1.2.4.3.1 Enabling and Displaying the App Process a Chargeable Item

Before pricing specialists can use the their app Process a Chargeable Item in the Cockpit user interface, a user administrator must create a dedicated and specialized SAP CC user as a service user.

Enable and Display the App

The charging process in the Core Server system requires an SAP CC user with the **Message Charging Client** role.

Therefore, a service user for the Cockpit must be created, granted with the Message Charging Client role:

1. As an SAP CC user administrator, refer to the Users section of the SAP CC 2021 Primary Help for Core Tool to:
   - Create a new service user granted with the Message Charging Client role.
   - Check if this user needs credentials according to the settings of the MESSAGE_API_AUTHENTICATION_REQUIRED system parameter.
2. As an SAP CC system administrator, create a backup of the `sap_cc_cockpit_<NAME>.properties` configuration file of the Cockpit (where `<NAME>` corresponds to the name of the deployed Cockpit web application).
3. Add the 4 following settings to the configuration file:
   - `msg.api.login`
   - `msg.api.password`
   - `msg.api.timeout`

   of the newly created technical user in the initial configuration file of the Cockpit user interface.
4. Save your modifications.
5. Inform your SAP CC users that they can reopen their app Process a Chargeable Item or refresh their web browsers.

Once done, any individual user of the Cockpit user interface with the **Marketing** role can use the app Process a Chargeable Item.

Related Information

Process a Chargeable Item

3.1.2.4.3.2 Enabling the Billable Items Immediate Loading Feature in the App Process a Chargeable Item

Before pricing specialists can use the function in their app Process a Chargeable Item, an SAP CC system administrator must enable the feature in the installed Cockpit web application.

When several Cockpit user interfaces are deployed and installed in your integrated SAP CC system landscape, you can choose which user interface is able to load the billable items into SAP Convergent Invoicing.

Prerequisites

- The relevant Core Server system of SAP Convergent Charging is already configured for billable items immediate loading (See Enabling the Billable Items Immediate Loading [page 75]). The deployed Cockpit web application connects to this Core Server system.

Enable the Feature in the App

As an SAP CC system administrator:

1. Modify [page 38] the configuration file of the deployed Cockpit web application.
2. Set up the `charging.output.item.immediate.loading.enabled` setting to **`true`**.
3. Save your modifications.
4. In a production SAP system landscape, back up the modified configuration file.
5. Inform your SAP CC users that they can reopen their app Process a Chargeable Item or refresh their web browsers.

Related Information

Enabling the Billable Items Immediate Loading [page 75]

Convergent Charging Cockpit

Process a Chargeable Item

Editing and Processing a Chargeable Item

3.1.2.4.3.3 Disabling the Automatic Synchronization of Pricing Catalogs in Rater and Bulkloader Instances

Disable the Feature in the App

As an SAP CC system administrator:

1. Modify [page 38] the configuration file.
2. Set up the `charging.catalog.sync.enabled` setting to **`false`**.
3. Inform your SAP CC users that they can reopen their apps or refresh their web browsers.

3.1.2.4.4 Java Virtual Machine (JVM)

The architecture of SAP Convergent Charging relies on a set of Java-based applications whose execution require a Java Runtime Environment (JRE). The Cockpit back-end application runs in a Java process whose execution can be configured to fit specific requirements.

In the environment of the Cockpit back-end application, the JVM options are configured in the settings for the Java Web Server system. Cockpit is deployed on this system and this system runs Cockpit as a webapp.

The SAP installer tool for SAP CC has not set these values.

As an IT administrator, you configure and test the JVM options.

1. Configure the JVM heap: You can manage the initial size of the heap memory (`Xms`) and maximum size of the heap memory (`Xmx`). By default, 1 GB are set.
2. Configure other settings.
3. Tune some settings: See Tuning the Cockpit Webapp and Apache Tomcat Server To Handle Large Files.

> → Remember
>
> If the SAP CC Cockpit user interface (web application) needs to work with large files and big data, then you need to adapt the system. See Tuning the Cockpit Webapp and Apache Tomcat Server To Handle Large Files.
>
> For example, the pricing specialists who configure some pricing catalogs for the chargeable services, can configure data tables with more than 100,000 rows.

References

Configuration Options [page 42]

Procedures

Modifying JVM Options [page 42]

Tuning the Cockpit Webapp and Apache Tomcat Server To Handle Large Files

Configuring the JVM Heap [page 43]

Related Information

Tuning the Cockpit Webapp and Apache Tomcat Server To Handle Large Files

3.1.2.4.4.1 Configuration Options

Available configuration options depend on the installed JVM and Java Web Server system that runs the Cockpit back-end application. Refer to the official product documentation.

See Configuration Options [page 158] as a reference for SAP CC Core Server.

Verify that these JVM options are valid for your Java Web Server.

3.1.2.4.4.2 Modifying JVM Options

If you need to reconfigure these JVM options, then change the configuration of the Java web server that runs SAP CC Cockpit: the Apache Tomcat Server system.

Refer to the official product documentation.

Modifying JVM Options of Apache Tomcat Server Running SAP CC Cockpit

Linux Operating Systems

To configure the JVM options:

1. Create or modify a `setenv.sh` file in the `<APACHE_TOMCAT>`/bin directory.
   Make a backup copy if it already exists.
2. Add the following JVM options to your `setenv.sh` file:

```
export CATALINA_OPTS="$CATALINA_OPTS <JVM_OPTION_1>"
export CATALINA_OPTS="$CATALINA_OPTS <JVM_OPTION_2>"
```

3. In the <APACHE_TOMCAT>/bin directory, run the command:

```
./catalina.sh run
```

4. Monitor the log file and check that the Apache Tomcat Server system started with the JVM options specified in the modified or new `setenv.sh` file.
5. Verify the configurations using the following command:

```
ps -ef | grep java
```

Microsoft Windows Operating Systems

To configure the JVM options:

1. Create or modify a `setenv.bat` file in the <APACHE_TOMCAT>\bin directory. Make a backup copy if it already exists.
2. Add the following JVM options to your `setenv.bat` file:

```
set "JAVA_OPTS=%JAVA_OPTS% <JVM_OPTION_1> <JVM_OPTION_2>"
```

3. In the <APACHE_TOMCAT>\bin directory, run the command:

```
catalina.bat
```

4. Monitor the log file and check that the Apache Tomcat Server system started with the JVM options specified in the modified or new `setenv.bat` file.

See Also

Configuring the JVM Heap [page 43]

3.1.2.4.4.3 Configuring the JVM Heap

Linux Operating System

To configure the JVM Heap:

1. Create or modify a `setenv.sh` file in the <APACHE_TOMCAT>/bin directory. Make a backup copy.
2. Add the following `-Xmx` and `-Xms` JVM options to your `setenv.sh` file:

```
export CATALINA_OPTS="$CATALINA_OPTS -Xms1G"
export CATALINA_OPTS="$CATALINA_OPTS -Xmx1G"
```

3. In the <APACHE_TOMCAT>/bin directory, run the command:

```
./catalina.sh run
```

4. Monitor the log file and check that the Apache Tomcat Server system started with the JVM options specified in the modified or new `setenv.sh` file.
5. Verify the configurations using the following command:

```
ps -ef | grep java
```

Microsoft Windows Operating System

To configure the JVM heap:

1. Create or modify a `setenv.bat` file in the `<APACHE_TOMCAT>\bin` directory. Make a backup copy.
2. Add the following -Xmx and -Xms JVM options to your `setenv.bat` file:

```
set "JAVA_OPTS=%JAVA_OPTS% -Xms1G -Xmx1G"
```

3. In the `<APACHE_TOMCAT>\bin` directory, run the command:

```
catalina.bat
```

4. Monitor the log file and check that the Apache Tomcat Server system started with the JVM options specified in the modified or new `setenv.bat` file.

3.1.3 Configuring the Command-Line Applications

Keywords [page 44]

Preliminary Notes [page 44]

Description [page 45]

Configuring Setup Tool [page 45]

Configuring Config Tool [page 45]

3.1.3.1 Keywords

UI, user interface, Setup Tool, Config Tool

3.1.3.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- Installing and Launching Setup Tool section of the SAP CC 2021 Installation and Maintenance Guide documentation
- Installing and Launching Config Tool section of the SAP CC 2021 Installation and Maintenance Guide documentation
- Installing and Launching CAT Tool section of the SAP CC 2021 Installation and Maintenance Guide documentation

3.1.3.3 Description

Some command-line applications of SAP Convergent Charging rely on a configuration file that needs to be modified to fit the configuration of your landscape.

You can use the following procedures to modify these configuration files and thus fit your specific needs:

- Configuring Setup Tool [page 45]
- Configuring Config Tool [page 45]

3.1.3.4 Configuring Setup Tool

The Setup Tool user interface directly connects to the back-end database, using connection parameters that are defined in the `boot.config` file available in the `config` folder of dispatcher instances.

For further information about the available parameters, refer to the Boot Configuration File [page 53] dedicated section afterwards.

3.1.3.5 Configuring Config Tool

The Config Tool user interface directly connects to the back-end database, using connection parameters that are defined in the `boot.config` file available in the `config` folder of dispatcher instances.

For further information about the available parameters, refer to the Boot Configuration File [page 53] dedicated section afterwards.

3.2 Configuring the Core Server System

> ⚠ Caution
>
> In case of an integrated *SAP Solution* scenario based on SAP S/4HANA Cloud Public Edition, all master data related to prepaid accounts are no longer available within SAP Convergent Charging (SAP CC).

In such a situation, some of the procedures described afterwards and dealing with such master data might be adapted.

As described in the SAP CC 2021 Application Help documentation, the Core Server system provides you with business features and technical mechanisms of which behavior can be configured to fit your specific business/technical requirements.

The behavior of these business features and technical mechanisms relies on system parameters and can be modified using different user interfaces. According to your needs, you may need to perform:

- **Immediate** (only for a predefined subset of system parameters) or **deferred** modifications (a restart of the instances is necessary)
- **Unitary** (a single parameter) or **mass** modifications (several parameters at the same time)

The following table summarizes the different possibilities in terms of modifications that can be performed on these business features and technical mechanisms:

			Immediate		Deferred	
Tool	**Type**	**Description**	**Unitary**	**Mass**	**Unitary**	**Mass**
Manage SAP CC System Parameters (in Convergent Charging Cockpit)	Web Application	The app Manage SAP CC System Parameters gives the possibility to search for and view one or multiple system parameter values for a targeted Core Server system. This performing web application allows you to inspect and change both persistent and runtime values quickly and efficiently.	■	■	■	■
Admin+	Console Application	Admin+ is a console application that gives the possibility to connect to local or remote instances of the Core Server system and modify their configuration parameters. This console application runs in interactive mode via commands/arguments and gives you results or prompts for new options.	■		■	
Config Tool	Command-Line User Interface	Config Tool is a command-line user interface that gives the possibility to modify some configuration parameters directly in the Core Database. This user interface provides you import and export commands to manage the configuration of the Core Server system.			■	■

→ Tip

Using Setup Tool

The Setup Tool is a command-line user interface (CLI) that gives the possibility to setup the Core Server system by directly modifying its back-end database (Core Database). This user interface provides you different commands to manage the business features and technical mechanisms of the Core Server system. You can also use Setup Tool to import or export configuration files containing system parameters.

Immediate Modifications [page 47]

When needed, some modifications can be immediate in the system and do not require a restart of an instance, a group of instances, or the complete system. Some features, functions, and mechanisms provide you with the immediate change capability.

Deferred Modifications [page 49]
Allowance Management [page 52]
Boot Configuration File [page 53]
Business Data Management [page 57]
Catalog Transport [page 70]
Chargeable Items Charging Process [page 72]
Chargeable Items Recharging Process [page 76]
Communications Channels [page 79]
Data Cache Management [page 82]
Data Files Generation (Item Files) [page 97]
Data Purging [page 153]
Java Virtual Machine [page 157]
JCo Destinations [page 161]
Logging and Tracing [page 167]
Master Data Management [page 177]
Notifications [page 206]
Security Certificates [page 219]
Startup Configuration Files [page 234]
System and Data Auditing [page 237]
Thread Dump Files Generation [page 242]
Third-Party Integration [page 244]
Users and User Management [page 245]
WS Destinations [page 260]

3.2.1 Immediate Modifications

When needed, some modifications can be immediate in the system and do not require a restart of an instance, a group of instances, or the complete system. Some features, functions, and mechanisms provide you with the immediate change capability.

Immediate changes are immediately taken into account.

Depending on the modified setting, some delay may be necessary before the change is effective.

Only few settings are candidate for immediate modifications.

Immediate modifications are temporary. As soon as an SAP CC system instance restarts, it automatically uses the persistent value that is defined for the instances of the same type. If you want to apply permanent modification, then mix immediate modifications and deferred modifications. See Deferred Modifications [page 49].

→ Remember

When you want to change a few settings and when one of the settings cannot be modified without a restart, then consider a procedure based on deferred modifications. See Deferred Modifications [page 49].

→ Tip

Consult the SAP CC documentation to see if your given modification can be immediate or requires a partial or complete restart of the instances composing your SAP system.

Example

A typical use is to change the log or trace generation settings for an instance in the system.

Immediate Modification with Cockpit and Its Dedicated App [page 48]

Immediate Modification with Admin+ [page 49]

3.2.1.1 Immediate Modification with Cockpit and Its Dedicated App

To perform an immediate modification of a given setting (a system parameter) related to a business or technical feature and a given function or mechanism, complete the following procedure:

1. Launch the Convergent Charging Cockpit user interface and log on to the distant Core Server system as the system administrator.
2. You know the system parameter or subset of parameters to change. You know the target instances of the modification.
3. Open the app Manage SAP CC System Parameters.
4. Modify the runtime values of the system parameter using the app.
   See Configuring Parameter Values.
5. Check your system changes.
   Various methods are possible. It depends on the changed function or mechanism.

Related Information

Cockpit
Manage SAP CC System Parameters

3.2.1.2 Immediate Modification with Admin+

> **Note**
>
> Sometimes, you must apply a predefined sequence of changes. For example, if you want to modify the logging and tracing policy. See Immediate Modifications Using Admin+ [page 172].

To perform an immediate modification of a given setting (a system parameter) related to a business or technical feature and a given function or mechanism, complete the following procedure:

1. Launch the Admin+ user interface and log on to the distant Core Server system as the system administrator.
2. You know the system parameter or subset of parameters to change. You know the target instances of the modification.
3. Modify the runtime values of the system parameter using the `set` command with the `memory` scope.

   > **Example**
   >
   > See a few examples [page 172] in this documentation.

4. Check your system changes.
   Various methods are possible. It depends on the changed function or mechanism.

Related Information

Admin+

3.2.2 Deferred Modifications

In the environment of SAP Convergent Charging, you configure or reconfigure your running SAP system (Core Server). Several methods are possible.

Deferred Modification with Cockpit and Its Dedicated App [page 50]

Deferred Modification with Admin+ [page 50]

Deferred Modification with Config Tool [page 51]

Deferred Modification with Setup Tool [page 51]

Deferred Modification Using the Startup Configuration Files [page 51]

3.2.2.1 Deferred Modification with Cockpit and Its Dedicated App

To perform a deferred modification of a given setting (a system parameter) related to a business or technical feature and a given function or mechanism, complete the following procedure:

1. Launch the Convergent Charging Cockpit user interface and log on to the distant Core Server system as the system administrator.
2. You know the system parameter or subset of parameters to change. You know the target instances of the modification. In case of a deferred modification, you need to change the persistent value of a system parameter. This change applies to all instances of the same type.
3. Open the app Manage SAP CC System Parameters.
4. Modify the persistent value of the system parameter using the app.
   See Configuring Parameter Values.
5. Depending on your changes, restart the instances that are impacted by the modifications.
6. Check your system changes.
   Various methods are possible. It depends on the changed function or mechanism.

Related Information

Cockpit
Manage SAP CC System Parameters

3.2.2.2 Deferred Modification with Admin+

To perform a deferred modification of a given setting (a system parameter) related to a business or technical feature and a given function or mechanism, complete the following procedure:

1. Launch the Admin+ user interface and log on to the distant Core Server system as the system administrator.
2. You know the system parameter or subset of parameters to change. You know the target instances of the modification.
3.
4. Depending on your changes, restart the instances that are impacted by the modifications.
5. Check your system changes.
   Various methods are possible. It depends on the changed function or mechanism.

Related Information

Admin+

3.2.2.3 Deferred Modification with Config Tool

Config Tool is a command-line interface (CLI).

To perform a deferred modification of a given setting (a system parameter) related to a business or technical feature and a given function or mechanism, complete the following procedure:

For more information, refer to the Config Tool documentation or in-app help.

Related Information

Config Tool

3.2.2.4 Deferred Modification with Setup Tool

To perform a deferred modification of a given setting (a configuration set or data) related to a business or technical feature and a given function or mechanism, complete the following procedure:

For more information, refer to the Setup Tool documentation or in-app help.

Related Information

Setup Tool

3.2.2.5 Deferred Modification Using the Startup Configuration Files

To perform a deferred modification of a given setting (JVM options for example) related to a business or technical feature and a given function or mechanism, complete the following procedure:

See Modifying a Startup Configuration File [page 236].

Related Information

Configuration Options [page 236]

3.2.3 Allowance Management

Keywords [page 52]

Preliminary Notes [page 52]

Description [page 52]

Allowances Management with SAP CRM [page 52]

3.2.3.1 Keywords

Allowance, SAP CRM, integration

3.2.3.2 Preliminary Notes

For further information, refer to the Allowance Management feature description of the SAP CC 2021 Application Help documentation.

3.2.3.3 Description

As described in the SAP CC 2021 Application Help documentation, SAP CC gives the possibility to manage service add-ons for your charge plans and refill plans by modeling:

- The master data that define the allowances
- The interaction between allowances and charging or refilling logic, which define the usage of allowances

In addition to the allowances management capabilities provided by the Core Tool user interface, you can use the following procedure to configure allowances when SAP CC is integrated with SAP CRM.

3.2.3.4 Allowances Management with SAP CRM

If your landscape is integrated with SAP CRM, allowances can be created on demand or automatically by SAP CRM each time a provider contract is activated or modified.

To activate such functionality, you must configure SAP CC as described in the Integration Guide for Offer-to-Cash Based on BRIM Using SAP ERP.

For further information, refer to the “Allowances” section of this dedicated documentation available within the BRIM solution: help.sap.com/brim.

3.2.4 Boot Configuration File

Keywords [page 53]

Preliminary Notes [page 53]

Description [page 53]

Configuration Options [page 53]

Modifying the Boot Configuration File of an Instance [page 55]

3.2.4.1 Keywords

`boot.config`, SAP Profile Folder, database connection

3.2.4.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- SAP Central Repository section of the SAP CC 2021 Installation and Maintenance Guide
- Choosing System IDs section of the SAP CC 2021 Installation and Maintenance Guide

3.2.4.3 Description

For each instance of the Core Server system, a boot configuration file is defined in the SAP Profile Folder of the SAP Central Repository. Copied within the `config` folder of each deployed instance of the Core Server system during the instance startup phase, this `boot.config` file contains some parameters that are used to connect to the back-end database.

3.2.4.4 Configuration Options

The following table contains the list of parameters that you can change, grouped upon the following categories:

- Core Database communication settings
- Core Server communication settings

Parameter	Values	Type
Core Database Communication Settings		

SQLHELPER_JDBC_URI	JDBC URI of the Core database instance.
	For SAP HANA, the URI format is:
	`jdbc:sap://<primary_server>:<port>[;<secondary_server>:<port>][/?databaseName=<tenant_name>]`
	For Sybase ASE, the URI format is:
	`jdbc:sybase:Tds:<server>:<port>/<database_name>?<prop1>=<value1>&<prop2>=<value2>`
	For Oracle with "thin" driver, the URI format is:
	`- jdbc:oracle:thin:@//<server>:<port>/<ServiceName>`
	`- jdbc:oracle:thin:@<server>:<port>:<SID>`
	For MS SQL Server (IPv4), the URI format is:
	`jdbc:sqlserver://<server>:<port>;DatabaseName=<database_name>;sendStringParametersAsUnicode=true[;IntegratedSecurity=true]`
	For MS SQL Server (IPv4 or IPv6), the URI format is:
	`jdbc:sqlserver://;serverName=<server>;port=<port>;DatabaseName=<database_name>;sendStringParametersAsUnicode=true[;IntegratedSecurity=true]`
	For IBM DB2, the URI format is:
	`jdbc:db2://<server>:<port>/<database_name>:<prop1>=<value1>;<prop2>=<value2>`
		■

SQLHELPER_LOGIN	Login used for authenticating on the Core Database, that must correspond to an already defined user.	■
SQLHELPER_PASSWORD	Password of the previously specified login, used for authenticating on the Core Database.	■
Core Server Communication Settings		
SYSTEM_ID	SID of the Core Server system. You must set a value that corresponds to the system identifier used to at installation time. For further information, refer to the dedicated Choosing System IDs section available in the SAP CC 2021 Installation and Maintenance Guide documentation.	■
DISCOVERY_MULTICAST_ADDRESS	Multicast IP address (IPv4 or IPv6) used for UDP[23] discovery in case of database connection failure.	□
DISCOVERY_PORT	Port used for UDP discovery in case of database connection failure.	□

[23] User Datagram Protocol

Parameter	Values	Type
BOOT_DISPATCHER_LIST	CSV[24] list of IP addresses (or hostnames) that correspond to dispatcher instances that can be contacted in case of discovery failure.	☐

■ Mandatory ☐ Optional

3.2.4.5 Modifying the Boot Configuration File of an Instance

To modify the boot configuration file, execute the following procedure:

1. Edit the `boot.config` file, that is located in the SAP Profile Folder of the SAP Central Repository
2. Modify the relevant settings according to your needs
3. Save your modifications
4. To apply your modifications, restart the concerned instance of the Core Server system. During the startup phase, the modified boot.config file will be copied from the SAP Profile Folder to the **local** `config` folder of the instance (overwriting the existing `boot.config` file)

> ⓘ Note
>
> SAP SE recommends that you check the content of the `boot.config` file located in the **local** `config` folder of the instance, to ensure that your modifications have correctly been reflected.

> ⚠ Caution
>
> - If a property is defined more than once in the file, only the last value is considered
> - If the imported file contains several definitions for a same property, only the last one is kept

3.2.4.5.1 Modifying the Login and Password for the Database

For development or test purpose, you can change the login and password of the back-end database after installing SAP CC 2021. The SAP CC system handles this information in:

- The central boot configuration file
- The local boot configuration files
- Some system parameters stored in the back-end database

In a production system landscape, you must avoid using this procedure.

[24] Comma Separated Value

> ⚠ Caution
>
> Before doing the changes, you must stop all the instances of the Core Server system of SAP Convergent Charging.

Prerequisites

- The SAP CC Core Server system must be down when resetting the database login and password
- The new login and password are set in the database system

Procedure

1. Stop all the instances of the SAP CC Core Server system
2. Identify the installation folder of the primary dispatcher instance
3. Edit its local `boot.config` file in:
   - `<DRIVE>:\usr\sap\<SID>\CCD<SAP_INST_NB>\config\boot.config` (Microsoft Windows)
   - `/usr/sap\<SID>/CCD<SAP_INST_NB>/config/boot.config` (Linux and UNIX operating systems)
4. Change the entries related to the database credentials (SQLHELPER_LOGIN, SQLHELPER_PASSWORD) to the new login and password. The database password is not cyphered
5. Save and close the file
6. Use the Config Tool user interface to export all the system parameters to a temporary text file
7. Edit the exported file and change all the values of the parameters SQLHELPER_LOGIN and SQLHELPER_PASSWORD to the new database credentials. Set up the values with the plain text password
8. Import the file with Config Tool. This import operation cyphers both the values of the SQLHELPER_PASSWORD parameters and the value of the SQLHELPER_PASSWORD entry in the local boot configuration file
9. Delete the parameter export file after importing it
10. Open again the local `boot.config` file previously modified
11. Copy the password SQLHELPER_PASSWORD that is now cyphered
12. Edit and modify the reference `boot.config` file in the SAP Central Repository located in `/usr/sap/<SID>/SYS/profile/boot/`
13. Change the values to the new database login and new cyphered database password (see the SQLHELPER_LOGIN and SQLHELPER_PASSWORD entries)
14. Restart the instances of the Core Server system

Each instance of the Core Server system is started and establishes its permanent connection to the back-end database using the new database credentials (login and password). All the local boot configuration files are updated.

3.2.4.5.2 Modifying the Ports of the Core Server System

For development or test purpose, you can change the communication ports of the SAP CC Core Server system after installation.

Procedure

The Core Server instance map needs to be modified:

1. Stop all the instances of the SAP CC Core Server system
2. Export the Core Server system instance map, which path is:
   - `<DRIVE>:\usr\sap\<CORE_SYSTEM_ID>\CCD<SAP_INST_NB>\script\` (Microsoft Windows)
   - `/usr/sap\<CORE_SYSTEM_ID>/CCD<SAP_INST_NB>/script/` (Linux and UNIX operating systems)

   where `<SAP_INST_NB>` is the number assigned to the dispatcher, by executing the following Setup Tool command:

   ```
   setup instancemap export <pathToFile>
   ```
3. In this exported file, modify the ports with values in the range of port numbers from 1024 to 49151 and save and close the file.

   > **Note**
   >
   > Refer to the SAP Note 1673818.
4. Import this newly created file by executing the following Setup Tool command:

   ```
   setup instancemap import <pathToModifiedFile> -clean
   ```
5. Restart the instances of the Core Server system

3.2.5 Business Data Management

Keywords [page 58]

Preliminary Notes [page 58]

Description [page 58]

Managing Public Holidays [page 59]

Managing the Currencies Framework [page 62]

Managing Customer Management Areas [page 65]

Managing the Taxation Framework [page 65]

3.2.5.1 Keywords

Public holidays, currency, currencies, tax, taxation, EZTax, VAT[25], rate, rule

3.2.5.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- Public Holidays section of the Core Tool online help available in the SAP CC 2021 Primary Help for Core Tool documentation
- Tax Calculation feature description available in the SAP CC 2021 Application Help documentation
- Tax Calculation Process section of the SAP CC 2021 Application Help documentation
- Installing and Launching Core Tool, Installing and Launching Setup Tool and Installing and Launching Admin+ sections of the SAP CC 2021 Installation and Maintenance Guide documentation

3.2.5.3 Description

In addition to master data, your business also relies on business data that need to be configured. You can use the following procedure to manage the relevant business data and thus fit your specific needs:

- Managing Public Holidays [page 59]

As described in the SAP CC 2021 Application Help documentation, charging and refilling operations create transactions that need to be enriched with taxation information computed by SAP CC in accordance with the defined taxation settings.

You can use the following procedures to configure the settings related to currencies and taxation and thus fit your specific business needs:

- Managing the Currencies Framework [page 62], including:
  - Modifying ISO Currencies [page 63]
  - Synchronizing SAP Currencies [page 63]
  - Replicating SAP Currencies [page 64]
  - Reforming Currencies [page 64]
- Managing the Taxation Framework [page 65], including:
  - Enabling the VAT Framework [page 66]
  - Enabling the EZTax Framework [page 66]
  - Modifying VAT Rates [page 66]
  - Modifying VAT Rules [page 67]
  - Customizing VAT Rules [page 68]

[25] Value Added Tax

3.2.5.4 Managing Public Holidays

Public holidays represent business data that :

- Can be used within the Public Holidays Splitter and Public Holidays Comparator logic components
- Are part of the catalog cache (also called "offer-level" cache) handled by raters and updaters
- Are used at runtime by raters and updaters during charging and refilling operations, when processing the pricing logic configured in price plans, charging plans or refill logic
- Are defined within a dedicated file using an `XML` format defined in the following table:

Element	Sub element	Attribute	Description
holiday	description	code	Attribute used to identify the public holiday, that must be unique among the defined holidays
		lang	Language used for the description
		label	Description of the holiday in the specified language
	group	code	Name of the group of public holidays the currently defined holiday belongs to
	fixedDay	month	Name of the month. Possible values are: "January", "February", "March", "April", "May", "June", "July", "August", "September", "October", "November" or "December".
		day	Number of the day in the specified month. Possible values are integer values from 1 to 28, 29, 30 or 31 according to the concerned month.
	relativeDay	month	Name of the month. Possible values are: "January", "February", "March", "April", "May", "June", "July", "August", "September", "October", "November" or "December".
		weekday	Day of the week. Possible values are: "Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday" or "Sunday".
		position	Position of the weekday in the specified month. Possible values are: "First", "Second", "Third", "Fourth" or "Fifth".
	easterRelative	offset	Integer value use to define when the holiday occurs, relatively to Easter Sunday (which is used as the reference date).
	enumeratedDay		List of specified dates. Refer below in this table for further information about each enumerated date.

Element	Sub element	Attribute	Description
enumeratedDay	date	value	Concerned date, using the YYYY-MM-DD format, where: • YYYY corresponds to the year of the date • MM corresponds to the month of the date (from 1 to 12) • DD corresponds to the day of the date (from 1 to 31)

Example

Fixed Day

The following XML file defines the New Year's Day holiday. This public holiday:

- Is named "New Year's Day" in English, and "Nouvel An" in French
- Is available in 2 groups named "US" (United States) and "FR" (France)
- Occurs on the first day of January

```
<?xml version="1.0" encoding="UTF-8"?>
<holiday code="NewYearsDay">
  <description lang="en" label="New Year's Day" />
  <description lang="fr" label="Nouvel An" />
  <group code="US" />
  <group code="FR" />
  <fixedDay month="January" day="1" />
</holiday>
```

Example

Relative Day

The following XML file defines the Thanksgiving holiday. This public holiday:

- Is named "Thanksgiving" in both English and French
- Is only available in the "US" (United States) group
- Occurs on the fourth Thursday of November in the United States

```
<?xml version="1.0" encoding="UTF-8"?>
<holiday code="Thanksgiving">
  <description lang="en" label="Thanksgiving" />
  <description lang="fr" label="Thanksgiving" />
  <group code="US" />
  <relativeDay month="November" weekday="Thursday" position="fourth" />
</holiday>
```

Example

Easter-Relative Day

The following XML file defines the Easter Monday holiday. This public holiday:

- Is named "Easter Monday" in English, and "Lundi de Pâques" in French
- Is available in 2 groups named "US" (United States) and "FR" (France)
- Occurs one day after Easter Sunday

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<holiday code="EasterMonday">
  <description lang="en" label="Easter Monday" />
  <description lang="fr" label="Lundi de Pâques" />
  <group code="US" />
  <group code="FR" />
  <easterRelativeDay offset="1" />
</holiday>
```

Example

Easter-Relative Day

The following `XML` file defines the Good Friday holiday. This public holiday:

- Is named "Good Friday" in English, "Vendredi Saint" in French, and "Karfreitag" in German
- Is available in 3 groups named "BE" (Belgium), "FR" (France) and "DE" (Germany)
- Occurs 2 days before Easter Sunday

```
<?xml version="1.0" encoding="UTF-8"?>
<holiday code="GoodFriday">
  <description lang="en" label="GoodFriday" />
  <description lang="fr" label="Vendredi Saint" />
  <description lang="de" label="Karfreitag" />
  <group code="BE" />
  <group code="FR" />
  <group code="DE" />
  <easterRelativeDay offset="-2" />
</holiday>
```

Example

Enumerated Day

The following `XML` file defines the chinese New Year's Day holiday. This public holiday:

- Is named "Chinese New Year" in English and "Nouvel An Chinois" in French
- Occurs at a different date each year

```
<?xml version="1.0" encoding="UTF-8"?>
<holiday code="GoodFriday">
  <description lang="en" label="Chinese New Year" />
  <description lang="fr" label="Nouvel An Chinois" />
  <enumeratedDay>
  <date value="2015-02-19" />
  <date value="2016-02-08" />
  <date value="2017-01-28" />
  <date value="2018-02-16" />
  <date value="2019-02-05" />
  <date value="2020-01-25" />
  </enumeratedDay>
</holiday>
```

Note

By default, a list of predefined public holidays is embedded by SAP Convergent Charging. The `XML` file containing this default list of holidays is named `public_holidays.xml`, and is located in the `config` folder of the Core Tool user interface.

In case you need to modify the list of public holidays, execute the following procedure:

1. Launch the Core Tool user interface and log in with an individual user granted the "Marketing" role
2. Export the list of defined public holidays:
    1. From the menu bar in the main window, select *Tools* > *Public Holidays* > *Export*
    2. The *Save* dialog box opens
    3. Select a directory to which to export the list of public holidays
    4. In the *File name* field, type the name of the file
    5. Click *Save*
3. Modify the content of the created file using the previously provided description
4. Save your modifications
5. Import the list of modified public holidays:
    1. From the menu bar in the main window, select *Tools* > *Public Holidays* > *Import*
    2. The *Open* dialog box opens
    3. Open the directory in which the public holidays file is saved and select this file
    4. Click *Open*. A confirmation dialog box opens
    5. Click *Yes*

> ⚠ Caution
>
> Be careful when importing a new list of public holidays. After you import a new list, the old public holidays are replaced by the new ones, excepted those already used in the charges that were set up before importing the new list. In addition, the old public holidays that were already used appear dimmed (unavailable) in the new list, because:
>
> - They are no longer recognized by the SAP CC Core Server system
> - They are no longer active in the concerned charges

3.2.5.5 Managing the Currencies Framework

SAP Convergent Charging manages the following types of currencies:

- ISO[26] currencies, of which up-to-date list is provided at installation time
- SAP currencies, that are used in an integrated scenario with the SAP ERP/FI-CA component

These currencies are set up in master data that relate to both service provider and customers, within elements such as charges, charge plans, refill plans, refill logic, offers and subscriber accounts. You can use the following procedures to manage currencies within your landscape.

> ⓘ Note
>
> - You can set up reusable charges and refill logic using the multicurrency option. In this case, the final currency is then set up in the charge plans or offers. A unique charge can then be modeled and used in different charge plans with different currencies
> - In charge plans, you can set up the currency information for each charge customized in the plan. A specific currency parameter is automatically added by the Core Tool user interface, that gives you the possibility to:

[26] International Standards Organization

- Redefine this parameter with a new default value
- Link this parameter with an internal parameter defined in the charge plan
- Link this parameter with a parameter defined in the charge plan, that will be set up for each provider contract (in a CRM[27] application or external provisioning system)

- As far as refill plans are concerned, you cannot set up any currency in a refill plan. The final currency that will be used corresponds to the currency of the prepaid account to be refilled

3.2.5.5.1 Modifying ISO Currencies

> **Note**
>
> A default list of ISO[28] currencies is provided at installation time.

In case you need to modify the list of ISO currencies, execute the following procedure:

1. Launch the Core Tool user interface and log in with an individual user granted the “Marketing” role
2. Export the list of defined currencies:
    1. From the menu bar in the main window, select *Tools* > *Currencies* > *Export*
    2. The *Save* dialog box opens
    3. Select a directory where to export the list of currencies
    4. In the *File name* field, type the name of the file
    5. Click *Save*
3. Modify the content of the created file
4. Save your modifications
5. Import the list of modified currencies:
    1. From the menu bar in the main window, select *Tools* > *Currencies* > *Import*
    2. Open the directory containing the modified file and select this file
    3. Click *Open*. A confirmation dialog box opens
    4. Click *Yes*

> **Note**
>
> For further information about currencies import and export commands, refer to the Currencies section of the SAP CC 2021 Primary Help for Core Tool documentation.

3.2.5.5.2 Synchronizing SAP Currencies

SAP currencies correspond to currencies defined within the SAP ERP/FI-CA component of the SAP Business Suite. In an installation scenario integrated with SAP Convergent Invoicing, this list of currencies must be manually synchronized in SAP CC.

[27] Customer Relationship Management
[28] International Standards Organization

To synchronize the list of SAP currencies, execute the following procedure:

1. Launch the Core Tool user interface and log in with an individual user granted the “Marketing” role
2. From the menu bar of the main window, select *Tools* > *Currencies* > *Synchronize*.

> **Note**
>
> - When the synchronization of currencies succeeds, an information dialog box opens and displays the number of synchronized currencies
> - When the synchronization of currencies fails, an information dialog box opens and displays the reason of failure
> - For further information about the currencies synchronization command, refer to the Currencies section of the SAP CC 2021 Primary Help for Core Tool documentation

3.2.5.5.3 Replicating SAP Currencies

In an integrated *SAP Solution* scenario with SAP S/4HANA Cloud Public Edition, the list of SAP currencies defined in SAP Convergent Invoicing must be manually replicated in SAP Convergent Charging.

To replicate the list of SAP currencies in SAP CC, refer to the integration guides for BRIM.

3.2.5.5.4 Reforming Currencies

Reforming currencies consists in changing some characteristics of an existing, or replacing an existing currency by another one. The computed prices must then be converted by applying a given exchange rate.

SAP Convergent Charging gives the possibility to manage multiple currencies simultaneously, when the concerned currencies are used in different price plans and charged on different accounts. To switch from one currency to another without modifying the pricing implementation, no native mechanism is available. As a consequence, currencies reforming operations must be executed:

- During the initial implementation phase of SAP CC, when master data are configured. SAP SE recommends that you design your pricing logic independently from currencies, by:
  - Designing multicurrencies charges, offers and charge plans
  - Using reusable elements such as translation tables or pricing macros, that give the possibility to design 2 charges that use the same pricing logic, but that compute amounts with 2 different currencies
- During the maintenance phase of SAP CC, in accordance with other possible systems in case of an integrated scenario, in particular with third-party invoicing systems regarding:
  - Dates or slots for the currency switching time (for example to align invoices creation)
  - External accounts, whose information can be modified because of the currency switch
  - The impact of the currency switch operation, that may modify the switching policy (batch migration Vs on-the-fly migration)

3.2.5.6 Managing Customer Management Areas

- Editing the Customer Management Areas
- Deleting a Customer Management Area

See also:

- Creating JCo Destinations [page 165]
- Creating WS Destinations [page 261]

3.2.5.6.1 Creating Customer Management Areas

To create a Customer Management Area in the Core Server system, complete the following procedure:

1. Open your Core Tool user interface and log on.
2. View the area definitions:

   From the menu bar in the main window, select *Tools* > *Customer Management Areas* > *Edit*.
   The *Customer Management Areas* window opens.
3. In the configuration table, add the necessary rows.
4. Restart the dispatcher and the bulkloader instances in the Core Server system.
5. Verify if you must restart the rater instances in your Core Server system.

   If the system parameter JCO_CONNECTION_COUNT is set to 1 or more, then do restart all these rater instances.

3.2.5.7 Managing the Taxation Framework

SAP CC manages the following tax systems:

- Value Added Tax (VAT), with or without VAT rules
- Flat Tax, that consists of a constant marginal rate
- BillSoft EZTax, that is used to calculate taxes related to the U.S. Telco taxes

Tax information is used in master data that relate to both service provider and customers, within elements such as charges and subscriber accounts. You can use the following procedures to manage taxes within your landscape.

> ⚠ Caution
>
> - It is only possible to add a new tax system to the tax system configured at installation time
> - Never remove a tax system if a master data of the service provider or of the customers is configured to use this tax system

3.2.5.7.1 Enabling the VAT Framework

In case you did not select the VAT[29] tax system at installation time, you can use the following procedure to enable it:

1. Modify if necessary the VAT rates or VAT rules using the following dedicated procedures:
   - Modifying VAT Rates [page 66]
   - Modifying VAT Rules [page 67]
   - Customizing VAT Rules [page 68]
2. Using the Setup Tool user interface, execute the vatax importCountryTaxPolicies command, specifying the absolute path of the file containing the country tax policies to import
3. For the rater and updater instances of the Core Server system, modify the value of the `TAX_FRAMEWORK` system parameters to “VAT” or “VAT;EZTAX”, using Manage SAP CC System Parameters, Config Tool or Admin+
4. Restart all rater and updater instances of the Core Server system

3.2.5.7.2 Enabling the EZTax Framework

In case you did not select the EZtax tax system at installation time, you can use the following procedure to enable it:

1. Ensure that the Billsoft EZtax central repository `CC_EZTAX` exists within your landscape (it has been created at the installation time by the SAP installer of SAP CC
2. Using the Setup Tool user interface, execute the eztax importTSTypes command, specifying the absolute path of the file containing the EZtax transaction/service types
3. Edit the `default.fpl` file, that is located in the SAP Profile Folder of the SAP Central Repository, and:
   - Set the `tax.ezTax` parameter to "true"
   - Add the following parameter: `tax.eztaxRootFolder = <EZTAX_ROOT_FOLDER>`, where `<EZTAX_ROOT_FOLDER>` corresponds to the root folder where EZtax is installed
4. For the rater and updater instances of the Core Server system, modify the value of the `TAX_FRAMEWORK` system parameters to "EZTAX" or "VAT;EZTAX", using Manage SAP CC System Parameters, Config Tool or Admin+
5. Restart all rater and updater instances of the Core Server system
6. Install taxer instances in your landscape using the Adding Core Server Instances in a Multi-Hosts Landscape procedure available in the SAP CC 2021 Installation and Maintenance Guide documentation

3.2.5.7.3 Modifying VAT Rates

A list of VAT[30] rates is deployed by SAP installer at installation time when the VAT tax system is selected. You can use the following procedure to modify this default list:

[29] Value Added Tax

1. Using the Setup Tool user interface, execute the vatax exportVATRate command to export the VAT rates from the Core Database, specifying the absolute path of the file that will contain the exported VAT rates
2. Modify the content of the created file
3. Save your modifications
4. Using the Setup Tool user interface, execute the vatax importVATRate command to import the modified VAT rates into the Core Database, specifying the absolute path of the file containing the modified VAT rates
5. Use the Admin+ refresh_tax_cache command to trigger the update of the tax data caches dedicated to VAT rates of all the rater and updater instances of the Core Server system

> **Note**
>
> A default list of VAT rates is deployed at installation time in the `vat_rate.txt` file that is available in the `exe/uc/<OS>/CC_CORE_SERVER/tax/` subfolder of the SAP Central Repository, where `<OS>` corresponds to the operating system of the host.
>
> To import this default list, execute the step 4 of this procedure using this `vat_rate.txt` file.

3.2.5.7.4 Modifying VAT Rules

A list of VAT[31] rules is deployed by SAP installer at installation time when the VAT tax system is selected. You can use the following procedure to modify this default list:

1. Using the Setup Tool user interface, execute the vatax exportVATRules command to export the VAT rules from the Core Database, specifying the absolute path of the file that will contain the exported VAT rules
2. Modify the content of the created file

   > **Caution**
   >
   > You can modify EU VAT rules, but it is only possible to add customized VAT rules. For further information about VAT rule customization, refer to the Customizing VAT Rules [page 68] procedure afterwards.
3. Save your modifications
4. Using the Setup Tool user interface, execute the vatax importVATRules command to import the modified VAT rules into the Core Database, specifying the absolute path of the file containing the modified VAT rules
5. Use the Admin+ refresh_vat_rules_cache command to trigger the update of the tax data caches dedicated to VAT rules of all the rater and updater instances of the Core Server system

> **Note**
>
> A default list of VAT rules is deployed at installation time in the `vat_rules.txt` file that is available in the `exe/uc/<OS>/CC_CORE_SERVER/tax/` subfolder of the SAP Central Repository, where `<OS>` corresponds to the operating system of the host.
>
> To import this default list, execute the step 4 of this procedure using this `vat_rules.txt` file.

30 Value Added Tax
31 Value Added Tax

3.2.5.7.5 Customizing VAT Rules

When the VAT[32] tax system is enabled within your landscape, a list of VAT rules for EU VAT 2010 and EU VAT 2015 is deployed at installation time. In case you need to add specific VAT rules, you can use the following procedures to:

- Model your own VAT rules
- Create the modeled VAT rule

To model a specific VAT rule, consider the following guidelines:

- Determine the supplier zone, that corresponds to the reference for the service provider whose consumptions taxes must be calculated by SAP CC
- Determine the business categories (B2B or B2C) that are relevant for your VAT rule
- For each business category, determine the origin of the place of taxation that can be:
  - The supplier's place of establishment
  - The customer's place of establishment (B2B)
  - The customer's place of residence (B2C)
- For certain business requirements, the place of taxation depends on particular conditions that are not basically the places of establishment or residence. In this case, the adequate place must be dynamically determined at runtime by SAP CC according to the configuration of the pricing logic (see charges customized in charge plans or offers). You must then:
  - Configure the pricing logic in order to configure this place determination
  - Set up a price plan that defines a parameter or that computes a property containing at runtime the appropriate values, or you must ensure that the chargeable items received from the mediation system will contain this taxation place information
- Determine the default sources for the B2B and B2C places of taxation. The VAT 2010 directive defines the following places according to the business categories:
  - B2B services: Customer's place of establishment (the holder of the charged account in the subscriber account)
  - B2C services: Service provider's place of establishment

Example

You want to create a customized VAT rule to model the article 59b of the EU VAT 2010 directive that says: "Radio and television broadcasting services and telecommunications services, supplied by suppliers established in a third country to non-taxable customers in the EU, are taxable at the place where the private customer effectively consumes the service. Other general rules apply otherwise". To model this rule, consider the following configuration:

- **Reference Supplier Zone**: *Third Country*
- **Source for the B2B place of taxation**: *None*
- **Source for the B2C place of taxation**: *Place dynamically set by SAP Convergent Charging at runtime*. It must correspond to the effective place (country) of consumption of the customer service. The price plan must compute an appropriate property or the mediation system must provide this information in the chargeable items related to the service consumption
- **Default source for the B2B place of taxation**: *Customer's place of establishment* (the holder of the charged account in the subscriber account)

[32] Value Added Tax

- **Default source for the for the B2C place of taxation**: *Service provider's place of establishment*

Example

You want to create a customized VAT rule that gives you the possibility to finely configure the determination of the places of taxation within the pricing logic (in the price plans). To model this rule, consider the following configuration:

- **Reference Supplier Zone**: *Any Country*
- **Source for the B2B place of taxation**: *Place dynamically set by SAP Convergent Charging at runtime* (according to the configuration of the charges customized in charge plans or offers)
- **Source for the B2C place of taxation**: *Place dynamically set by SAP Convergent Charging at runtime* (according to the configuration of the charges customized in charge plans or offers)
- **Default source for the B2B place of taxation**: *Customer's place of establishment* (the holder of the charged account in the subscriber account)
- **Default source for the for the B2C place of taxation**: *Service provider's place of establishment*

Once your VAT rule has been modeled, use the following table to create the rule, that consists of a dedicated row:

- That must be inserted in the file containing VAT rules
- That respects the following syntax:
  `<RL_NAME>;<RL_DESCR>;<REF_SUPPLIER_ZONE>;<B2B_PLACE_OF_TAXATION_ORIG>;<B2C_PLACE_OF_TAXATION_ORIG>;<DEFAULT_B2B_PLACE_OF_TAXATION>;<DEFAULT_B2C_PLACE_OF_TAXATION>`

Property	Type	Description
`RL_NAME`	■	This property corresponds to the name of the VAT rule, used as a **unique** identifier of the VAT rule. This name is visible in the Core Tool when the power user configures the tax settings for each charge customized in a charge plan (or in an offer).
`RL_DESCR`	■	This property corresponds to the description of the VAT rule. This description is visible in the Core Tool when the power user configures the tax settings for each charge customized in a charge plan (or in an offer).
`REF_SUPPLIER_ZONE`	■	This property corresponds to the zone of the service provider. Possible values are: • 0: EU (European Union) • 1: TC (Third Country) • 2: AC (Any Country)

`B2B_PLACE_OF_TAXATION_ORIG`
 	
`B2C_PLACE_OF_TAXATION_ORIG`
	□	This property corresponds to the source for the place of taxation. Possible values are:
		• 0: Supplier's place of establishment (the service provider supplying the customer service to charge)
		• 1: Customer's place of establishment (the holder of the prepaid or external account that is charged by SAP CC)
		• 2: Place dynamically determined by SAP CCaccording to the configuration of the pricing catalog (see charges customized in charge plans or offers.)

Property	Type	Description
`DEFAULT_B2B_PLACE_OF_TAXATION`	■	This property corresponds to the default source for the place of taxation. Possible values correspond to the above listed values.
`DEFAULT_B2C_PLACE_OF_TAXATION`		

■ Mandatory ☐ Optional

> ### Example
>
> Considering the previously modeled VAT rules, the following rows must be created:
>
> - **Example 1:** ART59b;Rule description;1;;2;1;0
> - **Example 2:** CustomRule; Rule description;2;2;2;1;0

3.2.6 Catalog Transport

Keywords [page 70]

Preliminary Notes [page 70]

Description [page 71]

Managing Transport Destinations [page 71]

Scheduling Transport Requests Operations [page 72]

3.2.6.1 Keywords

Catalog transport, transport destination, change list, scheduler

3.2.6.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- Catalog Transport section of the SAP CC 2021 Application Help documentation
- Installing and Launching Setup Tool, Installing and Launching Admin+ and Securing Communications with the Core Server System sections of the SAP CC 2021 Installation and Maintenance Guide documentation
- `transportdestination importList`, `transportdestination exportList` and `certentry import` commands of the SAP CC Setup Tool User Interface documentation

3.2.6.3 Description

As described in the SAP CC 2021 Application Help documentation, SAP Convergent Charging gives the possibility to transport data of a given catalog from an SAP CC system to one or several other SAP CC system(s), using the Catalog Transport feature. This feature is based on change lists that can be transported to transport destinations that are defined within the Core Server system. You can use the following procedures to configure the Catalog Transport feature and thus fit your specific needs:

- Managing Transport Destinations [page 71]
- Scheduling Transport Requests Operations [page 72]

> ⚠ Caution
>
> If the Catalog Transport feature is used to transport change lists to secured SAP CC systems, please note that these remote SAP CC systems must respect the following constraints:
>
> - The external communications (relying on the XML over HTTP[33] or SOAP[34] over HTTP communication channels) of the dispatcher and updater instances must be configured with the "off" or "oneway" secure mode. **The "dual" authentication mode is not supported**
> - The certificates associated to the remote SAP CC systems must be valid for all the external communications

3.2.6.4 Managing Transport Destinations

By default, no transport destination is defined at installation time. In case you (or an SAP CC pricing specialist) want to transport change lists (and included master data) between SAP CC systems, it is thus necessary to configure transport destinations, each transport destination representing a target SAP CC system to which objects can be transported.

Configure the List of Transport Destinations

To configure a list of transport destinations on a given SAP CC system, execute the following procedure:

1. Using the Setup Tool user interface, execute the transportdestination exportList command, to export the list of transport destinations from the database to an `XML` file.

   > ⓘ Note
   >
   > In case the targeted transport destinations correspond to a secured landscape, you need to import the certificates of the secured SAP CC system, using the certentry import dedicated command. For further information, refer to the Securing Communications with the Core Server System section of the SAP CC 2021 Installation and Maintenance Guide documentation.

[33] HyperText Transfer Protocol
[34] Simple Object Access Protocol

2. Edit this newly created file and modify its content to add, modify, or remove transport destinations. For further information about the structure of transport destination files, refer to the dedicated XML schema accessible from the documentation of the transportdestination importList command.

   > **Note**
   >
   > In case of secured transport destination, specify the alias of the adequate certificate within the `trustedCertificateEntryAlias` parameter. When this alias is not specified, the certificates defined for the "ws" communication channel of the updaters will be used by default.

3. Save your modified XML file.
4. Using the Setup Tool user interface, execute the transportdestination importList command, to import the modified list of transport destinations from the `XML` file to the database.

   > **Note**
   >
   > No restart of the updaters is required to take the new destinations into account.

3.2.6.5 Scheduling Transport Requests Operations

Change lists can be transported to the defined transport destinations:

- Manually via the Core Tool user interface
- Automatically by a dedicated scheduler in the Core Server system

To configure the behavior of the dedicated scheduler, modify the value of the following system parameters [page 45] for the updater instances of the Core Server system:

- TRANSPORT_SCHEDULER_ENABLED, which gives the possibility to enable or disable the scheduler
- TRANSPORT_SCHEDULER_RECURRENCE, which defines the recurrence period of the scheduler

For further information about these system parameters, refer to their dedicated documentation available in the SAP CC 2021 System Parameter Reference documentation.

Related Information

Managing the Transport Requests

3.2.7 Chargeable Items Charging Process

Keywords [page 73]

Preliminary Notes [page 73]

Description [page 73]

Enabling the Session Database [page 73]

Limiting the Number of Charging Sessions [page 74]

Limiting the Period of Charging Operations [page 75]

Enabling the Billable Items Immediate Loading [page 75]

3.2.7.1 Keywords

Charging, Session Database, session-based

3.2.7.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- Chargeable Items Charging Process description of the SAP CC 2021 Application Help documentation
- Installing and Launching Admin+ section of the SAP CC 2021 Installation and Maintenance Guide documentation
- “Process: Charging and Rating” and “Process: Session-Based Charging” groups in the SAP CC 2021 System Parameter Reference documentation

3.2.7.3 Description

As described in the SAP CC 2021 Application Help documentation, SAP CC gives the possibility to charge incoming events using different execution modes that suit multiple businesses. These charging operations can concern both external or internal events, and lead to the generation of output data handled by external systems such as SAP Convergent Invoicing.

You can use the following procedures to manage the charging capabilities of SAP Convergent Invoicing:

- Enabling the Session Database [page 73], that is only relevant for landscapes implementing the Dual Database feature
- Limiting the Number of Charging Sessions [page 74], that can be used to limit the creation of an unexpected high number of charging sessions

3.2.7.4 Enabling the Session Database

In case your business implements the Dual Database feature, it is necessary to enable the Session Database once it has been installed within your landscape. To enable the use of the Session Database, execute the following procedure:

1. Stop the Core Server system
2. Modify the value of the following system parameters [page 45] of the rater instances of the Core Server system:
   - RATING_SESSION_SQLHELPER_CONNECTION_COUNT, which gives the possibility to specify a number of connections to open on an instance of the Session Database
   - RATING_SESSION_SQLHELPER_DB_INSTANCE_COUNT, specifying a positive number of Session Database instances that will be used by the raters
   - RATING_SESSION_SQLHELPER_JDBC_URI_DBI5 (where "5" corresponds to an example number of 5 RAC[35] nodes that are available for the Session Database), which defines the JDBC[36] URIs[37] for connecting to the Session Database
   - RATING_SESSION_SQLHELPER_LOGIN and RATING_SESSION_SQLHELPER_PASSWORD, which corresponds to the credentials used to authenticate on the Session Database
   - RATING_SESSION_UPDATE_POOL_SIZE, which defines the size of the pool containing the modifications queries performed on the Session Database
   - RATING_SESSION_UPDATE_MAX_BATCH_SIZE, which defines the maximum number of queries that can be handled by a given thread of the pool dedicated to modification queries performed on the Session Database
   - RATING_SESSION_SELECT_POOL_SIZE, which defines the size of the pool containing all the search queries performed on the Session Database
3. For each `RATING_SESSION_<X>` table that exists in the Core Database, execute the following command to remove the old data that are related to the session-based charging operations, considering that:
   - `<COREDB_USER>` corresponds to the name of the Core Database administrator
   - `<COREDB_PASSWORD>` corresponds to the password of the Core Database administrator

   ```
   sqlplus <COREDB_USER>/<COREDB_PASSWORD>
   truncate table RATING_SESSION_<X>
   ```
4. Restart the Core Server system

3.2.7.5 Limiting the Number of Charging Sessions

In case your business relies on charging operations performed using the session-based execution mode, you may want to prevent the creation of an unexpected high number of charging sessions.

To limit the number of charging sessions that can be started per subscriber account, modify the value of the following system parameter [page 45] of the rater instances of the Core Server system:

- RATING_SESSION_PER_SUBSCRIBER_ACCOUNT_LIMIT, which gives the possibility to specify the maximum number of charging sessions that can be started in parallel for each subscriber account

For further information about this system parameter, refer to its dedicated documentation available in the SAP CC 2021 System Parameter Reference.

[35] Real Application Cluster
[36] Java Database Connectivity
[37] Uniform Resource Identifier

3.2.7.6 Limiting the Period of Charging Operations

In case your business requires to plan the charging and the activation operations in the future, you can use the dedicated control function.

To enable this function, modify the value of the following system parameter [page 45] of the dispatcher instances of the Core Server system:

- PERIOD_LIMIT_ACTIVATION, which gives the possibility to specify a period (number of days), after the current system date, within which the charging and the activation operations will occur.

> **Note**
>
> By default, this control function is disabled in SAP CC

For further information about this system parameter, refer to its dedicated documentation available in the SAP CC 2021 System Parameter Reference.

3.2.7.7 Enabling the Billable Items Immediate Loading

In an integrated *SAP Solution* scenario with SAP Convergent Invoicing in its SAP system, your business scenario may require the *Billable Items Immediate Loading* feature in SAP Convergent Charging (as of SAP CC 2020 FPS 2) for manual charging operations.

> **Caution**
>
> The use of billable items immediate loading has a significant impact on rating performance. This feature should only be implemented if required for immediate billing purposes and as a complement but not as replacement of the standard bulk loading of billable items.

Prerequisites

- Customer management areas (CMAs) are defined in the integrated SAP system landscape and are configured in the SAP CC systems.
- Necessary JCo destinations are configured. See Creating JCo Destinations [page 165].
- Your integrated *SAP Solution* scenario is not based on SAP S/4HANA Cloud Public Edition with Convergent Invoicing.
- The feature is implemented in your SAP system landscape.
  - SAP Convergent Invoicing is ready to receive the raw billable items and corresponding consumption items in real time. Consult the integration guides and the SAP Note 3048461.
  - The client application or SAP Convergent Invoicing is able to trigger the immediate loading of billable items by sending single charging (or bundle charging) operation requests that include the itemImmediatelyLoaded XML element.
    The feature is dedicated to manual operations and for a low volume of operations. Mass use of such operations has a significant impact on global rating performance.

Procedure

As an SAP CC system administrator:

1. Log on to your administration tool.
2. Change the `JCO_CONNECTION_COUNT` system parameter for all the rater instances in the SAP CC Core Server system.
3. Restart these rater instances.
4. Check the log messages recorded by these instances. Verify that the log message com.sap.SCC.core_server.000734 is in the logs.

   If `com.sap.SCC.core_server.000735` is recorded, then verify the system settings (JCO_CONNECTION_COUNT system parameter, JCo destinations, and customer management areas).

> → Tip
>
> In the deployed Cockpit web applications, you can enable the feature in the apps Process a Chargeable Item. See Enabling the Billable Items Immediate Loading Feature in the App Process a Chargeable Item [page 40].

Related Information

JCO_CONNECTION_COUNT
Enabling the Billable Items Immediate Loading Feature in the App Process a Chargeable Item [page 40]
Log Messages
com.sap.SCC.core_server.000734
com.sap.SCC.core_server.000735
Application Help
Billable Items Immediate Loading

3.2.8 Chargeable Items Recharging Process

Keywords [page 77]

Preliminary Notes [page 77]

Description [page 77]

Configuring the Retention Period for Counter Snapshots [page 77]
During the project implementation phase, the integration team defines the valid value and resize the platforms.

Deactivating the Recharging Feature Based on Counter Snapshots [page 78]

Using SAP CI instead of BART Server for Recharging Operations [page 79]

3.2.8.1 Keywords

Recharging, rerating, counter snapshot, rerate file processor, retention period

3.2.8.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- Chargeable Items Rerating description of the SAP CC 2021 Application Help documentation
- Installing and Launching Setup Tool section of the SAP CC 2021 Installation and Maintenance Guide documentation
- "Process: Rerating" groups in the SAP CC 2021 System Parameter Reference documentation
- `deactivateRecharging` command of the SAP CC 2021 Setup Tool documentation

3.2.8.3 Description

As described in the SAP CC 2021 Application Help documentation, SAP CC gives the possibility to correct errors by executing recharging operations that cancel previous charging operations and perform new ones on identified sets of chargeable items. These recharging operations:

- Are executed by default in conjunction with the BART Server system
- Can be handled by SAP Convergent Invoicing in case your landscape is integrated with this SAP system

You can use the following procedures to manage the recharging capabilities of SAP CC:

- Configuring the Retention Period for Counter Snapshots [page 77]
- Deactivating the Recharging Feature Based on Counter Snapshots [page 78]
- Using SAP CI instead of BART Server for Recharging Operations [page 79]

3.2.8.4 Configuring the Retention Period for Counter Snapshots

During the project implementation phase, the integration team defines the valid value and resize the platforms.

By default, SAP Convergent Charging (SAP CC) gives the possibility to store counter snapshots during 40 days. For business, legal, or technical reason, you can change this retention period.

This retention feature of snapshots of the counters stored in provider contracts (in customer master data) is crucial. The same applies for snapshots of the counters stored in subscriptions.

To modify the retention period for counter snapshots in your SAP CC system landscape, execute the following procedure:

1. For the updater instances of the Core Server system, modify the value [page 45] of the COUNTER_SNAPSHOT_COUNT system parameter.
2. Restart the updater instances of the Core Server system.

> ⚠ Caution
>
> - SAP SE recommends that you do not change this retention period in a production landscape.
> - The retention period for counter snapshots must be lower than or equal to the retention period used for expired allowances.
>   For further information, refer to the Configuring the Allowances Purge Mechanism and Its Scheduler [page 154] section.
> - Changing the retention period may lead to a significant increase of the database size.

Related Information

migration increaseCounterSnapshotCount

3.2.8.5 Deactivating the Recharging Feature Based on Counter Snapshots

When the counter snapshot mechanism is enabled, counters are backed up regularly within counter snapshots that are used during recharging operations. In case you do not plan to perform recharging operations, you can deactivate the recharging feature (based on counter snapshots) and thus increase the reactivity of the Core Server system by optimizing your data storage.

> ⚠ Caution
>
> This procedure definitively deactivates the recharging capabilities (based on counter snapshots) of SAP Convergent Charging, and cannot be activated again. When performed, it will not be possible to execute any chargeable items recharging operation.

To deactivate the chargeable items recharging feature within your SAP CC landscape, execute the following procedure:

1. Using SAP CC 2021 Setup Tool, execute the migration deactivateRecharging command to disable the Counter Snapshots mechanism, and thus definitly deactivate the Chargeable Items Recharging feature
2. Restart the rater instances of the Core Server system
3. Restart the updater instances of the Core Server system
4. Once this command has ended, you can verify the following elements to ensure that the feature has been successfully deactivated:
   - The `com.sap.SCC.core_server.000632` log message (`The recharging feature is deactivated`) must be displayed in the starting log of both raters and updaters
   - The value of the `RECHARGING_STATUS` system indicator must be set to "off"

> **Note**
>
> The `COUNTER_SNAPSHOT` table of the Core Database contains the counter snapshots used for recharging purpose. As these counter snapshots will not be used anymore, you can empty this table to optimize your data storage. To ensure the integrity of the database structure for future updates, SAP SE recommends that you **do not delete** this table.

3.2.8.6 Using SAP CI instead of BART Server for Recharging Operations

In case your landscape has been installed using the "Integration Scenario set to Billing, Invoicing, and Storage of Consumption Data in SAP Convergent Invoicing" installation scenario, SAP installer automatically set SAP CI as the system in charge of performing the recharging operations.

If your landscape has been installed using another installation scenario, you can integrate SAP CC with SAP CI and use SAP CI for recharging operations of high volumes of chargeable items. For further information about this landscape migration operation, refer to the SAP CC 2021 Transition Guide (Activation of Consumption Item Management) documentation.

You use recharging operations to schedule recharging activities in SAP Convergent Charging and SAP Convergent Invoicing.

> **Note**
>
> In both landscapes, BART Server is still involved in the batch charging operations of chargeable items.

3.2.9 Communications Channels

Keywords [page 79]

Preliminary Notes [page 80]

Description [page 80]

Modifying the Database Connection Settings [page 80]

Modifying the Database Communication Settings: Number of Dedicated Connections [page 81]

Enabling the UDP Discovery Mechanism [page 82]

Securing the Communication Channels [page 82]

3.2.9.1 Keywords

Communication channels, securing, UDP[38]

3.2.9.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- "Communication Channel Security" section of the SAP CC 2021 Security Guide documentation
- Securing an SAP CC System Landscape and Installing and Launching Admin+ sections of the SAP CC 2021 Installation and Maintenance Guide documentation

3.2.9.3 Description

SAP Convergent Charging communications rely on different communication channels whose use depend on the concerned elements. You can use the following procedures to configure the settings related to communication channels and thus fit your specific needs or performance requirements:

- Modifying the Database Communication Settings: Number of Dedicated Connections [page 81]
- Enabling the UDP Discovery Mechanism [page 82]
- Securing the Communication Channels [page 82]

3.2.9.4 Modifying the Database Connection Settings

At startup, each instance of the SAP CC system temporarily connects to the database system specified in the `boot.config` configuration file.

Back-end database connection settings are collected from this database and used by an instance of the SAP CC Core Server system to connect to the back-end database(s) after the startup phase.

> **Note**
>
> Database connection settings specified in the `boot.config` configuration file may differ from the parameters values stored in the database.

Procedure

To edit your SAP Convergent Charging Core Server system database connection settings, apply the following procedure:

1. Stop all the Core Server instances.
2. Go to the `/usr/sap/<SYSTEM_ID>/SYS/profile/boot/` folder.
3. Make a backup of the `boot.config` file.

[38] User Datagram Protocol

4. Edit the `boot.config` file and modify the value of the following properties when relevant:
   - SQLHELPER_LOGIN
   - SQLHELPER_PASSWORD
   - SQLHELPER_JDBC_URI
5. Go to the `"config"` folder of one of the dispatcher instances. Use the folder: `"/usr/sap/<SID>/CCD<XX>/config/"`
6. Edit the `boot.config` file and modify the value of the following properties when relevant:
   - SQLHELPER_LOGIN
   - SQLHELPER_PASSWORD
   - SQLHELPER_JDBC_URI
7. Go to the `"script"` folder of the very same dispatcher instance: `"/usr/sap/<SID>/CCD<XX>/script/"`
8. Using Config Tool, export the Core Server parameters using the configuration export command:

```
./config.sh configuration export -login=<login> -password=<password>
parameters.xml all
```

9. Make a backup of the exported `parameters.xml` file.
10. Edit the `parameters.xml` file and edit the relevant database connectivity settings defined within the "Backends: Database Management Settings" parameters group for each Core Server instance type.
11. Using Config Tool, import the modified parameters.xml file using the configuration import command:

```
./config.sh configuration import -login=<login> -password=<password>
parameters.xml
```

12. Restart the SAP Convergent Charging Core Server system and check that database connectivity settings changes have been correctly taken into account.
    To do that, check the logs and traces.

See Also

Startup Configuration Files [page 234]

3.2.9.5 Modifying the Database Communication Settings: Number of Dedicated Connections

To communicate with the databases available in your SAP CC system landscape, it is necessary to define dedicated connections whose number depends on the enabled services.

To modify the number of connections used by each instance of the SAP system, modify the value of the following system parameters for each concerned instance:

- SQLHELPER_CONNECTION_COUNT, which defines the connections to open on the Core Database (valid for all the instances)

- RATING_SESSION_SQLHELPER_CONNECTION_COUNT, which defines the connections to open on the Session Database, when this database is used (only valid for rater instances)

> → Remember
>
> The value of this system parameter must be correlated to the total number of threads running in each instance of the SAP CC Core Server system.

For further information about these system parameters, refer to their dedicated documentation available in the SAP CC 2021 System Parameter Reference documentation.

3.2.9.6 Enabling the UDP Discovery Mechanism

SAP Convergent Charging gives to client applications and other deployed instances the possibility to detect the list of available dispatchers. Based on UDP[39], this proprietary discovery mechanism relies on multicast requests that are sent on the Messages over UDP communication channel.

If your network configuration support UDP communications, you can execute the following procedure to enable the UDP discovery mechanism within your SAP CC landscape:

1. Modify the value of the following system parameters [page 45] for the dispatcher instances of the Core Server system:
   - EXTERNAL_DISCOVERY_ENABLED, which gives the possibility to enable or disable the discovery mechanism
   - EXTERNAL_DISCOVERY_MULTICAST_ADDRESS, which represents the multicast IP address used by dispatchers to inform about their availability and their connection parameters
   - EXTERNAL_DISCOVERY_PORT, which represents the port of the defined multicast IP address
2. Restart the primary dispatcher
3. Restart the other dispatcher instances

3.2.9.7 Securing the Communication Channels

For security purposes, all communication channels (except the Messages over UDP[40] one) used to communicate within your SAP CC landscape can be secured.

For further information about the available securing procedures, refer to the dedicated Securing an SAP CC System Landscape section of the SAP CC 2021 Installation and Maintenance Guide documentation.

3.2.10 Data Cache Management

[39] User Datagram Protocol
[40] User Datagram Protocol

Keywords [page 83]
Preliminary Notes [page 83]
Description [page 83]
Determining the Size of the Cached Structures [page 84]
Modifying the Size of Cached Structures [page 85]
Modifying the Guiding Cache [page 86]
Modifying the Provisioning Cache [page 87]
Modifying the Session Cache [page 90]
Modifying the Shared Allowance Cache [page 91]
Modifying the SQL Statements Cache [page 93]
Enabling the Cache Warm-Up Mechanism [page 93]
Modifying the Network Data Transfer (NDT) Mechanism [page 95]
Configuring the Automated Refreshment of the Shared Allowance Caches [page 97]

3.2.10.1 Keywords

Cache, data cache, cached structure, cache size, cache warm-up, smart mode, normal mode, shared allowances, refresh, scheduler

3.2.10.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- Installing and Launching Admin+ section of the SAP CC 2021 Installation and Maintenance Guide documentation
- Caching Policies description of the SAP CC 2021 Application Help documentation
- `refresh_shared_allowance_cache` command of the SAP CC 2021 Primary Help for Admin+ documentation
- Data Cache Management Settings and Allowance Management Process groups in the SAP CC 2021 System Parameter Reference documentation

3.2.10.3 Description

As described in the SAP CC 2021 Application Help documentation, most of the instances of the Core Server system manage data within cached structures that are stored in memory. This memory is allocated and managed by each Java Virtual Machine (JVM) that runs an SAP CC system instance that composes the global system.

You can use the following procedures to manage some of these cached structures of your SAP CC system and running instances, and thus fit your specific needs or performance requirements:

- Determining the Size of the Cached Structures [page 84]
- Modifying the Size of Cached Structures [page 85]
- Modifying the Guiding Cache [page 86]
- Modifying the Provisioning Cache [page 87]
- Modifying the Session Cache [page 90]
- Modifying the Shared Allowance Cache [page 91]
- Modifying the SQL Statements Cache [page 93]
- Enabling the Cache Warm-Up Mechanism [page 93]
- Modifying the Network Data Transfer (NDT) Mechanism [page 95]
- Configuring the Automated Refreshment of the Shared Allowance Caches [page 97]

3.2.10.4 Determining the Size of the Cached Structures

To determine the size of a given cached structure, the best way consists in putting a set of objects in this cache and determining the average size of these objects. The recommended method consists in simulating the presence of 1,000 end customers of a service provider, and creating all the business objects that are relevant and necessary for your implementation:

- **Customer master data**:
  - 1,000 subscriber accounts
  - 1,000 provider contracts or subscriptions (and relevant accesses)
  - The relevant data tables (subscriber mapping tables and subscriber range tables)
- **Master data related to the service provider**: Relevant pricing objects (charge plans, offers and suboffers), charges with price plans and Charging Plan including pricing macros or data tables like tier or translation tables, mapping or range tables

To determine the size of a given cached structure, execute the following procedure:

1. Start an instance of each type (dispatcher, updater, guider, rater, and taxer when relevant) with the default values for sizing the cached structure.
2. Launch your favorite user interface [page 45] and identify as the administrator of the Core Server system.
3. Retrieve the `S1` value of the following system parameters for each concerned instance:
   - SUBSCRIPTION_CACHE_SIZE, which defines the memory allocated to store objects in the subscription cache
   - GUIDING_CACHE_SIZE, which defines the memory allocated to store objects in the access cache
   - SESSION_MEMORY_SIZE, which defines the memory allocated to store objects in the rating sessions cache
   - SHARED_ALLOWANCE_CACHE_SIZE, which defines the memory allocated to store objects in the shared allowances cache
4. Retrieve the `FM1` value of the following system indicators for each concerned instance:
   - SUBSCRIPTION_CACHE_FREE_MEMORY, which indicates the remaining free memory in the subscription cache

- GUIDING_CACHE_FREE_MEMORY, which indicates the remaining free memory in the access cache
- SESSION_FREE_MEMORY, which indicates the remaining free memory in the rating sessions cache
- SHARED_ALLOWANCE_CACHE_FREE_MEMORY, which indicates the remaining free memory in the shared allowances cache

5. Create **N0** typical objects (1,000 for example) that will be stored in each cached structure.
6. Restart the instances started in Step 1.
7. Re-execute Step 3 and Step 4, considering the retrieved values as S2 and FM2.
8. For each cached structure of each concerned instance, compute the consumed memory related to the created objects, taking into consideration that:
   - If the size of the cached structure has changed, apply the following formula: `(S2-FM2) - (S1-FM1)`
   - If the size of the cached structure has **not** changed, calculate the difference between the free memory available at the first launch and the free memory available at the second launch: `FM1 - FM2`
9. Divide the previous result by the number of objects you have created in the cached structure, and then multiply the result by the NF number of objects you plan to have in the future.
   To sum up, the size of the cached structure can be estimated using the following formula, depending whether the size of the cached structure has changed between the 2 launches, or not:
   - `((FM1-FM2) / N0) x NF`
   - `(((S2-FM2)-(S1-FM1)) / N0) x NF`
10. Add the result to the first computed size: You now have the size of your cached structure, that you can divide by the number of available instances supposed to manage this volume of objects.
11. Increase S2 and the total size of the JVM[41] with this number in order to prevent the memory allocated to the cached structure from being taken by another cached structure or by the JVM itself.

3.2.10.5 Modifying the Size of Cached Structures

To handle your business requirements, it is necessary to configure the different cached structures handled by the instances of the Core Server system. In case of a production system landscape, SAP SE recommends that you to use values that have previously been determined on a non-production landscape during test campaigns.

> ⚠ Caution
>
> The size of the cached structures handled by a given instance must not exceed the value of the `MaxDirectMemorySize` JVM[42] option. If this constraint is not respected, `OutOfMemory` errors will occur.

To configure the different cached structures, execute the following procedure:

1. For each concerned instance, modify the value of the relevant parameters [page 45], as described in the dedicated sections afterwards.
2. Restart the concerned instances.

[41] Java Virtual Machine
[42] Java Virtual Machine

3.2.10.6 Modifying the Guiding Cache

Also known as the *access cache*, the guiding cache is a data cache that contains the accesses data used for request routing purpose.

> → Remember
>
> The guiding cache is available in the guider instances of the Core Server system.

To configure the guiding cache, complete the following procedure:

1. Modify the value of the following system parameters [page 45] for the guider instances of the Core Server system:
   - GUIDING_CACHE_SIZE, which defines the total memory size reserved for the guiding cache in each guider instance of the Core Server system.

     > ⓘ Note
     >
     > To determine the size of the guiding cache, the best way consists in having a sample of objects contained in the cache and determine the average size of these objects.
     >
     > For further information, refer to the Determining the Size of the Cached Structures [page 84] section.

   - GUIDING_ENTRY_AVERAGE_SIZE, which defines the average size of the memory allocated for the storage of **one entry (values and key)** in the guiding cache.

     > ⓘ Note
     >
     > The appropriate value for this parameter can be found by retrieving the value of the `eavg` element in the last line of the value of the guider instance GUIDING_CACHE_STATUS indicator while the guiding cache is not empty. This value corresponds to the average size of the objects stored in the guiding cache, and should be used as the value for this parameter.
     >
     > The possible consequences of an incorrect value for this parameter are the following:
     > - If your parameter value is too small, objects may be removed from the guiding cache because there is not enough memory available in the cache (see the `vmen` element in the guider instance GUIDING_CACHE_STATUS indicator), leading to decreased performance.
     > - If your parameter value is too large, objects may also be removed from the guiding cache because there are not enough slots available in the cache (see the `vmap` element in the guider instance GUIDING_CACHE_STATUS indicator), leading to decreased performance.

   - GUIDING_CACHE_INSTANCES, which defines the maximum number of subcaches into the guiding cache, as the guiding cache is divided into smaller subcaches for reducing contention.

     > → Recommendation
     >
     > The value of this parameter is `computed` by default to the first prime number greater than 8 times the number of CPUs[43]. As a rule, there is no need to modify this default value.

2. For each guider instance in the Core Server system, check the value of the `MaxDirectMemorySize` JVM option defined in the `jstart.config` file.

[43] Central Processing Unit

> ⚠ Caution
>
> The size of the data caches handled by a given instance must not exceed the value of the `MaxDirectMemorySize` JVM option. If this constraint is not respected `OutOfMemory` errors will occur.

> → Recommendation
>
> The formula to compute the appropriate value for the `MaxDirectMemorySize` JVM option of each guider instance is: GUIDING_CACHE_SIZE + 1 GB

3. Restart the guider instances in the Core Server system.

3.2.10.7 Modifying the Provisioning Cache

Also known as the *subscription cache*, the provisioning cache is a data cache that contains the provisioning data (provider contracts, subscriptions, counters and subscriber accounts).

This provisioning cache is available in the rater and updater instances of the Core Server system.

> ⓘ Note
>
> The provisioning cache managed by updater instances does not contain counters data.

Modify the Provisioning Cache of the Rater Instances

To configure the provisioning cache for the rater instances of the Core Server system, execute the following procedure:

1. Modify the value of the following system parameters [page 45] for the rater instances of the Core Server system:
   - SUBSCRIPTION_CACHE_SIZE, which defines the total memory size reserved for the provisioning cache for each rater of the Core Server system.

     > ⓘ Note
     >
     > To determine the size of the provisioning cache for the rater instances of the Core Server system, the best way consists in having a sample of objects contained in the cache and determine the average size of these objects.
     >
     > For further information, refer to the Determining the Size of the Cached Structures [page 84] procedure.

   - SUBSCRIPTION_OBJECT_AVERAGE_SIZE, which defines the average size of of the memory allocated to the storage of one data object in the provisioning cache

     > ⓘ Note
     >
     > The appropriate value for this parameter can be found by retrieving the value of the `eavg` element in the last line of the value of the rater instance SUBSCRIPTION_CACHE_STATUS indicator while

the provisioning cache is not empty. This value corresponds to the average size of the objects stored in the provisioning cache, and should be used as the value for this parameter.

The possible consequences of an incorrect value for this parameter are the following:

- If your parameter value is too small, objects may be removed from the provisioning cache because there is not enough memory available in the cache (see the `vmem` element in the rater instance SUBSCRIPTION_CACHE_STATUS indicator), leading to decreased performance.
- If your parameter value is too large, objects may also be removed from the provisioning cache because there are not enough slots available in the cache (see the `vmap` element in the rater instance SUBSCRIPTION_CACHE_STATUS indicator), leading to decreased performance.

- SUBSCRIPTION_CACHE_INSTANCES, which defines the maximum number of subcaches into the provisioning cache, as the provisioning cache is divided into smaller subcaches for reducing contention.

→ Recommendation

The value of this parameter is `computed` by default to the first prime number greater than 8 times the number of CPUs[44]. As a rule, there is no need to modify this default value.

2. For each rater instance of the Core Server system, check the value of the `MaxDirectMemorySize` JVM option defined in the `jstart.config` file

⚠ Caution

The size of the data caches handled by a given instance must not exceed the value of the `MaxDirectMemorySize` JVM option. If this constraint is not respected `OutOfMemory` errors will occur.

→ Recommendation

The formula to compute the appropriate value for the `MaxDirectMemorySize` JVM option of each rater instance is: SUBSCRIPTION_CACHE_SIZE + SESSION_MEMORY_SIZE + SHARED_ALLOWANCE_CACHE_SIZE + 1Gb

3. Restart the rater instances in the Core Server system.

Modify the Provisioning Cache of the Updater Instances

To configure the provisioning cache for the updater instances of the Core Server system, execute the following procedure:

1. Modify the value of the following system parameters [page 45] for the updater instances of the Core Server system:
   - SUBSCRIPTION_CACHE_SIZE, which defines the total memory size reserved for the provisioning cache for each updater of the Core Server system

[44] Central Processing Unit

> **Note**
>
> To determine the size of the provisioning cache for the updater instances of the Core Server system, the best way consistsin having a sample of objects contained in the cache and determine the average size of these objects.
>
> For further information, refer to the Determining the Size of the Cached Structures [page 84] procedure.

- SUBSCRIPTION_OBJECT_AVERAGE_SIZE, which defines the average size of of the memory allocated to the storage of one data object in the provisioning cache

  > **Note**
  >
  > The appropriate value for this parameter can be found by retrieving the value of the `eavg` element in the last line of the value of the updater instance SUBSCRIPTION_CACHE_STATUS indicator while the provisioning cache is not empty. This value corresponds to the average size of the objects stored in the provisioning cache, and should be used as the value for this parameter.
  >
  > The possible consequences of an incorrect value for this parameter are the following:
  > - If your parameter value is too small, objects may be removed from the provisioning cache because there is not enough memory available in the cache (see the `vmem` element in the updater instance SUBSCRIPTION_CACHE_STATUS indicator), leading to decreased performance.
  > - If your parameter value is too large, objects may also be removed from the provisioning cache because there are not enough slots available in the cache (see the `vmap` element in the updater instance SUBSCRIPTION_CACHE_STATUS indicator), leading to decreased performance.

- SUBSCRIPTION_CACHE_INSTANCES, which defines the maximum number of subcaches into the provisioning cache, as the provisioning cache is divided into smaller subcaches for reducing contention.

  > **→ Recommendation**
  >
  > The value of this parameter is `computed` by default to the first prime number greater than 8 times the number of CPUs. As a rule, there is no need to modify this default value.

2. For each updater instance in the Core Server system, check the value of the `MaxDirectMemorySize` JVM option defined in the `jstart.config` file.

   > **⚠ Caution**
   >
   > The size of the data caches handled by a given instance must not exceed the value of the `MaxDirectMemorySize` JVM option. If this constraint is not respected `OutOfMemory` errors will occur.

   > **→ Recommendation**
   >
   > The formula to compute the appropriate value for the `MaxDirectMemorySize` JVM option of each updater instance is: SUBSCRIPTION_CACHE_SIZE + 1Gb

3. Restart the updater instances in the Core Server system.

For further information about these system parameters, refer to their dedicated documentation available in the SAP CC 2021 System Parameter Reference documentation.

3.2.10.8 Modifying the Session Cache

Also known as the *charging session cache*, the session cache is a data cache that contains the data and the history of all the charging sessions for all the subscriber accounts managed by a given rater instance.

The session cache is available in the rater instances of the Core Server system.

To configure the session cache, execute the following procedure:

1. Modify the value of the following system parameters [page 45] for the rater instances of the Core Server system:
   - SESSION_MEMORY_SIZE, which defines the memory reserved for charging sessions (data and history) into each rater instance of the Core Server

     > **Note**
     >
     > To determine the size of the session cache, the best way consists in having a sample of objects contained in the cache and determine the average size of these objects.
     >
     > For further information, refer to the Determining the Size of the Cached Structures [page 84] section.

   - SESSION_AVG_OBJECT_SIZE, which defines the average size of the memory allocated for the storage of **one charging session object** in the session cache.

     > **Note**
     >
     > The appropriate value for this parameter can be found by retrieving the value of the `eavg` element in the last line of the value of the rater instance SESSION_STATUS indicator while the session cache is not empty. This value corresponds to the average size of the objects stored in the session cache, and should be used as the value for this parameter.
     >
     > The possible consequences of an incorrect value for this parameter are the following:
     > - If your parameter value is too small, objects may be removed from the session cache because there is not enough memory available in the cache (see the `vmen` element in the rater instance SESSION_STATUS indicator), leading to decreased performance.
     > - If your parameter value is too large, objects may also be removed from the session cache because there are not enough slots available in the cache (see the `vmap` element in the rater instance SESSION_STATUS indicator), leading to decreased performance.

   - SESSION_AVG_ID_SIZE, which defines the average size of the memory allocated for the storage of **one charging session history object** in the session cache

     > **Note**
     >
     > The appropriate value for this parameter can be found by retrieving the value of the `eavg` element in the last line of the value of the rater instance SESSION_HISTORY_STATUS indicator while the session cache is not empty. This value corresponds to the average size of the objects stored in the session cache, and should be used as the value for this parameter.
     >
     > The possible consequences of an incorrect value for this parameter are the following:
     > - If your parameter value is too small, objects may be removed from the session cache because there is not enough memory available in the cache (see the `vmen` element in the rater instance SESSION_HISTORY_STATUS indicator), leading to decreased performance.

- If your parameter value is too large, objects may also be removed from the session cache because there are not enough slots available in the cache (see the `vmap` element in the rater instance SESSION_HISTORY_STATUS indicator), leading to decreased performance.

- SESSION_AVG_NB_PER_HISTORY, which defines the average number of sessions per subscriber in the charging session cache.

  > **Note**
  >
  > The value of this parameter should correspond to the number of charging sessions that are opened concurrently per subscriber account when using the session-based charging methods available in the stateful service client.

- SESSION_MEMORY_INSTANCES, which defines the maximum number of subcaches into the session cache, as the session cache is divided into smaller subcaches for reducing contention.

  > **→ Recommendation**
  >
  > The value of this parameter is `computed` by default to the first prime number greater than 8 times the number of CPUs[45]. As a rule, there is no need to modify this default value.

2. For each rater instance in the Core Server system, check the value of the `MaxDirectMemorySize` JVM option defined in the `jstart.config` file.

   > **⚠ Caution**
   >
   > The size of the data caches handled by a given instance must not exceed the value of the `MaxDirectMemorySize` JVM option. If this constraint is not respected `OutOfMemory` errors will occur.

   > **→ Recommendation**
   >
   > The formula to compute the appropriate value for the `MaxDirectMemorySize` JVM option of each rater instance is: SUBSCRIPTION_CACHE_SIZE + SESSION_MEMORY_SIZE + SHARED_ALLOWANCE_CACHE_SIZE + 1 GB

3. Restart the rater instances in the Core Server system.

3.2.10.9 Modifying the Shared Allowance Cache

The shared allowance cache is a data cache that contains the shared allowances (that represent allowances shared between subscriber accounts), and the distributed counters (that correspond to persistent counters of shared allowances).

This shared allowance cache is available in the rater instances of the Core Server system, and contains data that is shared among the running rater instances Core Server system.

> **Note**
>
> For performance reasons, the shared allowance cache can be refreshed:

[45] Central Processing Unit

- Automatically and regularly by a dedicated scheduler [page 97]
- On demand by using the refresh_shared_allowance_cache dedicated command of the Admin+ user interface

To configure the shared allowance cache for the rater instances in the Core Server system, complete the following procedure:

1. Modify the value of the following system parameters [page 45] for the rater instances of the Core Server system:
   - SHARED_ALLOWANCE_CACHE_SIZE, which defines the total memory size reserved for the shared allowance cache for each rater of the Core Server system.

     > **Note**
     >
     > To determine the size of the provisioning cache for the rater instances of the Core Server system, the best way consists in having a sample of objects contained in the cache and determine the average size of these objects.
     >
     > For further information, refer to the Determining the Size of the Cached Structures [page 84] procedure.

   - SHARED_ALLOWANCE_OBJECT_AVERAGE_SIZE, which defines the average size of the memory allocated to the storage of one data object in the shared allowance cache.

     > **Note**
     >
     > The appropriate value for this parameter can be found by retrieving the value of the `eavg` element in the last line of the value of the rater instance SHARED_ALLOWANCE_CACHE_STATUS indicator while the shared allowance cache is not empty. This value corresponds to the average size of the objects stored in the shared allowance cache, and should be used as the value for this parameter.
     >
     > The possible consequences of an incorrect value for this parameter are the following:
     > - If your parameter value is too small, objects may be removed from the provisioning cache because there is not enough memory available in the cache (see the `vmen` element in the rater instance SHARED_ALLOWANCE_CACHE_STATUS indicator), leading to decreased performance.
     > - If your parameter value is too large, objects may also be removed from the provisioning cache because there are not enough slots available in the cache (see the `vmap` element in the rater instance SHARED_ALLOWANCE_CACHE_STATUS indicator), leading to decreased performance.

   - SHARED_ALLOWANCE_CACHE_INSTANCES, which defines the maximum number of subcaches into the shared allowance cache, as the shared allowance cache is divided into smaller subcaches for reducing contention.

     > **→ Recommendation**
     >
     > The value of this parameter is `computed` by default to the first prime number greater than 8 times the number of CPUs[46]. As a rule, there is no need to modify this default value.

2. For each rater instance of the Core Server system, check the value of the `MaxDirectMemorySize` JVM option defined in the `jstart.config` file.

[46] Central Processing Unit

> ⚠ Caution
>
> The size of the data caches handled by a given instance must not exceed the value of the `XX:MaxDirectMemorySize` JVM option. If this constraint is not respected `OutOfMemory` errors will occur.

> → Recommendation
>
> The formula to compute the appropriate value for the `XX:MaxDirectMemorySize` JVM option of each rater instance is: SUBSCRIPTION_CACHE_SIZE + SESSION_MEMORY_SIZE + SHARED_ALLOWANCE_CACHE_SIZE + 1 GB

3. Restart the rater instances of the Core Server system.

For further information about these system parameters, refer to their dedicated documentation available in the SAP CC 2021 System Parameter Reference documentation.

3.2.10.10 Modifying the SQL Statements Cache

> ⚠ Caution
>
> SAP SE recommends you to avoid changing these parameters on a production landscape, except under the direct supervision of a support specialist from your SAP Support Team.

To modify the cached structure containing the SQL request statements, modify the value of the following system parameters [page 45] for each concerned instance:

- SQLHELPER_STATEMENT_CACHE_SIZE, which defines the memory allocated to store the cached objects and is used by all instances when communicating with the Core Database
- RATING_SESSION_SQLHELPER_STATEMENT_CACHE_SIZE, which defines the memory allocated to store the cached objects and is used by the rater instances when communicating with the Session Database

For further information about these system parameters, refer to their dedicated documentation available in the SAP CC 2021 System Parameter Reference documentation.

3.2.10.11 Enabling the Cache Warm-Up Mechanism

To increase the reactivity of the Core Server system during online and offline charging operations, the content of the cached structures can be pre-loaded from the Core Database and Session Database in order to avoid latency. These pre-loading operations rely on a cache warm-up mechanism that can be enabled or disabled to fit specific needs.

> ⓘ Note
>
> The cache warm-up mechanism is **disabled** by default.

To modify the configuration of the cache warm-up mechanism, execute the following procedure:

Modify the value of the following system parameters [page 45] for the rater and guider instances, taking into account the existing constraints between these parameters:

- CACHE_WARMUP_THREAD_COUNT, by defining a number of thread allocated to warm-up operations, that usually equals to the number of available CPUs[47] minus one
- SQLHELPER_CONNECTION_COUNT, by adding new connections to the Core Database
- Only for rater: RATING_SESSION_SQLHELPER_CONNECTION_COUNT, by adding new connections to the Session Database (when the Session Database is used within your landscape)

Example

To permanently disable the cache warm-up mechanism for all rater instances of the Core Server system, execute the following command:

```
set CACHE_WARMUP_THREAD_COUNT 0 all rater
```

`set SQLHELPER_CONNECTION_COUNT X all rater`, where X corresponds to the previous value of this parameter, decreased by the number of threads previously allocated to warm-up operations

Example

To permanently disable the cache warm-up mechanism for all guider instances of the Core Server system, execute the following command:

```
set CACHE_WARMUP_THREAD_COUNT 0 all guider
```

`set SQLHELPER_CONNECTION_COUNT X all guider`, where X corresponds to the previous value of this parameter, decreased by the number of threads previously allocated to warm-up operations

Example

To temporary allocate 4 threads to warm-up operations performed by a given instance of the Core Server system, execute the following commands:

`set SQLHELPER_CONNECTION_COUNT X memory <INSTANCE_ID>`, where X corresponds to the previous value of this parameter, increased by 4

```
set CACHE_WARMUP_THREAD_COUNT 4 memory <INSTANCE_ID>
```

Where `<INSTANCE_ID>` corresponds to the targeted rater or guider

Smart Warmup with the Smart Distribution Mode

When the cache warm-up mechanism is activated, it is possible to configure the Core Server system in order to activate a smart execution mode for the partitions distribution operations performed by the primary dispatcher. This execution mode gives the possibility to control the warm-up operations performed by the targeted instances during these partitions distribution operations, and thus to control the overall impact of warm-up operations on the online stateful charging operations.

To activate and configure this smart distribution mode, refer to the Distributing Partitions to guiders and raters Using a Smart Mode [page 310] section of this documentation.

[47] Central Processing Unit

3.2.10.12 Modifying the Network Data Transfer (NDT) Mechanism

To increase the reactivity of the Core Server system, the content of the cached structures can be transferred directly from one system instance to another using the network. These network transfer operations decrease the overall solicitation of the Core Database and Session Database, and decrease the duration of the warm-up operations performed by raters and guiders during partition redistribution.

> ⓘ Note
>
> **The network data transfer (NDT) mechanism is enabled by default.**
>
> Please note that the default setting is disabled as of SAP CC 2022 FPS 2 Patch Level 2. Check the system parameter values in your SAP system landscapes and determine if you need to change them. See Checking the Status of the Network Data Transfer (NDT) Mechanism.

As a system administrator, to modify the configuration of the network data transfer (NDT) mechanism, complete the following procedure:

1. Launch your favorite user interface [page 45] and log on to as the administrator of the Core Server system.
2. Modify the value of the following system parameters for each dispatcher instance of the Core Server system:
   - RATER_PARTITION_SWITCH_CACHE_TRANSFER_ENABLED, which gives the possibility to enable or disable the use of network data transfer operations when switching partitions handled by rater instances.
   - GUIDER_PARTITION_SWITCH_CACHE_TRANSFER_ENABLED, which gives the possibility to enable or disable the use of network data transfer operations when switching partitions handled by guider instances.
   - DATA_TRANSFER_REQUEST_HANDLER_QUEUE_SIZE, which gives the possibility to specify the maximum number of Class1 and Class2 requests that can be queued in the dispatcher instance.
   - DATA_TRANSFER_REQUEST_HANDLER_THREAD_COUNT, which gives the possibility to specify the number of threads dedicated to the processing of Class1 and Class2 requests in the dispatcher instance.
3. Modify the value of the following system parameters for each rater and/or guider instance of the Core Server system:
   - DATA_TRANSFER_CLASS1_QUEUE_SIZE and DATA_TRANSFER_CLASS2_QUEUE_SIZE, which give the possibility to specify the maximum number of respectively Class1 and Class2 requests that can be queued in the instance

- DATA_TRANSFER_CLASS1_THREAD_COUNT and DATA_TRANSFER_CLASS2_THREAD_COUNT, which give the possibility to specify the number of threads allocated to the instance for processing respectively Class1 and Class2 requests
- DATA_TRANSFER_CLASS1_RESPONSE_HANDLER_THREAD_COUNT and DATA_TRANSFER_CLASS2_RESPONSE_HANDLER_THREAD_COUNT, which give the possibility to specify the number of threads allocated to the instance for processing responses to respectively Class1 and Class2 requests
- DATA_TRANSFER_CLASS2_MAX_SEND_THROUGHPUT, which gives the possibility to specify the maximum network bandwidth that the instance can use to respond to Class2 requests
- DATA_TRANSFER_CLASS2_MAX_RECEIVE_THROUGHPUT, which gives the possibility to specify the maximum network bandwidth that the instance can use to receive responses to Class2 requests
- DATA_TRANSFER_CLASS2_MAX_MESSAGE_SIZE, which gives the possibility to specify the maximum size of messages that can be sent by the instance when responding to Class2 requests
- DATA_TRANSFER_CLASS1_TIMEOUT and DATA_TRANSFER_CLASS2_TIMEOUT, which give the possibility to specify the maximum time (in seconds) that the instance can take to respond to respectively Class1 and Class2 requests

4. Restart the instances that relate to your system parameter changes.
5. **For expert system administrators**, you can consider some optimization and fine-tuning. See Fine-Tuning the Network Data Transfer (NDT) Mechanism [page 96].

For further information about these system parameters, refer to their dedicated documentation available in the SAP CC 2021 System Parameter Reference documentation.

3.2.10.12.1 Fine-Tuning the Network Data Transfer (NDT) Mechanism

Procedure

> ⚠ Caution
>
> Fine-tuning may put the system at risk. Only expert system administrators can consider these optimizations in production SAP system landscapes.

The Network Data Transfer (NDT) mechanism relies on two different types of messages (requests and responses) used by both rater instances and guider instances:

- Class1 messages correspond to data transfer requests and responses performed during priority operations such as real-time operations, administrations queries during data transfer operations, and so on
- Class2 messages correspond to data transfer requests and responses performed during nonpriority operations such as data transfer operations in a context of cache warm-up operations

To optimize your SAP CC Core Server system, you can fine-tune some system parameters. For optimization purpose, as an expert system administrator, you can control the behavior of these messages during data transfer operations by configuring:

1. The resources allocated to rater instances and guider instances when processing Class1 or Class2 messages (number of threads, size of queues):
   - DATA_TRANSFER_CLASS1_THREAD_COUNT
   - DATA_TRANSFER_CLASS2_THREAD_COUNT
   - DATA_TRANSFER_CLASS1_QUEUE_SIZE
   - DATA_TRANSFER_CLASS2_QUEUE_SIZE
2. The bandwidth allocated to rater instances and guider instances when dealing with Class2 messages, to avoid interfering with real-time services like stateful charging:
   - DATA_TRANSFER_CLASS2_MAX_RECEIVE_THROUGHPUT
   - DATA_TRANSFER_CLASS2_MAX_SEND_THROUGHPUT
3. The size of the Class2 messages, to avoid overloading the network:
   - DATA_TRANSFER_CLASS2_MAX_MESSAGE_SIZE

Test and adjust your changes in quality assurance SAP system landscapes.

3.2.10.13 Configuring the Automated Refreshment of the Shared Allowance Caches

Procedure

To modify the configuration of the scheduler in charge of refreshing the cached structure containing the shared allowances, execute the following procedure:

1. Launch the Cockpit or Admin+ user interface and identify as the administrator of the Core Server system.
2. For each dispatcher instance in the Core Server system, modify the value of the following parameters:
   - SHARED_ALLOWANCE_CACHE_REFRESH_SCHEDULER_ENABLED, which gives the possibility to enable or disable the execution of the scheduler.
   - SHARED_ALLOWANCE_CACHE_REFRESH_SCHEDULER_RECURRENCE, which defines the recurrence period of the scheduler.
3. Restart the dispatcher instances in the Core Server system.

3.2.11 Data Files Generation (Item Files)

Keywords [page 98]

Preliminary Notes [page 98]

Description [page 98]

Customizing the Charging Output Integration Framework (CIF) [page 99]

As a "Customizing" activity for expert implementers, you can fine configure the generations of output items from your SAP Convergent Charging system (Core Server).

Modifying the Item/Data Files Processor [page 138]

Modifying the Versions of Data Files Generated by Raters [page 144]

Enabling the File Name Sequencing for Data Files Generated by Raters [page 144]

Enabling the Line Numbering for Data Files Generated by Raters [page 145]

Enabling the Archive Files Cleaning for Data Files Generated by Raters [page 145]

Enabling the Precreation Policy for Data Files Generated by the Rater and Bulkloader Instances [page 146]

Modifying the Version of Error Files Generated by Bulkloaders [page 147]

Enabling the File Name Sequencing for Error Files Generated by Bulkloaders [page 147]

Enabling the Recording of File Metadata [page 148]

Enabling the Event Date Shifting Mechanism for Data Files Generated by the Rater Instances [page 148]

Enabling the Billable Item Validation for Data Files Generated by Rater Instances [page 149]

Enabling the Billable Item Validation for Data Files Generated by Bulkloader Instances (Cloud) [page 150]

Executing a Charging Plan for Consumption Item Mapping [page 150]

Requesting Verbose Logs for Creation of Billable Items (BITs) - Scenario with SAP S/4HANA Cloud Public Edition [page 152]

3.2.11.1 Keywords

Data files, error files, bulkloader

3.2.11.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- Data Files Generation and Data Files Bulk Loading Process description available in the SAP CC 2021 Application Help documentation
- Output Data Generation Process and Output Data Transfer Process (Billable Items Processing) system parameter group description available in the SAP CC 2021 System Parameter Reference documentation

3.2.11.3 Description

During the execution of the different processes handled by SAP Convergent Charging, data files are generated for multiple purposes such as integration with third-party systems, error management, notifications, logging, tracing, and so on. You can use the following procedures to configure the generation of the data files and thus fit your specific needs:

- Modifying the Item/Data Files Processor [page 138]
- Modifying the Versions of Data Files Generated by Raters [page 144]
- Enabling the File Name Sequencing for Data Files Generated by Raters [page 144]
- Enabling the Line Numbering for Data Files Generated by Raters [page 145]
- Enabling the Archive Files Cleaning for Data Files Generated by Raters [page 145]
- Enabling the Precreation Policy for Data Files Generated by the Rater and Bulkloader Instances [page 146]
- Modifying the Version of Error Files Generated by Bulkloaders [page 147]
- Enabling the File Name Sequencing for Error Files Generated by Bulkloaders [page 147]
- Enabling the Recording of File Metadata [page 148]
- Enabling the Event Date Shifting Mechanism for Data Files Generated by the Rater Instances [page 148]
- Enabling the Billable Item Validation for Data Files Generated by Rater Instances [page 149]
- Enabling the Billable Item Validation for Data Files Generated by Bulkloader Instances (Cloud) [page 150]
- Requesting Verbose Logs for Creation of Billable Items (BITs) - Scenario with SAP S/4HANA Cloud Public Edition [page 152]

3.2.11.4 Customizing the Charging Output Integration Framework (CIF)

As a "Customizing" activity for expert implementers, you can fine configure the generations of output items from your SAP Convergent Charging system (Core Server).

> → Remember
>
> This section is dedicated to expert implementers who are members of the implementation project team. In this *Customizing* activity, you design, configure, and tune the Charging Output Integration Framework (CIF) of SAP Convergent Charging (SAP CC).

The *CIF* (*Charging output Integration Framework*) is dedicated to the management of all output items produced by SAP Convergent Charging:

- Charged transactions (used by the old SAP CC *Transaction Integration Framework (TIF)*)
- Charged items
- Billable items (a model owned by SAP Convergent Invoicing)
- Chargeable items
- Consumption items (a model owned by SAP Convergent Invoicing)

Several processings can be configured to manage not only the transactions produced by the rating engine, but also the events involved in the generation of the usage transactions (for example allowing the "implementer" to configure the storage of the chargeable items' data in a data storage provided by SAP Convergent Invoicing).

Those processings are implemented by some processors that are configured and combined into `XML` configuration files.

As output items are managed by rater and instances, 2 CIF[48] configurations have to be defined: one for the rater instances and another one for the bulkloader instances.

[48] Charging output Integration Framework

The configurations are described in two configuration files and then stored in the Core Database thanks to the Setup Tool user interface by using the `cif import` command. Enter the two commands below:

```
setup cif import rater <raterConfigFile.xml>
setup cif import bulkloader <bulkloaderConfigFile.xml>
```

> → Tip
>
> The configurations can be exported from the Core Database system by using the `cif export` command.

> ⚠ Caution
>
> Importing configuration files changes the parameter values stored in the back-end database system (SAP CC Core Database). There is no `undo` command. SAP SE recommends that you first export your configuration files to a backup file that can be used to restore the SAP CC system.

The change of these configurations requires restarting the rater and bulkloader instances to consider the new behaviors.

The two configuration files can then be considered as logic that handle all the SAP CC output items (including the items in error).

Moreover, the system administrator can choose the operations to be applied on the generated output items. Indeed, according to their specifications, the processors can propose different operations to read, write, validate, store, route, modify, and transform the SAP CC output items.

3.2.11.4.1 Output Item Pipelines

Regarding the processings of the output items, the following pipelines are supported, and some of them can be mixed according to the type and the value of the items.

Performed by Rater Instances	Performed by Bulkloader Instances
1 - (rating context) → Charged Transaction **=>** Files (old SAP CC TIF[49] implementation. No more used but still available.)	
2 - (rating context) → Charged Item **=>** Files	
3 - (rating context) → Charged Item **=>** Files	-> Billable Item **=>** SAP Convergent Invoicing or SAP S/4HANA Cloud Public Edition
4 - Chargeable Item (+ additional fields) **=>** Files	
5 - Chargeable Item (+ additional fields) **=>** Files	-> Consumption Item **=>** SAP Convergent Invoicing

[49] Transaction Integration Framework

Legend:

- →: A transition between processors
- =>: A transition between processors that can use the CIF routing (chargeable items and charged items) or can benefit from the Customer Management Area (CMA) function.
  CMA can be used to route billable items and consumption items towards different landscapes connectable via SAP JCo[50] or via WS[51] to SAP CC according to a destination set for each customer account.

  > ### Example
  >
  > If a subscriber account A is initialized with a CMA's name X and a subscriber account B with a CMA name Y, then all the items from A will be sent to area X and all the items from B to the area Y if CMA is supported by the CIF configuration.

- ->: A transition between processor that additionally requires an item mapping definition to be processed.
  This mapping definition notion does not take part of the CIF framework but is implemented by the External item Integration Framework (EIF).

  > ### Note
  >
  > Each transformation (i.e. creating a billable item from a charged item or creating a consumption item from a chargeable item) requires a mapping definition and one or several processors to compute it (see the -> arrows in pipelines description above).
  >
  > A mapping definition is a table of correspondence between the property names of an item used as input by a processor, and the property names of another item that will be generated by the processor.
  >
  > Those mapping definitions are managed in SAP CC by using the Core Tool or the Cockpit user interfaces. However, some predefined mapping definitions are imported in SAP CC (and should not be modified in most cases) via the Setup Tool user interface.

> ### Note
>
> - **Pipeline 1:**
>   This pipeline is no longer used in SAP CC deployments (old SAP CC TIF) and the following example will be the unique example describing processors for handling charged transactions.
>   However, only this pipeline allows the user to implement its own transaction clearing processor (thanks to the wrapper processor). **But this can generate bad performance and transactional issues.**
>
>   > ### → Recommendation
>   >
>   > Please consult your SAP CC Support Team before starting such a development.

3.2.11.4.2 CIF Processors

To implement the previous pipelines [page 100], two groups of processors have to be considered:

[50] Java Connector
[51] Web Services

- One group is dedicated to transactions corresponding to the priced items from the SAP CC charging process (charged transactions, charged items, and billable items).
- The other group is dedicated to the usage events that have been used as usage input data to calculate prices (chargeable item and consumption item).

Among those two groups, some specialized subgroups of processors are dedicated to each type of items.

Furthermore, the CIF[52] transaction processor dedicated to the charged items (see `chargedItemTransactionProcessor` [page 112]) implements two permanent pipelines to process the postpaid and prepaid charged items separately.

In a similar way, two permanent pipelines are also implemented by the chargeable item transaction processor (see `chargeableItemTransactionProcessor`):

- One for the "charged" chargeable items
- Another one for the "acquired" chargeable items

Here is an overview about the processor structure implemented by the CIF:

- Transaction processor groups: `dispatchTransaction`, `transactionProcessorSequence`(*), `nopTransactionProcessor`

  > **Note**
  >
  > (*) However, there is an exception with `transactionProcessorSequence` which was initially designed only for transactions and which became the CIF root element when the event processors have been introduced in SAP CC (for compatibility reasons with previous configuration files. See the Processors [page 108] section).

  - Charged transaction group (deprecated old SAP CC Transaction Integration Framework framework): `transactionProcessorWrapper`
  - Charged item group: `chargedItemTransactionProcessor`
    With:
    - Postpaid Charged Item branch
    - Prepaid Charged Item branch
  - Billable item groups: `billableItemValidator`, `billableItemLoader`, `wsBillableItemLoader`
- Event processor groups:
  - Chargeable item group: `chargeableItemTransactionProcessor`
    With:
    - Charged Chargeable Item branch: the chargeable items on which the charging process has been applied
    - Acquired Chargeable Item branch: the chargeable items on which only the acquisition process has been performed
  - Consumption item groups: `consumptionItemValidator`, `consumptionItemLoader`

Those groups are independent and the processors belonging to transaction and event groups cannot be combined.

Some « generic » processors can be used by transaction and event pipelines.

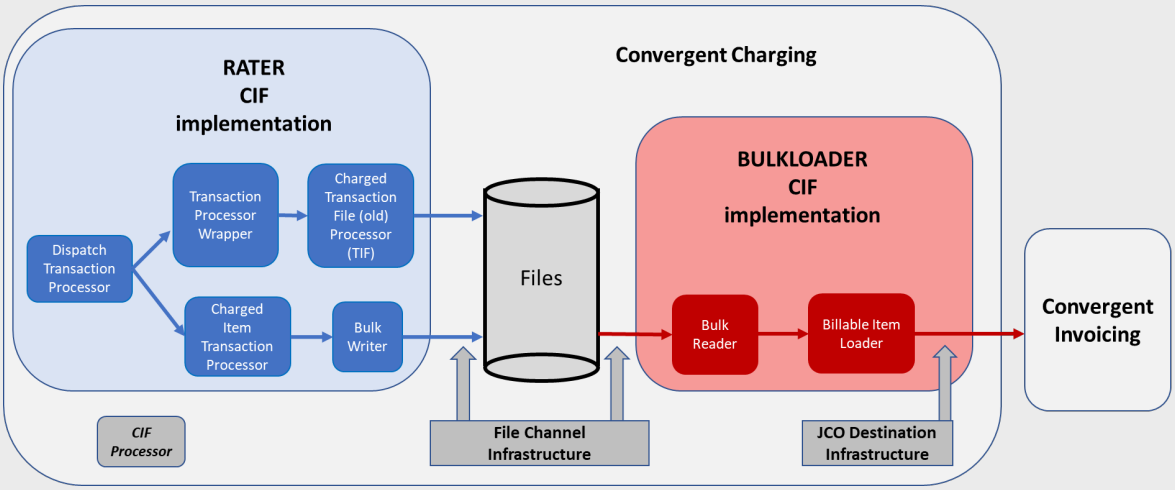
[52] Charging output Integration Framework

Generic Processors group: `itemSwitch`, `eventDateShifter`, `sequence`, `bulkWriter`, `bulkReader`, `customerManagementAreaRouting`, `errorHandling`, `multiJCODestination`, `multiDestination`.

Naturally, based on the role of each processor, some of them will be used by the rater instances and others by the bulkloader instances.

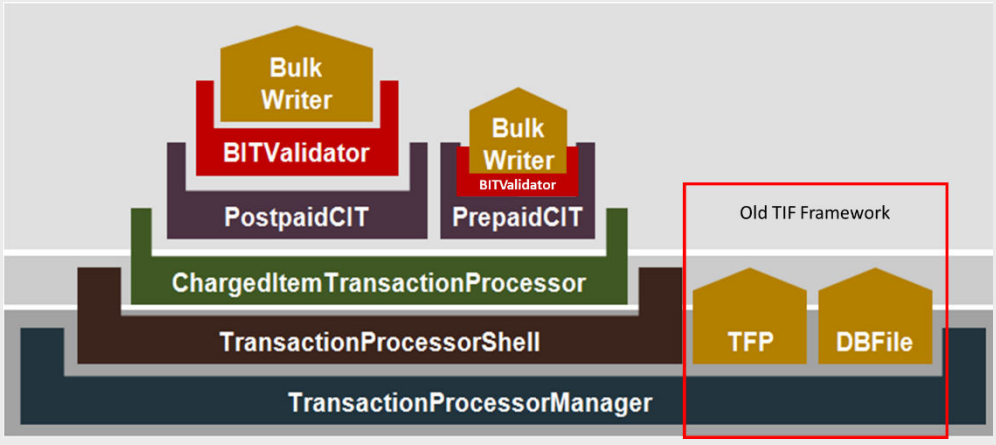
Example

The following schema details how the pipelines 1 and 3 are implemented with CIF processors:



Globally, considering an example of processes performed by a rater, the following sequence for managing the transactions occurs:

- Open postpaid charged item and prepaid charged item channels
- Check that each charged item can be mapped onto a billable item
- Write postpaid charged items into the channel dedicated to postpaid (in data files and error files for invalid items)
- Write prepaid charged items into the channel dedicated to prepaid (in data files and error files for invalid items)



Sample of a CIF configuration file used by raters:

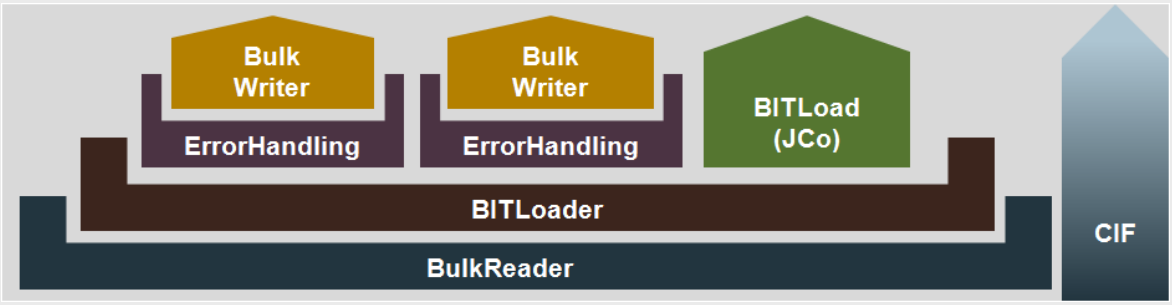
```
<chargedItemTransactionProcessor>
  <postpaidChargedItem>
    <billableItemValidator>
      <bulkWriter adminParameterPrefixName="POSTPAID_CIT_WRITER"
```

```
                    rootPath="CC_POSTPAID\Instance#@INSTANCE_ID@"
channelCount="2"
                    timeBasedFileRolloverPolicy="HOURLY"
                    maxSizeBasedFileRolloverPolicy="1M" deflate="true"
                    autoCommit="false" />
    </billableItemValidator>
  </postpaidChargedItem>
  <prepaidChargedItem>
    <billableItemValidator>
      <bulkWriter adminParameterPrefixName="PREPAID_CIT_WRITER"
                    rootPath="CC_PREPAID\Instance#@INSTANCE_ID@"
channelCount="2"
                    timeBasedFileRolloverPolicy="HOURLY"
                    maxSizeBasedFileRolloverPolicy="1M" deflate="true"
                    autoCommit="false" />
    </billableItemValidator>
  </prepaidChargedItem>
</chargedItemTransactionProcessor>
```

- `postpaidChargedItem` handles postpaid charged items
- `prepaidChargedItem` handles prepaid and refill charged items
- `billableItemValidator` checks that a charged item can be mapped onto a billable item
- `bulkWriter` writes charged items into files

Example

In a similar way, considering an example of processes performed by a bulkloader, the following sequence of actions occurs:




```
<bulkReader adminParameterPrefixName="POSTPAID_CIT_READER"
                    rootPath="CC_POSTPAID\Instance#@INSTANCE_ID@"
                    channelCount="2"
                    bufferSize="1000"
                    removeFile="true">
  <billableItemLoader adminParameterPrefixName="BIT_LOADER">
    <errorHandling errorType="invalidBillableItem">
      <bulkWriter
adminParameterPrefixName="BIT_LOADER_ERR_INVALID_BIT_WRITER"
                    rootPath="CC_CI_INVALID\Instance#@INSTANCE_ID@"
                    channelCount="2" timeBasedFileRolloverPolicy="HOURLY"
                    maxSizeBasedFileRolloverPolicy="1M"
                    deflate="false" autoCommit="true" />
    </errorHandling>
    <errorHandling errorType="communicationException">
      <bulkWriter
adminParameterPrefixName="BIT_LOADER_ERR_COM_EXCEPTION_WRITER"
                    rootPath="CC_CI_COM_EXCEPTION\Instance#@INSTANCE_ID@"
                    channelCount="2" timeBasedFileRolloverPolicy="HOURLY"
                    maxSizeBasedFileRolloverPolicy="1M"
                    deflate="false" autoCommit="true" />
    </errorHandling>
  </billableItemLoader>
```

```
</bulkReader>
```

- `bulkReader` loads postpaid charged items from files
- `billableItemLoader`maps charged items onto billable items, send them to SAP CI via JCo[53]
- `errorHandling`
- `bulkWriter`

3.2.11.4.3 File Channels and Destinations

The SAP CC implementer who configures does not need to care about how the data files or connections to SAP Convergent Invoicing are managed. Indeed, SAP Convergent Charging implements an infrastructure that provides the notions of **channels** and **destinations**. Only those notions are handled by the processors. The underlying management of files (including file naming, creation, closing, deletion, locking, errors), the JCo[54] connections to SAP Convergent Invoicing, and connections to WS[55] provided by SAP S/4HANA Cloud Public Edition are transparent for the user.

File Channel

A **File Channel** references a set of files used to manage the items (data files, error files, and control files). SAP CC implements some working directories in which the data files will be written by the rater instances and will be read by the bulkloader instances (however the bulkloader instances can also write data files too).

A "root" directory can be shared by all the rater and bulkloader instances but it is mandatory to use separated directories for each pair (rater, bulkloader) instances because only one instance at a time can read or write in a given directory.

For example, the `rootPath` attribute in the previous example (see `bulkWriter`), uses the variable `@INSTANCE_ID@` to create a working directory for each instance. Then, based on a `channelCount` attribute, the user can decide the number of data files opened simultaneously per processor instance, into which the processor is able to write/read items in parallel.

The two processors that use the file channels are mainly the `bulkWriter` [page 117] and `bulkReader` [page 120] processors.

Destinations

A **Destination** references a connector with SAP Convergent Charging (JCo) or a connector with SAP S/4HANA Cloud Public Edition (WS).

Destinations are used by the bulkloader instances in order to push the items into SAP Convergent Invoicing.

53 Java Connector
54 Java Connector
55 Web Services

Indeed, SAP Convergent Invoicing provides some connections based on the SAP JCo stack and SAP S/4HANA Cloud Public Edition provides some Web Services. That the reason why the CIF[56] requires to qualify the destination (JCo or WS) because SAP CC can be connected to SAP Convergent Charging and to SAP S/4HANA Cloud Public Edition billing service.

- The destination connected to SAP Convergent Invoicing (on premise) is called a JCo Destination. To create a JCo Destination in an SAP CC system, you can run the following Setup Tool command:

  ```
  setup jcodestination import -login=<login> -passwd=<pwd> <sapApplication
  (“CRM” ||“ERP”)> <destination name (unique)> <JCo destination from file>
  ```

  Refer to the JCo Destinations [page 161] section of this documentation for more information.
- The destination connected to Convergent Invoicing (in S/4HANA Cloud) landscape is called a WS Destination. To create a WS Destination in an SAP CC system, you can run the following Setup Tool command:

  ```
  setup wsdestination import -login=<login> -passwd=<pwd> <destination name
  (unique)> <WS destination from file>
  ```

  Refer to the WS Destinations [page 260] section of this documentation for more information.

A bulkloader instance requires at least a destination. However, the bulkloader instances can deal with multiple destinations by using the Customer Management Area (CMA) feature.

Only the destinations created at setup time are available and usable by the bulkloader instances.

> ⚠ Restriction
>
> It is not allowed in a CIF configuration, to simultaneously handle WS and JCo destinations. Indeed, these two destinations are used in two different integration scenarios.

3.2.11.4.4 Transition Between CIF Processors

Two types of transition can occur between CIF[57] processors: cascaded and sequenced.

- **Cascaded**: The result of a processor (parent) is transmitted to the children processors.
- **Sequenced**: Each processor listed in the sequence is executed independently from the previous one. However, each can use its parent output data.

> ❖ Example
>
> **cif.rater.config.xml**
>
> In this sample, cascaded processors are highlighted *in italic*, sequenced processors are **in bold**.

```
<transactionProcessorSequence>
  <chargedItemTransactionProcessor>
    <postpaidChargedItem>
  <eventDateShifter>
        <sequence>
          <billableItemValidator />
```

[56] Charging output Integration Framework
[57] Charging output Integration Framework

```
          <customerManagementAreaRouting>
            <bulkWriter adminParameterPrefixName="POSTPAID_CIT_WRITER"
rootPath="SAPCC_CIT/POSTPAID/Instance#@INSTANCE_ID@/@JCO_DESTINATION_NAME@"
channelCount="4"
              timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="true" deflateOutputFormat="GZIP"
storeFileInformation="true"
              fileVersion="2" lineNumbering="false"
fileSequenceId="CHARGED_ITEM" cleanArchivedFile="false"
preCreateFile="ROLLOVER" />
          </customerManagementAreaRouting>
        </sequence>
      </eventDateShifter>
    </postpaidChargedItem>
    <prepaidChargedItem>
  <eventDateShifter>
        <sequence>
          <billableItemValidator />
          <customerManagementAreaRouting>
            <bulkWriter adminParameterPrefixName="PREPAID_CIT_WRITER"
rootPath="SAPCC_CIT/PREPAID/Instance#@INSTANCE_ID@/@JCO_DESTINATION_NAME@"
channelCount="4"
              timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="true" deflateOutputFormat="GZIP"
storeFileInformation="false"
              fileVersion="2" lineNumbering="false"
fileSequenceId="CHARGED_ITEM" cleanArchivedFile="false"
preCreateFile="ROLLOVER" />
          </customerManagementAreaRouting>
        </sequence>
      </eventDateShifter>
    </prepaidChargedItem>
  </chargedItemTransactionProcessor>
</transactionProcessorSequence>
```

Each CIF processor is configured by setting values in their XML attributes.

Example

```
<bulkWriter adminParameterPrefixName="PREPAID_CIT_WRITER" rootPath="SAPCC_CIT/
PREPAID/Instance#@INSTANCE_ID@/@JCO_DESTINATION_NAME@" channelCount="4"
              timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="true" deflateOutputFormat="GZIP"
storeFileInformation="false"
              fileVersion="2" lineNumbering="false"
fileSequenceId="CHARGED_ITEM" cleanArchivedFile="false"
preCreateFile="ROLLOVER" />
```

It is possible to use in a processor some **variables**, depending on the instance by referencing them with `@VARIABLE_NAME@`.

The 3 variables that can be referenced are:

- `@INSTANCE_ID@`
- `@DESTINATION_NAME@`
- `@JCO_DESTINATION_NAME@`

See the examples in this documentation.

3.2.11.4.5 SAP CC Items and CIF Routing

Based on item names (as defined by their related item class), the user can select the items based on their origin and values of specified fields in order to route them on specific channels.

A processor must be defined for each route and the user must ensure that a pair (`bulkWriter`, `bulkReader`) must be defined for each channel.

The routing is handled by the `itemSwitch` processor. Its XML element defines a `switch/case/default` structure, with, for each case, an `itemCondition` based on the item name and a specific field value belonging to the item (see `itemSwitch` in Processors [page 108] section).

For each case, the user has to define a processor that will store the selected item in the related file channel.

3.2.11.4.6 Processors

> ⚠ Caution
>
> This section is dedicated to expert implementers who are members of the implementation project team. In this *Customizing* activity, you design, configure, and tune the Charging Output Integration Framework (CIF) of SAP Convergent Charging (SAP CC).

dispatchTransactionProcessor (old TIF framework) [page 109]

transactionProcessorWrapper (old TIF framework) [page 110]

transactionProcessorSequence [page 111]

nopTransactionProcessor [page 111]

chargedItemTransactionProcessor [page 112]

chargeableItemTransactionProcessor [page 113]

itemSwitch [page 114]

bulkWriter [page 117]

bulkReader [page 120]

eventDateShifter [page 124]

errorHandling [page 125]

billableItemValidator and consumptionItemValidator [page 127]

customerManagementAreaRouting [page 128]

multiJCoDestination and multiDestination [page 130]

billableItemLoader [page 133]

wsBillableItemLoader [page 135]

consumptionItemLoader [page 136]

3.2.11.4.6.1 dispatchTransactionProcessor (old TIF framework)

`dispatchTransactionProcessor` (old TIF framework)

Description

Handles the ChargedTransaction defined by the old transaction format (TIF[58]) and the ChargedItem (CIF) simultaneously.

CIF Usage

Rater instances only

> ⚠ Caution
>
> Reserved for old transaction format (TIF) - **DEPRECATED**

Attributes

None

Parent XML Element

None (root)

Example

```
<dispatchTransactionProcessor>
   <transactionProcessorWrapper class="a java class implementing the TIF API" />
   <transactionProcessorSequence>
      <chargedItemTransactionProcessor>
         ...
      </chargedItemTransactionProcessor>
      <chargeableItemTransactionProcessor>
         ...
      <chargeableItemTransactionProcessor>
   </transactionProcessorSequence>
<dispatchTransactionProcessor>
```

[58] Transaction Integration Framework

3.2.11.4.6.2 transactionProcessorWrapper (old TIF framework)

`transactionProcessorWrapper` (old TIF framework)

Description

Runs a specific transaction processor that handles the old TIF transaction format and implements the TIF transaction processor interface.

CIF Usage

Rater instances only

> ⚠ Caution
>
> Reserved for old transaction format (TIF) - **DEPRECATED**

The user can implement its own transaction processor by developing its own Java code. However, SAP SE informs users that the processor code will be executed into the Business and DB transaction created by the charging process. That means that the customized processor code can impact performance and the DB ACID principles implemented by SAP CC (deprecated).

Attributes

`class`: the class name of the TIF transaction processor

Parent XML Element

`dispatchTransactionProcessor` [page 109]

Example

```
<dispatchTransactionProcessor>
   <transactionProcessorWrapper class="a java class implementing the TIF API" />
   <transactionProcessorSequence>
      <chargedItemTransactionProcessor>
         ...
      </chargedItemTransactionProcessor>
      <chargeableItemTransactionProcessor>
         ...
      <chargeableItemTransactionProcessor>
   </transactionProcessorSequence>
<dispatchTransactionProcessor>
```

3.2.11.4.6.3 transactionProcessorSequence

> → Remember
>
> This section is dedicated to expert implementers who are members of the implementation project team. In this *Customizing* activity, you design, configure, and tune the Charging Output Integration Framework (CIF) of SAP Convergent Charging (SAP CC).

`transactionProcessorSequence`

Description

Used as root XML element (when `dispatchTransactionProcessor` element is not used) to sequence the execution of transaction processors, whatever the item types to process. The maximum number of processors to sequence is limited to 100.

CIF Usage

Rater instances

XML Attributes

None

Parent XML Elements

None (root) or `dispatchTransactionProcessor` [page 109]

Example

```
<transactionProcessorSequence
   <chargedItemTransactionProcessor>
      ...
   </chargedItemTransactionProcessor>
   <chargeableItemTransactionProcessor>
      ...
   <chargeableItemTransactionProcessor>
</transactionProcessorSequence
```

3.2.11.4.6.4 nopTransactionProcessor

> → Remember
>
> This section is dedicated to expert implementers who are members of the implementation project team. In this *Customizing* activity, you design, configure, and tune the Charging Output Integration Framework (CIF) of SAP Convergent Charging (SAP CC).

`nopTransactionProcessor`

Description

Empty processor.

CIF Usage

Rater instances

XML Attributes

None

Parent XML Element

None

3.2.11.4.6.5 chargedItemTransactionProcessor

> → Remember
>
> This section is dedicated to expert implementers who are members of the implementation project team. In this *Customizing* activity, you design, configure, and tune the Charging Output Integration Framework (CIF) of SAP Convergent Charging (SAP CC).

`chargedItemTransactionProcessor`

Description

Performs processings on "charged items".

Implements 2 default branches to route:

- Postpaid charged items: you specify the `postpaidChargedItem` XML element.
- Prepaid charged items: you specify the `prepaidChargedItem` XML element.

> ⓘ Note
>
> The prepaid route is optional because SAP Convergent Charging can be configured to not produce prepaid transactions.

CIF Usage

Rater instances

XML Attributes

None

Parent XML Element

`transactionProcessorSequence` [page 111]

Example

```
<transactionProcessorSequence>
   <chargedItemTransactionProcessor>
      <postpaidChargedItem>
         ...
      </postpaidChargedItem>
      <prepaidChargedItem>
         ...
      </prepaidChargedItem>
   </chargedItemTransactionProcessor>
</transactionProcessorSequence>
```

3.2.11.4.6.6 chargeableItemTransactionProcessor

> → Remember
>
> This section is dedicated to expert implementers who are members of the implementation project team. In this *Customizing* activity, you design, configure, and tune the Charging Output Integration Framework (CIF) of SAP Convergent Charging (SAP CC).

`chargeableItemTransactionProcessor`

Description

Performs processings on "chargeable items."

Implements 2 branches to route:

- Charged chargeable items: the chargeable items that have been processed by the charging process. You specify the `chargedChargeableItem` XML element.
- Acquired chargeable items: the chargeable items that have been processed by the acquisition process. You specify the `acquiredChargeableItem` XML element.

A processor is mandatory for both routes.

CIF Usage

Rater instances

XML Attributes

None

Parent XML Element

`transactionProcessorSequence` [page 111]

Example

```
<transactionProcessorSequence>
   <chargeableItemTransactionProcessor>
      <chargedChargeableItem>
         ...
      </chargedChargeableItem>
      <acquiredChargeableItem>
      ...
      </acquiredChargeableItem>
   </chargeableItemTransactionProcessor>
</transactionProcessorSequence>
```

3.2.11.4.6.7 itemSwitch

> → Remember
>
> This section is dedicated to expert implementers who are members of the implementation project team. In this *Customizing* activity, you design, configure, and tune the Charging Output Integration Framework (CIF) of SAP Convergent Charging (SAP CC).

`itemSwitch`

Description

Routes items towards specific channels based on their name and values of specified fields. For each case, the item is sent to a specific processor and, if no case is found, a default one is applied. This switcher is limited to 10 “switch/case” conditions.

The selection of the items is implemented thanks to the `itemFieldCondition` that parses all the items received by the parent processor. The below attributes allow the user to select the items.

CIF Usage

Rater instances

Attributes

- `ofClass`: the name of the item (defined by the item class) which will be parsed by the ItemSwitch processor. This attribute is set with the ID of the chargeable item class or with the ID of the charged item class corresponding to the items to filter.
- `named` (optional): the name of the field that must be checked to validate the case or not. Please note that the `ItemSwitch` processor does not check that this field name is defined by the corresponding class (ChargeableItemClass or the ChargedItemClass). However, if this attribute stays empty or does not exist, no condition on field will be applied. Then all items matching the `ofClass` attribute will be accepted by the condition.
- `equals` (optional): the value that must be compared to the field value. This is a string comparison so if the item field type is a number or a date, the `toString()` method is applied before the comparison. Empty value (“”) is accepted as a correct value for checking that the field value is empty or null.

> **Note**
>
> The field conditions are evaluated according to their declaration order. As soon as a condition is verified, the next ones are dropped.
>
> At last, the default case is applied if no condition is verified.

Parent XML Elements

`postpaidChargedItem`, `prepaidChargedItem`, `chargedChargeableItem`, `acquiredChargeableItem`

Example

cif.rater.config.xml

Charged items *in italic*, chargeable items **in bold**.

```
<?xml version="1.0" encoding="utf-8"?>
<transactionProcessorSequence>
   <chargedItemTransactionProcessor>
      <postpaidChargedItem>
<itemSwitch>
<!-- ========================== -->
<!-- VOI : BITTEXT50 = prepaid -->
<!-- ========================== -->
<case>
  <itemFieldCondition ofClass="VOI" named="BITTEXT50" equals="prepaid" />
<eventDateShifter>
<sequence>
<billableItemValidator />
<bulkWriter adminParameterPrefixName="POSTPAID_CIT_WRITER_01"
rootPath="SAPCC_CIT/POSTPAID/Instance#@INSTANCE_ID@/voi_bittext50_prepaid"
channelCount="2" timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="true" deflateOutputFormat="GZIP"
autoCommit="false" fileVersion="2" fileSequenceId="CHARGED_ITEM" />
</sequence>
</eventDateShifter>
</case>
<!-- ================================= -->
<!-- VOI : BITTEXT50 = <empty string> -->
<!-- ================================= -->
<case>
  <itemFieldCondition ofClass="VOI" named="BITTEXT50" equals="" />
<eventDateShifter>
<sequence>
<billableItemValidator />
<bulkWriter adminParameterPrefixName="POSTPAID_CIT_WRITER_02"
rootPath="SAPCC_CIT/POSTPAID/Instance#@INSTANCE_ID@/voi_bittext50_empty_string"
channelCount="2" timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="true" deflateOutputFormat="GZIP"
autoCommit="false" fileVersion="2" fileSequenceId="CHARGED_ITEM" />
</sequence>
</eventDateShifter>
</case>
<!-- ========================== -->
<!-- DAT : DAT = prepaid -->
<!-- ========================== -->
<case>
  <itemFieldCondition ofClass="DAT" named="BITTEXT50" equals="prepaid" />
<eventDateShifter>
<sequence>
<billableItemValidator />
<bulkWriter adminParameterPrefixName="POSTPAID_CIT_WRITER_03"
rootPath="SAPCC_CIT/POSTPAID/Instance#@INSTANCE_ID@/dat_bittext50_prepaid"
channelCount="2" timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="true" deflateOutputFormat="GZIP"
```

```
autoCommit="false" fileVersion="2" fileSequenceId="CHARGED_ITEM" />
</sequence>
</eventDateShifter>
</case>
<!-- ================= -->
<!-- DEFAULT ROUTE -->
<!-- ================= -->
<default>
<eventDateShifter>
<sequence>
<billableItemValidator />
<bulkWriter adminParameterPrefixName="POSTPAID_CIT_WRITER" rootPath="SAPCC_CIT/
POSTPAID/Instance#@INSTANCE_ID@"
channelCount="2" timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="true" deflateOutputFormat="GZIP"
autoCommit="false" fileVersion="2" fileSequenceId="CHARGED_ITEM" />
</sequence>
</eventDateShifter>
</default>
</itemSwitch>
</postpaidChargedItem>
<prepaidChargedItem>
<eventDateShifter>
<sequence>
<billableItemValidator />
<bulkWriter adminParameterPrefixName="PREPAID_CIT_WRITER" rootPath="SAPCC_CIT/
PREPAID/Instance#@INSTANCE_ID@" channelCount="2"
timeBasedFileRolloverPolicy="HOURLY" maxSizeBasedFileRolloverPolicy="1M"
deflate="true" deflateOutputFormat="GZIP" autoCommit="false"
fileVersion="2" fileSequenceId="CHARGED_ITEM" />
</sequence>
</eventDateShifter>
</prepaidChargedItem>
</chargedItemTransactionProcessor>
<!--
================================================================================ -->
<!-- Chargeable item / consumption items section -->
<!--
================================================================================ -->
<chargeableItemTransactionProcessor>
<chargedChargeableItem>
<itemSwitch>
<!-- ========================== -->
<!-- VOI_TICKET : text = prepaid -->
<!--Same destination than Charged Item when BITTEXT50 = prepaid -->
<!-- ========================== -->
<case>
    <itemFieldCondition ofClass="VOI_TICKET" named="user.text"
equals="prepaid" />
<sequence>
<consumptionItemValidator />
<bulkWriter adminParameterPrefixName="CHARGED_CI_WRITER_01" rootPath="SAPCC_CI/
CHARGED/Instance#@INSTANCE_ID@/voi_ticket_prepaid"
channelCount="2" timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="true" deflateOutputFormat="GZIP"
autoCommit="false" fileVersion="2" fileSequenceId="CHARGEABLE_ITEM" />
</sequence>
</case>
<!-- ================================= -->
<!-- VOI_TICKET : text = <empty string> -->
<!-- Same destination than Charged Item when VOI : BITTEXT50 = <empty string> -->
<!-- ================================= -->
<case>
    <itemFieldCondition ofClass="VOI_TICKET" named="user.text" equals="" />
<sequence>
<consumptionItemValidator />
<bulkWriter adminParameterPrefixName="CHARGED_CI_WRITER_02" rootPath="SAPCC_CI/
CHARGED/Instance#@INSTANCE_ID@/voi_ticket_empty_string"
```

```
channelCount="2" timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="true" deflateOutputFormat="GZIP"
autoCommit="false" fileVersion="2" fileSequenceId="CHARGEABLE_ITEM" />
</sequence>
</case>
<!-- ================================== -->
<!-- VOI_TICKET : UNKNOWN_ROUTING_FIELD = destination -->
<!-- Specific destination (Billable Items and Charged Items are separated in
different system -->
<!-- ================================== -->
<case>
<itemFieldCondition ofClass="VOI_TICKET" named="user.UNKNOWN_ROUTING_FIELD"
equals="ERP300" />
<sequence>
<consumptionItemValidator />
<bulkWriter adminParameterPrefixName="CHARGED_CI_WRITER_04" rootPath="SAPCC_CI/
CHARGED/Instance#@INSTANCE_ID@/voi_ticket_unknown_field_name"
channelCount="2" timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="true" deflateOutputFormat="GZIP"
autoCommit="false" fileVersion="2" fileSequenceId="CHARGEABLE_ITEM" />
</sequence>
</case>
<!-- ================= -->
<!-- DEFAULT DIRECTORY -->
<!-- ================= -->
<default>
<sequence>
<consumptionItemValidator />
<bulkWriter adminParameterPrefixName="CHARGED_CI_WRITER" rootPath="SAPCC_CI/
CHARGED/Instance#@INSTANCE_ID@" channelCount="2"
timeBasedFileRolloverPolicy="HOURLY" maxSizeBasedFileRolloverPolicy="1M"
deflate="true" deflateOutputFormat="GZIP" autoCommit="false"
fileVersion="2" fileSequenceId="CHARGEABLE_ITEM" />
</sequence>
</default>
    </itemSwitch>
</chargedChargeableItem>
<acquiredChargeableItem>
<sequence>
<consumptionItemValidator />
<bulkWriter adminParameterPrefixName="ACQUIRED_CI_WRITER" rootPath="SAPCC_CI/
ACQUIRED/Instance#@INSTANCE_ID@" channelCount="2"
timeBasedFileRolloverPolicy="HOURLY" maxSizeBasedFileRolloverPolicy="1M"
deflate="true" deflateOutputFormat="GZIP" autoCommit="false"
fileVersion="2" fileSequenceId="CHARGEABLE_ITEM" />
        </sequence>
      </acquiredChargeableItem>
   </chargeableItemTransactionProcessor>
</transactionProcessorSequence>
```

3.2.11.4.6.8 bulkWriter

> → Remember
>
> This section is dedicated to expert implementers who are members of the implementation project team. In this *Customizing* activity, you design, configure, and tune the Charging Output Integration Framework (CIF) of SAP Convergent Charging (SAP CC).

`bulkWriter`

Description

Generates item data files.

CIF Usage

Rater and bulkloader instances

XML Attributes

Note that the attributes below are translated into parameters by SAP Convergent Charging. Refer to the Output Data Generation Process section of the SAP CC 2021 System Parameter Reference documentation.

- `adminParameterPrefixName` represents the parameter name prefix set by the translation by SAP CC of each attribute (it replaces the "XXX" prefix below).
  For example: if `adminParameterPrefixName` = `POSTPAID_CIT_WRITER` then the other `XML` attributes are translated by SAP CC into persistent parameters with their related value: POSTPAID_CIT_WRITER_ROOT_PATH, POSTPAID_CIT_WRITER_CHANNEL_COUNT, ...
  See each parameter link below for more info.
- `rootPath`: contains the relative or absolute path of the directory containing the data files to create. See XXX_ROOT_PATH (BulkWriter) for more information.
- `channelCount`: represents the number of channels to use when writing the data files in a directory (cannot be lower than 1). See XXX_CHANNEL_COUNT (BulkWriter) for more information.
- `timeBasedFileRolloverPolicy`: defines the rolling time-based policy for the generated data files. It is triggered when system current date reaches time based rollover policy. See XXX_TIME_BASED_FILE_ROLLOVER_POLICY (BulkWriter) for more information.
- `maxSizeBasedFileRolloverPolicy`: defines the rolling size-based policy for the generated data files. It is triggered when the size of the file reaches a certain amount. See XXX_MAX_SIZE_BASED_FILE_ROLLOVER_POLICY (BulkWriter) for more information.
- `deflate` and `deflateOutputFormat`: define the compression policy (respecting the RFC 1951 specifications). See XXX_DEFLATE (BulkWriter) and XXX_DEFLATE_OUTPUT_FORMAT (BulkWriter) for more information
- `fileVersion`: represents the version to use when generating data files. See XXX_FILE_VERSION (BulkWriter) for more information.
- `lineNumbering`: writes the line numbers in the generated data files. See XXX_LINE_NUMBERING (BulkWriter) for more information.
- `fileSequenceId`: represents the identifier to use for sequencing the names of the generated data files. See XXX_FILE_SEQUENCE_ID (BulkWriter) for more information.
- `preCreateFile`: enables or disables the pre-creation of data files. See XXX_PRE_CREATE_FILE (BulkWriter) for more information.
- `cleanArchivedFile`: removes the uncommitted items from the generated data files. See XXX_CLEAN_ARCHIVED_FILE (BulkWriter) for more information.
- `storeFileInformation`: stores (or not) information about the generated files into the `CHARGED_ITEM_FILES` table of the Core Database. Default is true.

  > **Note**
  >
  > This attribute must be set to `false` with the SAP S/4HANA Cloud Public Edition configuration (ws mode) or with prepaid charged items. This way, file information used for rerating purpose is not stored

in the database, because rerating is not a supported feature in these configurations (ws mode or prepaid).

Parent XML Elements

`errorHandling`, `itemSwitch` (`case`), `sequence`, `customerManagementAreaRouting`, `chargedItemTransactionProcessor`, (`prepaidChargedItem` and `postpaidChargedItem`), `chargeableItemTransactionProcessor` (`chargedChargeableItem` and `aquiredChargeableItem`)

Example

From a rater CIF config:

```
<transactionProcessorSequence>
   <chargedItemTransactionProcessor>
      <postpaidChargedItem>
         <eventDateShifter>
            <sequence>
               <billableItemValidator />
               <customerManagementAreaRouting>
                  <bulkWriter adminParameterPrefixName="POSTPAID_CIT_WRITER"
rootPath="SAPCC_CIT/POSTPAID/Instance#@INSTANCE_ID@/@JCO_DESTINATION_NAME@"
channelCount="4"
                     timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="true" deflateOutputFormat="GZIP"
storeFileInformation="true" fileVersion="2" lineNumbering="false"
fileSequenceId="CHARGED_ITEM" cleanArchivedFile="false"
preCreateFile="ROLLOVER" />
               </customerManagementAreaRouting>
            </sequence>
         </eventDateShifter>
      </postpaidChargedItem>
      <prepaidChargedItem>
         <eventDateShifter>
            <sequence>
               <billableItemValidator />
               <customerManagementAreaRouting>
                  <bulkWriter adminParameterPrefixName="PREPAID_CIT_WRITER"
rootPath="SAPCC_CIT/PREPAID/Instance#@INSTANCE_ID@/@JCO_DESTINATION_NAME@"
channelCount="4"
                     timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="true" deflateOutputFormat="GZIP"
storeFileInformation="false" fileVersion="2" lineNumbering="false"
fileSequenceId="CHARGED_ITEM" cleanArchivedFile="false"
preCreateFile="ROLLOVER" />
               </customerManagementAreaRouting>
            </sequence>
         </eventDateShifter>
      </prepaidChargedItem>
   </chargedItemTransactionProcessor>
</transactionProcessorSequence>
```

From a bulkloader CIF config that works with the rater config above:

```
<sequence>
   <multiJCoDestination>
      <bulkReader adminParameterPrefixName="POSTPAID_CIT_READER"
rootPath="SAPCC_CIT/POSTPAID/Instance#@INSTANCE_ID@/@JCO_DESTINATION_NAME@"
channelCount="4"
         bufferSize="1000" removeFile="true">
         <billableItemLoader adminParameterPrefixName="POSTPAID_BIT_LOADER"
destination="@JCO_DESTINATION_NAME@" autoCommit="true">
            <errorHandling errorType="invalidBillableItem">
```

```
               <bulkWriter
adminParameterPrefixName="POSTPAID_CIT_ERR_INVALID_WRITER" rootPath="SAPCC_CIT/
POSTPAID/INVALID/Instance#@INSTANCE_ID@/@JCO_DESTINATION_NAME@" channelCount="4"
timeBasedFileRolloverPolicy="HOURLY" maxSizeBasedFileRolloverPolicy="1M"
deflate="false" storeFileInformation="true" fileVersion="2"
fileSequenceId="POSTPAID_CIT_ERR_INVALID"/>
            </errorHandling>
            <errorHandling errorType="communicationException">
               <bulkWriter
adminParameterPrefixName="POSTPAID_CIT_ERR_COM_EXCEPTION_WRITER"
rootPath="SAPCC_CIT/POSTPAID/COM_EXCEPTION/Instance#@INSTANCE_ID@/
@JCO_DESTINATION_NAME@" channelCount="4" timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="false" storeFileInformation="true"
fileVersion="2" fileSequenceId="POSTPAID_CIT_ERR_COM_EXCEPTION"/>
            </errorHandling>
         </billableItemLoader>
      </bulkReader>
   </multiJCoDestination>
   <multiJCoDestination>
      <bulkReader adminParameterPrefixName="PREPAID_CIT_READER"
rootPath="SAPCC_CIT/PREPAID/Instance#@INSTANCE_ID@/@JCO_DESTINATION_NAME@"
channelCount="4"
         bufferSize="1000" removeFile="true">
         <billableItemLoader adminParameterPrefixName="PREPAID_BIT_LOADER"
destination="@JCO_DESTINATION_NAME@" autoCommit="true">
            <errorHandling errorType="invalidBillableItem">
               <bulkWriter
adminParameterPrefixName="PREPAID_CIT_ERR_INVALID_WRITER" rootPath="SAPCC_CIT/
PREPAID/INVALID/Instance#@INSTANCE_ID@/@JCO_DESTINATION_NAME@" channelCount="4"
timeBasedFileRolloverPolicy="HOURLY" maxSizeBasedFileRolloverPolicy="1M"
deflate="false" storeFileInformation="false" fileVersion="2"
fileSequenceId="PREPAID_CIT_ERR_INVALID"/>
            </errorHandling>
            <errorHandling errorType="communicationException">
               <bulkWriter
adminParameterPrefixName="PREPAID_CIT_ERR_COM_EXCEPTION_WRITER"
rootPath="SAPCC_CIT/PREPAID/COM_EXCEPTION/Instance#@INSTANCE_ID@/
@JCO_DESTINATION_NAME@"  channelCount="4" timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="false" storeFileInformation="false"
fileVersion="2" fileSequenceId="PREPAID_CIT_ERR_COM_EXCEPTION"/>
            </errorHandling>
         </billableItemLoader>
      </bulkReader>
   </multiJCoDestination>
</sequence>
```

3.2.11.4.6.9 bulkReader

> → Remember
>
> This section is dedicated to expert implementers who are members of the implementation project team. In this *Customizing* activity, you design, configure, and tune the Charging Output Integration Framework (CIF) of SAP Convergent Charging (SAP CC).

`bulkReader`

Description

Reads the charged and chargeable item data files and to send data to an item processor that computes each data.

CIF Usage

Bulkloader instances

Requires an item processor as child element (most of time a `loader`)

Attributes

Note that the attributes below are translated into parameters by SAP Convergent Charging. Refer to the SAP CC 2021 System Parameter Reference documentation.

- `adminParameterPrefixName` represents the parameter name prefix set by the translation by SAP CC of each attribute (it replaces the "XXX" prefix below).
  By example: if `adminParameterPrefixName` = `POSTPAID_CIT_READER` then the other `XML` attributes are translated by SAP CC into persistent parameters with their related value: POSTPAID_CIT_READER_ROOT_PATH, POSTPAID_CIT_READER_CHANNEL_COUNT, ...
  See each parameter link below for more info.
- `rootPath`: contains the relative or absolute path of the directory containing the data files to read.
  See XXX_ROOT_PATH (BulkReader) for more information.
- `channelCount`: represents the number of channels to use when reading the data files in a directory (cannot be lower than 1). See XXX_CHANNEL_COUNT (BulkReader) for more information
- `bufferSize`: the number of items read before sending the set of items to the next processor.
  See XXX_BUFFER_SIZE (BulkReader) for more information.
- `removeFile`: used to specify if the data files used by the bulkloader instances should be kept after being completely read. See XXX_REMOVE_FILE (BulkReader) for more information

Parent XML Elements

`sequence`, `multiDestination`, `multiJCODestination`

Example

From a bulkloader CIF config that handles the charged items and chargeable items:

```
<sequence>
   <xmlMapping name="testSender"
class="com.highdeal.tests.cortex.tasks.trx.helpers.cif.GenericItemProcessorSender
Factory" />
      <bulkReader adminParameterPrefixName="POSTPAID_CIT_READER"
         rootPath="SAPCC_CIT/POSTPAID/Instance#@INSTANCE_ID@" channelCount="4"
bufferSize="1000" removeFile="true">
         <sequence>
            <billableItemLoader adminParameterPrefixName="POSTPAID_BIT_LOADER"
destination="CI_DESTINATION">
               <errorHandling errorType="invalidBillableItem">
                  <sequence>
                     <bulkWriter
adminParameterPrefixName="POSTPAID_CIT_ERR_INVALID_WRITER" rootPath="SAPCC_CIT/
POSTPAID/INVALID/Instance#@INSTANCE_ID@" channelCount="4"
```

```
                        timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="false" fileVersion="2"
fileSequenceId="POSTPAID_CIT_ERR_INVALID"
/>
<testSender label="chargedItemPostpaidErrorInvalid" />
</sequence>
</errorHandling>
<errorHandling errorType="communicationException">
<sequence>
<bulkWriter adminParameterPrefixName="POSTPAID_CIT_ERR_COM_EXCEPTION_WRITER"
rootPath="SAPCC_CIT/POSTPAID/COM_EXCEPTION/Instance#@INSTANCE_ID@"
channelCount="4" timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="false" fileVersion="2"
fileSequenceId="POSTPAID_CIT_ERR_COM_EXCEPTION"
/>
<testSender label="chargedItemPostpaidErrorCommunication" />
</sequence>
</errorHandling>
</billableItemLoader>
<testSender label="chargedItem" />
                        </sequence>
                        </bulkReader>
                            <bulkReader
adminParameterPrefixName="PREPAID_CIT_READER" rootPath="SAPCC_CIT/PREPAID/
Instance#@INSTANCE_ID@" channelCount="4" bufferSize="1000" removeFile="true">
                        <sequence>
                        <billableItemLoader
adminParameterPrefixName="PREPAID_BIT_LOADER" destination="CI_DESTINATION">
                        <errorHandling errorType="invalidBillableItem">
                        <sequence>
                        <bulkWriter
adminParameterPrefixName="PREPAID_CIT_ERR_INVALID_WRITER" rootPath="SAPCC_CIT/
PREPAID/INVALID/Instance#@INSTANCE_ID@" channelCount="4"
                        timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="false" fileVersion="2"
fileSequenceId="PREPAID_CIT_ERR_INVALID"
                        />
                        <testSender label="chargedItemPrepaidErrorInvalid" />
                        </sequence>
                        </errorHandling>
                        <errorHandling errorType="communicationException">
                        <sequence>
                        <bulkWriter
adminParameterPrefixName="PREPAID_CIT_ERR_COM_EXCEPTION_WRITER"
rootPath="SAPCC_CIT/PREPAID/COM_EXCEPTION/Instance#@INSTANCE_ID@"
                        channelCount="4" timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="false" fileVersion="2"
fileSequenceId="PREPAID_CIT_ERR_COM_EXCEPTION"
                        />
                        <testSender
label="chargedItemPrepaidErrorCommunication" />
                        </sequence>
                        </errorHandling>
                        </billableItemLoader>
                        <testSender label="chargedItem" />
                        </sequence>
                        </bulkReader>
                            <bulkReader
adminParameterPrefixName="CHARGED_CI_READER" rootPath="SAPCC_CI/CHARGED/
Instance#@INSTANCE_ID@" channelCount="4" bufferSize="1000" removeFile="true"
itemType="chargeableItem">

                        <sequence>
                        <consumptionItemLoader destination="CI_DESTINATION"
isCharged="true">
                        <errorHandling errorType="communicationException">
                        <sequence>
```

```
                        <bulkWriter
adminParameterPrefixName="CHARGED_CI_ERR_COM_EXCEPTION_WRITER"
rootPath="SAPCC_CI/CHARGED/COM_EXCEPTION/Instance#@INSTANCE_ID@"
                        channelCount="4" timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="false" fileVersion="2"
fileSequenceId="CHARGED_CI_ERR_COM_EXCEPTION"
                        />
                        <testSender
label="chargeableItemChargedErrorCommunication" />
                        </sequence>
                        </errorHandling>
                        <errorHandling errorType="invalid">
                        <sequence>
                        <bulkWriter
adminParameterPrefixName="CHARGED_CI_ERR_INVALID_WRITER" rootPath="SAPCC_CI/
CHARGED/INVALID/Instance#@INSTANCE_ID@" channelCount="4"
                        timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="false" fileVersion="2"
fileSequenceId="CHARGED_CI_ERR_INVALID"
                        />
                        <testSender label="chargeableItemChargedErrorInvalid" />
                        </sequence>
                        </errorHandling>
                        </consumptionItemLoader>
                        <testSender label="chargeableItem" />
                        </sequence>
                        </bulkReader>
                            <bulkReader
adminParameterPrefixName="ACQUIRED_CI_READER" rootPath="SAPCC_CI/ACQUIRED/
Instance#@INSTANCE_ID@" channelCount="4" bufferSize="1000" removeFile="true"
itemType="chargeableItem">

   <sequence>
      <consumptionItemLoader destination="CI_DESTINATION" isCharged="false">
         <errorHandling errorType="communicationException">
            <sequence>
               <bulkWriter
adminParameterPrefixName="ACQUIRED_CI_ERR_COM_EXCEPTION_WRITER"
rootPath="SAPCC_CI/ACQUIRED/COM_EXCEPTION/Instance#@INSTANCE_ID@"
channelCount="4" timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="false" fileVersion="2"
fileSequenceId="ACQUIRED_CI_ERR_COM_EXCEPTION"
/>
               <testSender label="chargeableItemAcquiredErrorCommunication" />
            </sequence>
         </errorHandling>
         <errorHandling errorType="invalid">
            <sequence>
               <bulkWriter
adminParameterPrefixName="ACQUIRED_CI_ERR_INVALID_WRITER" rootPath="SAPCC_CI/
ACQUIRED/INVALID/Instance#@INSTANCE_ID@" channelCount="4"
                  timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="false" fileVersion="2"
fileSequenceId="ACQUIRED_CI_ERR_INVALID"/>
               <testSender label="chargeableItemAcquiredErrorInvalid" />
            </sequence>
         </errorHandling>
      </consumptionItemLoader>
      <testSender label="chargeableItemAcquired" />
   </sequence>
</bulkReader>
```

3.2.11.4.6.10 eventDateShifter

> → Remember
>
> This section is dedicated to expert implementers who are members of the implementation project team. In this *Customizing* activity, you design, configure, and tune the Charging Output Integration Framework (CIF) of SAP Convergent Charging (SAP CC).

`eventDateShifter`

Description

Subtracts 1 second to the charged item transaction date. This feature allows the SAP CC implementer to adapt the charged transactions produced by SAP Convergent Charging (SAP CC) to the billing cycles defined in the billing system.

> → Remember
>
> Indeed, in SAP CC, a recurring period (for by example a monthly subscription fee) always includes the "from" datetime and excludes the "to" date/time. **By subtracting 1 second to the transactions, the implementer can assign a transaction generated at the end of the period (often the recurring fees) to the current invoice.**

CIF Usage

Rater instances

XML Attributes

None

Parent XML Elements

`postpaidChargedItem`, `prepaidChargedItem`, `sequence`, `case`

Example

```
<chargedItemTransactionProcessor>
   <postpaidChargedItem>
      <eventDateShifter>
         <billableItemValidator mode="WS">
            <customerManagementAreaRouting>
               <bulkWriter adminParameterPrefixName="POSTPAID_CIT_WRITER"
rootPath="SAPCC_CIT/POSTPAID/Instance#@INSTANCE_ID@/@DESTINATION_NAME@"
channelCount="4"
                  timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="true" deflateOutputFormat="GZIP"
fileVersion="2" storeFileInformation="false" cleanArchivedFile="true"
fileSequenceId="CHARGED_ITEM" preCreateFile="ROLLOVER" />
            </customerManagementAreaRouting>
         </billableItemValidator>
      </eventDateShifter>
   </postpaidChargedItem>
```

```
</chargedItemTransactionProcessor>
```

3.2.11.4.6.11 errorHandling

> → Remember
>
> This section is dedicated to expert implementers who are members of the implementation project team. In this *Customizing* activity, you design, configure, and tune the Charging Output Integration Framework (CIF) of SAP Convergent Charging (SAP CC).

`errorHandling`

Description

Handles errors that can be returned by some item processors. This element is triggered only if the type of the error matches the error type specified in the `errorType` attribute. If an error occurs, most of time the "error" item processor consists in writing the items in error into a dedicated channel reserved for errors.

CIF Usage

Bulkloader instances

XML Attributes

`errorType`: the type of the error that will trigger the processor. Supported values are:

- `communicationException`: when a communication error occured.
- `invalidBillableItem` (with `wsBillableItemLoader`, `billableItemLoader`): the billable item is rejected by the billing system because it is invalid (the GPART/VKONT is wrong, the item is malformed, ...).
- `invalid` (with `consumptionItemLoader`): the consumption item is rejected by SAP Convergent Invoicing because it is invalid (the GPART/VKONT is wrong, the item is malformed, ...).

Parent XML Elements

`wsBillableItemLoader` [page 135], `billableItemLoader`, `consumptionItemLoader`

Example

```
<sequence>
   <multiJCoDestination>
      <bulkReader ...>
         <billableItemLoader adminParameterPrefixName="POSTPAID_BIT_LOADER"
destination="@JCO_DESTINATION_NAME@" autoCommit="true">
            <errorHandling errorType="invalidBillableItem">
               <bulkWriter
adminParameterPrefixName="POSTPAID_CIT_ERR_INVALID_WRITER" rootPath="SAPCC_CIT/
POSTPAID/INVALID/Instance#@INSTANCE_ID@/@JCO_DESTINATION_NAME@"
channelCount="4"timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="false" storeFileInformation="true"
fileVersion="2" fileSequenceId="POSTPAID_CIT_ERR_INVALID"/>
            </errorHandling>
```

```
            <errorHandling errorType="communicationException">
               <bulkWriter
adminParameterPrefixName="POSTPAID_CIT_ERR_COM_EXCEPTION_WRITER"
rootPath="SAPCC_CIT/POSTPAID/COM_EXCEPTION/Instance#@INSTANCE_ID@/
@JCO_DESTINATION_NAME@"            channelCount="4"
timeBasedFileRolloverPolicy="HOURLY" maxSizeBasedFileRolloverPolicy="1M"
deflate="false" storeFileInformation="true" fileVersion="2"
fileSequenceId="POSTPAID_CIT_ERR_COM_EXCEPTION"/>
            </errorHandling>
         </billableItemLoader>
      </bulkReader>
   </multiJCoDestination>
   <multiJCoDestination>
      <bulkReader ...>
         <billableItemLoader adminParameterPrefixName="PREPAID_BIT_LOADER"
destination="@JCO_DESTINATION_NAME@" autoCommit="true">
            <errorHandling errorType="invalidBillableItem">
               <bulkWriter
adminParameterPrefixName="PREPAID_CIT_ERR_INVALID_WRITER" rootPath="SAPCC_CIT/
PREPAID/INVALID/Instance#@INSTANCE_ID@/@JCO_DESTINATION_NAME@" channelCount="4"
timeBasedFileRolloverPolicy="HOURLY" maxSizeBasedFileRolloverPolicy="1M"
deflate="false" storeFileInformation="false" fileVersion="2"
fileSequenceId="PREPAID_CIT_ERR_INVALID/>
            </errorHandling>
            <errorHandling errorType="communicationException">
               <bulkWriter
adminParameterPrefixName="PREPAID_CIT_ERR_COM_EXCEPTION_WRITER"
rootPath="SAPCC_CIT/PREPAID/COM_EXCEPTION/Instance#@INSTANCE_ID@/
@JCO_DESTINATION_NAME@"            channelCount="4"
timeBasedFileRolloverPolicy="HOURLY" maxSizeBasedFileRolloverPolicy="1M"
deflate="false" storeFileInformation="false" fileVersion="2"
fileSequenceId="PREPAID_CIT_ERR_COM_EXCEPTION"/>
            </errorHandling>
         </billableItemLoader>
      </bulkReader>
   </multiJCoDestination>
   <multiJCoDestination>
         <bulkReader ...>
            <errorHandling errorType="communicationException">
               <bulkWriter
adminParameterPrefixName="CHARGED_CI_ERR_COM_EXCEPTION_WRITER"
rootPath="SAPCC_CI/CHARGED/COM_EXCEPTION/Instance#@INSTANCE_ID@/
@JCO_DESTINATION_NAME@"          channelCount="4"
timeBasedFileRolloverPolicy="HOURLY" maxSizeBasedFileRolloverPolicy="1M"
deflate="false" storeFileInformation="true" fileVersion="2"
fileSequenceId="CHARGED_CI_ERR_COM_EXCEPTION"/>
            </errorHandling>
            <errorHandling errorType="invalid">
               <bulkWriter
adminParameterPrefixName="CHARGED_CI_ERR_INVALID_WRITER" rootPath="SAPCC_CI/
CHARGED/INVALID/Instance#@INSTANCE_ID@/@JCO_DESTINATION_NAME@" channelCount="4"
timeBasedFileRolloverPolicy="HOURLY" maxSizeBasedFileRolloverPolicy="1M"
deflate="false" storeFileInformation="true" fileVersion="2"
fileSequenceId="CHARGED_CI_ERR_INVALID"/>
            </errorHandling>
         </consumptionItemLoader>
      </bulkReader>
   </multiJCoDestination>
   <multiJCoDestination>
      <bulkReader ...>
         <errorHandling errorType="communicationException">
            <bulkWriter
adminParameterPrefixName="ACQUIRED_CI_ERR_COM_EXCEPTION_WRITER"
rootPath="SAPCC_CI/ACQUIRED/COM_EXCEPTION/Instance#@INSTANCE_ID@/
@JCO_DESTINATION_NAME@"            channelCount="4"
timeBasedFileRolloverPolicy="HOURLY" maxSizeBasedFileRolloverPolicy="1M"
deflate="false" storeFileInformation="true" fileVersion="2"
```

```
fileSequenceId="ACQUIRED_CI_ERR_COM_EXCEPTION"  />
         </errorHandling>
         <errorHandling errorType="invalid">
            <bulkWriter
adminParameterPrefixName="ACQUIRED_CI_ERR_INVALID_WRITER" rootPath="SAPCC_CI/
ACQUIRED/INVALID/Instance#@INSTANCE_ID@/@JCO_DESTINATION_NAME@" channelCount="4"
timeBasedFileRolloverPolicy="HOURLY" maxSizeBasedFileRolloverPolicy="1M"
deflate="false" storeFileInformation="true" fileVersion="2"
fileSequenceId="ACQUIRED_CI_ERR_INVALID" />
         </errorHandling>
         </consumptionItemLoader>
      </bulkReader>
   </multiJCoDestination>
</sequence>
```

3.2.11.4.6.12 billableItemValidator and consumptionItemValidator

> → Remember
>
> This section is dedicated to expert implementers who are members of the implementation project team. In this *Customizing* activity, you design, configure, and tune the Charging Output Integration Framework (CIF) of SAP Convergent Charging (SAP CC).

`billableItemValidator` **and** `consumptionItemValidator`

Description

Validates the billable and consumption items before their storage in a file channel. This processor validates the mapping definition by checking that all the properties of the billing/consumption items can be correctly set with their type constraints.

CIF Usage

Rater instances

XML Attributes

None

Parent XML Elements

`postpaidChargedItem`, `prepaidChargedItem`, `chargedChargeableItem`, `acquiredChargeableItem`, `sequence`, `eventDateShifter` [page 124]

Example

```
<transactionProcessorSequence>
   <chargedItemTransactionProcessor>
      <postpaidChargedItem>
         <eventDateShifter>
```

```
            <sequence>
               <billableItemValidator />
               <customerManagementAreaRouting>
                  <bulkWriter .../>
               </customerManagementAreaRouting>
            </sequence>
         </eventDateShifter>
      </postpaidChargedItem>
      <prepaidChargedItem>
         <eventDateShifter>
            <sequence>
               <billableItemValidator />
               <customerManagementAreaRouting>
                  <bulkWriter .../>
               </customerManagementAreaRouting>
            </sequence>
         </eventDateShifter>
      </prepaidChargedItem>
   </chargedItemTransactionProcessor>
   <chargeableItemTransactionProcessor>
      <chargedChargeableItem>
         <sequence>
            <consumptionItemValidator />
            <customerManagementAreaRouting>
               <bulkWriter .../>
            </customerManagementAreaRouting>
         </sequence>
      </chargedChargeableItem>
      <acquiredChargeableItem>
         <sequence>
            <consumptionItemValidator />
               <customerManagementAreaRouting>
                  <bulkWriter .../>
               </customerManagementAreaRouting>
         </sequence>
      </acquiredChargeableItem>
   </chargeableItemTransactionProcessor>
</transactionProcessorSequence>
```

Example with WS mode (it accepts an item processor)

```
<chargedItemTransactionProcessor>
   <postpaidChargedItem>
      <eventDateShifter>
         <billableItemValidator mode="WS">
            <customerManagementAreaRouting>
               <bulkWriter ... />
            </customerManagementAreaRouting>
         </billableItemValidator>
      </eventDateShifter>
   </postpaidChargedItem>
</chargedItemTransactionProcessor>
```

3.2.11.4.6.13 customerManagementAreaRouting

> → Remember
>
> This section is dedicated to expert implementers who are members of the implementation project team. In this *Customizing* activity, you design, configure, and tune the Charging Output Integration Framework (CIF) of SAP Convergent Charging (SAP CC).

`customerManagementAreaRouting`

Description

Retrieves the destination of the item in the related subscriber account. In an SAP CC system landscape with several SAP billing systems, SAP CC selects the relevant system by using the customer management area (CMA) information set in the subscriber account owning the provider contract that has generated the item.

CIF Usage

Rater instances

Attributes

None

Parent XML Elements

`postpaidChargedItem`, `prepaidChargedItem`, `chargedChargeableItem`, `acquiredChargeableItem`, `sequence`, `eventDateShifter` [page 124]

Example

```
<transactionProcessorSequence>
   <chargedItemTransactionProcessor>
      <postpaidChargedItem>
         <eventDateShifter>
            <sequence>
               <billableItemValidator/>
               <customerManagementAreaRouting>
                  <bulkWriter adminParameterPrefixName="POSTPAID_CIT_WRITER"
rootPath="SAPCC_CIT/POSTPAID/Instance#@INSTANCE_ID@/@JCO_DESTINATION_NAME@"
channelCount="4"
                  timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="true" deflateOutputFormat="GZIP"
storeFileInformation="true"
                  fileVersion="2" lineNumbering="false"
fileSequenceId="CHARGED_ITEM" cleanArchivedFile="false"
preCreateFile="ROLLOVER"/>
               </customerManagementAreaRouting>
            </sequence>
         </eventDateShifter>
      </postpaidChargedItem>
      <prepaidChargedItem>
         <eventDateShifter>
            <sequence>
               <billableItemValidator />
               <customerManagementAreaRouting>
                  <bulkWriter adminParameterPrefixName="PREPAID_CIT_WRITER"
rootPath="SAPCC_CIT/PREPAID/Instance#@INSTANCE_ID@/@JCO_DESTINATION_NAME@"
channelCount="4"
                  timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="true" deflateOutputFormat="GZIP"
storeFileInformation="false"
                  fileVersion="2" lineNumbering="false"
fileSequenceId="CHARGED_ITEM" cleanArchivedFile="false"
preCreateFile="ROLLOVER"/>
               </customerManagementAreaRouting>
            </sequence>
         </eventDateShifter>
      </prepaidChargedItem>
   </chargedItemTransactionProcessor>
</transactionProcessorSequence>
```

Another example with WS Destination:

```
<chargedItemTransactionProcessor>
   <postpaidChargedItem>
      <eventDateShifter>
         <billableItemValidator mode="WS">
            <customerManagementAreaRouting>
               <bulkWriter adminParameterPrefixName="POSTPAID_CIT_WRITER"
rootPath="SAPCC_CIT/POSTPAID/Instance#@INSTANCE_ID@/@DESTINATION_NAME@"
channelCount="4"
               timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="true" deflateOutputFormat="GZIP"
               fileVersion="2" storeFileInformation="false"
cleanArchivedFile="true" fileSequenceId="CHARGED_ITEM" preCreateFile="ROLLOVER"/>
            </customerManagementAreaRouting>
         </billableItemValidator>
      </eventDateShifter>
   </postpaidChargedItem>
</chargedItemTransactionProcessor>
```

3.2.11.4.6.14 multiJCoDestination and multiDestination

> → Remember
>
> This section is dedicated to expert implementers who are members of the implementation project team. In this *Customizing* activity, you design, configure, and tune the Charging Output Integration Framework (CIF) of SAP Convergent Charging (SAP CC).

`multiJCoDestination` and `multiDestination`

Description

Routes the items according to the associated Customer Management Area (CMA).

> ⓘ Note
>
> - The implementation of this processor will internally create as many routes as the number of destinations created in SAP CC and used by the CMA. This means that all the nested processors (`XML` sub elements) cannot reference a “constant” value in their destination attribute and only the `@JCO_DESTINATION_NAME@` (for `multiJCoDestination`) and `@DESTINATION_NAME@` (for `multiDestination`) variables are authorized as values in the destination attribute.
> - Moreover, as “multi” can also means “one” destination, the `multiDestination` element used for connections from SAP CC to SAP S/4HANA Cloud Public Edition must be used even if only one destination is used. It is the same remark with `multiJCoDestination` for the connections with SAP Convergent Invoicing. Anyway, such processors are not mandatory with only one billing system not using SAP S/4HANA Cloud Public Edition.

CIF Usage

Bulkloader instances

XML Attributes

None

Parent XML Elements

`postpaidChargedItem`, `prepaidChargedItem`, `chargedChargeableItem`, `acquiredChargeableItem`, `sequence`

Example

```
<sequence>
   <multiJCoDestination>
      <bulkReader adminParameterPrefixName="POSTPAID_CIT_READER"
rootPath="SAPCC_CIT/POSTPAID/Instance#@INSTANCE_ID@/@JCO_DESTINATION_NAME@"
channelCount="4" bufferSize="1000" removeFile="true">
         <billableItemLoader adminParameterPrefixName="POSTPAID_BIT_LOADER"
destination="@JCO_DESTINATION_NAME@" autoCommit="true">
            <errorHandling errorType="invalidBillableItem">
               <bulkWriter
adminParameterPrefixName="POSTPAID_CIT_ERR_INVALID_WRITER" rootPath="SAPCC_CIT/
POSTPAID/INVALID/Instance#@INSTANCE_ID@/@JCO_DESTINATION_NAME@" channelCount="4"
timeBasedFileRolloverPolicy="HOURLY" maxSizeBasedFileRolloverPolicy="1M"
deflate="false" storeFileInformation="true" fileVersion="2"
fileSequenceId="POSTPAID_CIT_ERR_INVALID"/>
            </errorHandling>
            <errorHandling errorType="communicationException">
               <bulkWriter
adminParameterPrefixName="POSTPAID_CIT_ERR_COM_EXCEPTION_WRITER"
rootPath="SAPCC_CIT/POSTPAID/COM_EXCEPTION/Instance#@INSTANCE_ID@/
@JCO_DESTINATION_NAME@" channelCount="4" timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="false" storeFileInformation="true"
fileVersion="2" fileSequenceId="POSTPAID_CIT_ERR_COM_EXCEPTION"/>
            </errorHandling>
         </billableItemLoader>
      </bulkReader>
   </multiJCoDestination>
   <multiJCoDestination>
<bulkReader adminParameterPrefixName="PREPAID_CIT_READER" rootPath="SAPCC_CIT/
PREPAID/Instance#@INSTANCE_ID@/@JCO_DESTINATION_NAME@" channelCount="4"
bufferSize="1000" removeFile="true">
<billableItemLoader adminParameterPrefixName="PREPAID_BIT_LOADER"
destination="@JCO_DESTINATION_NAME@" autoCommit="true">
<errorHandling errorType="invalidBillableItem">
<bulkWriter
adminParameterPrefixName="PREPAID_CIT_ERR_INVALID_WRITER" rootPath="SAPCC_CIT/
PREPAID/INVALID/Instance#@INSTANCE_ID@/@JCO_DESTINATION_NAME@"
channelCount="4" timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M"deflate="false" storeFileInformation="false"
fileVersion="2" fileSequenceId="PREPAID_CIT_ERR_INVALID"/>
</errorHandling>
<errorHandling errorType="communicationException">
<bulkWriter adminParameterPrefixName="PREPAID_CIT_ERR_COM_EXCEPTION_WRITER"
rootPath="SAPCC_CIT/PREPAID/COM_EXCEPTION/Instance#@INSTANCE_ID@/
@JCO_DESTINATION_NAME@"  channelCount="4" timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="false" storeFileInformation="false"
fileVersion="2" fileSequenceId="PREPAID_CIT_ERR_COM_EXCEPTION"/>
</errorHandling>
</billableItemLoader>
</bulkReader>
</multiJCoDestination>
<multiJCoDestination>
<bulkReader adminParameterPrefixName="CHARGED_CI_READER" rootPath="SAPCC_CI/
CHARGED/Instance#@INSTANCE_ID@/@JCO_DESTINATION_NAME@" channelCount="4"
bufferSize="1000" removeFile="true" itemType="chargeableItem">
<consumptionItemLoader destination="@JCO_DESTINATION_NAME@" isCharged="true"
autoCommit="true">
```

```
<errorHandling errorType="communicationException">
<bulkWriter adminParameterPrefixName="CHARGED_CI_ERR_COM_EXCEPTION_WRITER"
rootPath="SAPCC_CI/CHARGED/COM_EXCEPTION/Instance#@INSTANCE_ID@/
@JCO_DESTINATION_NAME@" channelCount="4" timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="false" storeFileInformation="true"
fileVersion="2" fileSequenceId="CHARGED_CI_ERR_COM_EXCEPTION"/>
</errorHandling>
<errorHandling errorType="invalid">
<bulkWriter
adminParameterPrefixName="CHARGED_CI_ERR_INVALID_WRITER" rootPath="SAPCC_CI/
CHARGED/INVALID/Instance#@INSTANCE_ID@/@JCO_DESTINATION_NAME@" channelCount="4"
timeBasedFileRolloverPolicy="HOURLY" maxSizeBasedFileRolloverPolicy="1M"
deflate="false" storeFileInformation="true" fileVersion="2"
fileSequenceId="CHARGED_CI_ERR_INVALID"/>
</errorHandling>
</consumptionItemLoader>
  </bulkReader>
            </multiJCoDestination>
            <multiJCoDestination>
               <bulkReader adminParameterPrefixName="ACQUIRED_CI_READER"
rootPath="SAPCC_CI/ACQUIRED/Instance#@INSTANCE_ID@/@JCO_DESTINATION_NAME@"
channelCount="4"
                  bufferSize="1000" removeFile="true" itemType="chargeableItem">
                  <consumptionItemLoader destination="@JCO_DESTINATION_NAME@"
isCharged="false" autoCommit="true">
            <errorHandling errorType="communicationException">
               <bulkWriter
adminParameterPrefixName="ACQUIRED_CI_ERR_COM_EXCEPTION_WRITER"
rootPath="SAPCC_CI/ACQUIRED/COM_EXCEPTION/Instance#@INSTANCE_ID@/
@JCO_DESTINATION_NAME@" channelCount="4" timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="false" storeFileInformation="true"
fileVersion="2" fileSequenceId="ACQUIRED_CI_ERR_COM_EXCEPTION"/>
            </errorHandling>
            <errorHandling errorType="invalid">
               <bulkWriter
adminParameterPrefixName="ACQUIRED_CI_ERR_INVALID_WRITER" rootPath="SAPCC_CI/
ACQUIRED/INVALID/Instance#@INSTANCE_ID@/@JCO_DESTINATION_NAME@" channelCount="4"
timeBasedFileRolloverPolicy="HOURLY" maxSizeBasedFileRolloverPolicy="1M"
deflate="false" storeFileInformation="true" fileVersion="2"
fileSequenceId="ACQUIRED_CI_ERR_INVALID"/>
            </errorHandling>
         </consumptionItemLoader>
      </bulkReader>
   </multiJCoDestination>
</sequence>
```

Example with SAP S/4HANA Cloud Public Edition landscapes (mandatory):

```
<multiDestination>
   <bulkReader adminParameterPrefixName="POSTPAID_CIT_READER"
rootPath="SAPCC_CIT/POSTPAID/Instance#@INSTANCE_ID@/@DESTINATION_NAME@"
channelCount="4" bufferSize="100" removeFile="true">
      <wsBillableItemLoader verbose="false">
         <errorHandling errorType="invalidBillableItem">
            <bulkWriter
adminParameterPrefixName="POSTPAID_CIT_ERR_INVALID_WRITER" rootPath="SAPCC_CIT/
POSTPAID/INVALID/Instance#@INSTANCE_ID@/@DESTINATION_NAME@" channelCount="4"
timeBasedFileRolloverPolicy="HOURLY" maxSizeBasedFileRolloverPolicy="1M"
deflate="false"  fileVersion="2" storeFileInformation="false"
fileSequenceId="POSTPAID_CIT_ERR_INVALID" />
         </errorHandling>
         <errorHandling errorType="communicationException">
            <bulkWriter
adminParameterPrefixName="POSTPAID_CIT_ERR_COM_EXCEPTION_WRITER"
rootPath="SAPCC_CIT/POSTPAID/COM_EXCEPTION/Instance#@INSTANCE_ID@/
@DESTINATION_NAME@" channelCount="4" timeBasedFileRolloverPolicy="HOURLY"
```

```
maxSizeBasedFileRolloverPolicy="1M" deflate="false" fileVersion="2"
storeFileInformation="false" fileSequenceId="POSTPAID_CIT_ERR_COM_EXCEPTION" />
         </errorHandling>
      </wsBillableItemLoader>
   </bulkReader>
</multiDestination>
```

3.2.11.4.6.15 billableItemLoader

> → Remember
>
> This section is dedicated to expert implementers who are members of the implementation project team. In this *Customizing* activity, you design, configure, and tune the Charging Output Integration Framework (CIF) of SAP Convergent Charging (SAP CC).

`billableItemLoader`

Description

Loads the billable items in SAP Convergent Invoicing (on premise) in its system (ERP or S/4HANA). In case of an error, the expert implementer can optionally specify an `errorHandling` processor to handle items in error.

CIF Usage

Bulkloader instances

XML Attributes

> ⓘ Note
>
> The attributes below are translated into system parameters by SAP Convergent Charging. You can view their values and change them with the SAP CC administrative tools.
>
> Refer to the SAP CC 2021 System Parameter Reference documentation.

- `adminParameterPrefixName` represents the internal parameter name prefix set by the interpretation by SAP CC of each attribute. It defines the appliance scope of values set in each attribute.
  For example: if `adminParameterPrefixName` = `POSTPAID_BIT_LOADER_CLIENT`, then the other `XML` attributes will be translated by SAP CC into internal parameters applicable to the postpaid charged item.
  Possible values are:
  - `POSTPAID_BIT_LOADER_CLIENT`
  - `PREPAID_BIT_LOADER_CLIENT`
- `destination` represents the name of the targeted JCo destination that is used to push the billable items. The `@JCO_DESTINATION_NAME@` variable must be set when the loader is used within a `multiJCoDestination` element.
- `autoCommit` gives the possibility to perform a single RFC[59] call for billable item loading and commit/rollback operation (set to `true` by default).

- `currencyAjustment` indicates how the currency precisions are chosen when sending billable items to SAP Convergent Invoicing (on premise).
  Refer to the XXX_CURRENCY_ADJUSTMENT (BillableItemLoader) system indicator for further information.

Parent XML Element

`bulkReader` [page 120]

Example

```
<sequence>
   <multiJCoDestination>
      <bulkReader adminParameterPrefixName="POSTPAID_CIT_READER"
rootPath="SAPCC_CIT/POSTPAID/Instance#@INSTANCE_ID@/@JCO_DESTINATION_NAME@"
channelCount="4"
      bufferSize="1000" removeFile="true">
         <billableItemLoader adminParameterPrefixName="POSTPAID_BIT_LOADER"
destination="@JCO_DESTINATION_NAME@" autoCommit="true">
            <errorHandling errorType="invalidBillableItem">
               <bulkWriter
adminParameterPrefixName="POSTPAID_CIT_ERR_INVALID_WRITER" rootPath="SAPCC_CIT/
POSTPAID/INVALID/Instance#@INSTANCE_ID@/@JCO_DESTINATION_NAME@" channelCount="4"
timeBasedFileRolloverPolicy="HOURLY" maxSizeBasedFileRolloverPolicy="1M"
deflate="false" storeFileInformation="true" fileVersion="2"
fileSequenceId="POSTPAID_CIT_ERR_INVALID"/>
            </errorHandling>
            <errorHandling errorType="communicationException">
               <bulkWriter
adminParameterPrefixName="POSTPAID_CIT_ERR_COM_EXCEPTION_WRITER"
rootPath="SAPCC_CIT/POSTPAID/COM_EXCEPTION/Instance#@INSTANCE_ID@/
@JCO_DESTINATION_NAME@" channelCount="4" timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="false" storeFileInformation="true"
fileVersion="2" fileSequenceId="POSTPAID_CIT_ERR_COM_EXCEPTION"/>
            </errorHandling>
         </billableItemLoader>
      </bulkReader>
   </multiJCoDestination>
   <multiJCoDestination>
      <bulkReader adminParameterPrefixName="PREPAID_CIT_READER"
rootPath="SAPCC_CIT/PREPAID/Instance#@INSTANCE_ID@/@JCO_DESTINATION_NAME@"
channelCount="4"
      bufferSize="1000" removeFile="true">
         <billableItemLoader adminParameterPrefixName="PREPAID_BIT_LOADER"
destination="@JCO_DESTINATION_NAME@" autoCommit="true">
            <errorHandling errorType="invalidBillableItem">
               <bulkWriter
adminParameterPrefixName="PREPAID_CIT_ERR_INVALID_WRITER" rootPath="SAPCC_CIT/
PREPAID/INVALID/Instance#@INSTANCE_ID@/@JCO_DESTINATION_NAME@" channelCount="4"
timeBasedFileRolloverPolicy="HOURLY" maxSizeBasedFileRolloverPolicy="1M"
deflate="false" storeFileInformation="false" fileVersion="2"
fileSequenceId="PREPAID_CIT_ERR_INVALID/>
            </errorHandling>
            <errorHandling errorType="communicationException">
               <bulkWriter
adminParameterPrefixName="PREPAID_CIT_ERR_COM_EXCEPTION_WRITER"
rootPath="SAPCC_CIT/PREPAID/COM_EXCEPTION/Instance#@INSTANCE_ID@/
@JCO_DESTINATION_NAME@" channelCount="4" timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="false" storeFileInformation="false"
fileVersion="2" fileSequenceId="PREPAID_CIT_ERR_COM_EXCEPTION"/>
            </errorHandling>
         </billableItemLoader>
      </bulkReader>
```

[59] Remote Function Call

```
   </multiJCoDestination>
</sequence>
```

3.2.11.4.6.16 wsBillableItemLoader

> → Remember
>
> This section is dedicated to expert implementers who are members of the implementation project team. In this *Customizing* activity, you design, configure, and tune the Charging Output Integration Framework (CIF) of SAP Convergent Charging (SAP CC).

`wsBillableItemLoader`

Description

Loads/pushes the billable items in SAP S/4HANA Cloud Public Edition. In case of an error, the expert implementer can optionally specify an `errorHandling` processor [page 125] to handle items in error.

CIF Usage

Bulkloader instances

XML Attribute

`verbose` gives the possibility to add verbose log messages in confirmation messages related to billable items creation operations in SAP S/4HANA Cloud Public Edition (`false` by default).

> ⓘ Note
>
> There is no `destination` attribute in this loader because this element must be always used within the `multiDestination` element and then, the destination is always retrieved thanks to the Customer Management Area (CMA) mechanism.

Parent XML Element

`bulkReader` [page 120] and must be always used within a `multiDestination` element with SAP S/4HANA Cloud Public Edition.

Example

```
<multiDestination>
   <bulkReader adminParameterPrefixName="POSTPAID_CIT_READER"
rootPath="SAPCC_CIT/POSTPAID/Instance#@INSTANCE_ID@/@DESTINATION_NAME@"
   channelCount="4" bufferSize="100" removeFile="true">
      <wsBillableItemLoader verbose="false">
         <errorHandling errorType="invalidBillableItem">
            <bulkWriter
adminParameterPrefixName="POSTPAID_CIT_ERR_INVALID_WRITER" rootPath="SAPCC_CIT/
POSTPAID/INVALID/Instance#@INSTANCE_ID@/@DESTINATION_NAME@" channelCount="4"
timeBasedFileRolloverPolicy="HOURLY" maxSizeBasedFileRolloverPolicy="1M"
```

```
deflate="false" fileVersion="2" storeFileInformation="false"
fileSequenceId="POSTPAID_CIT_ERR_INVALID" />
         </errorHandling>
         <errorHandling errorType="communicationException">
            <bulkWriter
adminParameterPrefixName="POSTPAID_CIT_ERR_COM_EXCEPTION_WRITER"
rootPath="SAPCC_CIT/POSTPAID/COM_EXCEPTION/Instance#@INSTANCE_ID@/
@DESTINATION_NAME@" channelCount="4" timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="false" fileVersion="2"
storeFileInformation="false" fileSequenceId="POSTPAID_CIT_ERR_COM_EXCEPTION" />
         </errorHandling>
      </wsBillableItemLoader>
   </bulkReader>
</multiDestination>
```

3.2.11.4.6.17 consumptionItemLoader

> → Remember
>
> This section is dedicated to expert implementers who are members of the implementation project team. In this *Customizing* activity, you design, configure, and tune the Charging Output Integration Framework (CIF) of SAP Convergent Charging (SAP CC).

`consumptionItemLoader`

Description

Loads consumption items in SAP Convergent Invoicing (on premise). In case of an error, the expert implementer can optionally specify an `errorHandling` processor to handle items in error.

CIF Usage

Bulkloader instances

XML Attributes

Please note that the XML attributes below are translated into parameters by SAP Convergent Charging. Please consult the SAP CC 2021 System Parameter Reference documentation.

- `destination` represents the name of the targeted JCo destination that is used for pushing the consumption items. The `@JCO_DESTINATION_NAME@` variable must be set when the loader is used within a `multiJCoDestination` element.
- `autoCommit` gives the possibility to perform a single RFC call for consumption item loading and commit/rollback operation (set to `true` by default).
- `isCharged` is used to configure the RFC call to SAP Convergent Invoicing in order to indicate whether the items being sent are charged or no.
  If true, only the consumption items issued from an SAP CC charging process will be selected.
  Otherwise, only the consumption items issued from an acquisition process is selected.

Parent XML Element

`bulkReader` [page 120]

Example

```
<sequence>
   <multiJCoDestination>
      <bulkReader adminParameterPrefixName="CHARGED_CI_READER"
rootPath="SAPCC_CI/CHARGED/Instance#@INSTANCE_ID@/@JCO_DESTINATION_NAME@"
channelCount="4"
      bufferSize="1000" removeFile="true" itemType="chargeableItem">
         <consumptionItemLoader destination="@JCO_DESTINATION_NAME@"
isCharged="true" autoCommit="true">
            <errorHandling errorType="communicationException">
               <bulkWriter
adminParameterPrefixName="CHARGED_CI_ERR_COM_EXCEPTION_WRITER"
rootPath="SAPCC_CI/CHARGED/COM_EXCEPTION/Instance#@INSTANCE_ID@/
@JCO_DESTINATION_NAME@" channelCount="4" timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="false" storeFileInformation="true"
fileVersion="2" fileSequenceId="CHARGED_CI_ERR_COM_EXCEPTION"/>
            </errorHandling>
            <errorHandling errorType="invalid">
               <bulkWriter
adminParameterPrefixName="CHARGED_CI_ERR_INVALID_WRITER" rootPath="SAPCC_CI/
CHARGED/INVALID/Instance#@INSTANCE_ID@/@JCO_DESTINATION_NAME@" channelCount="4"
timeBasedFileRolloverPolicy="HOURLY" maxSizeBasedFileRolloverPolicy="1M"
deflate="false" storeFileInformation="true" fileVersion="2"
fileSequenceId="CHARGED_CI_ERR_INVALID"/>
            </errorHandling>
         </consumptionItemLoader>
      </bulkReader>
   </multiJCoDestination>
   <multiJCoDestination>
      <bulkReader adminParameterPrefixName="ACQUIRED_CI_READER"
rootPath="SAPCC_CI/ACQUIRED/Instance#@INSTANCE_ID@/@JCO_DESTINATION_NAME@"
channelCount="4" bufferSize="1000" removeFile="true" itemType="chargeableItem">
         <consumptionItemLoader destination="@JCO_DESTINATION_NAME@"
isCharged="false" autoCommit="true">
            <errorHandling errorType="communicationException">
               <bulkWriter
adminParameterPrefixName="ACQUIRED_CI_ERR_COM_EXCEPTION_WRITER"
rootPath="SAPCC_CI/ACQUIRED/COM_EXCEPTION/Instance#@INSTANCE_ID@/
@JCO_DESTINATION_NAME@" channelCount="4" timeBasedFileRolloverPolicy="HOURLY"
maxSizeBasedFileRolloverPolicy="1M" deflate="false" storeFileInformation="true"
fileVersion="2" fileSequenceId="ACQUIRED_CI_ERR_COM_EXCEPTION"/>
            </errorHandling>
            <errorHandling errorType="invalid">
               <bulkWriter
adminParameterPrefixName="ACQUIRED_CI_ERR_INVALID_WRITER" rootPath="SAPCC_CI/
ACQUIRED/INVALID/Instance#@INSTANCE_ID@/@JCO_DESTINATION_NAME@" channelCount="4"
timeBasedFileRolloverPolicy="HOURLY" maxSizeBasedFileRolloverPolicy="1M"
deflate="false" storeFileInformation="true" fileVersion="2"
fileSequenceId="ACQUIRED_CI_ERR_INVALID"/>
            </errorHandling>
         </consumptionItemLoader>
      </bulkReader>
   </multiJCoDestination>
</sequence>
```

3.2.11.5 Modifying the Item/Data Files Processor

During charging and refilling operations, the running rater instances generate data files whose persistency is managed by a dedicated framework [page 99] which gives the possibility to implement the following file processors:

- **Transaction File** processor (deprecated), which creates a single file whose content can be customized.
- **Data File** processor (deprecated), which creates multiple data files which contain different data related to transactions.
- **Charged Item File, Refill Record File**, or **Chargeable Item File** processor, which give the possibility to create data files whose content corresponds to taxed & charged transactions transcribed into:
  - Chargeable items, during chargeable items acquisition and charging operations
  - Charged items, during chargeable items charging and prepaid accounts refilling operations
  - Refill records, during prepaid accounts refilling operations
- **Rerate File** processor, which gives the possibility to create data files whose content can be used by a third-party system for rerating purposes.

3.2.11.5.1 Enabling the Transaction Integration Framework (TIF) to Generate Transaction Files - DEPRECATED

> ⚠ Caution
>
> The use of the *Transaction File processor* is **deprecated as of release 2.0 of SAP CC**. See now: Enabling the Charging Output Integration Framework (CIF) Processor [page 139]
>
> To ensure compatibility with offers created through previous versions of SAP Convergent Charging, it is still possible to use it.
>
> For further information about this file processor, refer to the description of the Output Data Generation Process (Transaction File Processing) - Deprecated group available in the SAP CC 2021 System Parameter Reference documentation.

To enable the Transaction File processor, execute the following procedure:

1. Launch your favorite user interface [page 45] and log on as the administrator of the Core Server system.
2. For each rater instance of the Core Server system, modify the value of the TRANSAC_PROC_CLASS system parameter to the following value: `com.highdeal.tif.tfp.TransactionFileProcessor`
3. Restart the rater instances in the Core Server system.

3.2.11.5.2 Enabling the Transaction Integration Framework (TIF) to Generate Data Files - DEPRECATED

> ⚠ Caution
>
> The use of the *Data File processor* is **deprecated as of release 2.0 of SAP CC**. See now: Enabling the Charging Output Integration Framework (CIF) Processor [page 139]
>
> To ensure compatibility with offers created through previous versions of SAP CC, it is still possible to use it.
>
> For further information about this file processor, refer to the description of the Output Data Generation Process (DB File Processing) - Deprecated group available in the SAP CC 2021 System Parameter Reference documentation.

To enable the Data File processor, execute the following procedure:

1. Launch your favorite user interface [page 45] and log on as the administrator of the Core Server system.
2. For each rater of the Core Server system, modify the value of the TRANSAC_PROC_CLASS system parameter to the following value: `com.highdeal.tif.dbFile.TransactionFileProcessor`
3. Restart the rater instances in the Core Server system.

3.2.11.5.3 Enabling the Charging Output Integration Framework (CIF) Processor

To enable the Charging output Integration Framework (CIF) processor and generate data files containing either charged items, chargeable items, or refill records, complete the following procedure:

1. Launch your favorite user interface [page 45] and log on as the administrator of the Core Server system.
2. For each rater instance of your Core Server system, modify the value of the TRANSAC_PROC_CLASS system parameter to the following value: `com.highdeal.tif.chargeditem.TransactionProcessorShell`
3. Restart the rater instances in the Core Server system.

> ⓘ Note
>
> The CIF Processor is enabled by default if you chose to activate SAP CI during the installation.

Next Step

Configuring the CIF [page 140]

3.2.11.5.3.1 Configuring the CIF

Prerequisite

The feature is enabled. See Enabling the Charging Output Integration Framework (CIF) Processor [page 139].

Possible Settings

To fit specific needs such as data files path, multithreaded generation, rollover policy, file compression... you can configure the data file generation of all available channels (prepaid, postpaid, charged, and acquired) using the following parameters:

Setting (XML Element)	Description
`rootPath`	Writing location for generated files. For further information, refer to the description of the XXX_ROOT_PATH (BulkWriter) system parameter.
`channelCount`	Number of writing channels. For further information, refer to the description of the XXX_CHANNEL_COUNT (BulkWriter) system parameter.
`timeBasedFileRolloverPolicy` `maxSizeBasedFileRolloverPolicy`	Rolling policy for generated files. For further information, refer to the description of the XXX_TIME_BASED_FILE_ROLLOVER_POLICY (BulkWriter) and XXX_MAX_SIZE_BASED_FILE_ROLLOVER_POLICY (BulkWriter) system parameters.
`deflate` `deflateOutputFormat`	Compression policy, that respects the RFC 1951 specifications. For further information, refer to the description of the XXX_DEFLATE (BulkWriter) and XXX_DEFLATE_OUTPUT_FORMAT (BulkWriter) system parameters.
`storeFileInformation`	Flag used to store information about the generated files into the `CHARGED_ITEM_FILES` table of the Core Database **Note** In case of an installation scenario integrated with SAP S/4HANA Cloud Public Edition, this parameter is automatically set to false. SAP SE recommends that you do not modify this value.
fileVersion	Version of the generated data files. To modify this parameter, refer to the Modifying the Versions of Data Files Generated by Raters [page 144] dedicated procedure available afterwards

Setting (XML Element)	Description
`lineNumbering`	Flag used to write line numbers in the generated data files. To modify this parameter, refer to the Enabling the Line Numbering for Data Files Generated by Raters [page 145] dedicated procedure available afterwards
`cleanArchivedFile`	Flag used to enable or disable the removal of uncommitted items from the generated data files. To modify this parameter, refer to the Enabling the Archive Files Cleaning for Data Files Generated by Raters [page 145] dedicated procedure available afterwards.
`fileSequenceId`	Identifier used for sequencing the names of the generated data files. To modify this parameter, refer to the Enabling the File Name Sequencing for Error Files Generated by Bulkloaders [page 147] dedicated procedure available afterwards.
`preCreateFile`	Flag used to enable or disable the pre-creation of data files. To modify this parameter, refer to the Enabling the Precreation Policy for Data Files Generated by the Rater and Bulkloader Instances [page 146] dedicated procedure available afterwards.
`eventDateShifter`	To modify this mechanism, refer to the Enabling the Event Date Shifting Mechanism for Data Files Generated by the Rater Instances [page 148] dedicated procedure available afterwards.
`billableItemValidator`	To modify this mechanism, refer to the Enabling the Billable Item Validation for Data Files Generated by Rater Instances [page 149] dedicated procedure available afterwards.

→ Tip

Template files named `cif.rater.config.xml.sk` and `cif.bulkloader.config.xml.sk` are available in the `exe/uc/<OS>/CC_CORE_SERVER/config/<scenario>` subfolder of the SAP Central Repository, within subfolders whose name depends on the implementation and installation scenario:

- `sapci` in case of integrated scenario with SAP Convergent Invoicing in its system (SAP S/4HANA or SAP ERP)
- `s4hc` in case of integrated scenario with SAP S/4HANA Cloud Public Edition
- `standalone` otherwise

Procedure

- To configure the CIF[60] parameters for the rater instances, complete the following procedure:
  1. Using the Setup Tool user interface execute the cif export command to export the CIF configuration of the rater instances in the running SAP CC Core Server system.
  2. Edit the retrieved `cif.rater.config.xml` configuration file.
  3. Modify the relevant parameters.
  4. Save your modifications.
  5. Using the Setup Tool user interface execute the cif import command to import the modified CIF configuration for the rater instances of the Core Server system.
  6. Restart the rater instances in the Core Server system.
- To configure the CIF parameters for the bulkloader instances, complete the following procedure:

[60] Charging output Integration Framework

1. Using the Setup Tool user interface execute the cif export command to export the CIF configuration of the bulkloader instances in the running SAP CC Core Server system.
2. Edit the retrieved `cif.bulkloader.config.xml` configuration file.
3. Modify the relevant parameters.
4. Save your modifications.
5. Using the Setup Tool user interface execute the cif import command to import the modified CIF configuration for the bulkloader instances of the Core Server system.
6. Restart the bulkloader instances in the Core Server system.

3.2.11.5.4 Enabling the Rerating Integration Framework (RIF)

By default, the identifier of the rerate processor is set to `com.highdeal.rif.nop.NOPrerateProcessor`. This factory setting is used when:

- SAP Convergent Invoicing manages the rerating feature[61]
- BART Server manages the rerating feature but SAP Convergent Invoicing is not deployed in the SAP system landscape

In a non-standard implementation where the BART Server system manages the rerating feature and SAP Convergent Invoicing manages the reversal of generated billable items (BITs), it is necessary to redefine the rerate processor to `com.highdeal.rif.chargeditem.RerateProcessorShell`.

Another Java class can be customized during the implementation and integration time. The identifier of the rerate processor can then be set to the name of this class.

1. Launch your favorite user interface [page 45] and identify as the administrator of the Core Server system.
2. For each rater instance, modify the value of the RERATE_PROC_CLASS system parameter. The possible values are:
   - `com.highdeal.rif.nop.NOPrerateProcessor` (factory setting)
   - `com.highdeal.rif.chargeditem.RerateProcessorShell`
   - Another Java class name (customized integration)
3. Restart the rater instances of the Core Server system.

Next Step

Configuring the RIF [page 143]

[61] SAP Convergent Invoicing must be configured to manage both billable items and consumption items sent by SAP Convergent Charging

3.2.11.5.4.1 Configuring the RIF

Prerequisite

The feature is enabled. See Enabling the Rerating Integration Framework (RIF) [page 142].

Possible Settings

To fit specific needs such as data files path, multithreaded generation, rollover policy, file compression... you can configure the data file generation of rerating requests using the following parameters:

Setting (XML Element)	Description
`rootPath`	Writing location for generated files. For further information, refer to the description of the XXX_ROOT_PATH (BulkWriter) system parameter.
`channelCount`	Number of writing channels. For further information, refer to the description of the XXX_CHANNEL_COUNT (BulkWriter) system parameter.
timeBasedFileRolloverPolicy maxSizeBasedFileRolloverPolicy	Rolling policy for generated files. For further information, refer to the description of the XXX_TIME_BASED_FILE_ROLLOVER_POLICY (BulkWriter) and XXX_MAX_SIZE_BASED_FILE_ROLLOVER_POLICY (BulkWriter) system parameters.
deflate `deflateOutputFormat`	Compression policy, that respects the RFC 1951 specifications. For further information, refer to the description of the XXX_DEFLATE (BulkWriter) and XXX_DEFLATE_OUTPUT_FORMAT (BulkWriter) system parameters.
`preCreateFile`	Flag used to enable or disable the pre-creation of data files. To modify this parameter, refer to the Enabling the Precreation Policy for Data Files Generated by the Rater and Bulkloader Instances [page 146] dedicated procedure available afterwards.

→ Tip

Template files named `rif.rater.config.xml.sk` is available in the `exe/uc/<OS>/CC_CORE_SERVER/config/<scenario>` subfolder of the SAP Central Repository, within subfolders whose name depends on the installation scenario:

- `sapci` in case of integrated scenario with SAP Convergent Invoicing
- `s4hc` in case of integrated scenario with SAP S/4HANA Cloud Public Edition edition
- `standalone` otherwise

Procedure

To configure the RIF[62] parameters, complete the following procedure:

1. Using the Setup Tool user interface execute the rif export command to export the RIF configuration of the rater instances in the running SAP CC Core Server system.
2. Edit the retrieved `rif.rater.config.xml` configuration file.
3. Save your modifications.
4. Using the Setup Tool user interface execute the rif import command to import the modified RIF configuration for the rater instances of the Core Server system.
5. Restart the rater instances in the Core Server system.

3.2.11.6 Modifying the Versions of Data Files Generated by Raters

During charging and refilling operations, raters generate data files containing information about incoming events. To modify the version of these data files generated by raters, execute the following procedure:

1. Launch your favorite user interface [page 45]
2. Export the CIF[63] configuration of the rater instances using the `cif export` command
3. Edit the retrieved `rater.cif.xml` configuration file
4. For each `bulkwriter` XML element located under a **`charged`**`ItemTransactionProcessor` XML element, set the `fileVersion` attribute to **`1`** or **`2`**
5. For each `bulkwriter` XML element located under a **`chargeable`**`ItemTransactionProcessor` XML element, set the `fileVersion` attribute to **`1`** or **`2`**
6. Save your modifications
7. Import the modified CIF configuration for the rater instances using the `cif import` command
8. Restart the rater instances of the Core Server system

For more information about the CIF, refer to the Customizing the Charging Output Integration Framework (CIF) [page 99] section of the SAP CC 2021 Administration Guide [page 5].

3.2.11.7 Enabling the File Name Sequencing for Data Files Generated by Raters

To enable this file name sequencing for data files generated by raters, execute the following procedure:

1. Launch your favorite user interface [page 45] and log on as the administrator of the Core Server system.
2. Get the number of database connections currently set for **rater** instances using the following command:

```
get SQLHELPER_CONNECTION_COUNT persistent rater
```

[62] Rerating Integration Framework
[63] Charging output Integration Framework

3. Increase this number of connections by 1 using the following command:
   `set` SQLHELPER_CONNECTION_COUNT `<increased_value> persistent rater`
   Where `<increased_value>` corresponds to the number of database connections, increased by 1
4. Enable the sequencing for error files by executing the following command:
   `set` ITEM_FILE_SEQUENCE_ENABLED `true persistent rater`
5. Launch your favorite user interface [page 45].
6. Export the CIF[64] configuration of the rater instances using the `cif export` command.
7. Edit the retrieved `rater.cif.xml` configuration file.
8. For each `bulkwriter` XML element located under a **`charged`**`ItemTransactionProcessor` XML element, set the following attributes to the relevant values:
   - `fileVersion` to **`2`**
   - `fileSequenceId` to **`CHARGED_ITEM`**
9. For each `bulkwriter` XML element located under a **`chargeable`**`ItemTransactionProcessor` XML element, set the following attributes to the relevant values:
   - `fileVersion` to **`2`**
   - `fileSequenceId` to **`CHARGEABLE_ITEM`**
10. Save your modifications.
11. Import the modified CIF configuration for the rater instances using the `cif import` command.
12. Restart the rater instances in the Core Server system.

3.2.11.8 Enabling the Line Numbering for Data Files Generated by Raters

To enable the line numbering for data files generated by raters, execute the following procedure:

1. Launch your favorite user interface [page 45]
2. Export the CIF[65] configuration of the rater instances using the `cif export` command
3. Edit the retrieved `rater.cif.xml` configuration file
4. For each `bulkwriter` XML element, set the `lineNumbering` attribute to **`true`**
5. Save your modifications
6. Import the modified CIF configuration for the rater instances using the `cif import` command
7. Restart the rater instances of the Core Server system

3.2.11.9 Enabling the Archive Files Cleaning for Data Files Generated by Raters

SAP CC gives the possibility to activate an automatic cleaning mechanism in order to remove the uncommited items from data files, and the associated control files. To enable this automatic cleaning mechanism for data files generated by raters, execute the following procedure:

[64] Charging output Integration Framework
[65] Charging output Integration Framework

1. Launch your favorite user interface [page 45]
2. Export the CIF[66] configuration of the rater instances using the `cif export` command
3. Edit the retrieved `rater.cif.xml` configuration file
4. For each `bulkwriter` XML element, set the `cleanArchivedFile` attribute to **`true`**
5. Save your modifications
6. Import the modified CIF configuration for the rater instances using the `cif import` command
7. Restart the rater instances of the Core Server system

3.2.11.10 Enabling the Precreation Policy for Data Files Generated by the Rater and Bulkloader Instances

For optimization purpose, data files generated by rater and bulkloader instances can be asynchronously pre-created, instead of waiting for the associated channel to create these files on demand.

- To enable the precreation of data files generated by the rater instances, execute the following procedure:
  1. Launch your favorite user interface [page 45].
  2. Export the CIF[67] and RIF[68] configurations of the rater instances using the following commands:

     ```
     cif export rater rater.cif.xml
     rif export rater rater.rif.xml
     ```
  3. In production systems that are live, make backup copies.
  4. Edit the retrieved `rater.cif.xml` and `rater.rif.xml` configuration files.
  5. For each `bulkwriter` XML element, set the `preCreate` attribute to **`true`**.
  6. Save your modifications.
  7. Import the modified CIF and RIF configurations for the rater instances using the following commands:

     ```
     cif import rater rater.cif.xml
     rif import rater rater.rif.xml
     ```
  8. Restart the rater instances in the Core Server system.
- To enable the precreation of data files generated by the bulkloader instances, complete the following procedure:
  1. Launch your favorite user interface [page 45].
  2. Export the CIF and RIF configurations of the bulkloader instances using the following commands:

     ```
     cif export bulkloader bulkloader.cif.xml
     rif export bulkloader bulkloader.rif.xml
     ```
  3. In production systems that are live, make backup copies.
  4. Edit the retrieved `bulkloader.cif.xml` and `bulkloader.rif.xml` configuration files.
  5. For each `bulkwriter` XML element, set the `preCreate` attribute to **`true`**.
  6. Save your modifications.

[66] Charging output Integration Framework
[67] Charging output Integration Framework
[68] Rerating Integration Framework

7. Import the modified CIF and RIF configurations for the bulkloader instances using the following commands:

```
cif import bulkloader bulkloader.cif.xml
rif import bulkloader bulkloader.rif.xml
```

8. Restart the bulkloader instances in the Core Server system.

3.2.11.11 Modifying the Version of Error Files Generated by Bulkloaders

During charging and refilling operations, raters generate data files containing information about incoming events. These data files are taken into account by bulkloaders, in charge of converting the content of these files and sending them to third-party systems. When an error occurs during bulk loading operations, the bulkloaders generate error files whose content depends on the configured version.

To modify the version of error files used by bulkloaders, execute the following procedure:

1. Launch your favorite user interface [page 45]
2. Export the CIF[69] configuration of the bulkloader instances using the `cif export` command
3. Edit the retrieved `bulkloader.cif.xml` configuration file
4. For each `bulkwriter` XML element, set the `fileVersion` attribute to **`1`** or **`2`**
5. Save your modifications
6. Import the modified CIF configuration for the bulkloader instances using the `cif import` command
7. Restart the bulkloader instances of the Core Server system

3.2.11.12 Enabling the File Name Sequencing for Error Files Generated by Bulkloaders

The version 2 of error files generated by bulkloaders gives the possibility to add a sequence number to the names of the files. To enable this sequencing for error files, execute the following procedure:

1. Launch your favorite user interface [page 45] user interface and identify as the administrator of the Core Server system
2. Get the number of database connections currently set for **bulkloader** instances using the following command:
   `get` SQLHELPER_CONNECTION_COUNT `persistent bulkloader`
3. Increase this number of connections by 1 using the following command:
   `set` SQLHELPER_CONNECTION_COUNT `<increased_value> persistent bulkloader`
   Where `<increased_value>` corresponds to the number of database connections, increased by 1
4. Enable the sequencing for error files by executing the following command:
   `set` ITEM_FILE_SEQUENCE_ENABLED `true persistent bulkloader`
5. Launch your favorite user interface [page 45]

[69] Charging output Integration Framework

6. Export the CIF[70] configuration of the bulkloader instances using the `cif export` command
7. Edit the retrieved `bulkloader.cif.xml` configuration file
8. For each `bulkwriter` XML element located under a `bulkReader` XML element, set the following attributes to the relevant values:
   - `fileVersion` to **2**
   - `fileSequenceId` to a unique value (see XXX_FILE_SEQUENCE_ID (BulkWriter) parameter for further details)
9. Save your modifications
10. Import the modified CIF configuration for the bulkloader instances using the `cif import` command
11. Restart the bulkloader instances of the Core Server system

3.2.11.13 Enabling the Recording of File Metadata

To monitor the generation of data files at rater and bulkloader level, execute the following procedure:

1. Launch your favorite user interface [page 45] and identify as the administrator of the Core Server system
2. For each rater and bulkloader instance of the Core Server system, enable the service in charge of recording metadata related to data files generated by SAP CC by executing the following commands:
   ```
   set FILE_METADATA_RECORDING_ENABLED true persistent rater
   set FILE_METADATA_RECORDING_ENABLED true persistent bulkloader
   ```

   > **Note**
   >
   > For further information about this parameter, click on the link herafter: FILE_METADATA_RECORDING_ENABLED.

3. Modify if necessary the update interval related to the recording service by modifying the FILE_METADATA_UPDATE_INTERVAL system parameter.
4. Restart the rater and bulkloader instances of the Core Server system. Metadata related to data files generated during the execution of the different processes are now recorded in the Core Database. You can use the Analyze Item Files dedicated application available in the SAP CC Cockpit user interface to visualize and analyze these metadata.

3.2.11.14 Enabling the Event Date Shifting Mechanism for Data Files Generated by the Rater Instances

To enhance the integration with SAP Convergent Invoicing in its system, an event date shifting mechanism is available in SAP Convergent Charging (SAP CC) to match the behavior of both systems regarding the interval of event dates (inclusion, exclusion).

As a system administrator, to enable this mechanism and shift the event dates, complete the following procedure:

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1. Use the SAP CC 2021 Setup Tool user interface to export the CIF[71] configuration of the rater instances executing the following command:

```
cif export rater rater.cif.xml
```

2. Edit the retrieved `rater.cif.xml` configuration file.
3. Add the necessary `eventDateShifter` XML element in the processing channel, such as:
   - Around each `sequence` XML element that surrounds a `billableItemValidator` XML element
   - Around each `billableItemValidator` XML element that is not surrounded by a `sequence` XML element.
4. Save your modifications.
5. Import the modified CIF configuration for the rater instances using the following command:

```
cif import rater rater.cif.xml
```

6. Restart the rater instances in the Core Server system.

3.2.11.15 Enabling the Billable Item Validation for Data Files Generated by Rater Instances

To detect some possible errors during the generation of the billable items, SAP Convergent Charging (SAP CC) gives the possibility to perform a verification of the format of the charged items' fields.

→ Recommendation

- In case of charged item aggregation, SAP SE recommends that you disable the billable item validation to avoid losing charged items related to events which are already confirmed.
- The billable item validation mechanism may impact the global performance of your SAP system landscape. As a consequence, SAP SE recommends that you use this mechanism in validation landscapes to ensure the correct generation of billable items, and then disable it in production landscapes.

As a system administrator, to enable this validation mechanism, complete the following procedure:

1. Launch your favorite user interface [page 45].
2. Export the CIF[72] configuration of the rater instances using the `cif export` command.
3. Edit the retrieved `rater.cif.xml` configuration file
4. Add the necessary `billableItemValidator` XML element in the processing channel. SAP SE recommends that you put this element before the output operation within the data files, to avoid writing data that will later lead to errors
5. Save your modifications.
6. Import the modified CIF configuration for the rater instances using the `cif import` command.
7. Restart the rater instances in the Core Server system.

[71] Charging output Integration Framework
[72] Charging output Integration Framework

3.2.11.16 Enabling the Billable Item Validation for Data Files Generated by Bulkloader Instances (Cloud)

To detect some possible errors during the adaptation of billable items to the targeted web service, SAP CC gives the possibility to perform a verification of the billable items previously created by the rater instances in your SAP system.

As a system administrator, to enable this validation mechanism, complete the following procedure:

1. Launch your favorite user interface [page 45] and identify as the administrator of the Core Server system.
2. Export the CIF[73] configuration of the bulkloader instances using the following `cif export` command:

   ```
   cif export bulkloader bulkloader.cif.xml
   ```

3. Edit the retrieved `bulkloader.cif.xml` configuration file.
4. Add the necessary `billableItemValidator` XML element in the processing channel, and set a `mode` attribute to “WS”.
5. Save your modifications.
6. Import the modified CIF configuration for the bulkloader instances using the following command:
   `cif import bulkloader bulkloader.cif.xml`
7. Restart the bulkloader instances in the Core Server system.

3.2.11.17 Executing a Charging Plan for Consumption Item Mapping

From an existing SAP CC 5.0 SP 6 system running with the older consumption item mapping, execute the following procedure to upgrade to the new consumption item mapping that will retrieve GPART from the external account code.

Update Procedure:

1. Set the ACQUISITION_WITH_CHARGING_PLAN_ENABLED to TRUE.

   > ⚠ Caution
   >
   > For compatibility reasons, this parameter is set to `false` in the case of an update of SAP CC 5.0 from SP 5 (or lower) to SP 6, but set to `true` in case of a new installation.

2. Use the Setup Tool [page 45] user interface.
3. Export from the Core Server system the two files requested by the consumption item mapping.
   Execute the following commands:
   `setupTool eif exportDefaultMapping chargeableItem CC <sapcc_ci_additionalFields.xml>`
   `setupTool eif exportDefaultMapping chargeableItem CI <sapci_ci_defaultMapping.xml>`
   where `<sapcc_ci_additionalFields.xml>` and `<sapci_ci_defaultMapping.xml>` are the names you want to save the files as.

[73] Charging output Integration Framework

4. Create a backup of these two files.
5. Edit the `<sapcc_ci_additionalFields.xml>` file as follows:
    1. **Remove** the `subscriberAccountCode` variable:

```
<exportableItemMappingField name="default"
citcFieldName="default.subscriberAccountCode">
    <additionalInformation name="value" value="subscriberAccountCode"
type="string" />
</exportableItemMappingField>
```

    2. **Add** a new variable (if not already declared):

```
<exportableItemMappingField name="default"
citcFieldName="default.externalAccountCode">
    <additionalInformation name="value" value="externalAccountCode"
type="string" />
</exportableItemMappingField>
```

> **Note**
>
> If you want to complete the default mapping policy with the external account name, add another variable as above, replacing `externalAccountCode` by `externalAccountName`.

6. Save your modifications.
7. Edit the `<sapci_ci_defaultMapping.xml>` file by adding a new variable as follows:

```
<?xml version="1.0" encoding="UTF-8"?>
<exportableItemMapping externalSystemCode="CI">
  <exportableItemMappingField name="CITID"
citcFieldName="default.chargeableItemId" />
  <exportableItemMappingField name="CITIDTYPE"
citcFieldName="default.chargeableItemIdType" />
  <exportableItemMappingField name="SRCTAID"
citcFieldName="default.chargedItemSetIdStr" />
  <exportableItemMappingField name="SRCTATYPE"
citcFieldName="default.chargedItemSetIdType" />
  <exportableItemMappingField name="ACCESS_USERID"
citcFieldName="default.userServiceIdentifier" />
  <exportableItemMappingField name="ACCESS_SERVICEID"
citcFieldName="default.serviceIdentifier" />
  <exportableItemMappingField name="CONSUMDATE"
citcFieldName="default.consumptionDate" />
  <exportableItemMappingField name="GPART"
citcFieldName="default.externalAccountCode" />
  <exportableItemMappingField name="SUBAP"
citcFieldName="default.contractType" />
  <exportableItemMappingField name="VTREF"
citcFieldName="default.contractId" />
  <exportableItemMappingField name="CONTRACTITEMID"
citcFieldName="default.contractItemExternalId" />
  <exportableItemMappingField name="RATINGDATE"
citcFieldName="default.eventProcessingDate" />
</exportableItemMapping>
```

> **Note**
>
> If you want to complete the default mapping policy with the external account name, add another mapping line as above, replacing `GPART` by `VKONT` and `externalAccountCode` by `externalAccountName`.

8. Save your modifications.

9. **Stop the rater instances.**
   For further information about stopping SAP CC instances, refer to the Starting and Stopping the Server Systems section in the SAP CC 2021 Installation and Maintenance Guide documentation.
10. Control the termination of the rater instances using the dedicated app Analyze Item Files in the Convergent Charging Cockpit user interface.
    1. Wait until all the chargeable item data files acquired and charged have been processed by the bulkloader instances.
    2. Manage the potential error files.
11. Import the two previously created files. Execute the following commands:
    `setupTool eif importDefaultMapping chargeableItem <sapcc_ci_additionalFields.xml>`
    `setupTool eif importDefaultMapping chargeableItem <sapci_ci_defaultMapping.xml>`
    where `<sapcc_ci_additionalFields.xml>` and `<sapci_ci_defaultMapping.xml>` are the names you saved the files as.
12. Stop the Core Server system.
    For further information about stopping the Core Server, refer to the Starting and Stopping the Server Systems section in the SAP CC 2021 Installation and Maintenance Guide.
13. Start the updater and dispatcher instances of the Core Server system.
    For further information about starting SAP CC instances, refer to the Starting and Stopping the Server Systems section in the SAP CC 2021 Installation and Maintenance Guide.
14. Open the Core Tool user interface and refresh each existing consumption item mapping in SAP CC.
    Refer to the Modifying the Consumption Item Mapping section of the SAP CC 2021 Primary Help for Core Tool documentation.
    Check that the former subscriber account code has been replaced by the new external account code and that the mapping is the one expected.
15. Start the rater, guider, taxer and bulkloader instances of the Core Server system.

3.2.11.18 Requesting Verbose Logs for Creation of Billable Items (BITs) - Scenario with SAP S/4HANA Cloud Public Edition

In an integrated *SAP solution* scenario with SAP S/4HANA Cloud Public Edition, it is possible to ask SAP S/4HANA Cloud Public Edition to add verbose log messages to the confirmation messages related to the creation operations of billable items (BITs).

To request for these verbose log messages, execute the following procedure:

1. Launch the Admin+ user interface and identify as the administrator of the Core Server system.
2. For each concerned bulkloader instance, modify the value of the POSTPAID_WS_BIT_LOADER_VERBOSE system parameter (see the XXX_VERBOSE (WsBillableItemLoader) system parameters) in order to enable or disable the verbose mode for the log messages sent by SAP S/4HANA Cloud Public Edition.
   For further information about this system parameter, refer to its dedicated documentation available in the SAP CC 2021 System Parameter Reference documentation.

3.2.12 Data Purging

Keywords [page 153]

Preliminary Notes [page 153]

Description [page 153]

Configuring the Allowances Purge Mechanism and Its Scheduler [page 154]

Configuring the File Metadata Purge Mechanism and Its Scheduler [page 155]

Configuring the Product Usage Purge Mechanism and Its Scheduler [page 156]

3.2.12.1 Keywords

Purge, allowance, change list, file metadata, retention period, expired, scheduler, product usage metrics

3.2.12.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- Data Purging description of the SAP CC 2021 Application Help documentation
- Purging and Archiving section of the SAP CC 2021 Administration and Operation Guide documentation
- Installing and Launching Admin+ section of the SAP CC 2021 Installation and Maintenance Guide documentation

3.2.12.3 Description

As described in the SAP CC 2021 Application Help documentation, SAP Convergent Charging manages high volumes of data that is stored within the different databases (Core Database, Session Database and BART Database). To ensure performance, it is necessary to remove obsolete and/or useless data.

You can use this procedure to configure the purge of:

- Expired allowances whose expiration date exceeds a given retention period
- Metadata related to data files generated by SAP CC during the execution of the different processes
- Metrics related to the product usage

3.2.12.4 Configuring the Allowances Purge Mechanism and Its Scheduler

Purge operations related to expired allowances data can be executed:

- Automatically using a dedicated scheduler
- Manually using the `start_purge_allowances` command of the Admin+ user interface
- Automatically or manualy using an external triggering tool that implements some APIs of SAP CC.

To ensure the global performance of the SAP system, regular purges are required.

Allowance data is stored into the Coee Database system, in the ALLO table.

For automated purges with SAP Convergent Charging, the following table contains the list of system parameters and indicators you can use:

System Parameter	Description
ALLOWANCES_PURGE_RETENTION_PERIOD	Number of days until which expired allowances are kept into the database and cannot be deleted
ALLOWANCES_PURGE_THREAD_COUNT	Number of threads to use when performing the allowances purge operations
ALLOWANCES_PURGE_SCHEDULER_ENABLED	Enables/disables the scheduling of purge operations related to expired allowances
ALLOWANCES_PURGE_SCHEDULER_RECURRENCE	Execution recurrence of the dedicated scheduler
ALLOWANCES_PURGE_SCHEDULER_MAX_THROUGHPUT	Maximum number of charging contracts per second that the scheduler can consider when purging allowances

By default, the scheduler in charge of executing allowances purge operations is preconfigured:

- To be triggered every day at midnight
- To be executed using 1 thread per rate
- To purge the allowances of 10 charging contracts per second at the most
- To take into account a default retention period for allowances which must be the same than the retention period used for recharging purposes (set as 40 days by default)

Procedure

> **Note**
>
> Only one execution of the scheduler is possible at the same time.
>
> The date/time of this execution is recorded in the ALLOWANCES_PURGE_SCHEDULER_LAST_TRIGGER indicator, for information purpose.

> ⚠ Caution
>
> For performance purposes, the execution of the scheduler in charge of purging the expired allowances is linked to the scheduler used to perform the activation of provider contracts.
>
> When the execution of these two schedulers occurs at the same time, a merged operation is performed to immediately remove expired allowances of each activated provider contract. As this merged operation also shares the technical configured resources (threads and throughputs), **SAP SE recommends that you decouple the execution of the 2 schedulers if you detect any overload or latency.**

To modify the configuration of the purge operations of data that relate to expired allowances, execute the following procedure:

1. Launch your favorite user interface [page 45] and identify as the administrator of the Core Server system.
2. For each rater instance, modify the values of the following system parameters:
   - ALLOWANCES_PURGE_RETENTION_PERIOD
   - ALLOWANCES_PURGE_THREAD_COUNT

   Change both the persistent and the runtime values.
   You do not need to restart these instances in the system.
3. If you only modified the persistent value of the system parameters, you must restart the rater instances.
4. For each updater instance, modify the values of the following system parameters:
   - ALLOWANCES_PURGE_SCHEDULER_ENABLED
   - ALLOWANCES_PURGE_SCHEDULER_RECURRENCE
   - ALLOWANCES_PURGE_SCHEDULER_MAX_THROUGHPUT

   Change both the persistent and the runtime values.
   You do not need to restart these instances in the system.
5. If you only modified the persistent value of the system parameters, you must restart the updater instances.

3.2.12.5 Configuring the File Metadata Purge Mechanism and Its Scheduler

Purge operations related to metadata for data files (aka item files) can be executed:

- Automatically using a dedicated scheduler
- Manually using the `purge_file_metadata` command of the Admin+ user interface
- Automatically or manually using an external triggering tool that implements some APIs of SAP CC.

To ensure the global performance of the SAP system, regular purges are required.

For each data file (aka item file) generated by the SAP system, some metadata information is recorded and stored into the Core Database system, in the FILE_METADATA table.

For automated purges with SAP Convergent Charging, the following system parameters are available:

System Parameter	Description
FILE_METADATA_PURGE_SCHEDULER_ENABLED	Enables/disables the scheduler in charge of purging metadata related to data files generated by SAP CC

System Parameter	Description
FILE_METADATA_PURGE_SCHEDULER_RECURRENCE	Execution recurrence of the dedicated scheduler, which corresponds by default to an execution every day at midnight
FILE_METADATA_PURGE_RETENTION_PERIOD	Number of days until which file metadata whose reference date/time is exceeded must be taken into account by the dedicated scheduler in charge of purging file metadata

Preparation

Determine the custom values that suit your requirements for automated purges.

Procedure

1. Launch your favorite user interface [page 45] and identify as the administrator of the Core Server system.
2. For each updater instance, modify the values of the following system parameters:
   - FILE_METADATA_PURGE_SCHEDULER_ENABLED
   - FILE_METADATA_PURGE_SCHEDULER_RECURRENCE
   - FILE_METADATA_PURGE_RETENTION_PERIOD

   If you modified the runtime values, the changes are immediate.
   You do not need to restart the updater instances in the system.
3. If you only modified the persistent value of the system parameters, you must restart the updater instances.

3.2.12.6 Configuring the Product Usage Purge Mechanism and Its Scheduler

As described in the Recording SAP Product Usage Metrics [page 238] dedicated section, metrics related to SAP product usage are recorded in the Core Database by a dedicated scheduler of SAP Convergent Charging (SAP CC). In addition to the recording of these metrics, this scheduler is also responsible for executing the purge of metrics that exceed the retention period.

As a consequence, no manual purge operation is available. You can only tune the purge of product usage metrics by:

- Modifying the retention period for the recorded metrics through the METRIC_RECORD_TIME_WINDOW system parameter
- Enabling the scheduler in case it has been disabled through the METRIC_RECORDING_SCHEDULER_ENABLED system parameter

for each updater instance of the Core Server

Modify the Retention Period

Set up METRIC_RECORD_TIME_WINDOW to the persitent value that suits your needs. It defaults to 1,825 days that represents five years. Changes cannot be immediate.

Restart all the updater instances.

Enable the Scheduler Back

Set up METRIC_RECORDING_SCHEDULER_ENABLED to true.

Change all the persistent and runtime values so you do not need to restart the updater instances.

Disable

Set up METRIC_RECORDING_SCHEDULER_ENABLED to false.

Change all the persistent and runtime values so you do not need to restart the updater instances.

3.2.13 Java Virtual Machine

Keywords [page 157]

Preliminary Notes [page 158]

Description [page 158]

Configuration Options [page 158]

Modifying JVM Options [page 161]

Enabling the Generation of Java GC Logs [page 161]

3.2.13.1 Keywords

JVM[74] options, virtual machine, memory heap, garbage collection, buffers

[74] Java Virtual Machine

3.2.13.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- "JVM[75] option recommendations for SAP Convergent Charging" SAP Note 2006073 documentation
- "Generate Java GC Log for SAP Convergent Charging" SAP Note 1950168 documentation

3.2.13.3 Description

The architecture of SAP Convergent Charging relies on a set of Java-based applications whose execution require a Java Runtime Environment (JRE) that can be configured to fit specific requirements.

You can use this procedure to determine the different Java Virtual Machine (JVM) options that must be configured within the startup configuration files of each instance of the Core Server system.

> **Note**
>
> For further information on how to configure the startup configuration file, refer to the Startup Configuration Files [page 234] section of this documentation.

3.2.13.4 Configuration Options

The following table contains the list of JVM[76] options you can configure:

JVM Option	Description
JVM Settings for Garbage Collection	
`-XX:{+-}PrintGC`	Enables/disables the basic GC logging mode, which consists in recording 1 line for every Young Generation GC and for every Full GC
`-XX:{+-}PrintGCDetails`	Enables/disables the detailed GC logging mode, whose content depends on the algorithm used for garbage collection
`-Xloggc:<FULL_PATH>`	Defines the path and name of the logging file. Setting this option implicitly sets the -XX:+PrintGC and -XX:+PrintGCTimeStamps options.
`-XX:LogGcMaxFileSize=<SIZE>`	Defines the maximum size of a GC log file, expressed in bytes. If the log file exceeds this size, a new log file is created. A number of "old" log files are kept, according to the value of the -XX:LogGcMaxFileCount option
`-XX:LogGcMaxFileCount=<NUMBER>`	The maximum number of files to use for the GC log. A new GC log file is created when the JVM is restarted or when the log file size exceeds the size given by the -XX:LogGcMaxFileSize option

[75] Java Virtual Machine
[76] Java Virtual Machine

JVM Option	Description
`-XX:{+-}PrintGCCause`	Prints/hides information related to the reason of the garbage collection
`-XX:{+-}PrintGCTimeStamps`	Prints/hides the timestamp of the garbage collection
`-XX:{+-}PrintTenuringDistribution`	Prints/hides tenuring age information
`-XX:{+-}PrintPromotionFailure`	Prints/hides additional diagnostic information about promotion failure
`-XX:{+-}PrintFLSStatistics`	Prints/hides additional information about the fragmentation of the heap memory
JVM Settings for Garbage Collection and History	
`-XX:{+-}GCHistory`	Enables/disables the garbage collection history. When enabled, you can configure the way this history is stored by defining the other -XX:GCHistory options
`-XX:GCHistoryFilename=<FULL_PATH>`	Defines the path and name of the file used to store the information related to garbage collection history
`-XX:GCHistoryMaxFileSize=<SIZE >`	Defines the maximum size of a file containing information related to garbage collection history, expressed in bytes
`-XX:GCHistoryMaxFileCount=<NB_FILES>`	Defines the maximum number of files that are used for the garbage collection history, from 1 to 99. A new garbage collection history file is created when the JVM is restarted. **Note** The `.b<00..99>` extension is added for backup files (e.g. `sapjvm_gc.prf.b00`, which is the newest backup file).
`-XX:GCHistoryCompressionLevel=<LEVEL>`	Defines the compression level for garbage collection history files. • 0 means no compression • 9 is the maximum possible compression • 5 is the default value, which corresponds to medium compression and medium CPU[77] cost, but which is not recommended for SAP CC
JVM Settings for Performance and Stability	
`-Xmn<SIZE >`	Defines the size of the heap memory for the young generation, expressed in bytes. Append the letter k or K to indicate kilobytes, or m or M to indicate megabytes
`-Xms<SIZE >`	Defines the initial size of the heap memory, expressed in bytes. This value must be a multiple of 1024 greater than 1MB. Append the letter k or K to indicate kilobytes, or m or M to indicate megabytes. The default value is 2MB
`-Xmx<SIZE >`	Defines the maximum size of the heap memory, expressed in bytes. This value must be a multiple of 1024 greater than 2MB. Append the letter k or K to indicate kilobytes, or m or M to indicate megabytes. The default value is 64MB

[77] Central Processing Unit

JVM Option	Description
-Xss<SIZE >	Defines the size of the Java thread stack, expressed in bytes. Append the letter k or K to indicate KB, m or M to indicate MB, g or G to indicate GB. The default value depends on the virtual memory
-XX:{+-}UseParallelOldGC	Enables the use of the parallel compacting collector algorithm for GCs operations. Enabling this option implicitly enables the -XX:+UseParallelGC option
-XX:{+-}UseSplitVerifier	Enables the splitting of the verification process into two phases: type referencing (performed by the compiler) and type checking (performed by the JVM runtime). This option was deprecated in JDK 8, and verification is now split by default without a way to disable it
-XX:MaxDirectMemorySize=<SIZE >	Defines the maximum total size (in bytes) of the New I/O (the java.nio package) direct-buffer allocations. Append the letter k or K to indicate kilobytes, m or M to indicate megabytes, g or G to indicate gigabytes. By default, the size is set to 0, which means that the JVM chooses automatically the size for NIO direct-buffer allocations
-XX:CompileThreshold=<NUMBER>	Defines the number of interpreted method invocations before compilation. By default: • On server side, the JIT compiler performs 10000 method invocations to gather information for efficient compilation • On client side, the default setting is 1500 invocations
-XX:SurvivorRatio=<RATIO>	Defines the initial survivor space ratio that is used by the throughput garbage collector (which is enabled by the -XX:+UseParallelGC and/or -XX:+UseParallelOldGC options). By default, the initial survivor space ratio is set to 16
-XX:ConcGCThreads=<NUMBER>	Defines the number of threads used for concurrent GC. The default value depends on the number of CPUs that are available for the JVM
-XX:ParallelGCThreads=<NUMBER>	Defines the number of threads used for parallel garbage collection in the young and old generations. The default value depends on the number of CPUs that are available for the JVM
-XX:TargetSurvivorRatio=<NUMBER>	Defines the desired percentage of survivor space (from 0 to 100) used after young garbage collection. By default, this option is set to 90%
-XX:{+-}ScavengeBeforeFullGC	Enables garbage collection of the young generation before each full GC. This option is enabled by default and it is not recommended to disable it, as scavenging young generation before a full GC can reduce the number of objects reachable from the old generation space into the young generation space
-XX:{+-}UseAdaptiveSizePolicy	Enables the automatic sizing of the young generation and selection of an optimum survivor ratio to maximize performance during parallel GC
-XX:PromotedPadding=<NUMBER>	Defines the size of the buffer used for promotion failures
-XX:{+-}DisableExplicitGC	Disables the processing of calls to System.gc(). SAP SE recommends that you disable this option for updaters and dispatchers
-XX:{+-}TieredCompilation	Enables the tiered compilation
Other JVM Settings	
-Dsapcc.home	File path and name of the SAP CC Working Directory

JVM Option	Description
`-Dboot.config=<FULL_PATH>`	File path and name of the `boot.config` file
`-Dfile.encoding`	Encoding settings. Must be set to UTF-8
`-Dsun.nio.ch.useIPv6Stack`	Only required for MS Windows landscapes using IPv6
`-Dcom.sap.jvm.xerces.disallowDoctypeDecl.failEarly`	Must be set to `true`

3.2.13.5 Modifying JVM Options

If you need to reconfigure these JVM options, then change the startup configuration file (`jstart.config`). See Modifying a Startup Configuration File [page 236].

3.2.13.6 Enabling the Generation of Java GC Logs

Performance issues or timeout issues for one instance of your SAP Convergent Charging system may relate to a performance issue with the execution of the Java garbage collection (GC) process that is executed by the Java Virtual Machine (JVM) that runs an SAP CC system instance composing the SAP system.

To track any Java garbage collection performance issue, your support specialists can inspect detailed information available in recorded Java GC logs:

- Basic logs (Java GC history log)
- Advanced logs (Java verbose GC log)

Check the space for the Java GC logs in the file system (SAN, disks).

Complete the procedures detailed in SAP Note 1950168.

Related Information

Log Files of the Java Garbage Collection (GC)

3.2.14 JCo Destinations

Keywords [page 162]

Preliminary Notes [page 162]

Description [page 162]

Configuration Settings for JCo [page 162]
Creating JCo Destinations [page 165]
Maintaining a JCo Destination [page 166]
Setting Up the ERP Reference System [page 167]

3.2.14.1 Keywords

JCo[78] destination, RFC[79] over TCP/IP[80], SAP ERP reference, SAP CRM, SAP ERP/FI-CA

3.2.14.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- SAP JCo documentation available on SAP Help Portal
- Installing and Launching Setup Tool and Securing the RFC over TCP/IP Communications sections of the SAP CC 2021 Installation and Maintenance Guide documentation

3.2.14.3 Description

SAP Convergent Charging uses the SAP Java Connector (SAP JCo)[81] technology to communicate with the SAP CRM and SAP ERP/FI-CA systems, through the Remote Function Calls (RFCs) protocol provided by SAP.

You can use this procedure to manage the different JCo destinations in your SAP system landscapes.

3.2.14.4 Configuration Settings for JCo

The following table contains the list of configuration settings that you can change [page 166], grouped into the following categories:

- JCo[82] server settings
- JCo client settings
- Tuning settings

78 Java Connector
79 Remote Function Call
80 Transmission Control Protocol/Internet Protocol
81 Java Connector
82 Java Connector

- Security settings
- Load-balancing connections

Setting	Values
JCo Server Settings	
`jco.client.ashost` Host name or IP address of the SAP Application Server (or SAP Netweaver Gateway) → Remember This property is disallowed when using load balancing features.	By default, no value is set. You must set the value that corresponds to the targeted host.
`jco.client.sysnr` SAP System Number used to determine the gateway service → Remember This property is disallowed when using load balancing features.	By default, no value is set. The system number corresponds to the last two numbers of the port the targeted SAP System runs on.
JCo Client Settings	
`jco.client.client` Name of the SAP logon client	By default, no value is set. You can set any 3-digit value that corresponds to the targeted SAP System (preserving the leading zeros in case they appear in the number).
`jco.client.user` Logon used for authenticating on the remote SAP System	By default, no value is set. You must specify a service user that relates to a communication channel between SAP CC and an SAP system (SAP ERP, SAP CRM) that is available in your SAP system landscape.
`jco.client.passwd` Password of the previously specified logon, used for authenticating on the remote SAP System	By default, no value is set. You must set a password that respects the SAP password policy.
`jco.client.lang` Language of the remote SAP System	The default value is set to: **en** You can set any ISO two-character language code or SAP-specific single-character language code.
Tuning Settings	
`jco.destination.peak_limit` Maximum number of active connections that can be simultaneously created for a JCo destination	The default value is set to: **10** You can set any positive value.
`jco.destination.pool_capacity` Maximum number of idle connections kept open by the JCo destination	The default value is set to: **3** You can set any positive value. Setting this parameter to 0 disables the pooling of connections.

Setting	Values
Security Settings	
`jco.client.snc_mode` Enables/disables the use of Secured Network Communications (SNC)	The default value is set to: **0** Possible values are: • **0**: Not activated • **1**: Activated
`jco.client.snc_lib` Absolute path of the SAP Cryptographic Library that provides secured network connections	By default, no value is set. You must set a value that corresponds to the absolute path of the targeted library, such as: • For **MS Windows** operating systems: `C:\usr\sap\<SID>\SYS\exe\uc\<OS>\sapcrypto.dll` • For **Linux, HP-UX** and **Solaris** operating systems: `/usr/sap/<SID>/SYS/exe/uc/<OS>/libsapcrypto.so` • For **AIX** operating systems: `/usr/sap/<SID>/SYS/exe/uc/<OS>/libsapcrypto.o` Where: • `<SID>` corresponds to the system identifier of the concerned Core Server system • `<OS>` corresponds to the relevant operating system
`jco.client.snc_qop` Security level of the secured network connections	The default value is set to: **3** Possible values are: • **1**: Authentication only • **2**: Integrity protection • **3**: Privacy protection • **8**: Use the value from `snc/data_protection/use` on the SAP Application Server
`jco.client.snc_myname` Name of the secured network connection, that overrides the default secure network connection partner	By default, no value is set. You must set a value that corresponds to the SNC name for SAP Convergent Charging, such as: **`p:CN=SAPCC, O=<MyCompany>, C=US`** **Note** • Although this parameter is optional, SAP SE recommends that you set this value in order to ensure that a correct SNC name is used for the connection • The SNC name does not correspond to the Distinguished Name you used when creating the Personal Security Environment. The SNC name has the following syntax: **`p:<Distinguished_Name>`**.

Setting	Values
`jco.client.snc_partnername` SNC name of the Application Server, that acts os the SNC partner	By default, no value is set. You can find the Application Server's SNC name in the profile parameter named `snc/identity/as`
`jco.client.snc_sso` Single Sign-On protocol	The default value is set to: **1** Possible values are: • **0**: Single Sign-On protocol disabled • **1**: Single Sign-On protocol enabled Set the value to **0** as of JCo 3.0.9
Load-Balancing Connections	
`jco.client.mshost` Host name or IP address of the message server	By default, no value is set. You must set the value that corresponds to the targeted host.
`jco.client.msserv` Port number or logical name of the message server	The default value is set to: **sapms**`<sysid>`
`jco.client.r3name` SID of the system	By default, no value is set. Corresponds to the three-letter system ID of your backend system. This option is ignored if `jco.client.msserv` is set.
`jco.client.group` Name of the logon group of the messaging server (optional)	The default value is set to: **PUBLIC**

→ Remember

SAP Convergent Charging considers all JCo standard parameters. Consult the SAP Java Connector documentation for any other specific one: https://support.sap.com/en/product/connectors/jco.html

3.2.14.5 Creating JCo Destinations

To create a JCo[83] destination in the Core Server system, execute the following procedure:

1. In your SAP CC system landscape, connect to the host where a dispatcher instance is installed.
2. On this host, copy the `jco.destination.sk` file available in the `config/sapci` folder of the dispatcher instance in a new `jco.destination` file.
3. Edit this newly created file and modify the relevant settings to fit the targeted SAP system.

   For further information about the available configuration settings, refer to the Configuration Settings for JCo [page 162] section.

[83] Java Connector

4. In the Setup Tool user interface, execute the `jcodestination import` command to import these settings in the Core Database system.

   > ⚠ Caution
   >
   > Ensure that the JCo destination name does not contain spaces and/or special characters (especially slash and backslash).

5. Restart the dispatcher and the bulkloader instances in the Core Server system.
6. Verify if you must restart the rater instances in your Core Server system.

   If the system parameter JCO_CONNECTION_COUNT is set to 1 or more, then do restart all these rater instances.

Related Information

jcodestination import

3.2.14.6 Maintaining a JCo Destination

To maintain a JCo destination, still using the Setup Tool user interface:

1. List the existing JCo destinations by executing the `jcodestination list` command.
2. Export these JCo destinations settings by executing the `jcodestination export` command.
3. Edit the retrieved file and modify the relevant settings depending on your business or technical requirements.
4. Import the modified JCo destinations by executing the `jcodestination import` command.
5. To apply your changes, restart the dispatcher and the bulkloader instances in the Core Server system.
6. Verify if you must restart the rater instances in your Core Server system.

   If the system parameter JCO_CONNECTION_COUNT is set to 1 or more, then do restart all these rater instances.

Related Information

jcodestination list
jcodestination export
jcodestination import

3.2.14.7 Setting Up the ERP Reference System

SAP Convergent Charging communicates with a particular SAP ERP system in your landscape to synchronize some business or technical data (such as currencies, billable item (BIT) classes or consumption item classes). This reference ERP system corresponds to a JCo destination that is set at installation time by the SAP installer (in case of integrated installation scenario). In case you need to modify this initial configuration or refer to a new SAP ERP system, execute the following procedure:

1. In the Setup Tool, execute the `jcodestination list` command to retrieve the list of all the JCo destinations that are available in the Core Database system.
2. If the targeted SAP ERP system corresponds to one of the previously retrieved JCo destinations, execute the `jcodestination export` command to ensure that the configuration of this JCo destination is correct. If the targeted SAP ERP system does not correspond to one of the previously retrieved JCo destinations, or if the exported configuration does not fit, you must create the relevant JCo destination using the Creating JCo Destinations [page 165] section above.
3. Once the relevant JCo destination has been located or created, execute the `erpreferencesystem setJCoDestination` command, specifying the name of the JCo destination that must be used to communicate with the SAP ERP system.
4. Restart the dispatcher and bulkloader instances of the Core Server system.

3.2.15 Logging and Tracing

Keywords [page 167]

Preliminary Notes [page 167]

Description [page 168]

Configuration Options [page 169]

Modifying the Logging and Tracing Policy [page 171]

Integrating SAP CC Logs and Traces with Your Java Application [page 176]

3.2.15.1 Keywords

Log, trace, category, severity, domain, threshold, troubleshoot

3.2.15.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- Installing and Launching Admin+ section of the SAP CC 2021 Installation and Maintenance Guide documentation

- Installing and Launching Config Tool section of the SAP CC 2021 Installation and Maintenance Guide documentation
- Starting and Stopping the Server Systems section of the SAP CC 2021 Installation and Maintenance Guide documentation
- Enhanced Logging and Tracing Framework section of the SAP CC 2021 Application Help documentation
- “Troubleshooting: Logs and Traces” group in the SAP CC 2021 System Parameter Reference documentation
- Startup Configuration Files [page 234] configuration section of this documentation

This procedure uses the following variables:

- `<SAPCC_SYSADM_USERNAME>` and `<SAPCC_SYSADM_PASSWORD>`, which corresponds to the name and password of the administrator of your Core Server system ("admin" by default)
- `<INSTANCE_ID>`, which corresponds to the identifier of a concerned instance
- `<LEVEL>`, which corresponds to a possible level (DEBUG, PATH, INFO, WARNING, ERROR or FATAL)
- `<XML_FILE_PATH>`, which corresponds to an XML file containing the exported configuration of the Core Server system

3.2.15.3 Description

As described in the SAP CC 2021 Application Help documentation, SAP Convergent Charging provides logging and tracing features that give the possibility to generate log messages (for system administrators) and trace messages (for support specialists):

- According to defined levels such as severity, thresholds, domains, categories
- Using different output formats
- In multiple output destinations
- That can be viewed and analyzed in the SAP Management Console or SAP Microsoft Management Console applications

Mainly addressed to SAP CC administrators and support specialists, the generation of these log and trace messages can be configured in order to fit specific needs such as landscape monitoring, performance tracing, issue troubleshooting, and so on.

You use the following procedures to configure the logging and tracing capabilities of your Core Server system and thus fit your specific needs.

Related Information

Enhanced Logging and Tracing Framework

3.2.15.4 Configuration Options

Logging and tracing features can be configured:

- Through SAP CC system parameters
- Through dedicated options of the JVM[84]

The following table contains the list of available settings, including the correspondence between Core Server system parameters and JVM options (when exists):

Settings	Description	System Parameter	JVM Option
Logs (**business** processing)	Severity threshold for logs	LS_APP_SEVERITY	`ls.app.severity`
	Identifier(s) of the destination(s)	LS_APP_DESTINATION	`ls.app.destination`
Logs (**system** processing)	Severity threshold for logs	LS_SYS_SEVERITY	`ls.sys.severity`
	Identifier(s) of the destination(s)	LS_SYS_DESTINATION	`ls.sys.destination`
Traces	Severity threshold for traces	LS_TRC_SEVERITY	`ls.trc.severity`
	Identifier(s) of the destination(s)	LS_TRC_DESTINATION	`ls.trc.destination`
	Domain(s) for traces	LS_TRC_DOMAIN	`ls.trc.domain`
Console Destination	Enable/Disable	LDC_ENABLED	`ld.console.enabled`
	Output format	LDC_FORMATTER	`ld.console.formatter`
Fileset Destination #1	Enable/Disable	LDF1_ENABLED	`ld.file1.enabled`
	Output format	LDF1_FORMATTER	`ld.file1.formatter`
	Number of files available for the rotation	LDF1_FILESET_SIZE	`ld.file1.setsize`
	Maximum file size before rotation	LDF1_FILE_SIZE	`ld.file1.size`
	File path, name, and extension	LDF1_FILE_NAME	`ld.file1.name`
Fileset Destination #2	Enable/Disable	LDF2_ENABLED	`ld.file2.enabled`
	Output format	LDF2_FORMATTER	`ld.file2.formatter`
	Number of files available for the rotation	LDF2_FILESET_SIZE	`ld.file2.setsize`

[84] Java Virtual Machine

Settings	Description	System Parameter	JVM Option
	Maximum file size before rotation	LDF2_FILE_SIZE	`ld.file2.size`
	File path, name, and extension	LDF2_FILE_NAME	`ld.file2.name`
Fileset Destination #3	Enable/Disable	LDF3_ENABLED	`ld.file3.enabled`
	Output format	LDF3_FORMATTER	`ld.file3.formatter`
	Number of files available for the rotation	LDF3_FILESET_SIZE	`ld.file3.setsize`
	Maximum file size before rotation	LDF3_FILE_SIZE	`ld.file3.size`
	File path, name, and extension	LDF3_FILE_NAME	`ld.file3.name`
Fileset Destination #4	Enable/Disable	LDF4_ENABLED	`ld.file4.enabled`
	Output format	LDF4_FORMATTER	`ld.file4.formatter`
	Number of files available for the rotation	LDF4_FILESET_SIZE	`ld.file4.setsize`
	Maximum file size before rotation	LDF4_FILE_SIZE	`ld.file4.size`
	File path, name, and extension	LDF4_FILE_NAME	`ld.file4.name`
Fileset Destination #5	Enable/Disable	LDF5_ENABLED	`ld.file5.enabled`
	Output format	LDF5_FORMATTER	`ld.file5.formatter`
	Number of files available for the rotation	LDF5_FILESET_SIZE	`ld.file5.setsize`
	Maximum file size before rotation	LDF5_FILE_SIZE	`ld.file5.size`
	File path, name, and extension	LDF5_FILE_NAME	`ld.file5.name`
JCo[85] traces (**integrated** landscape)	Level of JCo traces, from 0 to 10	JCO_TRACE_LEVEL	*N/A*
	Level of JCo CPIC traces, from 0 to 3	JCO_CPIC_TRACE_LEVEL	*N/A*
	Enable/Disable JCo RFC[86] traces	JCO_JRFC_TRACE_LEVEL	*N/A*
	Path of the folder used to write trace files	JCO_TRACE_PATH	*N/A*

By default, logging and tracing features are configured to fit a productive landscape, which corresponds to the following configuration:

- The logging feature is enabled, with severity thresholds set to INFO
- The tracing feature is enabled, with severity thresholds set to ERROR

[85] Java Connector
[86] Remote Function Call

- Log and trace messages are generated in different files, that are available in the SAP Management Console or SAP Microsoft Management Console applications
- Generated files are stored in the working directory of each concerned instance
- File rotation is enabled

3.2.15.5 Modifying the Logging and Tracing Policy

To fit specific needs, you can modify the default configuration of the *Logging and Tracing* features for targeted instances. Regarding to this feature, some configuration settings can be immediate whereas some are not immediate and require a restart.

According to the modifications, you can perform:

- **Immediate** modifications:
  - Which do not need a restart of the targeted instances
  - Which can be temporary or permanent
  - Which can only be performed using:
    - The app Manage SAP CC System Parameters (in the SAP CC Cockpit user interface)
    - Or the Admin+ user interface
  - And which only concern severity thresholds and domains associated to:
    - A given instance
    - All instances of a given type
    - All instances of the Core Server system
- **Deferred** modifications:
  - Which need a restart of the targeted instances
  - Which are permanent
  - Which can be performed using:
    - The app Manage SAP CC System Parameters (in the SAP CC Cockpit user interface)
    - The Admin+ user interface (recommended in case of a few changes)
    - The Config Tool user interface (recommended in case of mass changes)
- For changes relating to destinations, storage location, file rotation rules, and format, SAP SE recommends that you update the startup configuration files. See Startup Configuration Files [page 234].

Related Information

Configuration Options [page 169]

3.2.15.5.1 Immediate or Deferred Modifications Using Manage SAP CC System Parameters

With the app Manage SAP CC System Parameters of the SAP CC Cockpit, you can search for and view some system parameter values that are defined in a running SAP CC Core Server system. You quickly and efficiently inspect and change the persistent values and the runtime values.

For more information, refer to the Manage SAP CC System Parameters section of the SAP CC 2021 Primary Help for Cockpit documentation.

3.2.15.5.2 Immediate Modifications Using Admin+

In order to debug your SAP Convergent Charging system (Core Server), use this procedure to tune the generation and recording of trace messages by the system or by specific instances in the system.

Procedure

> ⚠ Caution
>
> In a live production system, contact your SAP Support Team for troubleshooting instructions and supervision.

> ⓘ Note
>
> When tuning the trace generation, **change the tracing domains first, the trace severity threshold second.**
>
> **At the end, always reset the severity threshold to its default value for the trace messages: ERROR.**

To perform immediate modification of the severity level threshold for traces and recorded tracing domains, complete the following procedure:

1. Launch the Admin+ user interface and identify as the system administrator of the Core Server system.
2. Depending on your troubleshooting investigation, determine and configure the tracing domains for either the system or specific instances in the system.

   > → Remember
   >
   > In a production system landscape, SAP Support Team provides you with the relevant domains.

   Modify the runtime values of the LS_TRC_DOMAIN system parameter using the `set` command with the `memory` scope:

Typical Technical Operation	Procedure
Change the selection of tracing domains for **all the instances of the same type** in the system. For example, all the bulkloader instances.	`set LS_TRC_DOMAIN JCO memory bulkloader`
Change the selection of tracing domains for **one specific instance in the system**. For example, `rater#1`.	`set LS_TRC_DOMAIN CACHE,MSG memory rater#1`
Change the selection of tracing domains for **two (or more) specific instances** in the system. For example, `rater#3` and `bulkLoader#3`.	`set LS_TRC_DOMAIN MSG,JCO memory rater#3 bulkLoader#3`
Change the selection of tracing domains for the **complete system**.	`set LS_TRC_DOMAIN SQL memory`

> → Tip
>
> With the `set` command, you can specify a group of system instances by entering their instance IDs and instance types. See the primary help for more information.

3. Enable the generation and recording of debug trace messages in the selected instance, complete system, or specific instances.

   By default, the threshold is set to ERROR (only the trace messages associated to errors are generated). In SAP Convergent Charging, the DEBUG threshold for traces is essential for the cause analysis by the SAP Support Team organization and our SAP labs.

   Modify the values of the LS_TRC_SEVERITY system parameter with the `memory` scope:

Typical Technical Operation	Procedure
Change the trace severity threshold for **all the instances of the same type** in the system. For example, all the bulkloader instances.	`set LS_TRC_SEVERITY DEBUG memory bulkloader`
Change the trace severity threshold for **one specific instance in the system**. For example, `rater#1`.	`set LS_TRC_SEVERITY DEBUG memory rater#1`
Change the trace severity threshold for **two (or more) specific instances** in the system. For example, `rater#3` and `bulkLoader#3`.	`set LS_TRC_SEVERITY DEBUG memory rater#3 bulkLoader#3`

Typical Technical Operation	Procedure
Change the trace severity threshold for the **complete system**.	`set LS_TRC_SEVERITY DEBUG memory`

4. The reconfigured instances generate and record trace messages accordingly.
5. Monitor the traces and continue your troubleshooting. For example, try to reproduce an issue.
6. **As soon as possible, stop this debug mode.** Important trace messages may be lost or erased in case of rotating trace files.
   Reset the trace severity threshold to the default value for all instances in your Core Server system, enter the following command:

   ```
   reset LS_TRC_SEVERITY all
   ```

   Otherwise, you can use the `set` command:

   ```
   set LS_TRC_SEVERITY ERROR all
   ```

> ⚠ Caution
>
> Do not forget to reset the trace severity threshold to its default value: ERROR.

For more information, refer to the SAP CC 2021 Primary Help for Admin+ documentation.

3.2.15.5.3 Deferred Modifications Using Admin+

To perform deferred modification of a given setting related to logging and tracing features, execute the following procedure:

1. Launch the Admin+ user interface and identify as the administrator of the Core Server system.
2. Modify the configuration of logs and/or traces, using the **`set`** command with the `persistent` scope. See Configuration Options [page 169] for the list of system parameters.
3. Restart the concerned instances of the Core Server system.

> Example
>
> - To permanently increase the severity of traces to DEBUG for a given instance of the Core Server system, enter the following command:
>   `set LS_TRC_SEVERITY DEBUG persistent <INSTANCE_ID>`
> - To change the number of files generated in destination#1 and apply this modification to all the instances of the Core Server system, enter the following command:
>   `set LDF_FILESET_SIZE <NB_FILE> persistent all`

3.2.15.5.4 Deferred Modifications Using Config Tool

> **Note**
>
> As of the release 4.1 of SAP CC, the configuration of the logging and tracing mechanism relating to the Core Server system has changed. It is now possible to write all logging and tracing information in the same location (before and after connecting to the back-end database).
>
> For backward compatibility reasons, the existing logging and tracing configuration is not modified during the upgrade process. You must use the procedure below to benefit from this enhancement.

To perform deferred modification of a given setting related to logging and tracing features, execute the following procedure:

1. Launch the Config Tool user interface using the configuration export command in order to export the whole configuration of your SAP CC Core Server system into an `XML` file:
   `config configuration export <XML_FILE_PATH> all`
2. Open the newly created `XML` file, and for each instance type (rater, updater, guider, bulkloader, taxer, dispatcher), mark down the values of each parameter that relate to the configuration of the logging and tracing features. For a complete list of parameters to analyze, refer to the Configuration Options [page 169] section above.
3. For each analyzed system parameter:
   - Set the corresponding JVM[87] option to the same retrieved value
   - Remove the `value` XML attribute of the system parameter from the configuration file
4. Update the following JVM options:
   - `-Dls.trc.severity=PATH`
   - `-Dls.trc.domain=SQL,NET,HB`
5. Remove the following JVM option:
   - `-Dld.console.enabled=true`
6. Edit the Startup Configuration File of each concerned instance and set the previously determined values for the different JVM options. For further information, refer to the Startup Configuration Files [page 234] section
7. Execute the `configuration import` command in order to import the content of the `XML` file containing the modified configuration of your Core Server system:
   `config configuration import <XML_FILE_PATH>`
8. Restart the concerned instances of the Core Server system

> **Example**
>
> Considering the following list of system parameters for a given instance type:
>
> - `LDF1_ENABLED = true`
> - `LDF1_FILE_NAME = work/log/@INSTANCE_ID@.log`
> - `LDF1_FILE_SIZE 4`
> - `LDF1_FILESET_SIZE 200M`
> - `LDF1_FORMATTER LIST`

[87] Java Virtual Machine

- LS_APP_DESTINATION **LDF1**
- LS_SYS_DESTINATION **LDF1**
- LS_TRC_DESTINATION **LDF2**

The corresponding JVM options to set are:

- `-Dld.file1.enabled=true`
- `-Dld.file1.name=work/log/@INSTANCE_ID@.log`
- `-Dld.file1.size=4`
- `-Dld.file1.setsize=200M`
- `-Dld.file1.formatter=LIST`
- `-Dls.app.destination=LDF1`
- `-Dls.sys.destination=LDF1`
- `-Dls.trc.destination=LDF2`
- `-Dls.trc.severity=PATH`
- `-Dls.trc.domain=SQL,NET,HB`

3.2.15.5.5 Deferred Modifications Using the Startup Configuration Files

To perform deferred modification of a given setting related to logging and tracing features, complete the following procedure:

1. Edit the startup configuration file [page 236] of each concerned SAP CC system instance.
   For more information, refer to the Startup Configuration Files [page 234] section.
2. Update the value of each JVM option determined using the list of parameters available in the Configuration Options [page 169] section.
3. Restart the concerned instances in your Core Server system.

Related Information

Modifying a Startup Configuration File [page 236]

3.2.15.6 Integrating SAP CC Logs and Traces with Your Java Application

In case you need to interface your Java-based application to the SAP CC logging and tracing framework, you need to fulfill the following requirements:

- You must use the SAP JVM[88]

- Your source code interacts with the `sap.com~tc~logging~java.jar` SAP CC library, that needs to be included in the classpath of your application
- Your source code interacts with the `core_client.jar` SAP CC library, that contains public APIs[89] you can use within your application, in particular the following packages:
  - `com.highdeal.cnd.message`, used for communications with SAP CC performed over the Messages over TCP/IP[90] communication channel
  - `com.highdeal.hci.http`, used for communications with SAP CC performed over the XML[91] over HTTP[92] communication channel
- Your client application contains a configuration file that defines the relevant system parameters of SAP CC as defined in the Configuration Options [page 169] section above

3.2.16 Master Data Management

Keywords [page 177]

Preliminary Notes [page 177]

Description [page 178]

Configuring Master Data for the Service Provider [page 178]

Configuring Master Data for the Customer of the Service Provider [page 203]

3.2.16.1 Keywords

Product Modeling, chargeable item class, refill item class, charged item class, Refill Record Class, refill logic, reusable charge, counter name dictionary, spending status description, charge plan, refill plan, monitoring plan, shared counters, provider contract, subscription, access, batch rating group assignment, subscriber account, subscriber mapping table, subscriber range table, offer

3.2.16.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- Master Data section of the SAP CC 2021 Application Help documentation
- Installing and Launching Core Tool section of the SAP CC 2021 Installation and Maintenance Guide documentation

88 Java Virtual Machine
89 Application Programming Interface
90 Transmission Control Protocol/Internet Protocol
91 eXtended Markup Language
92 HyperText Transfer Protocol

- “Roles and Authorizations” section of the SAP CC 2021 Security Guide documentation

3.2.16.3 Description

As described in the SAP CC 2021 Application Help, SAP Convergent Charging manages different sets of master data whose availability depends on the implemented model that can be:

- **Provider contracts and subscriptions**, that corresponds to the initial model which was based on subscriptions, but that is no more supported except in upgraded landscapes containing related master data
- **Provider contracts only**, that corresponds to a new model designed to facilitate the integration with the SAP ERP/FI-CA component of the SAP Business Suite, within which commercial products are configured through charging units available from SAP CC charge plans.

For a complete list of master data associated to each available model, refer to the Master Data section of the SAP CC 2021 Application Help documentation. You can use the following procedure to configure the different master data related to the service provider and thus fit your specific business needs.

3.2.16.4 Configuring Master Data for the Service Provider

To configure the master data related to the service provider and its chargeable services, execute the following procedure:

1. Launch the Core Tool user interface and identify as an SAP CC user granted the "Marketing" role.
2. As a pricing specialist, you configure the service pricing.

   Create a pricing catalog that suits your business needs, design, and modeling.
3. Configure the necessary pricing elements in this catalog. Set up the relevant master data according to your needs, using the following information:

Master Data	Description
(reusable) Charge	**Core Tool menu**: *File* > *New* > *Charge* Charges are used to manage fees related to the use of a customer service. Each charge applies to a chargeable item class, and includes: • A price plan • A charge plan

Master Data	Description
Charge Plan	**Core Tool menu**: *File* > *New* > *Charge Plan* Charge plans represent the charging conditions for accessing or consuming goods or services of a service provider. Each charge plan consists of : • One or multiple charges • A set of parameters used to configure the charge(s) • A release status
Chargeable Item Class	**Core Tool menu**: *File* > *New* > *Chargeable Item Class* Chargeable items are sent to SAP CC by network elements for charging purpose. Each chargeable item represents a set of properties used: • During the execution of the price plan that is associated to the different charges related to the class of the chargeable item. • By a third-party billing system during billing and invoicing operations

Master Data	Description
Charged Item Class	**Core Tool menu**: *File* > *New* > *Charged Item Class* Charged items represents items generated by SAP CC during charging operations. Each charged item is associated to a class that defines the properties to output, whose values can be generated by price plans or on-the-fly computed. The following table contains the list of properties that can be generated by SAP CC, that you can also use when configuring: • A charge customized in a charge plan or in an offer • An allowance logic customized in an allowance plan

Property	Description
`Account Type Code`	Type of account charged by SAP CC. Possible values are: • 0: Prepaid account • 1: External (postpaid) account
`Account Type Is Prepaid`	Boolean used to indicate that the account charged by SAP CC is a prepaid account
`Account Type Name`	Textual representation of the type of account charged by SAP CC. Possible values are: • "prepaid" • "postpaid"
`Allowance Context`	Type of event performed on the allowance and that generated the charged item. Possible values are: • "C", when the event corresponds to a one-shot trigger performed on an allowance creation event • "U" otherwise
`Allowance Event Class Name`	Class name of the allowance event that was sent to the allowance. **Note** This property is empty for one-shot and recurring triggers.
`Allowance Share Identifier`	Unique identifier used to group and access to shared allowances.
`Allowance Unique Identifier`	Unique identifier of the allowance.
`Allowance Validity Start Date`	Start date of the validity period of the allowance
`Allowance Validity End Date`	End date of the validity period of the allowance

Master Data	Description

Property	Description
`Amount (Excluding Tax)`	Amount of the charged item computed by SAP CC during the charging operation, without any tax amount. **Note** This amount equals to the base amount when the charge customized in a charge plan (or in an offer) is configured to deal with a net amount.
`Base Amount`	Reference amount of the charged transaction used by SAP CC to compute the tax amount and generate the final amounts (amount excluding tax, total amount including tax). This amount can be net or gross, according to the configuration of the charge customized in a charge plan (or in an offer) that has been triggered during the charging operation. **Note** • To manage tax in an external system, consider the *VAT Transaction Gross Price Flag* and the *Tax Is Included* fields • In case the EZtax tax system is selected, SAP CC considers the net amount
`Base Rated Amount`	Reference amount, computed during the execution of a price plan of a master charge, and used by SAP CC during the execution of the price plan of a dependent charge (when exists).
`Chargeable Items Are Exported`	Boolean used to inform whether the operation which generated the charged item was also configured to generate chargeable items, or not
`Charge Activation Cancelation Date`	Cancelation date of the subscription, only available for cancelling a one-shot rate within a charge
`Charge Activation Code`	Identifier of the activated charge in a provider contract or subscription. **Note** This identifier can also correspond to the identifier of the refill logic in a provider contract
`Charge Activation Effective Date`	Creation date of the subscription, only available for one-shot rates containing an activation event

Master Data	Description

Property	Description
`Charge Component Name`	Type of the rate component that is activated in the price plan, that can be: • Usage Rates • Recurring Rates • One-Shot Rates
`Charge Date`	Date of the charged event (usage, recurring or one-shot) or refill event
`Charge Reference`	In case the charging operation was based on an external reference, this property corresponds to the identifier of this external reference
`Charge Reference Name`	In case the charging operation was based on an external reference, this property corresponds to the name of this external reference, set in the charging plan of the activated charge
`Charge Type Code`	Numeral code used to identify the charge. Possible values are: • 0: Closing • 1: Refill charge • 2: Internal charge • 3: External charge • 4: Not charged
`Charge Type Name`	Textual representation of the type of charge charged by SAP CC. Possible values are: • "closing" • "refillCharge" • "internalCharge" • "externalCharge" • "notCharged"
`Chargeable Item Class Name`	Class name of the chargeable item that was charged by SAP CC, only available for usage charging operations
`Charged Amount is a Credit`	Boolean used to indicate whether the amount of the charged transaction is a credit amount, or not
`Charged Amount is a Debit`	Boolean used to indicate whether the amount of the charged transaction is a debit amount, or not

Master Data	Description

Property	Description
`Charged Item Set Identifier` or `Refill Record Set Identifier`	Identifier of a set of charged items or refill records generated by SAP CC. **Note** A charged item set represents a charged item coming from a master charge plus charged items possibly coming from the dependent charges.
`Charged Tax Amount`	Tax amount of the charged transaction, that is an alias of the `Tax Amount` property
`Charged Tax Code`	Tax code of the charged transaction, that is an alias of the `Tax Code` property
`Charged Tax Detail`	
`Charged Tax Status Code`	**Deprecated** (use `Tax Status Code` instead) Textual representation of the tax status of the charged transaction, that is an alias of the `Tax Status Code` property. Possible values are: • "tsNoTax": No tax was applied • "tsForInfo": The computed tax information is given for information purpose • "tsApplied": The tax was applied • "tsTaxExempted": Tax exemption • "tsBuyerSubjectToPay": The buyer is responsible for reversing the tax to the tax authority Deprecated values: • "tsTaxMissedForInfo": Information is missing for the data related to the tax • "tsTaxMissedForApply": The tax information was missing to apply the tax • "tsUnknown": The tax status is unknown
`Charged Tax Status Identifier`	**Deprecated** (use `Tax Status Identifier` instead) Numerical code of the tax status of the charged transaction, that is an alias of the `Tax Status Identifier` property.

Master Data	Description

Property	Description
	Possible values are: • 0: No tax was applied • 1: The computed tax information is given for information purpose • 2: The tax was applied • 3: Tax exemption • 4: The buyer is responsible for reversing the tax to the tax authority Deprecated values: • -1: Information is missing for the data related to the tax • -2: The tax information was missing to apply the tax • -3: The tax status is unknown
`Charged Total Amount (incl. Tax)`	**Deprecated** (use `Total Amount` instead) Total amount of the charged transaction, including tax
`Charged Transaction is a Prepaid Debit`	Boolean used to indicate whether the charged transaction concerns a charging operation related to a prepaid account, or not
`Charged Transaction is a Prepaid Refill`	Boolean used to indicate whether the charged transaction concerns a refilling operation related to a prepaid account, or not
`Charging Component Name`	Name of the charging component
`Charging Detail Field`	
`Contract Identifier`	Identifier of the provider contract, whose value depends on the context: • In a charge plan, this identifier corresponds to the identifier of the provider contract which generated the charged item • In an allowance plan, this identifier corresponds to the identifier of the provider contract owning the allowance which generated the charged item
`Contract Item External Identifier`	External identifier set by an external system (or by the SAP CRM system in case of integrated scenario), retrieved from the external code set in a contract item stored within SAP CC

Master Data	Description

Property	Description
`Contract Item Identifier`	Identifier set by an external system (or by the SAP CRM system in case of integrated scenario), retrieved from the name (or code) set in a contract item stored within SAP CC
`Contract Type`	Type of contract which generated the charged item. Possible values are: • "P", for provider contracts • "S", for subscriptions
`Currency`	ISO 4217 (three-letter) code of the charged transaction currency
`End of Rating Recurring Period`	End date of the period related to a recurring rate, that can be different from the `End of Rating Reference Recurring Period` property that is used when the recurring rate is prorated
`End of Rating Reference Recurring Period`	End date of the reference period related to a recurring charge
`Event Date`	Date of the rated event, that equals to the consumption date of a usage rate or to the triggering date of the recurring rates or one-shot rates
`Event Processing Date`	Date when the event which generated the charged item was processed by SAP CC
`Excluded Last Day of the Rating Recurring Period`	
`External Account Code`	Identifier of the postpaid account to be charged, that is set in the external account of the related subscriber account
`External System Code`	Identifier of the external system to be charged (in case of postpaid account), that is set in the external account of the related subscriber account
`First Dispatch Code`	Dispatch code #1, which corresponds to the name of the property chosen as the first criteria when filtering External Account transactions
`Name of Rate Component Used`	
`Offer Code`	Identifier (name) of the activated offer concerned by the charging operation
`Overrun From`	Identifier of the prepaid account which called the overrun, that is: • Empty for the first overrun

Master Data	Description

Property	Description
	• Only available in the context of prepaid account overrun
`Overrun To`	Identifier of the prepaid account from which the overrun is called, that is: • Empty for the last overrun • Only available in the context of prepaid account overrun
`Prepaid Balance Amount`	Current amount of the prepaid account balance selected determined during the charging operation
`Prepaid or External Account Code`	Identifier of the account (prepaid or external postpaid) to charge
`Prerating Reservation Identifier`	**Deprecated** Identifier of the reservation in case of a prerating operation
`Price Plan Name`	Identifier (name) of the executed price plan of a charge
`Produced by a Master Charge Component`	Type of the charge (customized in a charge plan or in an offer) that contains the price plan triggered during the charging operation. Possible values are: • 1: Master charge • 2: Dependent charge
`Produced by an Allowance`	Boolean used to indicate whether the charged item was generated by an allowance, or not
`Rated Amount`	Amount of the rated transaction, before tax computation. This raw amount is computed by SAP CC when executing the price plan of a charge customized in a charge plan (or in an offer), and equals to the `Base Amount` property when no overrun exists. **Note** Do not use this field to set up the generation of charged transactions or charged items.
`Rated Amount is Prorated`	Boolean used to indicate whether the amount is prorated, or not
`Rated Amount is Refunded`	Boolean used to indicate whether the amount concerns a refund, or not

Master Data	Description

Property	Description
`Rated Boolean`	
`Rated Date`	
`Rated Number`	
`Rated String`	
`Rated Tax Amount`	**Deprecated** (use `Base Amount` instead) Tax amount computed during the execution of a (net) price plan, that is available in the rated transaction. **Note** • Do not use this property when the price plan is gross • Do not use this field to set up the generation of charged transactions or charged items
`Rated Tax Code`	Code of the tax
`Rated Tax Status Code`	**Deprecated** (use `Tax Status Code` instead)
`Rated Tax Status Identifier`	**Deprecated** (use `Tax Status Identifier` instead)
`Rated Total Amount`	**Deprecated** (use `Rated Amount` instead) Total amount, including tax, computed during the execution of a (net) price plan. **Note** • Do not use this property when the price plan is gross • Do not use this field to set up the generation of charged transactions or charged items
`Rating Activation Date`	Date when the recurring charge has been triggered during a charging operation
`Rating Consumption Date`	Consumption date of the chargeable item related to a usage charging operation
`Rating Event Origin`	Type of charged event.

Master Data	Description	
	Property	**Description**
		Possible values are: • "U" (Usage): The event corresponds to the usage of the consumed service, that is configured within usage rates or refills in the pricing logic • "R" (Recurring): The event corresponds to a periodic fee, that is configured within recurring rates or refills in the pricing logic • "O" (One-shot): The event corresponds to a one-off fee, that is configured within one-shot rates or refills in the pricing logic
	`Refill Amount`	
	`Refill Currency`	
	`Refill Date`	
	`Refill ISO Currency`	
	`Sender Job Identifier`	Identifier of the sender that sent a job of charging operations to SAP CC (for example, job identifiers are used in case of mass charging operations when SAP CI sends one or more jobs of charging operations to SAP CC)
	`Second Dispatch Code`	Dispatch code #2, which corresponds to the name of the property chosen as the second criteria when filtering External Account transactions
	`Service Identifier`	Identifier of the consumed customer service provided by a service provider, that is only available for usage charging operations
	`Service Provider Name`	Identifier of the service provider, that corresponds to the name of the catalog of the service provider
	`Service Provider Country Code`	ISO 3166 country code of the service provider according to the tax settings of the charge customized in a charge plan (or in an offer) executed at runtime
	`Session Rating Reservation Identifier`	Identifier of the session reservation, that is only available for session-based charging operations
	`Start of Rating Recurring Period`	Start date of the period related to a recurring rate, that can be different from the `Start of Rating Reference Recurring Period` used when a recurring charge is prorated

Master Data	Description

Property	Description
`Start of Rating Reference Recurring Period`	Start date of the reference period related to a recurring rate
`Subscriber Account Code`	Identifier of the subscriber account
`Subscriber Account Reference`	Reference of the subscriber account
`Subscriber Consumes Service (Client)`	Boolean used to inform about the relationship between the subscriber and the customer service. When this property is set to "true", the business transaction relates to a client relationship, which means that the subscriber consumed the service provided directly by the service provider
`Subscriber Provides Service (Partner)`	Boolean used to inform about the relationship between the subscriber and the customer service. When this property is set to "true", the business transaction relates to a partner relationship, which means that the subscriber consumed the service provided by a partner of the service provider
`Subscription Code`	Identifier of the triggered subscription
`Subscription Commitment Expiration Date`	Commitment date of the subscription
`Subscription Resumption Date`	Resumption date of the subscription
`Subscription Suspension Date`	Suspension date of the subscription
`Tax Amount`	Amount of tax applied and adjusted to a charged transaction, that is associated to the `Currency` property which contains the related currency
`Tax Code`	String representation of the tax code of a charged transaction (or charged item) assigned to a prepaid or external postpaid account. Possible values are: • "/`<COUNTRY_CODE>`/`<VAT_RATE_LEVEL>`#`<timestamp>`", for the VAT tax system, where: • `<COUNTRY_CODE>` is an ISO 3166 country code (2-digit) • `<VAT_RATE_LEVEL>` is a value from 0 to 5 • "/RAW/`<RATE_ID>`", for the Flat tax system, where `<RATE_ID>` is the identifier of the flat tax rate defined in the charge

Master Data	Description

Property	Description
	• "NO_TAX", when no tax system is defined in the charge or in the subscriber account, or when the system configuration is erroneous • "EZTAXED", for the EZtax tax system used for U.S telco taxes
`Tax Determination Type`	Information about the tax, used by the billing system. Possible values, that depends on the tax status, are: • "00", default value available when the tax status is "BUYER_SUBJECT_TO_PAY", "TAX_EXEMPTED" or "NO_TAX" to inform that the tax status is applied and must not be calculated by the billing system • "01", available when the tax status is "FOR_INFO" to inform that the tax system is VAT, or Flat • "04", available when the tax status is "FOR_INFO" to inform that the tax system is EZtax
`Tax is Included in Total Amount`	Boolean used to inform whether the tax amount is included in the total amount, or not. This property is set to "true" when the tax status is set to "APPLIED" or "FOR_INFO"
`Tax Status Code`	Textual representation of the tax status. Possible values are: • "tsNoTax": No tax was applied • "tsForInfo": The computed tax information is given for information purpose • "tsApplied": The tax was applied • "tsTaxExempted": Tax exemption • "tsBuyerSubjectToPay": The buyer is responsible for reversing the tax to the tax authority Deprecated values: • "tsTaxMissedForInfo": Information is missing for the data related to the tax • "tsTaxMissedForApply": The tax information was missing to apply the tax

Master Data	Description

Property	Description
	• "tsUnknown": The tax status is unknown
`Tax Status Identifier`	Numerical code of the tax status. Possible values are: • 0: No tax was applied • 1: The computed tax information is given for information purpose • 2: The tax was applied • 3: Tax exemption • 4: The buyer is responsible for reversing the tax to the tax authority Deprecated values: • -1: Information is missing for the data related to the tax • -2: The tax information was missing to apply the tax • -3: The tax status is unknown
`Third Dispatch Code`	Dispatch code #3, which corresponds to the name of the property chosen as the third criteria when filtering External Account transactions
`Total Amount (Including Tax)`	Total amount of the charged transaction, tax included. This amount is used by SAP CC to debit the balance of the concerned prepaid account, or charge the concerned external postpaid account. **Note** The total amount equals to the base amount when the charge customized in a charge plan (or in an offer) is configured to gross.
`Triggering Contract Identifier`	Identifier of the provider contract which triggered the generation of the charged item via an allowance. The value of this property depends on the type of trigger: • For a one-shot trigger or a recurring trigger, the value corresponds to the identifier of the provider contract which owns the allowance • For an event-based trigger, the value corresponds to the identifier of the provider contract which sent an event to the allowance

Master Data	Description

Property	Description
`Triggering Contract Item External Identifier`	External identifier of the provider contract item which sent an event to the allowance generating the billable items through an Event-Based Trigger component. Otherwise, this property contains the external identifier of the parent provider contract item of the allowance
`Unrounded Rated Amount`	Amount computed by the charges, having up to 6 decimal places before being rounded according to the `TRANSACTION_PRECISION` system parameter. **Note** This property is required for the tax framework but may not be used in the charged item mapping.
`User Service Identifier`	Identifier of the end customer related to the usage of the consumed service
`VAT Business Entity`	Boolean used to inform about the business category (B2B or B2C) of the subscriber account relating to the charged account (prepaid or external postpaid). Possible values are: • "true", to inform the business category is B2B • "false", to inform the business category is B2C
`VAT Place of Taxation`	
`VAT Transaction Gross Price Flag`	Boolean used to inform whether prices computed during the execution of a price plan are net, or gross. Possible values are: • "true", for gross prices • "false", for net prices
`VAT Transaction Rate Code`	Code of the tax rate level, that is configured in the charge customized in a charge plan (or in an offer) when the invoicing tax system is EU VAT. Possible values are: • "0": Normal • "1": Intermediary

Master Data	Description

Property	Description
	• "2": Reduced • "3": Super reduced • "4": Zero • "5": Special • "-1": No tax
`VAT Transaction Raw Rate`	Flat tax rate configured in the charge customized in a charge plan (or in an offer), that is only available when the customized charge is configured to use the "Flat" tax system
`VAT Transaction Raw Rate ID`	Identifier of the flat tax rate as configured in the charge customized in a charge plan or in an offer, that is only available when the customized charge is configured to use the "Flat" tax system.

> **Note**
>
> Some properties can be used both in charge plans and allowance plans, while other properties can only be used in charge plans or in allowance plans.
>
> The following properties can only be used in allowance plans:
>
> - Allowance Event Class Name
> - Allowance Unique Identifier
> - Allowance Validity Start Date
> - Allowance Validity End Date
> - Triggering Contract Identifier
>
> The following properties can be used in both charge plans and allowance plans:
>
> - Account Type Code
> - Account Type Name
> - Account Type is Prepaid
> - Chargeable Item Unique Identifier
> - Chargeable Item Unique Identifier Type
> - Charged Item or Refill Record Set Identifier
> - Charged Item or Refill Record Set Identifier Type
> - Charged Item or Refill Record Set identifier
> - Contract Identifier
> - Contract Type
> - Currency
> - End of Rating Recurring Period

Master Data	Description
	• Event Date • Event Processing Date • External Account Code • External Account Name • ISO Currency • Prepaid Account Name • Prepaid or External Account Code • Produced by an Allowance • Sender Job Identifier • Session Rating reservation Identifier • Start of Rating Recurring Period • Subscriber Account Code The other properties can only be used in charge plans
(shared) Counters	Counters represent numeric variables that you can define and declare within a reusable charge, and reuse within other master data. SAP CC provides you with 2 types of counters: • Persistent counters, that are saved in the Core Database • Transient counters, that are only stored in memory If you want to share a counter between several items in a provider contract, you may need to set up some specific configurations in the catalog of the service provider. By default, such a shared counter is initialized with the zero value. If you want to design other initial values, you must create a logic that will initialize the counter with the value of a parameter redefined in each provider contract. To initialize such shared counters, you can: • Create and set up a dedicated charge plan containing a customized charge that includes a **recurring rate** component charged at the start of the period • Create and set up a dedicated charge plan containing a customized charge that includes a **one-shot rate** component triggered by an Activation event (at the creation of the related contract item) • Configure the refill logic of an existing refill plan with a customized charge that includes a **recurring refill** component • Configure the refill logic of an existing refill plan with a customized charge that includes a **one-shot refill** component triggered by an Activation event (at the creation of the contract item associated to this refill plan)
Counter Name Dictionary	**Core Tool menu**: *Tools* > *Counter Dictionary* > *Edit* The counter name dictionary defines a set of counter names that can be used in other master data such as charge plans or refill plans, when declaring counters

Master Data	Description
Monitoring Plan	**Core Tool menu**: *File* > *New* > *Monitoring Plan* Monitoring plans represents the generation policy of spending statuses, that include: • A list of spending status identifiers • The counters to monitor • A list of rules used to determine the relevant labels

Master Data	Description
Refill Item Class	**Core Tool menu**: *File* > *New* > *Refill Item Package* Refill items are sent to SAP CC by network elements for refilling purpose. Each refill item represents a set of properties used during the execution of a refill logic. The following table contains the list of properties you must add to the Refill Item Class, according to the data received from the SAP ERP system:

Property	Description
`Refill Date`	Date of the refill request
`Refill Amount`	Amount of the refill
`Refill Currency`	Currency of the refill
BKVID	Bank details ID
CRDID	Payment card ID
CGUID	Key for payment card supplement
CNTU1	Number executed (component 1)
CNTU2	Number executed (component 2)
CNTU3	Number executed (component 3)
CRDRS	Reason for free increase of prepaid balance
PPDOC	Posting document for prepaid account in the SAP ERP system
PPNOT	
PPRSN	Reason for balance change on prepaid account
PREPR	Reference number used for the refill
REFPK	Reference of the refill package
RFPCN	Number of refill packages
TULOC	Point of sale for prepaid credit
TUMTH	Refill method
TUSE1	Refillable unit (component 1)
TUSE2	Refillable unit (component 2)
TUSE3	Refillable unit (component 3)
XRVRF	Decrease of prepaid credit or units

> **Note**
>
> Consult the customizing of the SAP ERP system to determine if other fields are present in the refill items sent to SAP CC to trigger the refills.

Master Data	Description
Refill Logic	**Core Tool menu**: *File* > *New* > *Refill Logic* Refill logic is used by SAP CC to manage refill operations related to the use of a customer service. This master data can be customized at refill plan level. **Note** • For a request for refill, you must add two `Usage Refill` components in the refill logic • For an automated periodic refill, you must define some `Recurring Refill` components in the refill logic • For an automated initial refill, you must define a `One-Shot Refill` component in the refill logic A refill logic requires the creation of counters and a translation table both dedicated to the refill management based on refill packages. The following table contains the list of properties that can be generated by SAP CC:
Refill Plan	**Core Tool menu**: *File* > *New* > *Refill Plan* Refill plans are used to credit the prepaid accounts of customers. Each refill plan consists of: • A refill logic • A set of parameters used to configure the refill logic • A release status

Property	Description
`computed_TUSE1`	Refillable unit ID
`computed_TUSE2`	
`computed_TUSE3`	
`computed_CNTU1`	Refillable unit value
`computed_CNTU2`	
`computed_CNTU3`	
`computed_CNTU1_REQ`	Requested number
`computed_CNTU2_REQ`	
`computed_CNTU3_REQ`	
`computed_PPAMOUNT_REQ`	Requested change of credit
`computed_CRDRS`	Reason for free increase of prepaid balance
`computed_PPRSN`	Reason for balance change onj prepaid account
`BASE AMOUNT`	Final value of the prepaid balance

Master Data	Description
Refill Record Class	**Core Tool menu**: *File* > *New* > *Refill Record Class* A refill record class represents a list of properties that must be included in each refill record output by SAP CC during refilling operations. The following table contains the list of properties that can be generated by SAP CC and can be expected by SAP ERP/FI-CA:

Property	Description
SRCATYPE	Constant (default value: CCCIT)
SRCTAID	Default property calculated by SAP CC
SUBPROCESS	Node
BITTYPE	Node
PREPAID	Account type is prepaid
PPREFILL	Charged transaction is a prepaid refill
VKONT	External account Name in SAP CC (postpaid)
GPART	External account ID or Prepaid account ID in SAP CC
VTREF	Provider contract ID in SAP CC
VTPOS	
SUBAP	Contract type
PPACC	Prepaid account ID in SAP CC
BILL_BASEDATE	
BILL_FIRST	
BITDATE	Default property calculated by SAP CC
BITTIME	Default property calculated by SAP CC
BITDATE_FROM	Start date of refill recurring period
BITDATE_TO	Stop date of refill recurring period
BIT_AMOUNT	Final refill amount computed by the execution of the relevant refill logic
BIT_CURR	Currency of the refill
TAX_INCLUDED	Tax is included in total amount
TAX_DET_TYPE	Constant (default value: 01)
EXT_TAX_DATE	
EXT_TAX_ID	VAT[93] transaction rate code

[93] Value Added Tax

Master Data	Description

Property	Description
BITREF32	
PY_GROUP	
BITTEXT50	Property from input data (refill item)
BKVID	Property from input data (refill item)
CNTU1	Particular property computed during the execution of the refill logic
CNTU1_REQ	Particular property computed during the execution of the refill logic
CNTU2	Particular property computed during the execution of the refill logic
CNTU2_REQ	Particular property computed during the execution of the refill logic
CNTU3	Particular property computed during the execution of the refill logic
CNTU3_REQ	Particular property computed during the execution of the refill logic
CRDID	Payment card ID
CRDRS	Particular property computed during the execution of the refill logic
PCARD_GUID32	Property from input data (refill item)
PPAMOUNT_REQ	Particular property computed during the execution of the refill logic
PPDOC	Posting document for prepaid account in the SAP ERP system
PPRSN	Reason for balance change on prepaid account
PREPR	Reference number used for the refill
REFPK	Reference of the refill package
RFPCN	Number of refill packages
TULOC	Point of sale for prepaid credit
TUMTH	Refill method
TUSE1	Refillable unit (component 1)
TUSE2	Refillable unit (component 2)
TUSE3	Refillable unit (component 3)
XRVRF	Decrease of prepaid credit or units

Master Data	Description
	Once the refill record class is configured, you must associate it to a billable item mapping.
Billable Item Mapping	**Core Tool menu**: *File* > *New* > *Billable Item Mapping* The billable item mapping consists in mapping charged item classes or refill record classes onto billable item classes defined in a particular SAP ERP/FI-CA system that is the ERP reference system in the SAP system landscape (SAP CI). **Note** After the billable item class is assigned to the `released as productive` status, you can change it in SAP Convergent Invoicing by adding optional fields at the end of the structure. In some business cases, it is necessary to take these changes into account to allow SAP CC to generate the new version of billable items.

Master Data	Description
Spending Status Description	**Core Tool menu**: *File* > *New* > *Spending Status Description* A spending status description represents a list of named labels that are used when modeling monitoring plans, each spending status being exposed to an external policy control or monitoring system. You can manage it by configuring in parallel a *Spending Status Monitoring*. The following table contains the list of parameters that can be handled [page 45] by SAP CC:

System Parameter	Description
SPENDING_STATUS_REPORT_SEND_TIMEOUT	Sets up the maximum time (in ms) to be used by the rater instance for sending the spending status report
SPENDING_STATUS_REPORT_RESEND_ATTEMPTS	Sets up the maximum number of resend attempts of the spending status report
SPENDING_STATUS_REPORT_RESEND_INTERVAL	Sets up the minimum time interval (in seconds) between two resend attempts of the spending status report
MULTICAST_SERVER_THREAD_COUNT	Sets up the number of threads reserved for processing the multicast service requests
STATEFUL_SERVICE_THREAD_COUNT	Sets up the number of threads allocated to the handling of stateful operations
STATELESS_SERVICE_THREAD_COUNT	Sets up the number of threads reserved for the Stateless Message Service (support of the online stateless rating services)
BATCH_SERVICE_THREAD_COUNT	Sets up the number of threads reserved for processing the Batch Message Service of a rater instance
STATEFUL_SERVICE_SQL_UPDATE_POOL_SIZE	Sets up the size of the pool allocated to modification operations and data files generation related to stateful requests
CACHE_WARMUP_THREAD_COUNT	Sets up the number of threads reserved for processing the cache warm-up
RENEW_RESERVATION_RESEND_THREAD_COUNT	Sets up the number of processing threads dedicated to resending renew reservation notifications
ACTIVATION_THREAD_COUNT	Sets up the number of threads reserved for performing the activation process in rater instance (see recurring and one-shot rate activation)
CLEANUP_THREAD_COUNT	Sets up the number of threads reserved for performing the cleanup process into each rater instance
SPENDING_STATUS_REPORT_RESEND_THREAD_COUNT	Sets up the number of processing threads dedicated to resending spending status report notifications

Master Data	Description
Offer	**Core Tool menu**: *File* > *New* > *Offer* Offers are master data that belong to the old data model, and that represent set of charges customers can subscribe to. SAP CC provides you with 2 types of offers: • Simple offers, that contain one or multiple charges • Complex offers, that contains one or multiple suboffers with the addition of zero, one or multiple charges

> **Note**
>
> Master data related to the service provider are grouped within a pricing catalog that can be transported to another landscape. For further information, refer to the Catalog Transport [page 70] dedicated section available in this documentation.

3.2.16.4.1 Removing the References to the Chargeable Item Packages of Chargeable Item Classes

> **Caution**
>
> **As of release 4.0 for SAP Convergent Charging, the chargeable item packages are revoked.**
>
> As an application administrator you must prepare the removal of this master data in the system. Then you remove the links between chargeable item packages and chargeable item classes. Contact the SAP CC pricing specialists who configure and maintain pricing catalogs for some chargeable services to monetize.

Because this step requires restarting some instances in the SAP system, we strongly recommend that you perform this procedure when the workload of the SAP Convergent Charging system is the lowest.

Task

Remove the references to the chargeable item packages of chargeable item classes. It ensures that the identifiers of the chargeable item classes are unique between all the chargeable item classes of the system.

Procedure

1. Generate the chargeable item package migration report.

1. As an application administrator, execute the command with the Setup Tool user interface. in an Operating System (OS) console:

   ```
   ./setup migration generateCIPackageReport ./report.xml
   ```

2. In this system console, enter your administrator logon and password at the prompt.
3. Check on the console that the report has been successfully exported.

   > **Note**
   >
   > If the report cannot be exported because some invalid charges (aka `charge components`) have been found, first fix the listed charges with the relevant SAP CC pricing specialists, and start the generation of the report again. In master data, several pricing catalogs may be impacted for different pricing specialists. Contact your SAP Support Team when needed.

2. Copy the final report, for example `report.to.execute.xml`
3. Open the copied report with a text editor (an XML editor is recommended) and, as described in the header of the XML file, provide the name of the new chargeable item classes when requested.
4. Save your changes.
5. Stop the updater instances in the running SAP CC system (Core Server).
6. Execute the modified report with Setup Tool.
   1. In a system console, execute the following command:

      ```
      ./setup migration executeCIPackageReport ./report.to.execute.xml
      ```

   2. Enter your administrator logon and password at the prompt.
   3. Check on the console that the report has been successfully exported.

      > **Note**
      >
      > If the report cannot be executed, read the error message displayed on the console, try to fix the problem, and then execute again the report. Contact your SAP Support Team.

7. Restart the updater instances in the SAP CC system.

3.2.16.5 Configuring Master Data for the Customer of the Service Provider

To configure the master data related to the customers of the service provider, execute the following procedure:

1. Launch the Core Tool user interface and identify as a user granted the "CSR" role
2. Set up the relevant master data according to your needs, using the following information:

Master Data	Description
Access	**Core Tool menu:** *File* > *New* > *Access* Accesses are used to grant end customers the access and use of services provided by the service provider. Each access includes: • The technical identifier of the consumed service (Service ID) • The technical identifier under which an end customer consumes the service (User Service ID)
Provider contract	**Core Tool menu:** *File* > *New* > *Provider Contract* **Web Service:** `contractProvisioning` Provider contracts represent the business agreement between the service provider and a given customer, taking into account some specific conditions. Each provider contract refers to a unique subscriber account, and contains all the information that is necessary to execute charging (or refilling) operations on this account. **Note** • Necessary accesses are automatically created by SAP CC in background • For development, test, or training purposes, you can manually create a provider contract via the Core Tool user interface. In case of an integrated landscape with the SAP CRM system, SAP SE recommends that you use the dedicated Web Service to synchronize this master data with all the relevant systems • If you installed the BART Server system within your landscape to manage batch charging operations, you can modify the default policy related to batch rating group assignment in order to fit your specific needs. For further information, refer to the Implementing the Batch Rating Group Assignment Mechanism [page 205] dedicated procedure afterwards
Subscriber account	**Core Tool menu:** *File* > *New* > *Subscriber Account* **Web Service:** `suacProvisioning` Subscriber accounts are used to store the information related to the account and balance of each customer of the service provider. **Note** For development, test, or training purposes, you can manually create a subscriber account via the Core Tool user interface. In case of an integrated landscape with the SAP CRM system, SAP SE recommends that you use the dedicated Web Service to synchronize this master data with all the relevant systems.

Master Data	Description
Subscriber mapping table	**Core Tool menu:** *File* > *New* > *Subscriber Mapping Table* **Web Service:** `subscriberMappingTableManagement` Subscriber mapping tables are part of the subscriber accounts, and contain specific data related to customers and represented as mapping tables. **Note** • Subscriber mapping tables are not available in subscriptions • A contract item can link several subscriber mapping tables • A given subscriber mapping table can be reused within different contract items in different provider contracts • For development, test, or training purposes, you can manually create a subscriber mapping table via the Core Tool user interface. In case of an integrated landscape with the SAP CRM system, SAP SE recommends that you use the dedicated Web Service to synchronize this master data with all the relevant systems
Subscriber range table	**Core Tool menu:** *File* > *New* > *Subscriber Range Table* **Web Service:** `subscriberRangeTableManagement` **Note** • Subscriber range tables are not available in subscriptions • A contract item can link several subscriber range tables • A given subscriber range table can be reused within different contract items in different provider contracts • For development, test, or training purposes, you can manually create a subscriber range table via the Core Tool user interface. In case of an integrated landscape with the SAP CRM system, SAP SE recommends that you use the dedicated Web Service to synchronize this master data with all the relevant systems
Subscription	**Core Tool menu:** *File* > *New* > *Subscription* Subscriptions represent the charging agreement conditions between the service provider and a given customer, taking into account some specific conditions. Each subscription can be: • **Online**, which means that they can be charged in real-time mode • **Offline**, which means that they can only be charged in batch mode • **Hybrid**, which means that they can be charged in both batch or real-time mode according to the consumed service

3.2.16.5.1 Implementing the Batch Rating Group Assignment Mechanism

When the BART Server is deployed in a landscape, the created provider contracts dedicated to offline charging operations are by default balanced on the defined batch rating groups. In case this assignment policy does

not fit your business requirements, you can use the following procedure to modify this default behavior by developing a Java class that implements your own assignment policy:

1. Develop your specific Java class by using the `com.highdeal.batchgroup.contract` package available in the Core SDK and:
   - Implementing the `IContractBatchRatingGroupAssignmentPolicy` interface
   - Overriding the `getBatchratingGroupAssignment(…)` method
2. Create a `JAR` file containing this newly created single class, and copy this `JAR` file to the `exe/uc/<OS>/CC_CORE_SERVER/jars/` subfolder of the SP Central Repository, where `<OS>` correspond to the operating system of the Core Server
3. Launch your favorite user interface [page 45] and identify as a user granted the "CSR" role
4. Modify the value of the BATCH_RATING_GROUP_ASSIGNMENT_POLICY_CLASS system parameter for each updater instance, specifying the name of your own Java class that must be used when assigning batch rating groups, using the following syntax: `<package_name>.<class_name>`
   For further information about this system parameter, refer to its dedicated documentation available in the SAP CC 2021 System Parameter Reference documentation.
5. Update the startup configuration file of each updater instance by adding the created `JAR` file to the classpath. For further information, refer to the Startup Configuration Files [page 234] dedicated procedure afterwards.
6. Restart the updater instances of the Core Server system.

> **Note**
>
> To test that your modifications have been correctly taken into account, you can trigger the creation of a provider contract from the SAP CRM system or from the Core Tool user interface and check the assigned batch rating group.

3.2.16.5.2 Enabling the Optimization of Provider Contracts

The provider contracts in master data are optimized to reduce their memory footprint and improve performances. Some versions of the provider contracts are merged or removed to decrease the number of versions. To optimize the master data management, execute the following procedure:

1. Launch your favorite user interface [page 45] and identify as the administrator of the Core Server system
2. For each updater instance, modify the value of the CONTRACT_PACKING_ENABLED system parameter to **`true`**
3. Restart all the instances of the Core Server system

3.2.17 Notifications

Keywords [page 207]

Preliminary Notes [page 207]

Description [page 207]

Available Notifications [page 207]

Configuring the Notification Policy [page 211]

Enabling the Notification Policy [page 217]
You want to enable the notification policy for your SAP system (a policy for the rater instances and another policy for the bulkloader instances).

Handling SAP CC Notification in your Java Application [page 217]

3.2.17.1 Keywords

Notification group, notification policy, monitoring, alert

3.2.17.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- SAP CC 2021 Notification Reference documentation
- `com.highdeal.notification` Java package and `NotifBook` API descriptions, available in the Java/XML API Reference (Core Library) documentation

3.2.17.3 Description

For monitoring and alerting purposes, SAP Convergent Charging (SAP CC) gives the possibility to generate notifications at business and administrative levels. These notifications are organized into notification groups, and rely on a dedicated framework you can integrate with your client application/system.

You can use the following procedures to configure the management of notifications within your SAP CC landscape and thus fit your specific needs:

- Configuring the Notification Policy [page 211]
- Enabling the Notification Policy [page 217]
- Handling SAP CC Notification in your Java Application [page 217]

3.2.17.4 Available Notifications

Notifications are organized into the following groups:

- **ABM** group, which contains the notifications triggered during the charging operations when:
  - The balance of a prepaid account or of a credit limit goes below a specified threshold

- An alert has been configured in the price plan or charging plan of a charge that has been activated within a provider contract or a subscription
- A prepaid account has expired

- **Administrative** group, which contains the notifications related to the connections between the different running elements (instances, databases, and so on)

The following table contains the list of available notifications, including their belonging group:

Notification			Group	
ID	Level	Name, Pretty Name, and Description	ABM	Admin.
0	inform	`AMOUNT_ALERT(amountAlert)` Sent when a prepaid account or a credit limit balance goes below the amount threshold of the alert, this notification provides the following information: • The service provider of the subscriber account. It is an ID of the pricing catalog of the service provider company. • The identifier of the subscriber account • The related balance type (prepaid or credit limit) • The identifier of the balance for which an amount alert has been triggered • The current amount of the balance with its currency • The message of the alert as configured in the subscriber account • The amount threshold of the alert These alerts are configured for each subscriber account by the external CRM[94] or the provisioning system during the customer data provisioning process (in an integrated installation scenario).	■	□

[94] Customer Relationship Management

Notification			Group	
ID	Level	Name, Pretty Name, and Description	ABM	Admin.
1	inform	`EXPIRATION_ALERT (expirationAlert)` Sent when the expiration alert of a prepaid account is triggered, this notification provides the following information: • The service provider of the subscriber account • The identifier of the subscriber account • The identifier of the balance for which an amount alert has been triggered • The message of the alert as configured in the subscriber account • The state related to the alert • The date when the state change occurs • The period when the alert is triggered around the change date These alerts are configured for each subscriber account by the external CRM or the provisioning system during the customer data provisioning process (in an integrated installation scenario).	■	□
2	inform	`INSTANCE_UP (instanceUp)` The instance of the Core Server is running.	□	
3	inform	`INSTANCE_DOWN (instanceDown)` The instance of the Core Server has been manually stopped. **Note** This notification may not be sent in case of instance failure.	□	■
4	failure	`DB_LINK_DOWN (dbLinkDown)` The instance of the Core Server has lost its connection with the database, due to the following possible reasons: • The database is stopped (accidentally or voluntarily) • The network interface is down • The instance cannot contact the database host because of a network problem (such as a firewall configuration) **Note** This notification is not sent when the instance cannot contact the database because of a wrong configuration.	□	
5	inform	`DB_LINK_UP (dbLinkUp)` The instance of the Core Server has opened new database connections.	□	■

Notification			Group	
ID	Level	Name, Pretty Name, and Description	ABM	Admin.
6	failure	`DB_FAILURE` (`dbFailure`) The database returned a failure exception. See the "message" argument of the notification for further details about the reason of the problem.	☐	■
7	fatal	`OUT_OF_MEMORY` (`outOfMemory`) The instance of the Core Server has been stopped because its JVM has thrown an `OutOfMemory` exception. **Note** This notification may not be sent in case of insufficient JVM[95] memory.	☐	■
8	inform	`PNR_SCHEDULER_WAKESUP` (`pnrSchedulerWakesUp`) The scheduler of the updater instance wakes up and starts a new activation cycle.	☐	■
9	inform	`PNR_SCHEDULER_SLEEPS` (`pnrSchedulerSleeps`) The scheduler of the updater instance is sleeping and may be waiting for its next activation cycle.	☐	■
10	warn	`PNR_SCHEDULER_INTERRUPTED` (`pnrSchedulerInterrupted`) The execution of the scheduler of the updater instance has been interrupted. The environment of the instance prevents the scheduler to complete its cycle (the activation failed, a problem occurred at memory level, and so on), according to a reason available in the notification details.	☐	■
11	security	`AUTHENTICATION_FAILURE` (`authenticationFailure`) Sent when an XML[96] message has been received without a valid authentication (unknown login or wrong password), this notification provides the following information: • The specified login (or "no login" when no login is specified) • The reason of the authentication failure (unknown login or wrong password)	☐	■

[95] Java Virtual Machine
[96] eXtended Markup Language

Notification			Group	
ID	Level	Name, Pretty Name, and Description	ABM	Admin.
12	inform	`USER_ALERT (userAlert)` Sent when a user defined alert is triggered, this notification provides the following information: • The code that identifies the user alert • The message describing the user alert • The service provider of the subscriber account • The code of the subscriber account • A list of properties that gives additional information about the user alert • Use the Core Tool user interface to configure your own alerts by using the Alert Triggering Operator in a reusable charge defined for the service provider. • Use the Cockpit user interface to configure your own alerts by using the Notification Triggering Operator in a reusable charge defined for the service provider.	■	□
13	inform	`RENEW_RESERVATION (renewReservation)` For internal use only.	■	□
14	inform	`SPENDING_STATUS_REPORT (spendingStatusReport)` For internal use only.	■	□
15	inform	`NO_ACCESS (noAccess)` Sent when the No Access Function component is triggered, this notification provides the following information: • The values of each property selected in the No Access Function component • The message defined in the No Access Function component	■	□

3.2.17.5 Configuring the Notification Policy

To fit specific needs, it is possible to define a notification policy that relies on XML-formatted configuration files that include a sequence of notification publishers. Each notification publisher gives the possibility to filter a subset of notifications that must be sent to a given output destination.

> → Remember
>
> This section is dedicated to **implementers** who are members of the implementation and integration project team. In this *Customizing* activity, you design, configure, and test the notification management provided by SAP Convergent Charging (SAP CC).

→ Tip

To ease the configurations, you can export the current settings from your SAP system or use the delivered templates (`notification_policy.rater.xml.sk` and `notification_policy.bulkloader.xml.sk`). View these `*.SK` skeleton files and do your settings.

Configure the XML Files

Design the notification policy that suits your business and technical needs. When required, design a policy for the rater instances in your SAP system and another policy for the bulkloader instances.

Create your `XML` configuration files with the `notificationPolicy` XML element as the root element. Open your files. Add the expected notification publishers as previously designed.

Filters

To configure the filtering of notifications by the SAP system, you can use one of the following XML elements:

XML Element	Nested XML Element	XML Attribute	Description
`filters`	`all`	`accept`	**Optional** attribute used to enable or disable all notifications Possible values are: "true" or "false"
	`range`	`lower`	Lower identifier of a range of notifications
		`upper`	Upper identifier of a range of notifications
	`set`	`id`	Identifier of a notification
	`name`	`pattern`	Regular expression that defines the list of notifications
	`prettyName`	`pattern`	Regular expression that defines the list of notifications
	`level`	`lowerLevel`	Lower level of notifications to send Possible values are: "quiet", "inform", "warn", "security", "alert", "failure", or "fatal"
`nameFilter`		`pattern`	Regular expression that defines the list of notifications to filter according to their name
`argumentFilter`		`name`	Name of the argument available in the notification, that must be used as a filter
		`pattern`	Regular expression that defines the value of the argument

Example

The following `XML` file defines a notification policy that filters the `USER_ALERT` and `NO_ACCESS` notifications:

```
<?xml version="1.0" encoding="UTF-8"?>
<notificationPolicy>
    <filters>
        <set id="12"/>
        <set id="15"/>
    </filters>
</notificationPolicy>
```

Example

The following `XML` file defines a notification policy that filters the notifications whose ID is in the range [2,6], which means:

- `INSTANCE_UP`
- `INSTANCE_DOWN`
- `DB_LINK_UP`
- `DB_LINK_DOWN`
- `DB_FAILURE`

```
<?xml version="1.0" encoding="UTF-8"?>
<notificationPolicy>
    <filters>
        <range lower="2" upper="6"/>
    </filters>
</notificationPolicy>
```

Example

The following `XML` file defines a notification policy that filters all notifications:

- Containing a "sample" argument whose value is "value"
- Containing a "sample2" argument whose value is "value2"

```
<?xml version="1.0" encoding="UTF-8"?>
<notificationPolicy>
    <sequence>
        <argumentFilter name="sample" pattern="value" />
        <argumentFilter name="sample2" pattern="value2" />
    </sequence>
</notificationPolicy>
```

Output Locations

Once filters have been defined, it is then necessary to define output locations, that can be:

- The **network layer**, used to publish notifications to one or many custom client applications (or systems), that you must implement using our Java API[97] (refer to the SAP CC 2021 Web Services Documentation (SOAP) documentation).

> → Remember
>
> `USER_ALERT` notifications cannot be received by JCo[98] destinations, so the `network` attribute is the only way to handle them.

- **JCo destinations**, used to publish `AMOUNT_ALERT` and `EXPIRATION_ALERT` notifications to a JCo destination.
- **Customer Management Areas (CMAs)**, used to send notifications to the relevant SAP CRM or SAP ERP system according to information available in the concerned subscriber account.

The following table contains the possible XML elements that you can use to specify output locations:

XML Element	XML Attribute	Description
`network`	`groups`	This optional attribute gives the possibility to specify a group containing a subset of notifications to send on the network. When not specified, all previously filtered notifications are sent. Possible values are: "abm", "admin", or "abm,admin"
`JCo`	`destination`	Name of the JCo destination defined in the system
	`function`	Name of the JCo function to use
`multiJCoDestination`	`selectedApplication`	SAP system to use for sending notifications
`customerManagementAreaRouting`	`selectedApplication`	SAP system to use for sending notifications

> ✤ Example
>
> The following `XML` files both define a notification policy that:
>
> - Filters the `INSTANCE_UP` and `INSTANCE_DOWN` notifications.
> - Sends these notifications on the network.

```
<?xml version="1.0" encoding="UTF-8"?>
<notificationPolicy>
    <filters>
        <set id="2" />
        <set id="3" />
    </filters>
    <network />
</notificationPolicy>
```

```
<?xml version="1.0" encoding="UTF-8"?>
<notificationPolicy>
    <sequence>
        <nameFilter pattern="INSTANCE_UP">
            <network />
        </nameFilter>
        <nameFilter pattern="INSTANCE_DOWN">
            <network />
        </nameFilter>
    </sequence>
```

[97] Application Programming Interface
[98] Java Connector

```
</notificationPolicy>
```

Example

The following `XML` files both define a notification policy that:

- Filters the `AMOUNT_ALERT` (whose property "sample" is set to "value") and `EXPIRATION_ALERT` notifications.
- Sends the `AMOUNT_ALERT` notifications on the network.
- Sends the `EXPIRATION_ALERT` notifications:
  - On the network
  - To the "FCT" function of the "dest" JCo destination, writing possible errors into dedicated files according to a given policy defined within a `bulkWriter` element

You set up two configuration files as the following samples.

Notification policy for the rater instances:

```
<?xml version="1.0" encoding="UTF-8"?>
<notificationPolicy>
    <sequence>
        <nameFilter pattern="AMOUNT_ALERT">
            <argumentFilter name="sample" pattern="value">
                <network />
            </argumentFilter>
        </nameFilter>
        <nameFilter pattern="EXPIRATION_ALERT">
            <sequence>
                <JCo destination="dest" function="FCT">
                    <bulkWriter
                        adminParameterPrefixName="NOTIF_WRITER"
                        rootPath="SAPCC_NOTIF/ERROR/Instance#@INSTANCE_ID@/
@JCO_DESTINATION_NAME@"
                        channelCount="4"
                        timeBasedFileRolloverPolicy="HOURLY"
                        maxSizeBasedFileRolloverPolicy="1M"
                        deflate="false"
                        fileVersion="2"
                        fileSequenceId="NOTIF_WRITER_EXCEPTION"
                    />
                </JCo>
                <network />
            </sequence>
        </nameFilter>
    </sequence>
</notificationPolicy>
```

Notification policy for the bulkloader instances:

```
<?xml version="1.0" encoding="UTF-8"?>
<notificationPolicy>
    <nameFilter pattern="AMOUNT_ALERT">
        <argumentFilter name="sample" pattern="value">
           <network />
        </argumentFilter>
        <nameFilter pattern="EXPIRATION_ALERT">
            <sequence>
               <JCo destination="dest" function="FCT">
                    <bulkWriter
                         adminParameterPrefixName="NOTIF_WRITER"
                         rootPath="SAPCC_NOTIF/ERROR/Instance#@INSTANCE_ID@/
@JCO_DESTINATION_NAME@"
```

```
                        channelCount="4"
                        timeBasedFileRolloverPolicy="HOURLY"
                        maxSizeBasedFileRolloverPolicy="1M"
                        deflate="false"
                        fileVersion="2"
                        fileSequenceId="NOTIF_WRITER_EXCEPTION"
                    />
                </JCo>
                <network />
            </sequence>
        </nameFilter>
    </nameFilter>
</notificationPolicy>
```

Example

For the bulkloader instances, you can set up the following notification policy:

```
<?xml version="1.0" encoding="UTF-8"?>
<notificationPolicy>
   <nameFilter pattern="^AMOUNT_ALERT$|^EXPIRATION_ALERT$">
      <multiJCoDestination>
         <JCo destination="@JCO_DESTINATION_NAME@">
            <bulkWriter
               adminParameterPrefixName="NOTIF_WRITER"
               rootPath="SAPCC_NOTIF/ERROR/Instance#@INSTANCE_ID@/
@JCO_DESTINATION_NAME@"
               channelCount="4"
               timeBasedFileRolloverPolicy="HOURLY"
               maxSizeBasedFileRolloverPolicy="1M"
               deflate="false"
               fileVersion="2"
               fileSequenceId="NOTIF_WRITER_EXCEPTION"
            />
         </JCo>
      </multiJCoDestination>
      <network />
   </nameFilter>
</notificationPolicy>
```

Next Steps

Enable the feature [page 217] in SAP CC and implement the reception of the SAP CC notifications by your client application or system. See Handling SAP CC Notification in your Java Application [page 217].

3.2.17.6 Enabling the Notification Policy

You want to enable the notification policy for your SAP system (a policy for the rater instances and another policy for the bulkloader instances).

Prerequisites

Your notification policy is designed and a few XML files are configured. For example, a first file for the rater instances and a second file for the bulkloader instances. See Configuring the Notification Policy [page 211].

Procedure

To enable your specific notification policy defined within an `XML` configuration file, complete the following procedure:

1. Open a system console in the directory that contains the executable of the Setup Tool user interface.
2. Execute the `notificationpolicy import` command, specifying the absolute path of the `XML` configuration file containing your notification policy.
3. Restart the concerned instances of the Core Server system.
4. If your running system is composed of bulkloader instances, then use this procedure to enable the notification policy in these instances.
5. Restart the bulkloader instances in the Core Server system.

Next Step

See Handling SAP CC Notification in your Java Application [page 217].

Related Information

Setup Tool

3.2.17.7 Handling SAP CC Notification in your Java Application

If you developed a Java client application within which you want to handle the notifications generated by your Core Server system, you need to use the `com.highdeal.notification` package available in the Core SDK in order to:

- Implement the `NotificationHandler` Java interface
- Instantiate the `NotificationServiceClient` object

3.2.17.7.1 Implementing the NotificationHandler Java Interface

As described in the Java/XML API Reference (Core Library) documentation, the NotificationHandler interface always invokes the `handleNotification` method that provides the following information:

- **systemName**, that corresponds to the SID that contains the instance which sent the notification
- **instanceID**, that corresponds to the identifier of the instance which sent the notification
- **timestamp**, that corresponds to the date when the notification has been sent
- **uid**, that corresponds to the unique identifier of the notification class
- **args**, that corresponds to the arguments of the notification

To retrieve information related to an handled notification, consider the following sample code that you can adapt to fit your business requirements:

```
public void handleNotification
 (String sid, InstanceId iid, long timestamp, int uid, String... args) {
    NotificationDescription desc = NotifBook.searchNotificationDescription(uid);
    if (desc == null) {
           // Should never occur
           System.out.println("Unknown notification");
           return;
    }
    String notifName = desc.getName();
    String notifPrettyName = desc.getPrettyName();
    NotificationLevel notifLevel = desc.getSeverityLevel();
    String[] notifArgName = desc.getAdditionalInfoKeys();
    ...
```

> ⚠ Caution
>
> The thread invoking the handler is the thread that listens to the socket. As a consequence, this thread must be quickly freed to avoid losing further notifications.

3.2.17.7.2 Implementing the NotificationServiceClient Object

Once you implemented the NotificationHandler interface, it is necessary to perform the following operations within you Java application:

- Instantiate an `InstanceMap` object using the `searchInstanceMap` method of the `NotificationServiceClient` class
- Instantiate a `NotificationServiceClient` object and:
  - Initialize this object using the `setNotificationHandler` method
  - Prepare the connection to the relevant SAP CC system using the `addAllConnections` method

- Connect this object to start receiving notifications from the relevant SAP CC system

To receive and handle notifications within your Java application, consider the following sample code that you can adapt to fit your business requirements:

```
NotificationHandler handler = <YOUR_NOTIFICATIONHANDLER_IMPLEMENTATION>
InstanceMap map = NotificationServiceClient.searchInstanceMap(serverUrl, login,
password);
client = new NotificationServiceClient();
client.setNotificationHandler(handler);
client.addAllConnections(map);
client.connect(null, 10000);
```

3.2.18 Security Certificates

See Also

Securing Communications with the Core Server System

Maintaining the Secure Configuration of SAP CC Core Server [page 219]
During the lifecycle of your SAP Convergent Charging system (Core Server), you maintain an existing secure configuration of this SAP server system based on certificates.

Exporting an Existing Certificate from the SAP CC Core Server System [page 233]

Configuring an SAP CC Client to Trust an SAP CC Core Server System [page 234]

3.2.18.1 Maintaining the Secure Configuration of SAP CC Core Server

During the lifecycle of your SAP Convergent Charging system (Core Server), you maintain an existing secure configuration of this SAP server system based on certificates.

During your SAP Convergent Charging (SAP CC) system lifecycle you may need to maintain an existing SAP CC Core Server secure configuration involving:

- An SAP CC Core Server **key entry**
  - Either a couple private key / certificate
  - Or a keystore (containing the certificate and its private key)
- An SAP CC Core Server **certificate entry** (an SAP CC Client Certificate)

> ### Example
>
> For example, you can activate TLS on the updater and dispatcher instances in the SAP CC Core Server system for the SAP CC Web Services both for SOAP and REST.

Maintaining a Key Entry for a Core Server System [page 220]

Maintaining a Certificate Entry for a Core Server System [page 226]

3.2.18.1.1 Maintaining a Key Entry for a Core Server System

Maintaining an SAP CC Core Server system **key entry** consists in the following steps:

1. Identify the alias of the targeted key entry
2. Unlink the targeted key entry
3. Remove the targeted key entry
4. Implement a new secure configuration to renew the targeted key entry

3.2.18.1.1.1 Identify the Alias of the Targeted Key Entry

To maintain an SAP CC Core Server key entry, you first need to know the alias of the targeted key entry.

Use the Setup Tool user interface to execute the `keyentry listLinks` command to display all the linked key entries and identify the alias of the key entry that you want to maintain.

Result

Once you have identified the alias of the targeted key entry you can unlink the key entry from all the declared services. See Unlink the Targeted Key Entry [page 220].

3.2.18.1.1.2 Unlink the Targeted Key Entry

Once you have identified the alias of the targeted key entry you can unlink the key entry from all the declared services.

Use the Setup Tool user interface to execute the `keyentry unlink` command to unlink the key entry from all the declared services.

Result

Once the Key entry is unlinked, you may want to remove it from the SAP CC Core Server system. See Delete the Targeted Key Entry from the SAP CC Core Server System [page 221].

3.2.18.1.1.3 Delete the Targeted Key Entry from the SAP CC Core Server System

Once you have identified the alias of the targeted key entry you can delete the key entry from the SAP CC Core Server system.

Use the Setup Tool user interface to execute the `keyentry delete` command to delete the key entry from the SAP CC Core Server system.

> ⚠ Caution
>
> Deleting a Key entry will automatically unlink it from all the declared services if any.

Result

Once the Key entry has been deleted from the SAP CC Core Server system, you may want to implement a new secure configuration. See Implement a New Secure Configuration to Renew the Targeted Key Entry [page 221].

3.2.18.1.1.4 Implement a New Secure Configuration to Renew the Targeted Key Entry

To secure an SAP CC Core Server communication service, follow these steps:

1. Create a private key/certificate or create a keystore containing the certificate and its private key
2. Import the private key and certificate or import the keystore into the SAP CC Core Server system
3. Link the imported private key and certificate or the imported keystore to the targeted services
4. Configure the SAP CC Core Server instance map to enable the required secure mode on the targeted services
5. Restart the SAP CC Core Server system to apply the changes

> ⓘ Note
>
> After you completed this procedure, you may want to reconfigure an SAP CC Client to use the freshly secured communication service.
>
> Now that the targeted secure configuration is implemented on the SAP CC Core Server system side, you can implement the security changes for:
>
> - SAP CC Cockpit
> - SAP CC Admin+
> - SAP CC Core Tool
> - CTS+ Plugin

3.2.18.1.1.4.1 Create a Private Key and Certificate

To secure an SAP CC Core Server communication service, you will need a private key and certificate.

> ⓘ Note
>
> The following constraints apply:
>
> - The private key must be in the PKCS#8 format encoded in DER.
> - The certificate must be in the X.509v3 format encoded in DER.

> ✤ Example
>
> Keys and certificates can be generated by any SSL tool.
>
> For example, see the SAP KBA 2845357 that provides you with commands based on OpenSSL.

Result

Next, you will need to import the private key and certificate into the SAP CC Core Server system. See Import the Private Key and Certificate [page 222].

3.2.18.1.1.4.1.1 Import the Private Key and Certificate

To use the created private key and certificate, you need to import them as a key entry into the SAP CC Core Server system.

> → Remember
>
> The following constraints apply:
>
> - The private key must be in the PKCS#8 format encoded in DER.
> - The certificate must be in the X.509v3 format encoded in DER.

Use the Setup Tool user interface to execute the `keyentry import`keyentry import command to **import the Private Key and Certificate as a Key entry** into the SAP CC Core Server system.

> ⚠ Caution
>
> Make sure the chosen alias is appropriate.
>
> For example, a Key entry that is valid for the year 2019 and that will be used on the HTTP service of the dispatcher instances:
>
> - An appropriate alias would be: dispatcher_http_2019
> - An irrelevant alias would be: alias1

Result

Once the private key and certificate have been successfully imported in the SAP CC Core Server system, you need to link them to the targeted service or services. See Link the Imported Private Key and Certificate (or the Imported Keystore) [page 224].

3.2.18.1.1.4.2 Create a Keystore

To secure an SAP CC Core Server communication service, you will need a keystore containing the certificate and its private key.

> ### Note
>
> The following constraints apply:
>
> - The keystore must be in the PKCS#12 format.
> - As the keystore contains the certificate and its private key:
>   - The private key must be in the PKCS#8 format encoded in DER.
>   - The certificate must be in the X.509v3 format encoded in DER.

1. Create a certificate and its private key. See Create a Private Key and Certificate [page 222].
2. Create a keystore.

Result

Next, you will need to import the keystore in the SAP CC Core Server system. See Import the Keystore [page 223].

3.2.18.1.1.4.2.1 Import the Keystore

To use the created keystore, you need to import it as a key entry into the SAP CC Core Server system.

> ### Note
>
> The Keystore must be in the PKCS#12 format

Use the Setup Tool user interface to execute the `keyentry import_p12`keyentry import_p12 command to **import the keystore as a key entry** into the SAP CC Core Server system.

Result

Once the Keystore has been successfully imported in the SAP CC Core Server system, you need to link it to the targeted service or services. See Link the Imported Private Key and Certificate (or the Imported Keystore) [page 224].

3.2.18.1.1.4.3 Link the Imported Private Key and Certificate (or the Imported Keystore)

The couple (private key/certificate) or the keystore has been imported as a key entry into the SAP CC Core Server system. You can now link this key entry to the communication services that you want to secure.

Use the SAP CC 2021 Setup Tool user interface to execute the `keyentry link` command to link the key entry to one or several services.

You specify the instances that compose the Core Server system and the services.

→ Recommendation

SAP SE strongly recommends to:

- Avoid the use of the keyword `all` for the instances, for any services except the `internal` service.
- Avoid the use of the keyword `all` for the services, whatever the instances.
- Be careful when using instance IDs in your Key links, as it makes the secure configuration trickier to set up and maintain.

⚠ Caution

Only one key entry can be taken into account for a given instance and service. If several key entries exist for a given instance and service, **the most specific key link is chosen**:

- An instance ID is more specific than an instance type.
- An instance type is more specific than the all keyword.
- A service name is more specific than the all keyword.

In addition, it is not possible to link two different key entries to the same instance/service couple: If a key link exists for a given instance and service, **linking another key entry to the exact same instance and service will replace the existing key link**.

For example, considering the following key links:

```
Key links:
                alias    instance      service
"myAlias_Updater_HTTP"   updater       http
 "myAlias_Updater_ALL"   updater       all
     "myAlias_ALL_ALL"   all           all
```

Note:

- In this example, the **myAlias_Updater_HTTP** key entry will be chosen over the **myAlias_Updater_ALL** and **myAlias_ALL_ALL** key entries for the HCI communications (service code: `http`) of the updater instances.
- The **myAlias_Updater_ALL** key entry will be chosen over the myAlias_ALL_ALL key entry for the Web Service communications (service code: `ws`) of the updater instances.

Result

Once the key entry has been correctly linked to the targeted service or services, you need to configure the SAP CC Core Server instance map to enable the one-way or dual secure mode on the corresponding targeted service or services. See Reconfigure the Instance Map [page 225].

3.2.18.1.1.4.4 Reconfigure the Instance Map

To enable a secure mode on one or more communication services of the SAP CC Core Server system, you need to configure the instance map accordingly.

To modify the SAP CC Core Server instance map configuration, apply the following procedure:

1. Use the Setup Tool user interface to execute the `instancemap export` command to export the instance map from the back-end database to a text file to modify.
2. Open the file with the exported instance map in a text editor.
3. Edit the values of the secure mode for the targeted communication services (`HCISecure`, `WSSecure`, `ExternalSecure`, or `InternalSecure`) according to your needs.
   Possible values are:
   - `off`: No security
   - `oneway`: The one-way secure mode
   - `dual`: The dual secure mode (mutual authentication)

   > ⓘ Note
   >
   > For the `InternalSecure` settings, the only possible values are `off` or `dual`.

   "HCI" corresponds to the `http` service (HCI = HTTP Communication Interface). "External" relates to the *Message TCP* ultra-speed technical interface. "Internal" relates to the interinstance communications.

   > → Remember
   >
   > Make sure that every instance of the same type shares the exact same `<XXX>Secure` mode settings.
4. Use the Setup Tool user interface to execute the `instancemap import` command to import the new version of the instance map from the modified file into the back-end database.

Result

Once the SAP CC Core Server Instance Map has been successfully updated according to your needs, you must restart the SAP CC Core Server system for changes to be applied. See Restart the SAP CC Core Server System to Your Apply Security Changes [page 226].

3.2.18.1.1.4.5 Restart the SAP CC Core Server System to Your Apply Security Changes

The SAP CC Core Server system must be restarted for the newly implemented secure configuration to be applied.

To restart the SAP CC Core Server system, refer to the *Starting and Stopping the Servers* section available in the *SAP CC Installation Guide*.

Result

Now that the targeted secure configuration is implemented on the SAP CC Core Server side, you may want to configure an SAP CC Client to use the freshly secured communication service.

3.2.18.1.2 Maintaining a Certificate Entry for a Core Server System

Maintaining an SAP CC Core Server **certificate entry** consists in the following steps:

1. Identify the alias of the targeted certificate entry
2. Unlink the targeted certificate entry
3. Remove the targeted certificate entry
4. Implement a new secure configuration to renew the targeted certificate entry

3.2.18.1.2.1 Identify the Alias of the Targeted Certificate Entry

To maintain an SAP CC Core Server certificate entry, you first need to know the alias of the targeted certificate entry.

Use the Setup Tool user interface to execute the `certentry listLinks` command to display all the linked certificate entries and identify the alias of the certificate entry that you want to maintain.

Result

Once you have identified the alias of the targeted certificate entry you can unlink the client certificate from all the declared services. See Unlink the Targeted Certificate Entry [page 227].

3.2.18.1.2.2 Unlink the Targeted Certificate Entry

Once you have identified the alias of the targeted certificate entry you can unlink the client certificate from all the declared services.

Use the Setup Tool user interface to execute the `certentry unlink` command to unlink the client certificate from all the declared services.

Result

Once the Certificate entry is unlinked, you may want to remove it from the SAP CC Core Server system. See Delete the Targeted Certificate Entry from the SAP CC Core Server System [page 227].

3.2.18.1.2.3 Delete the Targeted Certificate Entry from the SAP CC Core Server System

Once you have identified the alias of the targeted certificate entry you can delete the certificate entry from the SAP CC Core Server system.

> ⚠ Caution
>
> Deleting a certificate entry will automatically unlink it from all the declared services if any.

Use the Setup Tool user interface to execute the `certentry delete` command to delete the certificate entry from the SAP CC Core Server system.

Result

Once the certificate entry has been deleted from the SAP CC Core Server system, you may want to implement a new secure configuration. See Implement a New Secure Configuration to Renew the Targeted Certificate Entry [page 228].

3.2.18.1.2.4 Implement a New Secure Configuration to Renew the Targeted Certificate Entry

In dual mode, both Client and Server authenticate each other to ensure that both parties involved in the communication are trusted. Both parties share their public certificates to each other.

To secure communications between an SAP CC Core Server system and an SAP CC Client application (or system) in dual mode, you need to:

1. Secure the communication in one-way mode: Make the SAP CC Core Server system shares its certificate with the SAP CC Client application (or system)
2. Enhance from one-way to dual mode: Make the SAP CC Client application (or system) shares its certificate with the SAP CC Core Server system

3.2.18.1.2.4.1 Secure the Communication in One-Way Mode

In one-way mode, only SAP CC Client applications (and systems) validate the SAP CC Core Server system to ensure that it receives data from the intended SAP CC Core Server system.

For implementing one-way mode, the SAP CC Core Server system shares its certificates with the SAP CC Client applications (and systems).

To secure the communication between an SAP CC Core Server and an SAP CC Client in one-way mode, you need to:

1. Secure the corresponding communication service(s) on the SAP CC Core Server side in one-way mode.

   > ### Note
   >
   > If the corresponding communication services are already secure on the SAP CC Core Server side, you may want to export the corresponding certificates for the next SAP CC Client configuration step.
   >
   > See Exporting an Existing Certificate from the SAP CC Core Server System [page 233].

2. Reconfigure the SAP CC Client applications (and systems) to use a secure connection using the SAP CC Core Server certificates.
   Every SAP CC Client application (or system) needs to trust the SAP CC Core Server system.
3. Check that the one-way secure mode is properly implemented.

Result

The one-way secure mode is properly implemented. As a maintenance operation, you can now improve the security of your SAP system landscape by implementing the dual secure mode that provides mutual authentication. See Enhance from the One-Way to the Dual Mode [page 230].

3.2.18.1.2.4.1.1 Secure the Corresponding Communication Services on the SAP CC Core Server System Side in One-Way Mode

Apply the procedures described in the section: Implement a New Secure Configuration to Renew the Targeted Key Entry [page 221]

3.2.18.1.2.4.1.2 Configure an SAP CC Client to Trust the SAP CC Core Server System

To be able to establish a connection on the communication services that have been secured in one-way mode on the SAP CC Core Server side, an SAP CC Client must be configured to trust the SAP CC Core Server system using the SAP CC Core Server certificates.

1. Configuring SAP CC Admin+ to Trust SAP CC Core Server
2. Configuring SAP CC Core Tool to Trust SAP CC Core Server

3.2.18.1.2.4.1.3 Check That the One-Way Secure Mode Is Properly Set

The secure configurations of both the SAP CC Core Server system and the SAP CC Client (application or system) are now complete.

You can now check that the one-way secure mode is properly set for all installed components in your SAP CC system landscape.

1. SAP CC Cockpit:
2. SAP CC Admin+:
   See Checking That the One-Way Secure Mode Is Properly Set for Admin+
3. SAP CC Core Tool:
   See Checking That the One-Way Secure Mode Is Properly Set for Core Tool
4. SAP CC CTS+ Plugin:
   Now that the SAP CC CTS+ Plugin is configured to trust the SAP CC Core Server, it is time to test that one-way secure mode is correctly set.
   Run the "IS-CC" deploy process in "CONNECT" mode with the proper user to test if the deployment is possible.

> → Recommendation
>
> Once the one-way mode is properly set up, you may want to improve security by enhancing the secure mode configuration from one-way to dual.

3.2.18.1.2.4.2 Enhance from the One-Way to the Dual Mode

Once the one-way mode is correctly set up, you can improve security by moving from one-way to dual mode.

Enhancing from one-way to dual mode consists in updating the secure configuration to make the SAP CC Client (applications and systems) share their certificate with the SAP CC Core Server system.

This enhancement is completed through the following steps for each targeted client application and system:

1. Create the SAP CC Client private key, certificate, and the corresponding keystore
2. Configure the SAP CC Client to use this created keystore (containing the private key and certificate)
3. Import the SAP CC Client certificate into the SAP CC Core Server system
4. Link the imported SAP CC Client certificate to the targeted SAP CC Core Server service or services
5. Check that the dual secure mode is properly implemented

Result

The dual secure mode is properly implemented with mutual authentication between the SAP CC Core Server system and SAP CC Client application (or system). Consider other SAP CC Clients.

3.2.18.1.2.4.2.1 Create the SAP CC Client Private Key, Certificate, and the Corresponding Keystore

For an SAP CC Client to be trusted by an SAP CC Core Server system, you will need to create a keystore containing the SAP CC Client private key and certificate.

> ⚠ Caution
>
> In SAP CC environments, the following constraints apply:
>
> - The **private key must be in the PKCS#8 format encoded in DER**.
> - The **certificate must be in the X.509v3 format encoded in DER**.
> - The **keystore must be in the PKCS#12 format**.

1. Create the SAP CC Client private key, certificate, and the corresponding keystore.

> ✤ Example
>
> Keys and certificates can be generated by any SSL tool.
>
> For example, see the SAP KBA 2845357 that provides you with commands based on OpenSSL.

Result

Once the keystore is created, you will need to configure the SAP CC Client to use it. See Configure the SAP CC Client to Use the Created Keystore [page 231].

3.2.18.1.2.4.2.2 Configure the SAP CC Client to Use the Created Keystore

An SAP CC Client must take into account its own keystore in order to use it when communicating with an SAP CC Core Server system in a dual secured mode.

To enable this keystore, it is necessary to declare it in the SAP CC Client configuration.

1. SAP CC Cockpit
2. SAP CC Admin+
   See Configure Admin+ to Declare the Keystore.
3. SAP CC Core Tool
   See Configure Core Tool to Declare the Keystore.
4. SAP CC CTS+ Plugin
5. Client Application or System based on SAP CC APIs

3.2.18.1.2.4.2.3 Import the SAP CC Client Certificate into the SAP CC Core Server System

To give the possibility for an SAP CC Core Server system to trust an SAP CC Client application, it is necessary to import the certificate of the SAP CC Client application into the SAP CC Core Server system.

Use the Setup Tool user interface to execute the `certentry import` command to import the certificate of the SAP CC Client application into the SAP CC Core Server system.

Linux:

```
./setup.sh certentry import <alias> <certificate>
```

MS Windows:

```
setup.bat certentry import <alias> <certificate>
```

Where:

- **`<alias>`** is the alias name of the certificate entry
- **`<certificate>`** is the file that contains the certificate of the SAP CC Client application (it must be in the X509v3 format encoded in DER)

> ⚠ Caution
>
> Make sure the chosen alias is appropriate.

For example, a certificate from an installed SAP CC Admin+ user interface that is valid for the year 2019 and that will be used on the HTTP service of all the dispatcher instances in the system:

- An appropriate alias would be: `admin+_dispatcher_http_2019`
- An irrelevant alias would be: `client_alias1`

Result

Next, you need to link the imported certificate to the relevant service or services.

3.2.18.1.2.4.2.4 Link the Imported SAP CC Client Certificate to the Targeted SAP CC Core Server Services

For an SAP CC Core Server system to trust an SAP CC Client application (or system), it is necessary to link the imported certificate of the SAP CC Client application (or system) to the relevant SAP CC Core Server service or services.

Use the Setup Tool user interface to execute the `certentry link` command to link an imported certificate to one or several services.

Example

To link a certificate entry to the `http` service of all supported instances considering the certificate entry alias is **`core_tool_http`**, execute the following set of commands:

```
./setup.sh certentry link core_tool_http dispatcher http
```

```
./setup.sh certentry link core_tool_http updater http
```

or do it all at once with one of the following command:

```
./setup.sh certentry link core_tool_http dispatcher,updater http
```

```
./setup.sh certentry link core_tool_http all http
```

Result

You can now test the configuration. See Check That the Dual Secure Mode Is Properly Set [page 233].

3.2.18.1.2.4.2.5 Check That the Dual Secure Mode Is Properly Set

Improvements to the secure configuration of both the SAP CC Client (application or system) and the SAP CC Core Server system are now complete.

You can now check that the dual secure mode is properly set for all installed components in your SAP CC system landscape.

1. SAP CC Cockpit:
2. SAP CC Admin+:
   Apply Check That the Dual Secure Mode Is Properly Set for Admin+
3. SAP CC Core Tool:
   Apply Check That the Dual Secure Mode Is Properly Set for Core Tool
4. SAP CC CTS+ Plugin:
   Now that the secure configurations of SAP CC CTS+ Plugin and Core Server have been improved, it is time to test that dual secure mode is correctly set.
   Run the "IS-CC" deploy process in "CONNECT" mode with the proper user to test if the deployment is possible.

3.2.18.2 Exporting an Existing Certificate from the SAP CC Core Server System

If a communication service is already secure on the SAP CC Core Server system side, you may want to export the corresponding certificate to configure an SAP CC Client application (or system).

Exporting an existing SAP CC Core Server certificate requires to know the corresponding key entry alias.

1. To identify the key entry alias, use the Setup Tool user interface and execute the keyentry listLinks command to list all the linked key entries.

   ✤ Example

   You will obtain a result like below:

   ```
   Key links:
   alias                          instance        service
   "my_dispatcher_http_alias"     dispatcher      http
   "my_dispatcher_internal_alias" dispatcher      internal
   Key link list successful
   ```

2. With the list, find the target alias.
3. Once you have identified the correct key entry alias, use the Setup Tool user interface again to execute the keyentry export command to export the corresponding private key and certificate.
   Microsoft Windows operating systems: Microsoft Windows:

   ```
   setup.bat keyentry export <alias> <private_key_filename>
   <certificate_filename>
   ```

Linux and UNIX operating systems: Linux:

```
./setup.sh keyentry export <alias> <private_key_filename>
<certificate_filename>
```

Where:

- <alias> is the alias name of the key entry
- <private_key_filename> is the file that will contains the exported private key
- <certificate_filename> is the file that will contains the exported certificate

4. Delete the exported private key file.

> ⚠ Caution
>
> Do not forget to suppress the exported private key file as you will not use it.

Once you have successfully exported the certificate, you can use it to configure the SAP CC Client application or system. For example, the SAP CC Admin+ user interface

3.2.18.3 Configuring an SAP CC Client to Trust an SAP CC Core Server System

To be able to establish a connection on the communication services that have been secured in one-way mode on the SAP CC Core Server side, an SAP CC Client must be configured to trust the SAP CC Core Server system using the SAP CC Core Server certificates.

This SAP CC Client is:

- A tool of SAP CC such as Admin+, Cockpit, or Core Tool
- An SAP application, system, engine, or solution
- A client application or system (in your SAP system landscape) that connects to the SAP CC Core Server system

See Securing Communications with the Core Server System.

3.2.19 Startup Configuration Files

Keywords [page 235]

Preliminary Notes [page 235]

Description [page 235]

Configuration Options [page 236]

Modifying a Startup Configuration File [page 236]

3.2.19.1 Keywords

startup, jstart, instance, JVM[99] option, classpath, libpath

3.2.19.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- Document Definitions section of the SAP CC 2021 Installation and Maintenance Guide documentation

This procedure uses the following variables:

- `<SYSTEM_ID>`, which corresponds to the identifier of your system
- `<INSTANCE_NUMBER>`, which corresponds to the global number of the concerned instance, which is different from the identifier of the instance (such as "Dispatcher#1")
- `<INSTANCE_NAME>`, which corresponds to the name of the concerned instance
- `<INSTANCE_DIRECTORY>`, which corresponds to the directory of an SAP CC instance (CC<D/U/G/R/B/T><INSTANCE_NUMBER>)
- `<HOSTNAME>`, which corresponds to the name of the host containing the concerned instance

3.2.19.3 Description

In the environment of SAP Convergent Charging(SAP CC), each instance of the Core Server system uses a startup configuration file to initiate its startup process. These configuration files are:

- Stored in the SAP Profile Folder in the SAP Central Repository of an SAP system.
- Copied from the SAP Profile Folder to the `config` folder of each deployed instance during the instance startup phase.

You configure the startup configuration file of each instance that is available in your SAP CC system and thus fit your specific needs.

References

Configuration Options [page 236]

[99] Java Virtual Machine

Technical Operations

Modifying a Startup Configuration File [page 236]

3.2.19.4 Configuration Options

The following table contains the list of settings that can be configured within startup configuration files:

Settings	Parameter	Description
JVM Options	`<INSTANCE_NAME>.javaParameters`	This parameter includes all the JVM options used by the JRE[100] to execute an SAP CC Instance. For further information, refer to the Java Virtual Machine [page 157] section of this documentation. Each option has the same prefix: `-D` or `-X`
Java class path of customized functions	`<INSTANCE_NAME>.classPath`	This parameter is used to specify all the specific Java classes that must be loaded by the instance on startup.
Library path of customized functions	`<INSTANCE_NAME>.libPath`	This parameter is only relevant when the `.classPath` parameter is modified and when the related Java classes require native libraries.

> ⚠ Caution
>
> Do not modify other parameters.

3.2.19.5 Modifying a Startup Configuration File

To modify the startup configuration file of a given instance composing your SAP Convergent Charging system (Core Server), complete the following procedure:

1. Edit the `jstart.config` file of the concerned instance, that is located in the following subdirectory of the SAP Central Repository: `profile/jstart/<SYSTEM_ID>_<INSTANCE_DIRECTORY>_<HOSTNAME>`

   > ✤ Example
   >
   > For example, the directory name of a **dispatcher** instance with the SAP instance number **03** is "CCD03".

   > → Recommendation
   >
   > SAP SE strongly recommends that you back up the `jstart.config` file before editing its content.

[100] Java Runtime Environment

2. Modify the relevant settings according to your needs.
3. Save your modifications.
4. Restart the concerned instance of the Core Server system to apply your modifications.
   During the startup phase of the instance, the modified `jstart.config` file is copied from the **SAP Profile Folder** to the **local** `config` folder of the instance, overwriting the existing local `config/jstart.config` file.

> → Recommendation
>
> SAP SE recommends that you check the content of the `jstart.config` file located in the **local** `config` folder of the instance, to ensure that your modifications have correctly been reflected.

3.2.20 System and Data Auditing

Keywords [page 237]

Preliminary Notes [page 237]

Description [page 238]

Measuring Product Usage [page 238]

Data Object Auditing [page 240]

3.2.20.1 Keywords

Auditing, recording, usage measurement, usage metric, change log, snapshot, user operation

3.2.20.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- System and Data Auditing section of the SAP CC 2021 Application Help documentation
- "System and Data Auditing: Product Usage Measurement Settings" and "System and Data Auditing: Data and Operation Auditing Settings" groups in the SAP CC 2021 System Parameter Reference documentation
- `metriclist`, `purge_object_change_logs` and `purge_user_operations` commands of the SAP CC 2021 Primary Help for Admin+ documentation
- Installing and Launching Admin+ and Installing and Launching Core Tool sections of the SAP CC 2021 Installation and Maintenance Guide documentation
- "Installation of Diagnostics Agents" SAP Note 1365123 documentation

3.2.20.3 Description

For system control purpose, SAP Convergent Charging provides detailed information about the usage of the SAP CC product and the tracking of modifications performed on master data and related user operations.

You can use the following procedures to configure the product usage measurement and master data auditing mechanisms available within your SAP CC landscape and thus fit your specific needs:

- Recording SAP Product Usage Metrics [page 238]
- Measuring Customer Usage in SAP Solution Manager [page 239]
- User Operation Auditing [page 240]
- Object Snapshot Recording [page 241]

3.2.20.4 Measuring Product Usage

In terms of product usage measurement, the SAP CC Core Server system gives the possibility to:

- Record metrics dedicated to the business usage of the SAP CC product
- Provide some customer usage information to SAP Solution Manager

  Recording SAP Product Usage Metrics [page 238]

  Measuring Customer Usage in SAP Solution Manager [page 239]

3.2.20.4.1 Recording SAP Product Usage Metrics

SAP Convergent Charging (SAP CC) is delivered with a set of metrics dedicated to the measurement of its business usage, providing information such as:

- The volume of master data configured within the system
- The volume of transactional data generated by the system during business operations

These metrics are recorded in the Core Database system by a dedicated scheduler whose execution depends on a policy that you can modify to fit specific needs.

> ⓘ Note
>
> This scheduler is also responsible for purging out-of-date data on a daily basis. As this scheduler is executed once a day, 1 day of recorded metrics are deleted at each automated purge operation.

To modify the behavior of the scheduler in charge of recording metrics related to business usage, execute the following procedure:

1. Launch your favorite user interface [page 45] and identify as the administrator of the Core Server system.
2. Modify the value of the following system parameters for each **updater** instance of the Core Server system:
   - METRIC_RECORDING_SCHEDULER_ENABLED, which gives the possibility to enable or disable the scheduler

- METRIC_RECORDING_DAY_TIME, which defines the time when the metrics are recorded
- METRIC_RECORD_TIME_WINDOW, which defines the retention period for recorded metrics. Once this period is reached, the scheduler in charge of recording metrics deletes the recorded metrics that are older than this period.
  As this scheduler is executed once a day, 1 day of recorded metrics are deleted at each purge operation.

For further information about these system parameters, refer to their dedicated documentation available in the SAP CC 2021 System Parameter Reference documentation.

> **Note**
>
> The recorded business usage metrics can be browsed using:
>
> - The Display Usage Metrics app provided by the Cockpit user interface
> - The Metrics view available in the Core Tool user interface

3.2.20.4.2 Measuring Customer Usage in SAP Solution Manager

In an integrated scenario with SAP Solution Manager, SAP CC can communicate with the SAP SMD Agent to provide information related to customer usage. The following procedures give the possibility to configure both SAP Convergent Charging and SAP Solution Manager for customer usage measuring purpose.

Configuring SAP CC for customer usage measurement

To communicate with SAP Solution Manager, SAP CC must fulfill the following requirements:

- SAP Solution Manager must be installed within your landscape
- SAP CC must be registered in SAP Solution Manager as a technical system. For further information, refer to the “Managed Systems” section of the SAP Solution Manager documentation: http://support.sap.com/solutionmanager
- An SMD Agent must be installed on each machine hosting a Core Server system. For further information, refer to the dedicated SAP Note 1365123 documentation

Once these requirements are fulfilled, execute the following procedure:

1. Enable the scheduler in charge of recording metrics related to business usage. For further information, refer to the Recording SAP Product Usage Metrics [page 238] dedicated procedure described above
2. Launch the Core Tool user interface and log in with an individual user granted the "User Administrator" role
3. Create a technical user named **`SM_COLL_<SID>`**, where `<SID>` corresponds to the SID of the concerned Core Server system. This user must not be granted any role. The password of this used will be used afterwards when configuring the SAP Solution Manager side

Configuring SAP Solution Manager for customer usage measurement

To communicate with SAP CC, execute the following procedure:

1. Launch the SMD Agent Administration user interface and identify as an administrator
2. Click the *Application Configuration* tab

3. Select *Agent Applications* > *com.sap.smd.agent.application.global.configuration*
4. For each updater of the concerned Core Server system:
   - Fill the *Select scope* field with the name of the machine hosting the updater
   - Click the *Edit* button
   - Add the `<SID>/sapj2ee/com/user` property
   - Fill this property with the name of the **`SM_COLL_<SID>`** user created in Step 1
   - Add the `<SID>/sapj2ee/com/pwd` property
   - Fill this property with the password of the **`SM_COLL_<SID>`** user created in Step 1
   - Add the `<SID>/sapj2ee/msgserver/httpport` property
   - Fill this property with the HTTP[101] port number of the updater
5. Save your modifications

3.2.20.5 Data Object Auditing

SAP Convergent Charging (SAP CC) gives the possibility to track and view the changes that are performed on master data everytime a user operation is executed. **Enabled by default**, this *Data Object Auditing* feature **cannot be deactivated** in your SAP Convergent Charging system landscape.

In terms of master data and user operations' audit, the SAP CC Core Server system gives the possibility to record the operations [page 240] that are executed by the SAP CC users, and the resulting modifications on the associated master data [page 241].

User Operation Auditing [page 240]

Object Snapshot Recording [page 241]

3.2.20.5.1 User Operation Auditing

SAP Convergent Charging gives the possibility to track and view some user operations executed by the SAP CC users and performed on the SOAP[102] over HTTP[103] and XML[104] over HTTP communication channels. Enabled by default with all available domains activated, SAP SE recommends that you do not disable this User Operation Auditing feature which enriches the auditing capabilities of your SAP Convergent Charging landscape.

> ⚠ Caution
>
> This feature increases the volume of data that is stored in the `USER_OPERATION` table of the Core Database, which may significantly impact the performance of the whole Core Server system. SAP SE recommends that you regularly perform purge operations to optimally maintain your Core Server system. For further information about such purge operations, refer to the `purge_user_operations` command of the Admin+ user interface.

101 HyperText Transfer Protocol
102 Simple Object Access Protocol
103 HyperText Transfer Protocol
104 eXtended Markup Language

To modify the behavior of the User Operation Auditing feature, change the value of the following system parameters [page 45] for all the dispatcher, updater, guider, rater, bulkloader and taxer instances of the Core Server system:

- AUDIT_DOMAIN, which gives the possibility to specify a subset of user operations domains to audit
- AUDIT_ACTIVATION, which gives the possibility to enable or disable the recording of user operations

For further information about these system parameters, refer to their dedicated documentation available in the SAP CC 2021 System Parameter Reference documentation.

> ⓘ Note
>
> Enabling the User Operation Auditing feature requires a restart of the concerned instances. Disabling the feature does not necessary require a restart of the concerned instances, depending whether permanent and/or runtime values are modified. If the runtime values are set to false, the feature is immediately disabled, but the pending mass operations continue to be audited until they end.

3.2.20.5.2 Object Snapshot Recording

To get additional details about the modifications made on master data, SAP Convergent Charging provides an Object Snapshot Recording feature that is **enabled by default with all available domains activated** and that gives the possibility to record snapshots of the modified master data.

> ⚠ Caution
>
> This feature increases the volume of data that is stored in the `OBJECT_SNAPSHOT` table of the Core Database, which may significantly impact the performance of the whole Core Server system. SAP SE recommends that you regularly perform purge operations to optimally maintain your Core Server system. For further information about such purge operations, refer to the `purge_object_change_logs` command of the Admin+ user interface.

To modify the behavior of the Object Snapshot Recording feature, modify the value of the following system parameters [page 45] for each updater instance:

- AUDIT_OBJECT_DOMAIN, which gives the possibility to specify a subset of user operations domains to audit
- AUDIT_OBJECT_ACTIVATION, which gives the possibility to enable or disable the recording of user operations

For further information about these system parameters, refer to their dedicated documentation available in the SAP CC 2021 System Parameter Reference documentation.

> ⓘ Note
>
> Enabling the Object Snapshot Recording feature requires a restart of the updaters. Disabling the feature does not necessary require a restart of the updaters, depending whether permanent and/or runtime values are modified. If the runtime values are set to false, the feature is immediately disabled.

3.2.21 Thread Dump Files Generation

Keywords [page 242]

Preliminary Notes [page 242]

Description [page 242]

Configuration Options [page 243]

Enabling the Generation of Thread Dump Files [page 243]

3.2.21.1 Keywords

Thread dump, troubleshoot

3.2.21.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- Installing and Launching Admin+ section of the SAP CC 2021 Installation and Maintenance Guide documentation
- `dump` command of the SAP CC 2021 Primary Help for Admin+ documentation
- Thread Dump File Generation Troubleshooting Settings group in the SAP CC 2021 System Parameter Reference documentation

3.2.21.3 Description

In addition to logging and tracing capabilities, SAP Convergent Charging gives the possibility to generate thread dump files, which contain information about the execution context of the different threads that are executed by the instances of the Core Server system.

In case a problem occurs during the execution of a business or technical operation, or if the execution exceeds some defined thresholds, the related instance generates these dump files automatically.

> → Tip
>
> Mainly addressed to support specialists, the generation of these thread dump files can be configured in order to fit specific needs mainly related to issue troubleshooting.

You can use this procedure [page 243] to configure the generation of thread dump files in your running SAP CC system and thus fit your specific needs.

3.2.21.4 Configuration Options

The following table lists the SAP CC system parameters that you can use to configure the generation of thread dump files by an instance in the SAP CC Core Server system:

System Parameter	Description
Function Enablement	
THREAD_DUMP_ENABLED	Enables/disables the generation of thread dump files
Function Options	
THREAD_DUMP_MINIMUM_INTERVAL	Minimum time interval (in seconds) between the generation of two thread dump files
THREAD_DUMP_OUTPUT_PATH	Storage location of the thread dump files
THREAD_DUMP_RETENTION_PERIOD	Retention period (in days) for thread dump files
Function Triggers	
THREAD_DUMP_THRESHOLD_CPU_TIME	Threshold (in ms) relating to CPU[105] time above which a thread dump file must be generated
THREAD_DUMP_THRESHOLD_OUTPUT_ITEMS	Threshold relating to the number of output items generated by a thread above which a thread dump file must be generated
THREAD_DUMP_THRESHOLD_RATING_CONTEXTS	Threshold relating to the number of rating contexts used by a thread above which a thread dump file must be generated
THREAD_DUMP_THRESHOLD_RFC_TIME	The threshold (in milliseconds) relating to the duration of an RFC executed within a thread above which a thread dump file must be generated
THREAD_DUMP_THRESHOLD_SQL_TIME	Threshold (in ms) relating to the duration of an SQL[106] query executed within a thread above which a thread dump file must be generated
THREAD_DUMP_TRIGGER_PERIOD	Period (in seconds) used by the scheduled trigger after which a thread dump file must be generated

3.2.21.5 Enabling the Generation of Thread Dump Files

To enable the thread dump files mechanism for a set of instances of the Core Server system, execute the following procedure:

1. For each concerned instance, set up the value [page 45] of the THREAD_DUMP_ENABLED system parameter to `true`.
2. For each concerned instance, modify the value of the other configuration options [page 243] according to your business or technical needs.

[105] Central Processing Unit
[106] Structured Query Language

3.2.22 Third-Party Integration

Keywords [page 244]

Preliminary Notes [page 244]

Description [page 244]

Modifying the SAP SLD Connection Settings [page 244]

Modifying the Introscope Agent Settings (Agent Profile) [page 245]

3.2.22.1 Keywords

SLD[107], payload

3.2.22.2 Preliminary Notes

For further information, refer to the following SAP CC documentation:

- "Update of SAP System Component Repository in SLD" SAP Note 669669 documentation

3.2.22.3 Description

The Core Server system can be integrated with third-party applications.

You can use the following procedure to configure the settings related to the integration with the SAP System Landscape Directory system.

3.2.22.4 Modifying the SAP SLD Connection Settings

To send payload information to the SAP System Landscape Directory (SLD) system, it is necessary to configure some settings related to the integration with third-party systems. To modify this configuration, execute the following procedure:

1. Launch your favorite user interface [page 45] and identify as the administrator of the Core Server system
2. For each dispatcher instance, modify the values of the following system parameters:
   - SLD_URL, which corresponds to the URL[108] of the SLD Data Supplier Service in the SAP SLD system, that must respect the following format: `http://<SLD_HOST>:<SLD_PORT>/sld/ds`

[107] System Landscape Directory

- SLD_USER, which gives the possibility to specify the name (or identifier) of the SAP SLD user that must be used by the Core Server system to log on the SAP SLD system
- SLD_PASSWORD, which corresponds to the password of the specified SAP SLD user
- SLD_SCHEDULER_ENABLED, which gives the possibility to enable or disable the scheduler in charge of sending information to the SAP SLD system
- SLD_SCHEDULER_RECURRENCE, which gives the possibility to modify the recurrence of the scheduler, configured by default to send information to the SAP SLD system every day at 1:00 AM

For further information about these system parameters, refer to their dedicated documentation available in the SAP CC 2021 System Parameter Reference.

3.2.22.5 Modifying the Introscope Agent Settings (Agent Profile)

SAP installer always preinstalls the CA APM Introscope Java Bytecode Instrumentation Agent in each SAP CC Core Server system.

You can modify the configurations of the Introscope Agent according to your needs by consulting the SAP Note 1453216.

> **Note**
>
> If you implement some customizations, do not forget to manage the backups of these configurations as part of the backup and restore strategy for SAP Convergent Charging.
>
> Your custom settings relate to the *Software Configuration* category. Back up the corresponding files.

Related Information

Making a Backup Copy of the Software Configurations for the Introscope Agents
Backup and Restore

3.2.23 Users and User Management

Keywords [page 246]

Preliminary Notes [page 246]

Description [page 246]

Separating Service and Individual Users [page 247]

Configuring the Security Policy for User Passwords [page 248]

108 Uniform Resource Locator

Managing User Sessions within User Interfaces [page 251]

Creating User Accounts [page 252]

Granting Roles to SAP CC User Accounts [page 253]

Resetting the Emergency User [page 253]
In some given emergency situations, you must reset the emergency user of SAP Convergent Charging (SAP CC).

Users Management via External Systems [page 254]

3.2.23.1 Keywords

User, password, security policy, user isolation, user session, concurrent edition, role, right, SCIM[109].

3.2.23.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- User Administration and Authentication section of the SAP CC 2021 Security Guide documentation
- User Session Management section of the SAP CC 2021 Application Help documentation
- Installing and Launching Admin+ and Installing and Launching Core Tool sections of the SAP CC 2021 Installation and Maintenance Guide documentation
- "Security Management Settings" and "User Management Settings" groups in the SAP CC 2021 System Parameter Reference documentation
- Users section of the SAP CC 2021 Primary Help for Core Tool documentation
- `list_clients`, `disconnect_client`, `purge_object_change_logs` and `purge_user_operations` commands of the SAP CC 2021 Primary Help for Admin+ documentation
- SCIM API documentation

3.2.23.3 Description

SAP Convergent Charging provides proprietary mechanisms to manage users and associated concepts such as password policy, user roles, working sessions, and so on.

You can use the following procedures to configure the management of users within your SAP CC landscape and thus fit your specific needs:

- Separating Service and Individual Users [page 247]
- Configuring the Security Policy for User Passwords [page 248]

[109] System for Cross-domain Identity Management

- Managing User Sessions within User Interfaces [page 251]
- Creating User Accounts [page 252]
- Resetting the Emergency User [page 253]
- Users Management via External Systems [page 254]

3.2.23.4 Separating Service and Individual Users

As described in the SAP CC 2021 Security Guide documentation, SAP Convergent Charging (SAP CC) defines two types of users:

- **Individual users**, whose passwords are encrypted and must be regularly changed.
- **Service users**, whose passwords are encrypted (with possible lower encryption level for performance reasons) and never expire.

SAP Convergent Charging gives the possibility to **isolate service users from individual users**. When this *User Isolation* mechanism is enabled:

- Only **service** users can use the operations provided by the Web Services (WS) and the Message Client technical interfaces.
- Only **individual** users can use the SAP CC user interfaces and the operations provided by the HCI[110] technical interface.

> ⓘ Note
>
> As of SAP CC 4.1 SP1, the *User Isolation* mechanism is enabled by default at installation time for security reasons. It is the recommended system configuration and you need to consider how to turn it on.

> → Recommendation
>
> When you update your SAP system landscape from SAP CC 4.1 SP0 or when you upgrade from a previous release (4.0, 3.0, or 2.0), consider improving the security by implementing the user isolation as a new security measure and test and complete this procedure: Separating Service and Individual Users [page 247].
>
> See also: User Isolation

Procedure

To enable or disable the isolation between individual users and service users, complete the following procedure:

1. Launch your favorite user interface [page 45] and identify as the administrator of the Core Server system.
2. For each updater instance of the Core Server system, modify the value of the USER_ISOLATION_ENABLED system parameter in order to enable or disable the isolation of users.

[110] HTTP Communication Interface

3. Change both the persistent and runtime values so you do not need to restart the updater instances in your running system.

> → Tip
>
> In a development system landscape, you can disable the user isolation to simplify your developments and integration activities.

3.2.23.5 Configuring the Security Policy for User Passwords

Every SAP CC user defined in your SAP CC system landscape is protected by a password. The *Password Management Policy* feature allows you to configure your security policy for the user passwords to fit specific security requirements.

When the password management policy is enabled (default installation), SAP CC also gives the possibility to modify the following options that make up this securing policy:

Policy Option	Description
Minimum length	The PASSWORD_MANAGEMENT_MINIMUM_LENGTH system parameter specifies the minimum length for passwords. ⚠ Caution Setting this system parameter to 0 means that no minimum length is required for passwords.
Complexity	The PASSWORD_MANAGEMENT_COMPLEXITY system parameter specifies the complexity rules that passwords must respect.
Maximum duration	The PASSWORD_MANAGEMENT_EXPIRATION_DELAY system parameter specifies the number of days after which the password of an individual user expires and must be changed. ⚠ Caution Setting this system parameter to 0 means that passwords never expire.
Maximum idle duration	The PASSWORD_MANAGEMENT_MAX_IDLE_DELAY system parameter specifies the maximum number of days until which a user must logon before being automatically locked. Setting this system parameter to 0 means that users are not locked even if they never logon.
Reuse cycle	The PASSWORD_MANAGEMENT_REUSE_CYCLE system parameter specifies the number of different values that must be used for password before reusing a given value. Setting this system parameter to 0 means that there is no limit regarding the reuse of passwords.
Reuse delay	The PASSWORD_MANAGEMENT_REUSE_DELAY system parameter specifies the number of days until which a previously used password can be reused. Setting this system parameter to 0 means that previously used passwords can be reused.

Policy Option	Description
Modification interval	The PASSWORD_MANAGEMENT_CHANGE_DELAY system parameter specifies the minimum number of days between 2 password modifications. Setting this system parameter to 0 means that passwords can be changed whenever you want.
Failed login attempts	The PASSWORD_MANAGEMENT_FAIL_LIMIT system parameter specifies the number of successive failed logon attempts until which the user is automatically locked. Setting this system parameter to 0 means that logon attempts are not limited.
SHA-256 hash rounds	Gives the possibility to configure the complexity of encryption applied to passwords by specifying the number of steps of the algorithm that is applied to hash passwords before storing them in the Core Database. The PASSWORD_HASH_ROUNDS_FOR_INDIVIDUAL_USERS and PASSWORD_HASH_ROUNDS_FOR_SERVICE_USERS system parameters specify the number of hash rounds applied to password hashing for respectively individual users and service users. **⚠ Caution** This number of steps has a performance impact on communications with SAP CC, during which passwords are hashed using the same number of steps as the number used for storing passwords in the database. By default, this number of steps is set to 10.000 for both individual and service users. This value ensures a good protection for passwords of individual users, but increase the latency on the Web Services (WS) and HCI technical interfaces by 10 milliseconds approximately. To reduce this performance impact, SAP SE recommends using a lower value for service users (PASSWORD_HASH_ROUNDS_FOR_SERVICE_USERS) such as 1.000 rounds or 100 rounds: • For 1.000 rounds, the latency is expected to increase of 1 ms approximately. • For 100 rounds, the increase of latency is nonmeasurable. **Fine-tune the values of the PASSWORD_HASH_ROUNDS_FOR_SERVICE_USERS system parameter to optimize the security and performance of the communications with external systems.** **ⓘ Note** Changing the value of these parameters is not taken into account immediately. It only affects the users when their password is modified.

Modify the Password Management Policy

At any time, you can change the options for the password management policy.

To modify the password management policy for your SAP Convergent Charging system landscape, complete the following procedure:

1. Launch the Cockpit or Admin+ user interface and identify as the system administrator of the SAP CC Core Server system.

2. For each updater instance composing the Core Server system, check that the value of the PASSWORD_MANAGEMENT_ENABLED system parameter is set to `TRUE`.
3. For each updater instance, modify the value of the system parameters that correspond to the policy options you need to modify.
   For further information about the possible system parameters, refer to their dedicated documentation available in the SAP CC 2021 System Parameter Reference documentation and user assistance.
4. Restart these instances of the running Core Server system.

Disable the Password Management Policy

To disable the password management policy, complete the following procedure:

1. Launch the Admin+ or the Cockpit user interface and identify as the administrator of the Core Server system.
2. For each updater instance composing your Core Server system, modify the value of the PASSWORD_MANAGEMENT_ENABLED system parameter in order to disable the password management policy. Set up the values to `FALSE`.
3. Restart these instances of the running Core Server system.
4. After the restarts, your changes are available for all the instances of this system and for all the user interfaces, other systems, applications, and solutions that communicate with this Core Server system.

Enable the Password Management Policy Again

In this situation, you previously disabled the password management policy in your SAP system and want to enable it again.

> **Note**
>
> When the password management policy is enabled again, every individual user whose password has been modified will have to modify again this password, by himself at the next logon to an SAP CC user interface.

1. As a first step, configure the options.
2. Then enable the function in your SAP system again.
   For each updater instance composing the Core Server system, set up the value of the PASSWORD_MANAGEMENT_ENABLED system parameter to `TRUE`.

3.2.23.5.1 Fine-Tuning the Password Encryption Mechanism

Depending on your security level requirements, expected performance for communications with SAP Convergent Charging, and capacities of your SAP system, you optimize the processing of the encryption of user passwords.

Determine the values that are relevant for your implementation scenario:

- PASSWORD_HASH_ROUNDS_FOR_INDIVIDUAL_USERS
- PASSWORD_HASH_ROUNDS_FOR_SERVICE_USERS

See details here: Configuring the Security Policy for User Passwords [page 248]

3.2.23.6 Managing User Sessions within User Interfaces

User sessions within the Core Tool and BART Tool user interfaces of SAP Convergent Charging (SAP CC) can be controlled in terms of:

- Concurrent edition within the Core Tool, used to prevent multiple SAP CC users working on the same master data
- Multiple accesses to a given user interface by a given individual user
- Security regarding inactivity between the user interface and the Core Server system

To enable or disable the management of user sessions within user interfaces, execute the following procedure:

1. Launch your favorite user interface [page 45] and identify as the administrator of the Core Server system.
2. For each updater instance, modify the value of the USER_SESSION_ENABLED system parameter in order to enable or disable the management of user sessions.
   For further information about this system parameter, refer to its dedicated documentation available in the SAP CC 2021 System Parameter Reference documentation.
3. Restart the updater instances of the Core Server system.

When the management of user sessions is enabled, SAP CC also gives the possibility to modify the following options that make up this management policy:

Policy Option	Description
Simultaneous connections	The USER_SESSION_SESSION_LIMIT_PER_USER_AND_TOOL system parameter specifies the maximum number of simultaneous connections that a given user can make on a given GUI[111]. When this limit is reached, further connections are refused. **Note** Setting this system parameter to 0 means that a given user can connect as many times as he wants to a given GUI.

[111] Graphical User Interface

Policy Option	Description
Inactivity period	The USER_SESSION_VALIDITY_PERIOD system parameter specifies the maximum period during which a user session can be left inactive, which means that there is no interaction between the GUI and Core Server. **Note** Setting this system parameter to 0 means that user sessions never expire. **→ Recommendation** The objects being edited in the Core Tool user interface are not protected against concurrent edition any longer when the session expires. SAP SE recommends that you increase the value of this parameter if SAP CC users need to work on the same objects without saving them for a long time.

→ Tip

In case of user session failure, you can use the `search_user_session` and `delete_user_session` commands of the Admin+ user interface. When a session is deleted, the objects opened in "edition" mode during this user session are released and can be modified within another session.

See: SAP CC 2021 Primary Help for Admin+

Modify the Management Policy for User Sessions

At any time, you can change the options of the management policy for user sessions.

To modify the management policy for user sessions, execute the following procedure:

1. Launch your favorite user interface [page 45] and identify as the administrator of the Core Server system.
2. For each updater instance in the system, modify the value of the system parameters that correspond to the policy options you need to modify.
   For further information about the concerned system parameters, refer to their dedicated documentation available in the SAP CC 2021 System Parameter Reference documentation.
3. Restart the updater instances of the Core Server system.

3.2.23.7 Creating User Accounts

Both individual users and service users defined in SAP Convergent Charging are:

- Protected by a password that respects a security policy you can configure to fit specific needs. For further information, refer to the Configuring the Security Policy for User Passwords [page 248] dedicated section above
- Assigned to roles that represent access rights to sets of master data domains

Using Core Tool

As a user administrator, use the Core Tool user interface to create and maintain your SAP CC users.

To create users within your SAP Convergent Charging system landscape, execute the following procedure:

1. Launch the Core Tool user interface and identify as the administrator of the Core Server system or as a user granted the "User Administrator" role.
2. Create the relevant users, assigning them the adequate roles.

3.2.23.8 Granting Roles to SAP CC User Accounts

Using Core Tool

As a user administrator or a system administrator, use the Core Tool user interface to grant roles to SAP CC user accounts.

See Granting Roles to a User.

3.2.23.9 Resetting the Emergency User

In some given emergency situations, you must reset the emergency user of SAP Convergent Charging (SAP CC).

> ⚠ Caution
>
> In case you created users in an incorrect way that prevents you from logging on to any SAP CC user interface, you can reset the emergency user. This user corresponds to the first super administrator user created at installation time, with whom you will be able to log on again and perform adequate modifications.
>
> For more information, refer to the dedicated SAP Note 1890952. **This documentation is released for SAP Customers and SAP Partners. This emergency recovery procedure is confidential.**

> ⚠ Caution
>
> This user account must be temporary.
>
> As soon as the emergency user is reset, do manage all the relevant SAP CC user accounts and **delete this emergency user**. This security measure is fundamental.

3.2.23.10 Users Management via External Systems

External systems can **provision individual users** in SAP Convergent Charging (SAP CC) with the implementation of the System for Cross-domain Identity Management (SCIM) API of SAP Convergent Charging.

> ✤ Example
>
> See SCIM.

If you are using a third-party SCIM-compliant identity management system, please refer to its official documentation for details on its use.

To check if your running SAP Convergent Charging system provides SCIM services, get the values of the SCIM_MANAGEMENT_ENABLED system parameter in the updater instances that compose the system.

> → Remember
>
> All the conditions and restrictions related to the management of users and user accounts apply. Refer to the Users chapter of the SAP CC documentation for more information.

> ⚠ Restriction
>
> - Only a system administrator can create, modify, lock/unlock, or delete a system administrator account.
> - If a system administrator is assigned to a catalog, this administrator can only manage user accounts assigned to the same catalog.
> - A system/user administrator cannot edit, or delete his/her own account.

User Provisioning

Consult the official documentation of the SCIM-based system that can create your individual SAP CC users.

SCIM API and Specifications

The SAP Convergent Charging SCIM API enables you to manage users and their group assignments. If the SAP CC users are centrally managed in an external system, such as SAP Identity Authentication Service, this API can be used to integrate with the external system for user provisioning.

The SCIM 2.0 version at least is required, which is defined by two RFCs:

- RFC 7643, which defines the core schema related to models used by SCIM 2.0
- RFC 7644, which defines the (HTTP-based) protocol used by SCIM 2.0

> ⚠ Caution
>
> To use the SCIM services in SAP Convergent Charging, a service user must be previously created with the role ADMIN USER. This user must be present in each REST request header with an *Authentication* field set to `Basic Authentication`.
>
> For more information about authentication, refer to the User Administration and Authentication section in the SAP CC 2021 Security Guide.

For a matter of security, the SCIM_MANAGEMENT_ENABLED system parameter must be set to `true`.

> → Remember
>
> - This system parameter cannot be modified while SAP Convergent Charging is running.
> - The default value is false.
> - A TLS/SSL connection for Web Services is mandatory between the client and the updater, because without TLS, the updater ignores the updater system parameter and just logs a message about it. Then, the updater runs normally, but all SCIM endpoints just return 404, without any explanation.

Supported SCIM Resources

SAP Convergent Charging supports the following SCIM resources:

- User
- Group

User Attributes

The following table describes the mapping between SCIM User Attributes and SAP CC User Attributes:

SCIM User Attributes	SAP CC User Attributes
SCIM Common Attributes	
`id`	`oid`
`externalId`	`login`
`active` > ⓘ Note > For the read requests and creation/replace responses, this attribute is always returned with a value computed at the current system date/time of the SAP CC Core Server system.	• `non locked` • system date between `validFrom` and `validTo`
`meta.created`	`operation date`

SCIM User Attributes	SAP CC User Attributes
SCIM Core Attributes	
`username`	`login`
`groups`	`permissions`

> **Note**
>
> - SAP CC users created via SCIM are individual users.
> - SAP CC users are created with an access to all pricing catalogs for chargeable services and products.
> - SAP CC user IDs are case sensitive.

Groups and Group Attributes

To implement the SCIM groups, new groups are defined and are mapped onto SAP CC roles as follows:

SCIM User Group Name	SAP CC User Role
`pricing_specialist`	MARKETING + CSR
`technical_support`	REMOTE SUPPORT
`system_administrator`	ADMIN + BARTADMIN
`user_administrator`	USER_ADMIN
`process_manager`	PROCESS_MANAGER

The following table shows the settings of the group attributes:

SCIM Group Attributes	SAP CC Supported Values
SCIM Common Attributes	
`id`	• `pricing_specialist` • `technical_support` • `system_administrator` • `user_administrator` • `process_manager`
`externalId`	• `pricing_specialist` • `technical_support` • `system_administrator` • `user_administrator` • `process_manager`
`meta.created`	`epoch(s)`
SCIM Core Attributes	

SCIM Group Attributes	SAP CC Supported Values
`displayName`	• `pricing_specialist` • `technical_support` • `system_administrator` • `user_administrator` • `process_manager`

> → Remember
>
> The authentication method is **Basic** or **OAuth 2.0** standard.

Service Provider Configuration Schema

SCIM Server Configuration Attributes	Values
`patch`	`"supported":false`
`bulk`	`"supported":false`
`paging`	`"supported":false`
`filter`	`"supported":false` > ⓘ Note > For the `User` resource, filtering is supported only with: > • the `eq` operation > • the mandatory attributes `externalId` and `userName`
`changePassword`	`"supported":true`
`sort`	`"supported":false`
`etag`	`"supported":false`
`authenticationSchemes` • `type` • `name` • `description`	 • `Httpbasic` • `HTTP Basic` • `Basic Authentication`

Operations

Only Users and Groups are proposed as resources by the SCIM API of SAP Convergent Charging (SAP CC). The following operations are available:

SCIM 2.0 Operation	REST HTTP Method / URI	User	Group
`Create`	POST `https://<HOST_NAME>:<HOST_PORT>/rest/scim/v2/<resource>`	■	□
`Delete`	DELETE `https://<HOST_NAME>:<HOST_PORT>/rest/scim/v2/<resource>/<id>`	■	□
`Read`	GET `https://<HOST_NAME>:<HOST_PORT>/rest/scim/v2/<resource>/<id>`	■	■
`Replace`	PUT `https://<HOST_NAME>:<HOST_PORT>/rest/scim/v2/<resource>/<id>`	■	□
`Search (ReadAll)`	GET `https://<HOST_NAME>:<HOST_PORT>/rest/scim/v2/<resource>`	■	■

■: Available □: Unavailable

Where:

- `<HOST_NAME>` is the name or IP address (IPv4 or IPv6) of the machine that hosts an updater instance of the SAP CC system (default: localhost)
- `<HOST_PORT>` is the communication port number of the machine that hosts the updater instance of the SAP CC system (default: 9080)

> **Note**
>
> Only individual users are visible.
>
> Therefore:
>
> - If an ID corresponding to a service user in SAP CC results from a getting / searching operation, a `404 NOT FOUND` message is returned.
> - If an ID already exists as a service user in SAP CC, the same ID for an individual user is not supported by a creating / replacing / deleting operation. In such a case, a `403 FORBIDDEN` message is returned.

User Validity Period

As of SAP CC 2020 FPS 1, the SCIM API of SAP CC is enhanced with the IPSUser custom schema allowing to define a validity period for each individual user.

This schema is available at the following location:

```
https://<HOST_NAME>:<HOST_PORT>/rest/scim/v2/Schemas/
urn:sap:cloud:scim:schemas:extension:custom:2.0:IPSUser
```

Where:

- <HOST_NAME> is the name or IP address (IPv4 or IPv6) of the machine that hosts an updater instance of the SAP CC system (default: `localhost`)
- <HOST_PORT> is the communication port number of the machine that hosts the updater instance of the SAP CC system for the Web Services (default: `9080`)

Example

```
{
  "schemas": [
    "urn:sap:cloud:scim:schemas:extension:custom:2.0:IPSUser",
    "urn:ietf:params:scim:schemas:core:2.0:User"
  ],
  "externalId": "demo",
  "userName":"demo",
  "password": "foobar",
  "active": true,
  "groups": [
    {
      "value": "pricing_specialist"
    }
  ],
  "urn:sap:cloud:scim:schemas:extension:custom:2.0:IPSUser": {
    "validFrom": "2020-10-01T00:00:00Z",
    "validTo": "2021-01-01T00:00:00Z"
  }
}
```

Configuration Endpoints

The following table summarizes the service provider configuration endpoints:

Endpoint	REST HTTP Method / URI
`/ServiceProviderConfig`	GET `https://<HOST_NAME>:<HOST_PORT>/rest/scim/v2/ServiceProviderConfig`
`/Schemas`	GET `https://<HOST_NAME>:<HOST_PORT>/rest/scim/v2/Schemas`
`/ResourceTypes`	GET `https://<HOST_NAME>:<HOST_PORT>/rest/scim/v2/ResourceTypes`

Where:

- <HOST_NAME> is the name or IP address (IPv4 or IPv6) of the machine that hosts an updater instance of the SAP CC system (default: localhost)
- <HOST_PORT> is the communication port number of the machine that hosts the updater instance of the SAP CC system for Web Services (default: 9080)

3.2.24 WS Destinations

Keywords [page 260]

Preliminary Notes [page 260]

Description [page 260]

Configuration Options [page 260]

Creating WS Destinations [page 261]

3.2.24.1 Keywords

WS[112] destination, SOAP[113] over HTTP[114], SAP S/4HANA Cloud Public Editionfor Contract Accounting and Invoicing, BIT[115], Billable Item

3.2.24.2 Preliminary Notes

For further information, refer to the following SAP CC documentation:

- Installing and Launching Setup Tool section of the SAP CC 2021 Installation and Maintenance Guide documentation

3.2.24.3 Description

SAP Convergent Charging SOAP[116] over HTTP[117] communication channel to send billable items to SAP S/4HANA Cloud Public Edition.

You can use this procedure to manage the different WS[118] destinations within your landscape.

3.2.24.4 Configuration Options

The following table contains the list of parameters that you can change:

112 Web Services
113 Simple Object Access Protocol
114 HyperText Transfer Protocol
115 Billable Item
116 Simple Object Access Protocol
117 HyperText Transfer Protocol
118 Web Services

Parameter	Values
`service.url` The URL[119] of the "Convergent Invoicing Billable Items - Create - Request (Single)" web service used to create billable items	You must set the valid HTTPS URL that corresponds to the targeted web service
`service.certentry.alias` The alias of the certificate entry containing the TLS[120] certificate used by the web service	This parameter is optional. You can set a valid alias of a certificate entry imported using the "certentry" target of the Setup Tool user interface
`auth.http.basic.login` The login used to authenticate to the web service	You must set a valid login accepted by the web service
`auth.http.basic.password` The password used to authenticate to the web service	You must set a valid password accepted by the web service
`auth.keyentry.alias` The alias of the key entry containing the TLS key pair to use to connect to the web service	This parameter is optional. You can set a valid alias of a key entry imported using the "keyentry" target of the Setup Tool user interface
`request.timeout` The maximum time (expressed in milliseconds) for establishing connections and receiving responses from the web service	This parameter is **optional** and is set by default to: **60000**. You can set any positive value.
`response.timeout` The maximum time (expressed in milliseconds) for receiving confirmation responses from the web service	This parameter is **optional** and is set by default to: **60000**. You can set any positive value.

3.2.24.5 Creating WS Destinations

To create a WS[121] destination, execute the following procedure:

1. Connect to the host of your landscape where a dispatcher is installed
2. Launch your favorite user interface [page 45] and identify as the administrator of the Core Server system
3. Execute the `wsdestination list` command to retrieve the list of WS destinations that are available in the Core Database
4. If the targeted web service corresponds to one of the previously retrieved WS destinations, execute the `wsdestination export` command to ensure that the configuration of this WS destination is correct. If the targeted web service does not correspond to one of the previously retrieved WS destinations, or if the exported configuration does not fit, you must create the relevant WS destination by:
   - Copying the `ws.destination.sk` file available in the `config/s4hc` folder of the dispatcher instance

[119] Uniform Resource Locator
[120] Transport Layer Security
[121] Web Services

- Editing this newly created file and modifying the relevant parameters to fit the targeted web service. For further information about the available parameters, refer to the Configuration Options [page 260] section above

5. Once the WS destination has been correctly set up, execute the `wsdestination import` command to import these parameters in the Core Database
6. Restart the concerned instances of the Core Server system

3.3 Configuring the BART Server System

Acquisition Sessions and Consolidation [page 262]

BART Server Configuration File [page 266]

Communication Channels [page 272]

Java Virtual Machine [page 275]

Logging and Tracing [page 276]

Magic Number Pattern [page 278]

Monitoring and Event Broadcasting [page 279]

Rating Sessions [page 283]

Rerating Sessions [page 284]

Security Certificates [page 285]

Third-Party Integration [page 285]

BART Database [page 287]

3.3.1 Acquisition Sessions and Consolidation

Keywords [page 263]

Preliminary Notes [page 263]

Description [page 263]

Modifying the Acquisition and Consolidation Settings [page 263]

Acquiring CDRs with the Java API [page 264]

Acquiring CDRs with an IEC [page 265]

3.3.1.1 Keywords

CDR[122], acquisition session, consolidation cache

3.3.1.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- Chargeable Items Acquisition section of the SAP CC 2021 Application Help documentation
- Chargeable Items Consolidation Process section of the SAP CC 2021 Application Help documentation

3.3.1.3 Description

As described in the SAP CC 2021 Application Help documentation, the BART Server system provides acquisition and consolidation features for incoming CDRs[123], that you can configure using the following procedure

3.3.1.4 Modifying the Acquisition and Consolidation Settings

To configure the acquisition and consolidation operations performed by the BART Server system, refer to the Modifying the BART Server Configuration File and Restarting the System [page 271] section below and modify the following settings of the BART Server:

- `acquisition.session.max.number`, that gives the possibility to specify the maximum number of simultaneous acquisition sessions that can run at a given time
- `acquisition.session.monitoring.period`, that corresponds to the period between 2 generations of "acquisition session status" events. For further information, refer to the Monitoring and Event Broadcasting [page 279] dedicated section below
- `consolidation.cache.enable`, that gives the possibility to enable or disable the consolidation cache used to temporary store additional information related to CDRs[124]
- `consolidation.cache.size`, that corresponds to the size of the consolidation cache

> **Note**
>
> Chargeable Item Acquisition is also possible with the SAP CC Core Server. Implement your acquisition process by using the `com.highdeal.cnd.message.BatchServiceClient` Java class.

[122] Call Detail Record (Telecommunications industry), or more generally Consumption Detail Record
[123] Call Detail Record (Telecommunications industry), or more generally Consumption Detail Record
[124] Call Detail Record (Telecommunications industry), or more generally Consumption Detail Record

3.3.1.5 Acquiring CDRs with the Java API

Network elements or applications can send CDRs[125] to the BART Server system and manage the related acquisition sessions. The BART Server then acquires these incoming CDRs, and sends them as chargeable items to the Core Server system. You can develop your own application using the Java APIs[126] available in the BART[127] Software Development Kit (SDK) and:

- Implementing the `CDRServiceClient` Java class of the `com.highdeal.bart.message` Java package, in case of communications performed via the Packets over TCP/IP[128] communication channel
- Implementing the Java classes of the `com.highdeal.bart.hci` Java package, in case of communications performed via the XML[129] over HTTP[130] communication channel (HTTP Communication Interface)

For further information, refer to the `com.highdeal.bart.message` and `com.highdeal.bart.hci` Java packages descriptions available in the Java/XML API Reference (BART Library) documentation, that also include the documentation of the concerned XSD fragments.

Example

The following Java snippet defines a client application that communicates using the Packets over TCP/IP communication channel for handling incoming CDRs:

```
CDRServiceClient collector;
try { collector = new CDRServiceClient("BART_HOST",2009); }
catch (Exception e) {
  System.out.println("Cannot initialize BART Collector client -> "+e);
  e.printStackTrace();
}
// Building up the acquisition session
AcquisitionSessionModel session = new AcquisitionSessionModel(...)
try {session = collector.startAcquisitionSession(session);}
catch (InvalidAcquisitionException e)
{System.out.println("InvalidAcquisitionException -> "+e);}
catch (ServerUnavailableException e)
{System.out.println("OperationFailureException -> "+e);}
catch (CommunicationFailureException e)
{System.out.println("OperationFailureException -> "+e);}
catch (Throwable t) {System.out.println("Throwable -> "+t); }
System.out.println("StartAcquisition() completed");
// Building up some CDRs
Vector cdrs = new Vector();
cdrs.add(new CDRModel(...)
try {result = collector.acquireCDR(cdrs, session);}
catch (InvalidCDRException e) {System.out.println("InvalidCDRException ->
"+e);}
catch (InvalidAcquisitionException e)
{System.out.println("InvalidAcquisitionException -> "+e);}
catch (ServerUnavailableException e)
{System.out.println("OperationFailureException -> "+e);}
catch (CommunicationFailureException e)
{System.out.println("OperationFailureException -> "+e);}
catch (Throwable t) {System.out.println("Throwable -> "+t);}
try {session = collector.stopAcquisitionSession(session);}
```

125 Call Detail Record (Telecommunications industry), or more generally Consumption Detail Record
126 Application Programming Interface
127 Batch Acquisition and Rating Toolset
128 Transmission Control Protocol/Internet Protocol
129 eXtended Markup Language
130 HyperText Transfer Protocol

```
catch (InvalidAcquisitionException e)
{System.out.println("InvalidAcquisitionException -> "+e);}
catch (ServerUnavailableException e)
{System.out.println("OperationFailureException -> "+e);}
catch (CommunicationFailureException e)
{System.out.println("OperationFailureException -> "+e);}
catch (Throwable t) {System.out.println("Throwable -> "+t);}
System.out.println("StopAcquisition() completed");
```

Example

The following Java snippet defines a client application that communicates using the XML over HTTP communication channel for handling incoming CDRs:

```
HttpMessageSender sender =  ....
StartAcquisitionSessionOp op = new StartAcquisitionSessionOp();
String description = "Test snippet";
String source = "BART Reference";
AcquisitionSessionModel session =
     new AcquisitionSessionModel(source, description,

AcquisitionSessionModel.ACQUIRE_AND_DEDUPLICATE_MODE);
op.setAcquisitionSession(session);
MessageOriginator originator = new MessageOriginator();
// Setting the name and password of a user granted the right to create a
session in the BART Server
originator.setName("USER");
originator.setAuthentication(new SimpleMessageAuthentication("PASSWORD"));
MessageHeader header = new MessageHeader();
header.setOriginator(originator);
MessageBody body = new MessageBody();
body.addOperation(op);
MessageEnvelope envelope = new MessageEnvelope();
envelope.setBody(body);
envelope.setHeader(header);
try {
  MessageEnvelope result = sender.send(envelope);
  StartAcquisitionSessionResult opRes =
     (StartAcquisitionSessionResult)result.getBody().getResult(0);
  session = opRes.getAcquisitionSession();
}
catch (CommunicationFailureException e)
{System.out.println("CommunicationFailureException: "+e);}
catch (InvalidAcquisitionException e)
{System.out.println("InvalidAcquisitionException -> "+e);}
catch (OperationFailureException e)
{System.out.println("OperationFailureException -> "+e);}
System.out.println("StartAcquisition() completed");
```

3.3.1.6 Acquiring CDRs with an IEC

An IEC (Import/Export Connector) application can send CDRs[131] to the BART Server system and manage the related acquisition sessions. The BART Server then acquires these CDRs, and sends them as chargeable items to the Core Server system. You can customize this IEC application using some Java tasks that will apply to the concerned data.

[131] Call Detail Record (Telecommunications industry), or more generally Consumption Detail Record

For further information, refer to the Scenarios section of the CAT Tool online help documentation.

3.3.2 BART Server Configuration File

Keywords [page 266]

Preliminary Notes [page 266]

Description [page 266]

Configuration Settings for BART Server [page 267]

Modifying the BART Server Configuration File and Restarting the System [page 271]

3.3.2.1 Keywords

`bart.config`

3.3.2.2 Preliminary Notes

For further information, refer to the following SAP CC documentation:

- Starting and Stopping the Server Systems section of the SAP CC 2021 Installation and Maintenance Guide documentation

3.3.2.3 Description

As described in the SAP CC 2021 Installation and Maintenance Guide documentation, the configuration of the BART Server system is located in the `bart.config` file:

- This configuration file is available in the `config` folder of the BART Server system already deployed on a dedicated host in your SAP system landscape.
- The file content is grouped upon the following sections:
  - Database connectivity
  - Logs and traces
  - Connection to the Core Server system
  - Connection to the BART Server system
  - Business operations settings

  The content and sections change and depend on your installed software version. Check the file to see the details. You can also view the skeleton file (*.CONFIG.SK) that is delivered.

You can use this generic procedure to configure the different settings of the BART Server configuration and thus fit your specific needs.

- Modifying the BART Server Configuration File and Restarting the System [page 271]
- Configuration Settings for BART Server [page 267]

3.3.2.4 Configuration Settings for BART Server

The following table contains the list of settings that can be configured within the configuration file of the BART Server system. For further information about the possible values of each setting, refer to the documentation available within the `bart.config` file itself:

Category	Description	BART Server Setting
Database Connectivity		
Connection	JDBC URI of the BART Database	`db.SQLHELPER_JDBC_URI`
	Name of the BART Database user	`db.SQLHELPER_LOGIN`
	Password of the BART Database user	`db.SQLHELPER_PASSWORD`
	CSV list of additional properties for SQL connections	`db.SQLHELPER_JDBC_PROPERTIES`
	Number of database connections to open	`db.SQLHELPER_CONNECTION_COUNT`
Retry policy	Maximum number of request retries	`db.SQLHELPER_MAX_REDO`
	Duration (in seconds) between 2 reconnection attempts	`db.SQLHELPER_RECONNECT_INTERVAL`
	Maximum number of reconnection attempts	`db.SQLHELPER_RECONNECT_ATTEMPTS`
Technical settings	Size of the statement cache	`db.SQLHELPER_STATEMENT_CACHE_SIZE`
	Unique ID for batch reservation	`db.SQLHELPER_ID_INTERVAL`
Logs and Traces		
Logs (**business** processing)	Severity threshold for logs	`ls.app.severity`
	Identifier(s) of the destination(s)	`ls.app.destination`
Logs (**system** processing)	Severity threshold for logs	`ls.sys.severity`
	Identifier(s) of the destination(s)	`ls.sys.destination`
Traces	Severity threshold for traces	`ls.trc.severity`
	Identifier(s) of the destination(s)	`ls.trc.destination`
	Domain(s) for traces	`ls.trc.domain`
Console Destination	Enable/Disable	`ld.console.enabled`
	Output format	`ld.console.formatter`

Category	Description	BART Server Setting
Fileset Destination #1	Enable/Disable	`ld.file1.enabled`
	Output format	`ld.file1.formatter`
	Number of files available for the rotation	`ld.file1.setsize`
	Maximum file size before rotation	`ld.file1.size`
	File path, name, and extension	`ld.file1.name`
Fileset Destination #2	Enable/Disable	`ld.file2.enabled`
	Output format	`ld.file2.formatter`
	Number of files available for the rotation	`ld.file2.setsize`
	Maximum file size before rotation	`ld.file2.size`
	File path, name, and extension	`ld.file2.name`
Fileset Destination #3	Enable/Disable	`ld.file3.enabled`
	Output format	`ld.file3.formatter`
	Number of files available for the rotation	`ld.file3.setsize`
	Maximum file size before rotation	`ld.file3.size`
	File path, name, and extension	`ld.file3.name`
Fileset Destination #4	Enable/Disable	`ld.file4.enabled`
	Output format	`ld.file4.formatter`
	Number of files available for the rotation	`ld.file4.setsize`
	Maximum file size before rotation	`ld.file4.size`
	File path, name, and extension	`ld.file4.name`
Fileset Destination #5	Enable/Disable	`ld.file5.enabled`
	Output format	`ld.file5.formatter`
	Number of files available for the rotation	`ld.file5.setsize`
	Maximum file size before rotation	`ld.file5.size`
	File path, name, and extension	`ld.file5.name`
Configuration of the BART Server system		
Configuration	IP address or host name of the listening BART Server	`bart.collector.server.hostname`
	Collector port of the listening BART Server	`bart.collector.server.port`
	Number of threads for the listening BART Server	`bart.collector.server.threads`
	HTTP[132] port numberBART Server	`bart.http.server.port`

[132] HyperText Transfer Protocol

Category	Description	BART Server Setting
	Number of threads for BART's HTTP server	`bart.http.server.threads`
Securing the servers	HTTP server security and encryption	`bart.http.secure`
	Client authentication required or not when HTTP server is secured	`bart.http.secure.client_auth`
	Collector server security and encryption	`bart.collector.secure`
	Client authentication required or not when collector server is secured	`bart.collector.secure.client_auth`
	List of protocol names that are enabled when performing encrypted communications using the SSL[133]/TLS[134] protocol for the technical interfaces Message TCP and HCI[135]	`tls.protocols`
	List of cipher suites names that are enabled when performing encrypted communications using the SSL/TLS protocol for the technical interfaces Message TCP and HCI	`tls.cipher_suites`
Connection to the Core Server system		
Connection	CSV list of dispatchers to contact	`cc.dispatcher.boot.http.url.list`
	Name of the technical user used to connect to the Core Server system	`cc.login`
	Password of the specified technical user	`cc.password`
	Activation or deactivation of the authentication on the message API with the previous credentials	`message.api.authentication.enabled`
Third-party integration		
Connection to the SAP SLD[136] system	URL[137] of the targeted SAP SLD system	`bart.sld.url`
	Name of the SAP SLD user	`bart.sld.user`
	Password of the SAP SLD user	`bart.sld.password`
Business Settings		
Acquisition	Maximum number of parallel acquisition sessions	`acquisition.session.max.number`
	Polling period (in milliseconds) for generating events	`acquisition.session.monitoring.period`
Rerating	Number of threads used for rerating operations, each subscription being rerated by a dedicated thread	`rerating.pool.size`

133 Secure Socket Layer
134 Transport Layer Security
135 HTTP Communication Interface
136 System Landscape Directory
137 Uniform Resource Locator

Category	Description	BART Server Setting
	Maximum number of rerating requests that can be queued when rerating subscriptions	`rerating.queue.size`
	Specifies whether CDRs[138] to select for rerating purpose must have the STATUS_RATED status, or not	`rerating.limited_to_rated_status`
	Maximum number of subscriptions (by batch rating group) that can be rerated as the same time within a rerating session	`rerating.max_number_subscriptions`
Rating and charging	Maximum number of parallel rating sessions	`rating.session.max.number`
	Polling period (in milliseconds) for generating events	`rating.session.monitoring.period`
	Maximum number of concurrent threads for each rating session	`rating.session.threads`
	Size of the CDR commit block	`rating.cdr.commit.block`
	Specifies whether the batch charge API[139] must be used, or not	`rating.cdr.batch_charge`
	Maximum number of CDRs in a given batch charge operation	`rating.cdr.charge.block`
	Name of a specific property to add in the chargeable item during rating operations and containing the rating date	`rating.cdr.rating_date.property_name`
	Name of a specific property to add in the chargeable item during rating operations and containing a unique identifier	`rating.cdr.unique_id.property_name`
	Source of the unique identifier to add in the chargeable item during rating operations	`rating.cdr.unique_id.source`
	Specifies whether CDRs must be re-consolidated before being rated	`rating.consolidate.before.rating`
	Specifies whether CDRs selected from the BART Database must be rated, or simply used for benchmarking purpose	`rating.bench_mode`
	Absolute path of the file used to store CDRs that have been rated but not committed due to a database failure	`sql.failures.log.file.name`
	Enables/disables the cached structure used for consolidation operations	`cache.consolidation.enable`
	Size of the cached structure used for consolidation operations	`cache.consolidation.size`
Job management	Maximum number of running jobs	`scheduler.job.pool.size`

[138] Call Detail Record (Telecommunications industry), or more generally Consumption Detail Record
[139] Application Programming Interface

Category	Description	BART Server Setting
	Polling period (in milliseconds) for generating events	`scheduler.job.monitoring.period`
Bulk operations	Number of threads that can simultaneously handle requests concerning bulk operations	`bart.bulk.threads`
	Number of objects to select within the BART Database for each transaction	`bart.bulk.searchFetchSize`
	Number of objects modified by each commit for each database transaction	`bart.bulk.modificationFetchSize`
Technical Settings		
XML[140] validation	Enables/disables the validation of the XML schema	`bart.xml_schema_validation`
Report handling	Number of report handlers	`report.handler.number`
	Java class name of the handler	`report.handler.<n>.classname`
	Absolute path of the configuration file to use when instantiating the report handler	`report.handler.<n>.properties`
	List of topics (separated by ";") the report handler must subscribe to	`report.handler.<n>.topics`
Magic number computation	Enables/disables the deduplication operations of CDRs within the BART Database	`magic_number.database_deduplicate`
	Number of magic numbers to use	`magic_number.number`
	Name of the CDR to which the magic number <n> should be associated	`magic_number.<n>.item_name`
	List of fields that must be concatenated to build the magic number	`magic_number.<n>.fields`
Monitoring	Enables/disables the monitoring capabilities of the BART Database	`bart.monitoring`
	Maximum number of monitoring listeners	`bart.monitoring.clients.max.number`

3.3.2.5 Modifying the BART Server Configuration File and Restarting the System

> ⚠ Caution
>
> An invalid modification of this file can prevent the SAP CC BART Server system from starting, or make it unstable.
>
> Ensure that you understand the consequences of your modification, contact SAP Support for further information.

[140] eXtended Markup Language

Modify the Configuration File and Restart the System

As a system administrator, to modify the configuration file of the BART Server system, complete the following procedure:

1. Find the `bart.config` file that is located in the `config` folder of the deployed BART Server system (on its dedicated host).
2. In a production SAP system landscape, back up this configuration file.
3. Edit this text file in a text editor.
4. Modify the relevant settings according to your technical or business needs. See Configuration Settings for BART Server [page 267] as a reference.
5. Save your modifications.
6. In a production SAP system landscape, back up the modified configuration file.
7. Restart (stop then start again) the BART Server system and check its successful startup.

> ⚠ Caution
>
> - If a setting is defined more than once in the file, only the last value is considered by the SAP system.
> - If the imported file contains several definitions for a same setting, only the last one is kept.

3.3.2.5.1 Modifying the Ports of the BART Server System

For development or test purpose, you can change the communication ports of the SAP CC BART Server system after installation.

Procedure

1. Edit the local `bart.config` file:
   - `<DRIVE>:\usr\sap\<BART_SYSTEM_ID>\CAB<SAP_INST_NB>\config\bart.config` (Microsoft Windows)
   - `/usr/sap\<BART_SYSTEM_ID>/CAB<SAP_INST_NB>/config/bart.config` (Linux and UNIX operating systems)
2. Modify the ports that need top be changed
3. Save and close the modified file
4. Restart the BART Server system

3.3.3 Communication Channels

Keywords [page 273]

Preliminary Notes [page 273]

Description [page 273]

Modifying the Database Communication Settings [page 273]

Modifying the Core Server Connection Settings [page 274]

Modifying the Communication Settings for Acquisition and Rerating Operations [page 274]

Securing the Communication Channels [page 275]

3.3.3.1 Keywords

Monitoring, event, broadcasting, report file handler, UDP[141]

3.3.3.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- Chargeable Items Acquisition and Chargeable Items Rerating features descriptions, available in the SAP CC 2021 Application Help documentation
- Enhanced Logging and Tracing Framework section of the SAP CC 2021 Application Help documentation

3.3.3.3 Description

Communications with the BART Server system rely on different communication channels whose use depend on the concerned elements.

You can use the following procedures to configure the settings related to communication channels and thus fit your specific needs or performance requirements:

- Modifying the Database Communication Settings [page 273]
- Modifying the Core Server Connection Settings [page 274]
- Modifying the Communication Settings for Acquisition and Rerating Operations [page 274]
- Securing the Communication Channels [page 275]

3.3.3.4 Modifying the Database Communication Settings

To communicate with the BART Database, refer to the Modifying the BART Server Configuration File and Restarting the System [page 271] section above and modify the following settings of the BART Server:

[141] User Datagram Protocol

- `logon`, that corresponds to the name of the BART Database user
- `password`, that corresponds to the password of the specified user
- `database URI`, that corresponds to the URI[142] of the BART Database
- `sql.failures.log.file.name`, that corresponds to the absolute path of the file used to store information related to non-committed CDR[143] modifications in case of BART Database failure. This SQL[144] failure file is used on startup by the BART Server: to commit these non-committed CDRs

3.3.3.5 Modifying the Core Server Connection Settings

To communicate with the Core Server system, refer to the Modifying the BART Server Configuration File and Restarting the System [page 271] section above and modify the following settings of the BART Server:

- `cc.login`, that corresponds to the name of the technical user used to connect to the Core Server system
- `cc.password`, that corresponds to the password of the specified technical user
- `cc.dispatcher.boot.http.url.list`, that corresponds to a CSV[145] list of URLs[146] (including port number) relating to dispatcher instances.

> **Note**
>
> - Use HTTPS[147] URLs in case of secured communications
> - When the security is activated, the BART Server system cyphers the passwords specified in the `bart.config` configuration file

3.3.3.6 Modifying the Communication Settings for Acquisition and Rerating Operations

To perform acquisition and recharging operations using the BART Server system, it is necessary to configure some settings related to communications performed over the Packets over TCP/IP[148] communication channel. To modify this configuration, refer to the BART Server Configuration File [page 266] section above and modify the following settings of the BART Server:

- `bart.collector.server.hostname`, that corresponds to the IP address or name of the machine hosting the BART Serverr system
- `bart.collector.server.port`, that corresponds to the port number of the BART Server collector service
- `bart.collector.server.threads`, that corresponds to the number of threads that can simultaneously handle requests concerning the collector service of the BART Server

142 Uniform Resource Identifier
143 Call Detail Record (Telecommunications industry), or more generally Consumption Detail Record
144 Structured Query Language
145 Comma Separated Value
146 Uniform Resource Locator
147 HyperText Transfer Protocol Secured
148 Transmission Control Protocol/Internet Protocol

- `rerating.pool.size`, that corresponds to the size of the pool dedicated to rerating requests
- `rerating.queue.size`, that corresponds to the number of rerating requests handled by the rerating service

3.3.3.7 Securing the Communication Channels

For security purposes, communication channels used to communicate with the BART Server system can be secured. For further information about the available securing procedures, refer to the dedicated Securing an SAP CC System Landscape section of the SAP CC 2021 Installation and Maintenance Guide documentation.

3.3.4 Java Virtual Machine

Keywords [page 275]

Preliminary Notes [page 275]

Description [page 276]

Modifying JVM Options [page 276]

3.3.4.1 Keywords

JVM[149] options, virtual machine, memory heap, garbage collection, buffers

3.3.4.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- "JVM[150] option recommendations for SAP Convergent Charging" SAP Note 2006073 documentation
- "Generate Java GC Log for SAP Convergent Charging" SAP Note 1950168 documentation

149 Java Virtual Machine
150 Java Virtual Machine

3.3.4.3 Description

The architecture of SAP Convergent Charging relies on a set of java-based applications whose execution require a Java Runtime Environment. The BART Server system is a Java process whose execution can be configured to fit specific requirements.

To determine and modify the different JVM[151] options for the BART Server system, refer to the Java Virtual Machine [page 157] procedure available for the instances of the Core Server system, and apply the relevant modifications on the startup file of the BART Server system.

3.3.4.4 Modifying JVM Options

If you need to reconfigure these JVM options, then change the startup configuration file (`jstart.config`). See Modifying a Startup Configuration File [page 236].

3.3.5 Logging and Tracing

Keywords [page 276]

Preliminary Notes [page 276]

Description [page 277]

Configuration Options [page 277]

Modifying the Logging and Tracing Policy [page 277]

3.3.5.1 Keywords

Log, trace, category, severity, domain, threshold, troubleshoot

3.3.5.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- Installing and Launching BART+ section of the SAP CC 2021 Installation and Maintenance Guide documentation

[151] Java Virtual Machine

- Enhanced Logging and Tracing Framework section of the SAP CC 2021 Application Help documentation
- Logging and Tracing Settings group in the SAP CC 2021 System Parameter Reference documentation

3.3.5.3 Description

As described in the SAP CC 2021 Application Help documentation, SAP Convergent Charging provides logging and tracing features that give the possibility to generate both trace and log messages:

- According to defined levels such as severity, thresholds, domains, categories
- Using different output formats
- In multiple output destinations

Mainly addressed to administrators and support specialists, the generation of these log and trace messages can be configured in order to fit specific needs such as landscape monitoring, performance tracing, issue troubleshooting, and so on. You can use this procedure to configure the logging and tracing capabilities of your BART Server system and thus fit your specific needs.

3.3.5.4 Configuration Options

Logging and tracing features of the BART Server system can be configured to fit specific needs. By default, logging and tracing features are configured to fit a productive landscape, which corresponds to the following configuration:

- The logging feature is enabled, with severity thresholds set to `INFO`
- The tracing feature is enabled, with severity thresholds set to `ERROR`
- Log and trace messages are generated in different files, that are available in SAP Management Console or SAP Microsoft Management Console applications
- Generated files are stored both in the console and in the `log` directory of the BART Server system
- File rotation is enabled

3.3.5.5 Modifying the Logging and Tracing Policy

To fit specific needs, you can modify the default configuration of the logging and tracing features.

According to the modifications, you can perform:

- **Immediate** modifications, that are temporary and that can only be performed using the `set` command of the BART+ user interface. For further information, refer to its dedicated online help available in the SAP CC 2021 Primary Help for BART+ documentation
- **Deferred** modifications, that consist in modifying the BART Server configuration file as described in the Modifying the BART Server Configuration File and Restarting the System [page 271] dedicated section

3.3.6 Magic Number Pattern

Keywords [page 278]

Description [page 278]

Setting Up a Pattern for Magic Numbers [page 278]

3.3.6.1 Keywords

Magic number, deduplicate

3.3.6.2 Description

To identify and tag duplicated CDRs[152], the BART Server system manages sequences of magic numbers that are used to uniquely identify a CDR within the BART Database. Each magic number can be made up with the following information:

- The CDR name
- The consumption date, that represents the usage date of the customer service
- The user identifier (USID), that represents the technical identifier of the end customer
- The service identifier (SID), that represents the technical identifier of the customer service
- And all the other data, such as call duration, callee, caller, usage type, service type, and so on... whose availability depends on your business requirements

You can use this procedure to define the pattern for magic numbers used by the BART Server system for each type of CDR, and thus fit your specific needs.

3.3.6.3 Setting Up a Pattern for Magic Numbers

To set up a pattern for magic numbers used during deduplicating operations, refer to the Modifying the BART Server Configuration File and Restarting the System [page 271] section above and modify the following settings of the BART Server:

- `magic_number.database_deduplicate`, that gives the possibility to enable or disable the CDR[153] deduplication feature provided by the BART Server system
- `magic_number.number`, that corresponds to the number of magic number patterns to use, which shall be similar to the number of CDR types

[152] Call Detail Record (Telecommunications industry), or more generally Consumption Detail Record
[153] Call Detail Record (Telecommunications industry), or more generally Consumption Detail Record

- `magic_number.<n>.item_name`, that corresponds to the name of the CDR to which the magic number pattern is associated
- `magic_number.<n>.fields`, that corresponds to the list of fields that must be concatenated to build the magic number, each field corresponding to:
  - A standard property of the associated CDR
  - An additional information available in the associated CDR, that is prefixed by the `item.` string

> ✤ Example
>
> Considering a CDR named “SAMPLE_CDR” containing:
>
> - The `consumption_date`, `user_id` and `service_id` standard properties
> - The “ZZ_USAGE_TYPE” and “ZZ_SERVICE_TYPE” additional information, that are used during batch charging operations to price the usage of the customer service
>
> You can then define the following magic number pattern using the following configuration:
>
> ```
> magic_number.database_deduplicate=true
> magic_number.number=1
> magic_number.1.item_name=SAMPLE_CDR
> magic_number.1.fields=consumption_date;user_id;service_id;item.ZZ_USAGE_TYPE;i
> tem.
> ZZ_SERVICE_TYPE
> ```

3.3.7 Monitoring and Event Broadcasting

Keywords [page 279]

Preliminary Notes [page 279]

Description [page 280]

Modifying the Monitoring Settings [page 281]

Modifying the Report File Settings [page 282]

3.3.7.1 Keywords

Monitoring, event, broadcasting, report file handler, UDP[154]

3.3.7.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

[154] User Datagram Protocol

- Chargeable Items Acquisition Process and Chargeable Items Rerating Processsections of the SAP CC 2021 Application Help documentation
- Enhanced Logging and Tracing Framework section of the SAP CC 2021 Application Help documentation

3.3.7.3 Description

The BART Server system can broadcast events through XML[155] messages that are generated according to a defined frequency on the network. The following table contains the list of supported events:

Topic	Event	Description
Acquisition	Acquisition session start	This event is triggered each time a new acquisition session is started through the mediation chain. This event mainly contains the definition of the session being created
	Acquisition session status	This event is triggered at a regular frequency (depending on the BART Server settings) and describes the elapsed time of the session, the number of newly acquired CDRs[156] and the number of duplicated CDRs
	Acquisition session stop	This event is triggered each time an acquisition session is complete, and contains the total duration of the session, the number of acquired CDRs and the number of duplicated CDRs. When this event is triggered, a report file is generated, containing details about the acquisition session
Rating	Rating session start	This event is triggered each time a new rating session is started. This event mainly contains the definition of the session being created
	Rating session status	This event is triggered at a regular frequency (depending on the BART Server settings) and describes the elapsed time of the session, the number of rated CDRs and the number of erroneous CDRs

155 eXtended Markup Language
156 Call Detail Record (Telecommunications industry), or more generally Consumption Detail Record

Topic	Event	Description
	Rating session stop	This event is triggered each time a rating session is complete, and contains the total duration of the session, the number of rated CDRs and the number of erroneous CDRs. When this event is triggered, a report file is generated, containing details about the rating session
Job Execution	Job start	This event is triggered each time a job execution is started. This event mainly contains the definition of the job being started
	Job status	This event is triggered at a regular frequency (depending on the BART Server settings) and describes the elapsed time of the job execution
	Job stop	This event is triggered each time a job execution is complete. It contains the total duration of the job execution

Mainly addressed to administrators and support specialists, the generation of these events can be configured in order to fit specific needs such as landscape monitoring. You can use this procedure to configure the monitoring and event broadcasting capabilities of your BART Server system and thus fit your specific needs.

3.3.7.4 Modifying the Monitoring Settings

To adapt the BART Server system to specific monitoring needs, refer to the Modifying the BART Server Configuration File and Restarting the System [page 271] section above and modify the following settings of the BART Server:

- `bart.monitoring`, that gives the possibility to enable or disable the monitoring capabilities of BART Server
- `bart.monitoring.clients.max.number`, that corresponds to the maximum number of monitoring listeners
- `acquisition.session.monitoring.period`, that corresponds to the period between 2 generations of “acquisition session status” events described above
- `rating.session.monitoring.period`, that corresponds to the period between 2 generations of “rating session status” events described above
- `scheduler.job.monitoring.period`, that corresponds to the period between 2 generations of “job status” events described above
- `scheduler.job.pool.size`, that gives the possibility to specify the maximum number of jobs that can be active at a given time

3.3.7.5 Modifying the Report File Settings

Everytime a stop event relating to acquisition or rating session is triggered, the BART Server system generates a report file containing detailed information of the completed session.

Example

The following file represents a report related to an acquisition session:

```
CREATION DATE = Tue Nov 15 21:34:10 CET 2005
ACQUISITION DATE = 2005-11-15 21:34:10.717
ACQUISITION DURATION = 24 sec
ACQUISITION SOURCE = location_with_CDRs\source.txt
ACQUISITION DESCRIPTION = Sample acquisition
ACQUISITION MODE= Acquisition and de-duplicate mode
NB CDR ACQUIRED = 961
NB CDR DUPLICATE= 39
NB CDR CONSOLIDATE = 950
STATUS= ended
```

Example

The following file represents a report related to an rating session:

```
CREATION DATE = Tue Nov 29 17:53:32 CET 2005
RATING START DATE = 2005-11-29 17:53:32.303
RATING DURATION = 2 sec
RATING MAX CONSUMPTION DATE = 2005-11-29 17:53:32.287
RATING DESCRIPTION = null
RATING MODE= most mode
NB CDR RATED = 100
NB CDR IN ERROR= 0
STATUS= cleaned
```

The generation of report files relies on dedicated handlers in charge of listening to the adequate stop events. To configure this report file handling mechanism, refer to the Modifying the BART Server Configuration File and Restarting the System [page 271] section above and modify the following settings of the BART Server:

- `report.handler.number`, that gives the possibility to specify a number of file report handlers. Setting this parameter to 0 disables the reporting capabilities of BART Server
- `report.handler.<n>.classname`, that corresponds to the name of the Java class implementing the Event Handler Framework which describes how the events are handled
- `report.handler.<n>.properties`, that corresponds to the absolute path of the configuration file to use when instantiating the report handler
- `report.handler.<n>.topics`, that corresponds to the list of topics (separated by ";") the handler must subscribe to, considering the following possible topics: acquisition, rating and job

Once report handlers have been configured, you need to modify the configuration files specified in the `report.handler.<n>.properties` settings accordingly, using the execute the following procedure:

1. Edit the relevant configuration file, or create a copy from the default `report.config` file that is located in the config folder of the BART Server instance
2. Modify the following settings according to your needs:
   - `folder`, that corresponds to the absolute path of the folder containing the generated files

- `filename`, which gives the possibility to specify a prefix for the names of the generated files
- `fileSuffix`, which gives the possibility to specify a suffix for the names of the generated files

> **Note**
>
> The configuration of the default file handler leads to the generation of report files within the logs folder of the BART Server instance, that are named `report_<DATE>.txt`, where `<DATE>` corresponds to the date of the file generation.

3. Save your modifications

3.3.8 Rating Sessions

Keywords [page 283]

Preliminary Notes [page 283]

Description [page 283]

Modifying Settings Related to Rating Sessions Management [page 284]

3.3.8.1 Keywords

CDR[157], rating session, thread

3.3.8.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- Offline Charging section of the SAP CC 2021 Application Help documentation
- Description of the offline execution mode of the Chargeable Items Charging Process available in the SAP CC 2021 Application Help documentation

3.3.8.3 Description

As described in the SAP CC 2021 Application Help documentation, the BART Server system provides a batch rating feature used to rate acquired and consolidated CDRs[158]. This feature relies on rating sessions that you can configure using the following procedure.

[157] Call Detail Record (Telecommunications industry), or more generally Consumption Detail Record

3.3.8.4 Modifying Settings Related to Rating Sessions Management

To configure the management of rating sessions handled by the BART Server system, refer to the Modifying the BART Server Configuration File and Restarting the System [page 271] section above and modify the following settings of the BART Server:

- `rating.session.max.number`, that gives the possibility to specify the maximum number of simultaneous rating sessions that can run at a given time
- `rating.session.monitoring.period`, that corresponds to the period between 2 generations of "rating session status" events. For further information, refer to the Monitoring and Event Broadcasting [page 279] dedicated section above
- `rating.session.threads`, that corresponds to the maximum number of threads that can be allocated to a given rating session

3.3.9 Rerating Sessions

Keywords [page 284]

Preliminary Notes [page 284]

Description [page 285]

Modifying Settings Related to Rerating Sessions Management [page 285]

3.3.9.1 Keywords

CDR[159], rerating session, thread

3.3.9.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- Rerating Operations Driven by the BART Server System description of the Chargeable Items Rerating feature available in the SAP CC 2021 Application Help documentation
- Rerating Driven by the BART Server System description available in the Chargeable Items Rerating Process section of the SAP CC 2021 Application Help documentation

158 Call Detail Record (Telecommunications industry), or more generally Consumption Detail Record
159 Call Detail Record (Telecommunications industry), or more generally Consumption Detail Record

3.3.9.3 Description

As described in the SAP CC 2021 Application Help documentation, the BART Server system can be configured to work in conjunction with the Core Server system during rerating operations. You can use this business process to fix billing and pricing errors. You correct the consumption detail records or the pricing logic and you re-charge these service consumptions. Re-invoicing and re-billing is triggered. This feature relies on rerating sessions that you can configure using, the procedure described in the next section.

3.3.9.4 Modifying Settings Related to Rerating Sessions Management

To configure the management of rerating sessions handled by the BART Server system, refer to the Modifying the BART Server Configuration File and Restarting the System [page 271] section above and modify the following settings of the BART Server:

- `rerating.pool.size`, that gives the possibility to specify the number of threads used to rerate subscriptions
- `rerating.queue.size`, that corresponds to the maximum number of rerating requests that can be queued when rerating subscriptions

> ⚠ Caution
>
> The `rerating.pool.size` parameter must not exceed the maximum number of concurrent rating sessions defined though the `rating.session.max.number` parameter. Furthermore, the number of rating and rerating sessions running at the same time must also not exceed the maximum number of concurrent rating sessions.

3.3.10 Security Certificates

See Also

Securing Communications with the BART Server System

3.3.11 Third-Party Integration

Keywords [page 286]

Preliminary Notes [page 286]

Description [page 286]

Modifying the SAP SLD Connection Settings [page 286]

3.3.11.1 Keywords

SLD[160], payload

3.3.11.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- "Update of SAP System Component Repository in SLD[161]" SAP Note 669669 documentation

3.3.11.3 Description

The BART Server system can be integrated with third-party applications. You can use the following procedure to configure the settings related to the integration with the SAP System Landscape Directory (SLD) system.

3.3.11.4 Modifying the SAP SLD Connection Settings

To send payload information to the SAP SLD system during the startup sequence of the BART Server system, it is necessary to configure some settings related to the integration with third-party systems. To modify this configuration, refer to the "Modifying the BART Server configuration file" section above and modify the following settings of the BART Server:

- `bart.sld.url`, that corresponds to the IP address or name of the SLD Data Supplier Service in the targeted SAP SLD system, using the following format: `http://<SLD_HOST>:<SLD_PORT>/sld/ds`
- `bart.sld.user`, that corresponds to the name of the SAP SLD user
- `bart.sld.password`, that corresponds to the password of the specified user

160 System Landscape Directory
161 System Landscape Directory

3.3.12 BART Database

Keywords [page 287]

Preliminary Notes [page 287]

Description [page 287]

Modifying the Partitioning Settings [page 287]

3.3.12.1 Keywords

Database, retention period, partitioning

3.3.12.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- Database partitioning section of the SAP CC 2021 Application Help documentation
- "BART Database" dedicated online help available in the SAP CC 2021 Core Database Reference documentation

3.3.12.3 Description

In your SAP Convergent Charging (SAP CC) system landscape, BART Database is the back-end database of the BART Server system to store technical data. In order to improve performance of the running BART Server system regarding the storage of large number of CDRs[162], the BART Database is partitioned using a "Storage partitioning" method.

You can use the following procedures to modify the behavior of the BART Database regarding data partitioning, and thus fit your specific needs.

See Modifying the Partitioning Settings [page 287].

3.3.12.4 Modifying the Partitioning Settings

The storage partitioning of the BART Database can be adapted to improve performance during some operations such as data purging or archiving. According to your RDBMS technology, you can use the following procedures to configure:

[162] Call Detail Record (Telecommunications industry), or more generally Consumption Detail Record

- The number of days that CDRs[163] are kept in the BART Database
- The number of days or partitions that are created in advance
- How partitions are created (automatically or manually)
- The directories used to store tablespaces, filegroups, or segments

3.3.12.4.1 Partitioning Settings for Sybase ASE

The partitioning mechanism of a BART database running under a Sybase ASE RDBMS[164] relies on the following SQL[165] procedures:

Procedure	Description
CONFIGURE	This procedure gives DBA[166] the possibility to configure the global partitioning policy by filling in the `PARTITION_PURGE` table with the parameters specified as arguments
ADD_SEGMENT	This procedure gives DBA the possibility to create a new segment
REMOVE_SEGMENT	This procedure gives DBA the possibility to remove a segment which is no more used for storing new partitions
MANAGE_NEW_PARTITION	This procedure gives DBA the possibility to manually trigger the creation of new partitions, that are only created if needed. **Note** To create partitions automatically, you can create a job within Sybase ASE that calls the MANAGE_NEW_PARTITION procedure

To modify the default configuration of the partitioning mechanism, you must execute these procedures specifying arguments that correspond to your needs. The following table contains the list of available arguments you can use:

Procedure	Argument	Description
CONFIGURE	`partitionDays` (integer)	The number of days CDRs are kept in the BART database. Set by default to 40 days, this retention period cannot exceed 70 days.
	`partitionAdvanceDays` (integer)	The number of partitions to create in advance

163 Call Detail Record (Telecommunications industry), or more generally Consumption Detail Record
164 Relational Database Management System
165 Structured Query Language
166 DataBase Administrator

Procedure	Argument	Description
ADD_SEGMENT	`segment` (varchar)	Name of the segment to add, with a maximum of 256 characters
REMOVE_SEGMENT	`segment` (varchar)	Name of the segment to remove
MANAGE_NEW_PARTITION	`None`	None

3.3.12.4.2 Partitioning Settings for Oracle

The partitioning mechanism of a BART Database running under an Oracle RDBMS[167] relies on the following SQL[168] procedures:

Procedure	Description
`BART_ARCHIVE_PURGE.CONFIGURE`	This procedure gives DBA[169] the possibility to configure the global partitioning policy, by filling in the `PARTITION_PURGE` table with the parameters specified as arguments
`BART_ARCHIVE_PURGE.ADD_DIRECTORY`	This procedure gives DBA the possibility to create a new directory, resize an existing directory or reactivate a removed directory **Note** • Each directory name is unique and is respect the "BART_yyyymmdd" pattern, where yyyymmdd is a date • You must create one directory at least at installation • Resize the directory in case the initial size is exceeded
`BART_ARCHIVE_PURGE.CONFIGURE_TABLESPACE`	This procedure gives DBA the possibility to configure tablespace names
`BART_ARCHIVE_PURGE.REMOVE_DIRECTORY`	This procedure gives DBA the possibility to remove a directory which is no more used in order to store new tablespaces
`BART_ARCHIVE_PURGE.MANAGE_NEW_PARTITION`	This procedure gives DBA the possibility to manually trigger the creation of new partitions, that are only created if needed
`BART_ARCHIVE_PURGE.START_SCHEDULER`	This procedure activates the Oracle database scheduler for calling each day (at 2:00 AM), the MANAGE_NEW_PARTITION procedure

[167] Relational Database Management System
[168] Structured Query Language
[169] DataBase Administrator

Procedure	Description
BART_ARCHIVE_PURGE.STOP_SCHEDULER	This procedure stops the Oracle database scheduler used to execute the MANAGE_NEW_PARTITION regularly
BART_ARCHIVE_PURGE.RUN_SCHEDULER	This procedure is equivalent to the START_SCHEDULER procedure except that the first execution of the MANAGE_NEW_PARTITION procedure is immediate

To modify the default configuration of the partitioning mechanism, you must execute these procedures specifying arguments that correspond to your needs. The following table contains the list of available arguments you can use:

Procedure	Argument	Description
BART_ARCHIVE_PURGE.CONFIGURE	partitionDays (integer)	The number of days CDRs[170] are kept in the BART Database. Set by default to 40 days, this retention period cannot exceed 70 days
	partitionAdvanceDays (integer)	The number of partitions to create in advance
BART_ARCHIVE_PURGE.ADD_DIRECTORY	directory (varchar)	Path of the directory to create, with a maximum of 2.000 characters
	fileSize (varchar)	Initial size of the directory to create. Set by default to 1GB, this size cannot exceed 128GB
	extendSize (varchar)	The extension size of the directory, automatically handled by Oracle when necessary, and set by default to 250MB
BART_ARCHIVE_PURGE.CONFIGURE_TABLESPACE	tablespaceprefix (varchar)	The prefix of the tablespace name, set by default to "BART_"
	tablespacesuffix (varchar)	The suffix of the tablespace name
	datafilesuffix (varchar)	The suffix of the file containing the tablespace, set by default to ".dat"
BART_ARCHIVE_PURGE.REMOVE_DIRECTORY	directory (varchar)	The name of the directory to remove
BART_ARCHIVE_PURGE.MANAGE_NEW_PARTITION	None	None
BART_ARCHIVE_PURGE.START_SCHEDULER	None	None
BART_ARCHIVE_PURGE.STOP_SCHEDULER	None	None
BART_ARCHIVE_PURGE.RUN_SCHEDULER	None	None

[170] Call Detail Record (Telecommunications industry), or more generally Consumption Detail Record

3.3.12.4.3 Partitioning Settings for MS SQL Server

The partitioning mechanism of a BART Database running under a Microsoft SQL[171] Server RDBMS[172] relies on the following SQL procedures:

Procedure	Description
CONFIGURE	This procedure gives DBA[173] the possibility to configure the global partitioning policy by filling in the PARTITION_PURGE table with the parameters specified as arguments
ADD_DIRECTORY	This procedure gives DBA the possibility to create a new directory, resize an existing directory or reactivate a removed directory for storing filegroups **Note** • Each directory name is unique and is respect the "BART_yyyymmdd" pattern, where yyyymmdd is a date • You must create one directory at least at installation
REMOVE_DIRECTORY	This procedure gives DBA the possibility to remove a directory which is no more used in order to store new filegroups
MANAGE_NEW_PARTITION	This procedure gives DBA the possibility to manually trigger the creation of new partitions, that are only created if needed **Note** To create partitions automatically, you can create a job within SQL Server that calls the MANAGE_NEW_PARTITION procedure

To modify the default configuration of the partitioning mechanism, you must execute these procedures specifying arguments that correspond to your needs. The following table contains the list of available arguments you can use:

Procedure	Argument	Description
CONFIGURE	`partitionDays` (integer)	The number of days CDRs[174] are kept in the BART Database. Set by default to 40 days, this retention period cannot exceed 70 days.
	`partitionAdvanceDays` (integer)	The number of partitions to create in advance

171 Structured Query Language
172 Relational Database Management System
173 DataBase Administrator
174 Call Detail Record (Telecommunications industry), or more generally Consumption Detail Record

Procedure	Argument	Description
ADD_DIRECTORY	`directory` (varchar)	Path of the directory to create, with a maximum of 2.000 characters
	`fileSize` (varchar)	Initial size of the directory to create, set by default to 1GB
	`extendSize` (varchar)	The extension size of the directory, automatically handled by SQL Server when necessary, and set by default to 250MB
REMOVE_DIRECTORY	`directory` (varchar)	The name of the directory to remove
MANAGE_NEW_PARTITION	`None`	None

3.4 Configuring the Diameter Server System

> ⚠ Caution
>
> From SAP Convergent Charging SP 4, there is no further planned enhancements to the Diameter Server component. For further information, refer to SAP Note 2792189.

AVP Dictionary [page 292]

Communication Channels [page 294]

Diameter Server Configuration File [page 295]

Java Virtual Machine [page 299]

Logging and Tracing [page 300]

Multiple Services Credit-Control [page 303]

Security Certificates [page 305]

Service Dictionary [page 305]

Third-Party Integration [page 306]

3.4.1 AVP Dictionary

Keywords [page 293]

Preliminary Notes [page 293]

Description [page 293]

Modifying the AVP Dictionary [page 293]

3.4.1.1 Keywords

AVP[175] dictionary, XML[176], XSD

3.4.1.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- AVP Dictionary section of the SAP CC 2021 Application Help

3.4.1.3 Description

The Diameter Server system used an AVP[177] dictionary that is used to store all the AVP descriptions related to specific vendor network elements, that are not natively recognized by the Diameter Stack provided by the Traffix Systems company, and must thus be defined and described to be used during CCRs[178] and CCAs[179] mapping operations. You can use the following procedure to modify this configuration file and thus fit your specific business needs.

> ⚠ Caution
>
> The AVP dictionary is designed and customized by the project team during the implementation project phase. SAP SE recommends that system administrators do not change these settings.

3.4.1.4 Modifying the AVP Dictionary

To modify the AVP[180] dictionary of the Diameter Server system, execute the following procedure:

1. Stop the Diameter Server system
2. Backup the `avpDictionary.xml` file, that is located in the config folder of the Diameter Server instance
3. Edit the `avpDictionary.xml` file
4. Modify the relevant settings according to your needs
5. Save your modifications
6. Restart the Diameter Server system

175 Attribute-Value Pair
176 eXtended Markup Language
177 Attribute-Value Pair
178 Credit Control Request
179 Credit Control Answer
180 Attribute-Value Pair

3.4.2 Communication Channels

Keywords [page 294]

Preliminary Notes [page 294]

Description [page 294]

Securing the Communication Channels [page 294]

3.4.2.1 Keywords

Communication channels, securing

3.4.2.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- “Communication Channel Security” section of the SAP CC 2021 Security Guide documentation
- Securing an SAP CC System Landscape section of the SAP CC 2021 Installation and Maintenance Guide documentation

3.4.2.3 Description

Communications with the BART Server system rely on different communication channels whose use depend on the concerned elements. You can use the following procedure to secure these communication channels and thus fit your specific needs or performance requirements.

3.4.2.4 Securing the Communication Channels

For security purposes, all communication channels (except the Messages over UDP[181] one) used to communicate within your SAP CC landscape can be secured. For further information about the available securing procedures, refer to the dedicated Securing an SAP CC System Landscape section of the SAP CC 2021 Installation and Maintenance Guide documentation.

[181] User Datagram Protocol

3.4.3 Diameter Server Configuration File

Keywords [page 295]

Preliminary Notes [page 295]

Description [page 295]

Configuration Settings for Diameter Server [page 296]

Modifying the Diameter Server Configuration File [page 299]

3.4.3.1 Keywords

`serverConfig.xml`

3.4.3.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- Starting and Stopping the Server Systems section of the SAP CC 2021 Installation and Maintenance Guide documentation

3.4.3.3 Description

As described in the SAP CC 2021 Installation and Maintenance Guide documentation, the configuration of the Diameter Server system is located in the `serverConfig.xml` file:

- This configuration file is available in the `config` folder of the Diameter Server system already deployed on a dedicated host in your SAP system landscape.
- The file content is grouped upon the following sections:
  - License/keys settings
  - Diameter stack
  - Exchange capability information
  - Supported applications
  - Supported vendors
  - Peer discovery policy
  - Incoming messages handling
  - Routing table to handle proxiable requests
  - Security settings

- Charging stack
- Logging/tracing settings
- Third-party integration (with SAP SLD)
- TLS encryption (protocols, cypher suites)

The content and sections change and depend on your installed software version. Check the file to see the details.

- The XML format is basic and allows defining the necessary configuration settings for the SAP system. As the root XML element, the `properties` XML element contains the complete configuration.
  For a single-value setting, the XML element named `entry` is used with the XML attributes (`key` and `value`).
  For multivalue settings, the XML element `entry` with no XML attributes must be used. Sequences of `properties` XML elements encapsulate the different values specified by `entry` XML elements with the XML attributes (`key` and `value`).

  > **Example**
  >
  > For example: <entry key="TLSEnabledProtocols"> <properties> <entry key="TLSEnabledProtocol" value="TLSv1.1" /> </properties> <properties> <entry key="TLSEnabledProtocol" value="TLSv1.2" /> </properties> </entry>

You can use this generic procedure to configure some settings of the Diameter Server configuration and thus fit your specific needs.

- Modifying the Diameter Server Configuration File [page 299]
- Configuration Settings for Diameter Server [page 296]

3.4.3.4 Configuration Settings for Diameter Server

The following table contains the list of settings that can be configured within the configuration file of the Diameter Server system, including recommendations regarding their modification:

Setting and Category	Description	SAP CC Settings	Configurable
License Management			
`LicenseKey`	License key of the OpenBlox Java Diameter Stack used by the Diameter Server system	☐	○
Diameter Stack			
`URI`	IP address or name of the machine hosting the Diameter Server system	☐	○
`StackType`	Peer mode of the Diameter Server system	☐	☐
`MajorVersion`	Major version of the Diameter Base Protocol	☐	☐
`MinorVersion`	Minor version of the Diameter Base Protocol	☐	☐
Exchange Capability Information			

Setting and Category	Description	SAP CC Settings	Configurable
`Realm`	Domain name, used to determine whether messages can be satisfied locally or routed to another domain	☐	☐
`VendorId`	Identifier of the Diameter vendor	☐	☐
`ProductName`	Product name of the Diameter Server system	☐	☐
`FirmwareRevision`	Firmware revision, used to inform Diameter peers	☐	☐
Supported Applications			
`ApplicationId`	Identifier used to determine which Diameter application supports the messages	☐	☐
`ApplicationIdType`	Type of application	☐	☐
Supported Vendors			
`SupportedVendorID`	Identifier used in CER[182] and CEA[183] messages to inform peers that the sender supports the vendor-specific AVPs[184] defined by this identified vendor	☐	☐
Peer Discovery Policy			
`TWTimer`	Watchdog Timer (in seconds) used to control the frequency of DWR[185] messages. A small value fasters the detection of a peer failure, but increases the number of generated messages	☐	◊
`AcceptUnknownPeers`	Accept or refuse CERs from unknown peers	☐	◊
`CheckProtocolErrors`	Enable or disable the check of protocol errors	☐	◊
Incoming Messages Handling			
`OpenedSessionsMaximum`	Maximum number of opened sessions	☐	◊
`StateEventPoolCoreSize`		☐	◊
`StateEventPoolMaximumSize`		☐	◊
`StateEventHandlerQueueCapacity`		☐	◊
`StateEventPoolKeepAliveTime`		☐	◊
Routing Table to Handle Proxiable Requests			
`RealmName`	Primary key of the realm-based routing table	☐	☐
`RealmApplicationId`	Identifier of application, used as secondary key in routing table lookups	☐	☐

[182] Capabilities-Exchanging-Request
[183] Capabilities-Exchanging-Answer
[184] Attribute-Value Pair
[185] Device-Watchdog-Request

Setting and Category	Description	SAP CC Settings	Configurable
`RealmLocalAction`	Handling method for messages. The default value is set to: LOCAL Possible values are: LOCAL, RELAY, PROXY, or REDIRECT	☐	☐
Logging and Tracing			
`RootLoggerLeverl`	Severity threshold for logs, that is hot modifiable and thus does not require a restart of the Diameter Server system. The default value is set to: INFO Possible values are: OFF, SEVERE, WARNING, INFO, CONFIG, FINE, FINER, FINEST, or ALL	☐	■
Security Settings			
`SecurityId`	Identifier used during TLS negotiation between diameter peers	☐	■
`TLSKeyStoreFile`	Name of the file as keystore and containing private keys and certificates related to the Diameter Server system	☐	■
`TLSTrustStoreFile`	Name of the file used as truststore and containing private keys and certificates trusted by the Diameter Server system	☐	■
`TLSKeyStorePassword`	Passphrase of the keystore	☐	■
`TLSTrustStorePassword`	Passphrase of the truststore	☐	■
`TLSEnabledProtocol`	List of TLS protocols	☐	■
`TLSEnabledCipherSuite`	List of cypher suites	☐	■
Charging Stack			
`OCSHostName`	IP address or name of the machine hosting the Core Server system	■	○
`OCSPort`	External port of the dispatcher instance of the Core Server system that is used by the Diameter Server system to communicate over the Packets over TCP/IP[186] communication channel	■	○
`OCSUser`	Name of the technical user used to connect to the Core Server system	■	○
`OCSPassword`	Password of the specified technical user	■	○
`OCSSecured`	Enable or disable the securing of communications with the Core Server system	■	○

[186] Transmission Control Protocol/Internet Protocol

Setting and Category	Description	SAP CC Settings	Configurable
`OperationTimeout`	Timeout (in milliseconds) for charging requests **Note** Set by default to 1000ms, SAP SE recommends that you modify this value according to your business requirement.	■	■
Third-Party Integration			
`SLDURL`	URL of the targeted SAP SLD system	■	○
`SLDUser`	Name of the SAP SLD user	■	○
`SLDPassword`	Password of the SAP SLD user	■	○

■ Yes ☐ No ○ Yes, also during the installation ◊ Not recommended

3.4.3.5 Modifying the Diameter Server Configuration File

As a system administrator, to modify the configuration file of the Diameter Server system, execute the following procedure:

1. Stop the running Diameter Server system.
2. Find the `serverConfig.xml` file. It is is located in the `config` folder of the deployed Diameter Server system (on its dedicated host).
3. In a production SAP system landscape, back up this configuration file.
4. Edit this XML file in a text editor for example.
5. Modify the relevant settings according to your technical or business needs. See Configuration Settings for Diameter Server [page 296] as a reference.
6. Save your modifications.
7. In a production SAP system landscape, back up the modified configuration file.
8. Start the Diameter Server system again and check its successful startup.

3.4.4 Java Virtual Machine

Keywords [page 300]

Preliminary Notes [page 300]

Description [page 300]

3.4.4.1 Keywords

JVM[187] options, virtual machine, memory heap, garbage collection, buffers

3.4.4.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- "JVM option recommendations for SAP Convergent Charging" SAP Note 2006073 documentation
- "Generate Java GC Log for SAP Convergent Charging" SAP Note 1950168 documentation

3.4.4.3 Description

The architecture of SAP Convergent Charging relies on a set of java-based applications whose execution require a Java Runtime Environment. The Diameter Server system is a Java process whose execution can be configured to fit specific requirements. To determine and modify the different JVM[188] options for the Diameter Server system, refer to the “Java Virtual Machine” procedure available for the instances of the Core Server system, and apply the relevant modifications on the startup file of the Diameter Server system.

3.4.5 Logging and Tracing

Keywords [page 300]

Preliminary Notes [page 301]

Description [page 301]

Configuration Options [page 301]

Modifying the Logging Settings of the Diameter Server [page 303]

3.4.5.1 Keywords

Logging.properties, log, trace, category, severity, domain, threshold, troubleshoot

187 Java Virtual Machine
188 Java Virtual Machine

3.4.5.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- Starting and Stopping the Server Systems section of the SAP CC 2021 Installation and Maintenance Guide documentation

3.4.5.3 Description

As described in the SAP CC 2021 Application Help documentation, SAP Convergent Charging provides logging and tracing features that give the possibility to generate both trace and log messages:

- According to defined levels such as severity, thresholds, domains, categories
- Using different output formats
- In multiple output destinations

Mainly addressed to administrators and support specialists, the generation of these log and trace messages can be configured in order to fit specific needs such as landscape monitoring, performance tracing, issue troubleshooting, and so on. You can use this procedure to configure the logging and tracing capabilities of the Diameter Server system, that rely on the Java native logging framework whose configuration is located in the `logging.properties` file:

- That is available in the `config` folder of the Diameter Server instance
- Whose content is grouped upon the following sections:
  - Handlers definition
  - Handlers configuration
  - Default log level

3.4.5.4 Configuration Options

The following table contains the list of settings related to logging capabilities of the Diameter Server system that you can configure to fit your specific needs:

Settings	Description
Handlers Definition	
`handlers`	CSV (space delimited) list of handlers' class-names for the root logger, set by de-fault to: `java.util.logging.ConsoleHandler java.util.logging.FileHandler`
Default Log Level	

Settings	Description
`.level`	Default log level for every logger whose level is not explicitly defined, set by default to: CONFIG Possible values are: OFF, SEVERE, WARNING, INFO, CONFIG, FINE, FINER, FINEST or ALL
Console Handler Configuration	
`java.util.logging.ConsoleHandler.level`	Log level for the console output, set by default to: ALL Possible values are: OFF, SEVERE, WARNING, INFO, CONFIG, FINE, FINER, FINEST or ALL
`java.util.logging.ConsoleHandler.formatter`	Classname of the string formatter used for texts printed into the console
File Handler Configuration	
`java.util.logging.FileHandler.level`	Log level for the console output, set by default to: ALL Possible values are: OFF, SEVERE, WARNING, INFO, CONFIG, FINE, FINER, FINEST or ALL
`java.util.logging.FileHandler.formatter`	Classname of the string formatter used for texts printed into the console
`java.util.logging.FileHandler.pattern`	Pattern used to generate the ouput file name, that consists of a string including the following components which are replaced at runtime: • "/", that represents the local pathname separator • "%t", that represents the system temporary directory • "%h", that represents the value of the "user.home" system property • "%g", that represents the generation number used to distinguish rotated logs. If no "%g" field has been specified and if the file count is greater than one, the generation number will be added to the end of the generated filename, after a dot • "%u", that represents a unique number used to resolve conflicts • "%%", that can be used to translate a single percent sign "%"
`java.util.logging.FileHandler.limit`	Approximate maximum amount to write (in bytes) to a given file. Setting this system parameter to 0 means that the written amount is not limited
`java.util.logging.FileHandler.count`	Number of output files available for the rotation cycle
`java.util.logging.FileHandler.append`	Specifies whether the FileHandler should append content onto an existing output file, or not

3.4.5.5 Modifying the Logging Settings of the Diameter Server

To modify the logging settings of the Diameter Server system, execute the following procedure:

1. Stop the Diameter Server system
2. Backup the `logging.properties` file
3. Edit the `logging.properties` file, that is located in the config folder of the Diameter Server instance
4. Modify the relevant settings according to your needs
5. Save your modifications
6. Restart the Diameter Server system

3.4.6 Multiple Services Credit-Control

Keywords [page 303]

Preliminary Notes [page 303]

Description [page 303]

Implementing MSCC [page 304]

3.4.6.1 Keywords

MSCC[189], service dictionary, XML[190], XSD

3.4.6.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- Service Dictionary section of the SAP CC 2021 Application Help documentation
- Service Dictionary [page 305] section of this documentation

3.4.6.3 Description

As described in the SAP CC 2021 Application Help documentation, the service dictionary gives the possibility to register multiple services with different mappings, and thus implement the multiple-services credit control

189 Multiple-Services Credit Control
190 eXtended Markup Language

feature within the Diameter Server system. You can use the following procedure to modify the relevant settings of the service dictionary and thus fit your specific business needs regarding MSCC[191].

3.4.6.4 Implementing MSCC

The following table contains the list of AVPs[192] that rely to the MSCC[193] support:

XML[194] Element	Description
`msccServiceKeyDesc`	Determines which mapping must be applied to the CCR[195]
`msccServiceDesc`	Defines the mapping that must be used to transform CCRs into charging requests. Mandatory to activate the support of MSCC, this element contains one `msccMappingDesc` sub-element
`msccMappingDesc`	Indicates where the following sub-elements can be found in the CCR: • Mandatory sub-elements: • `eventNameDesc`, that corresponds to the name of the chargeable item class • `reservationIdDesc`, that corresponds to the identifier of the reservation • `unitPropertyDesc`, that corresponds to the unit property which is used to identify the property to inverse in case of inverse rating • Optional sub-elements: • `infoPropertyDesc`, that corresponds to a user property to map, when required. You can define a default value for this property in case this property is not available in the MSCC AVP[196]

To activate the support of MSCC, you must thus execute the Modifying the Service Dictionary [page 306] procedure and add at least one `msccServiceDesc` element to the service dictionary.

> **Note**
>
> - If a CCR contains several MSCC AVPs but no `msccServiceDesc` element is defined in the service dictionary, a `DIAMETER_UNABLE_TO_COMPLY` error will be sent
> - The `serviceDictionaryMSCC.xml` is provided in the examples folder of the Diameter Server instance as an example including MSCC elements. In this example, the “Rating-Group” AVP is used as key for the mapping and as reservation ID.

191 Multiple-Services Credit Control
192 Attribute-Value Pair
193 Multiple-Services Credit Control
194 eXtended Markup Language
195 Credit Control Request
196 Attribute-Value Pair

3.4.7 Security Certificates

See Also

Securing Communications with the Diameter Server System

3.4.8 Service Dictionary

Keywords [page 305]

Preliminary Notes [page 305]

Description [page 305]

Modifying the Service Dictionary [page 306]

3.4.8.1 Keywords

Service dictionary, AVP[197], XML[198], XSD

3.4.8.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- Service Dictionary section of the SAP CC 2021 Application Help documentation

3.4.8.3 Description

The Diameter Server system used a service dictionary that is used to store all the descriptions of each managed service, including the AVP[199] mapping between the Diameter messages and the charging operation messages exchanged between the Diameter Server and Core Server systems. You can use the following procedure to modify this configuration file and thus fit your specific business needs.

197 Attribute-Value Pair
198 eXtended Markup Language
199 Attribute-Value Pair

> ⚠ Caution
>
> The service dictionary is designed and customized by the project team during the implementation project phase. SAP SE recommends that system administrators do not change these settings.

3.4.8.4 Modifying the Service Dictionary

To modify the service dictionary of the Diameter Server system, execute the following procedure:

1. Stop the Diameter Server system
2. Backup the `serviceDictionary.xml` file, that is located in the config folder of the Diameter Server instance
3. Edit the `serviceDictionary.xml` file
4. Modify the relevant settings according to your needs
5. Save your modifications
6. Use the `serviceDictionary.xsd` XSD schema to ensure that your modifications are valid
7. Restart the Diameter Server system

3.4.9 Third-Party Integration

Keywords [page 306]

Preliminary Notes [page 306]

Description [page 307]

Modifying the SAP SLD Connection Settings [page 307]

3.4.9.1 Keywords

SLD[200]

3.4.9.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- “Update of SAP System Component Repository in SLD” SAP Note 669669 documentation

[200] System Landscape Directory

3.4.9.3 Description

The Diameter Server system can be integrated with third-party applications. You can use the following procedure to configure the settings related to the integration with the SAP System Landscape Directory (SLD) system.

3.4.9.4 Modifying the SAP SLD Connection Settings

To send payload information to the SAP SLD system during the startup sequence of the Diameter Server system, it is necessary to configure some settings related to the integration with third-party systems. To modify this configuration, refer to the Modifying the Diameter Server Configuration File [page 299] section above and modify the following settings of the Diameter Server:

- `SLDURL`, that corresponds to the IP address or name of the SLD Data Supplier Service in the targeted SAP SLD system, using the following format: `http://<SLD_HOST>:<SLD_PORT>/sld/ds`
- `SLDUser`, that corresponds to the name of the SAP SLD user
- `SLDPassword`, that corresponds to the password of the specified user

3.5 Configuring the Import/Export Connector System

Scenario Management [page 307]

Specific Actions [page 308]

3.5.1 Scenario Management

Keywords [page 307]

Preliminary Notes [page 308]

Description [page 308]

3.5.1.1 Keywords

CAT[201], scenario

[201] Connector Administration Tool

3.5.1.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- Installing and Launching CAT Tool section of the SAP CC 2021 Installation and Maintenance Guide documentation
- Scenarios section of the SAP CC 2021 Primary Help for CAT Tool (+IEC) documentation

3.5.1.3 Description

The CAT Tool user interface gives the possibility to model scenarios that represent sequences of basic actions executed by the Import/Export Connector system. For further information, refer to the Scenarios section of the SAP CC 2021 Primary Help for CAT Tool (+IEC) documentation, that contains details about:

- Scenario modeling
- Scenario building
- Scenario execution

3.5.2 Specific Actions

Keywords [page 308]

Preliminary Notes [page 308]

Description [page 309]

3.5.2.1 Keywords

Java function, customization, SDK

3.5.2.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- `com.highdeal.iec` Java package description, available in the SAP CC Java/XML API Reference (IEC Library) documentation

3.5.2.3 Description

The Import/Export Connector system can be customized by developing your own Java classes using the IEC SDK that gives the possibility to:

- Add new actions
- Customize the reporting capabilities of the Import/Export Connector system
- Customize the connection to the BART Server system

For further information, refer to the SAP CC Java/XML API Reference (IEC Library) documentation.

4 Fine-Tuning SAP Convergent Charging

Fine-tuning must be performed to further adjust the system settings to support specific business or technical requirements and optimize the overall performance.

- Fine-Tuning the Smart Distribution Mode [page 314]
- Fine-Tuning the Table Rollover Policy [page 318]

  Distributing Partitions to guiders and raters Using a Smart Mode [page 310]

  Modifying the Table Rollover Policy for Charging Sessions [page 317]

4.1 Distributing Partitions to guiders and raters Using a Smart Mode

Keywords [page 310]

Preliminary Notes [page 310]

Description [page 311]

Activating the Smart Distribution Mode [page 311]

Fine-Tuning the Smart Distribution Mode [page 314]

Protecting Raters and Guiders Against Warm-Up Overloading [page 315]

4.1.1 Keywords

cache, warm-up, distribution, partition, smart mode, online stateful, primary dispatcher, rater, guider, burst protection

4.1.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- Partitions Mapping Reorganization section of the SAP CC 2021 Application Help documentation
- Cache Warm-Up section of the SAP CC 2021 Application Help documentation
- Partitions Distribution and Warm-Up Settings group in the SAP CC 2021 System Parameter Reference documentation

4.1.3 Description

To increase the reactivity of the running Core Server system, rater and guider instances handle data within dedicated cached structures.

To avoid latency, a cache warm-up mechanism can be activated in order to retrieve this data from the Core Database and Session Database, and preload them into these cached structures. When this warm-up mechanism is activated, it is possible to configure the Core Server system in order to activate a smart execution mode for the partitions distribution operations performed by the primary dispatcher instance. This execution mode gives the possibility to control the warm-up operations performed by the targeted instances during these partitions distribution operations (executed when guider or rater instances start, shut-down, or fail), and thus to control the overall impact of warm-up operations on the online stateful charging operations.

4.1.4 Activating the Smart Distribution Mode

If the cache warm-up mechanism is activated in your SAP system, you configure the technical settings for the system instances:

- Rater
- Guider

Activate the Smart Distribution Mode for the Rater Instances

If the CACHE_WARMUP_THREAD_COUNT system parameter has been set to a positive value, the cache warm-up mechanism is activated. In this case, the primary dispatcher instance distributes partitions using the smart execution mode as soon as the number of running instances is greater than or equals the value of the RATER_PARTITION_SWITCH_MIN_INSTANCE_COUNT system parameter.

As the system administrator, you complete the procedure:

1. Tune and configure the following system parameters:
    1. RATER_PARTITION_SWITCH_WARMUP_TIMEOUT, which defines the maximum duration allocated to a partition warm-up operation performed by a rater instance in a smart distribution context. SAP SE recommends you to set up a value that corresponds to twice the average duration of the partition warm-up operations, that can be retrieved from the dedicated Introscope probes.
    2. RATER_PARTITION_SWITCH_CONCURRENT_PARTITIONS, which defines the maximum number of not-yet warmed partitions that can be simultaneously assigned to a rater for warming purpose.

    > **Example**
    >
    > Considering a Core Server system containing 4 rater instances. Setting the RATER_PARTITION_SWITCH_CONCURRENT_PARTITIONS system parameter to 3 implies that a maximum of 12 partitions can be simultaneously assigned by the primary dispatcher to the running rater instances for warming purpose.

3. RATER_PARTITION_SWITCH_DB_INSTANCE_CONCURRENT_PARTITIONS, which defines the maximum number of partitions per database instance that can be simultaneously assigned to the running rater instances for warming purpose.

   > **Example**
   >
   > Considering a landscape containing a Core Database made up with 3 database instances. Setting the RATER_PARTITION_SWITCH_DB_INSTANCE_CONCURRENT_PARTITIONS system parameter to 3 implies that a maximum of 9 partitions can be simultaneously assigned by the primary dispatcher to the running rater instances for warming purpose, whatever the number of running rater instances is.

> **Note**
>
> In case both RATER_PARTITION_SWITCH_CONCURRENT_PARTITIONS and RATER_PARTITION_SWITCH_DB_INSTANCE_CONCURRENT_PARTITIONS are defined, the minimum of the 2 corresponding number of partitions is used by the primary dispatcher when assigning partitions for warming purpose.

> **Example**
>
> Considering:
> - A Core Server system containing 5 raters, 1 of these raters shutting down
> - A Core Database made up with 3 database instances
> - The RATER_PARTITION_SWITCH_CONCURRENT_PARTITIONS system parameter set to 3
> - The RATER_PARTITION_SWITCH_DB_INSTANCE_CONCURRENT_PARTITIONS system parameter set to 3
>
> Then the number of partitions that can be simultaneously assigned by the primary dispatcher to the 4 running rater instances for warming purpose is: min((5-1)x3,3x3)=9 partitions.

2. Fine-tune the smart distribution mode for the Rater instances.
   See Fine-Tuning the Smart Distribution Mode [page 314]
3. Protect the Guider instances against warm-up overloading.
   See Protecting Raters and Guiders Against Warm-Up Overloading [page 315].

> **→ Remember**
>
> If the cache warm-up mechanism is not activated, no warm-up operation is performed by rater instances, which leads to additional latency that impacts the online stateful charging operations.

Activate the Smart Distribution Mode for the Guider Instances

If the CACHE_WARMUP_THREAD_COUNT system parameter has been set to a positive value, the cache warm-up mechanism is activated. In this case, the primary dispatcher instance distributes partitions using the smart execution mode as soon as the number of running instances is greater than or equals the value of the GUIDER_PARTITION_SWITCH_MIN_INSTANCE_COUNT system parameter.

As the system administrator, you complete the procedure:

1. Tune and configure the following system parameters:
    1. GUIDER_PARTITION_SWITCH_WARMUP_TIMEOUT, which defines the maximum duration allocated to a partition warm-up operation performed by a guider instance in a smart distribution context.
    SAP SE recommends you to set up a value that corresponds to twice the average duration of the partition warm-up operations, that can be retrieved from the dedicated Introscope probes.
    2. GUIDER_PARTITION_SWITCH_CONCURRENT_PARTITIONS, which defines the maximum number of not-yet warmed partitions that can be simultaneously assigned to a guider for warming purpose.

    > **Example**
    >
    > Considering a Core Server system containing 4 guider instances. Setting the GUIDER_PARTITION_SWITCH_CONCURRENT_PARTITIONS system parameter to 3 implies that a maximum of 12 partitions can be simultaneously assigned by the primary dispatcher to the running guider instances for warming purpose.

    3. GUIDER_PARTITION_SWITCH_DB_INSTANCE_CONCURRENT_PARTITIONS, which defines the maximum number of partitions per database instance that can be simultaneously assigned to the running guider instances for warming purpose.

    > **Example**
    >
    > Considering a landscape containing a Core Database made up with 3 database instances. Setting the GUIDER_PARTITION_SWITCH_DB_INSTANCE_CONCURRENT_PARTITIONS system parameter to 3 implies that a maximum of 9 partitions can be simultaneously assigned by the primary dispatcher to the running guider instances for warming purpose, whatever the number of running guider instances is.

    > **Note**
    >
    > In case both GUIDER_PARTITION_SWITCH_CONCURRENT_PARTITIONS and GUIDER_PARTITION_SWITCH_DB_INSTANCE_CONCURRENT_PARTITIONS are defined, the minimum of the 2 corresponding number of partitions is used by the primary dispatcher when assigning partitions for warming purpose.

    > **Example**
    >
    > Considering:
    > - A Core Server system containing 4 running guiders
    > - A Core Database made up with 3 database instances
    > - The GUIDER_PARTITION_SWITCH_CONCURRENT_PARTITIONS system parameter set to 2
    > - The GUIDER_PARTITION_SWITCH_DB_INSTANCE_CONCURRENT_PARTITIONS system parameter set to 3
    >
    > Then the number of partitions that can be simultaneously assigned by the primary dispatcher to the running guider instances for warming purpose is: min(4x2,3x3)=8 partitions.

2. Fine-tune the smart distribution mode for the Guider instances.
See Fine-Tuning the Smart Distribution Mode [page 314]
3. Protect the Guider instances against warm-up overloading.
See Protecting Raters and Guiders Against Warm-Up Overloading [page 315].

→ Remember

If the cache warm-up mechanism is not activated, no warm-up operation is performed by guider instances, which leads to additional latency that impacts the online stateful charging operations.

4.1.5 Fine-Tuning the Smart Distribution Mode

Fine-Tune the Smart Distribution Mode for the Guider Instances

As the system administrator, you complete the following procedure:

1. SAP SE recommends you to consider the following configuration as the minimum possible configuration:
   - GUIDER_PARTITION_SWITCH_CONCURRENT_PARTITIONS = 1
   - GUIDER_PARTITION_SWITCH_DB_INSTANCE_CONCURRENT_PARTITIONS = 1
   - GUIDER_PARTITION_SWITCH_MIN_INSTANCE_COUNT = (total number of running instances) - (acceptable number of down instances at a given time) + 1

   ✤ Example

   Considering:
   - A Core Server system containing 4 guider instances deployed on 2 hosts
   - An acceptable number of 1 host failure, similar to 2 guider instances failure

   Then the GUIDER_PARTITION_SWITCH_MIN_INSTANCE_COUNT system parameter can be set to (4-2)+1=3, which means that the smart execution mode will be used by the primary dispatcher:
   - Until more than 2 guider instances remain running
   - When a third guider will start
2. Then, you need to tune the values of both GUIDER_PARTITION_SWITCH_CONCURRENT_PARTITIONS and GUIDER_PARTITION_SWITCH_DB_INSTANCE_CONCURRENT_PARTITIONS system parameters according to the previously explained behavior.
   The objective is to determine the values that give the best ratio between:
   - The duration of a guider instance startup or shutdown operation
   - The impact of warm-up operations on the online stateful charging operations

   These values can be determined by analyzing the following performance and health metrics of the guider instances of the Core Server system:
   - SAP CC | Warm-Up | Partition Count, which shows the number of partitions assigned to a given instance
   - SAP CC | Warm-Up | Partition Count, which shows the number of warmed partitions of a given instance

Fine-Tune the Smart Distribution Mode for the Rater Instances

As the system administrator, you complete the following procedure:

1. SAP SE recommends you to consider the following configuration as the minimum possible configuration:
   - RATER_PARTITION_SWITCH_CONCURRENT_PARTITIONS = 1
   - RATER_PARTITION_SWITCH_DB_INSTANCE_CONCURRENT_PARTITIONS = 1
   - RATER_PARTITION_SWITCH_MIN_INSTANCE_COUNT = (total number of running instances) - (acceptable number of down instances at a given time) + 1

   > ✤ Example
   >
   > Considering:
   > - A Core Server system containing 6 rater instances deployed on 3 hosts
   > - An acceptable number of 1 host failure, similar to 2 rater instances failure
   >
   > Then the RATER_PARTITION_SWITCH_MIN_INSTANCE_COUNT system parameter can be set to (6-2)+1=4, which means that the smart execution mode will be used by the primary dispatcher:
   > - Until more than 4 rater instances remain running
   > - When a fifth rater will start

2. Then, you need to tune the values of both RATER_PARTITION_SWITCH_CONCURRENT_PARTITIONS and RATER_PARTITION_SWITCH_DB_INSTANCE_CONCURRENT_PARTITIONS system parameters according to the previously explained behavior.
   The objective is to determine the values that give the best ratio between:
   - The duration of a rater instance startup or shutdown operation
   - The impact of warm-up operations on the online stateful charging operations

   These values can be determined by analyzing the following performance and health metrics of the rater instances of the Core Server system:
   - SAP CC | Warm-Up | Partition Count, which shows the number of partitions assigned to a given instance
   - SAP CC | Warm-Up | Partition Count, which shows the number of warmed partitions of a given instance

4.1.6 Protecting Raters and Guiders Against Warm-Up Overloading

In case a rater or guider instance is added or removed from the list of running instances in the SAP system, the primary dispatcher instance redistributes the data partitions among the list of running raters and guiders for warming purposes. When the smart execution mode has been activated, it is also possible to configure the following system parameters in order to protect the raters and guiders against possible overloading due to warm-up operations:

- RATER_PARTITION_SWITCH_BURST_PROTECTION_INITIAL_DELAY, which defines the minimum delay (in seconds) between the first and second assignments of partition to each rater instance during a smart warm-up operation

- RATER_PARTITION_SWITCH_BURST_PROTECTION_DELAY, which defines the minimum delay (in seconds) between the assignment of partitions to each rater instance during a smart warm-up operation
- GUIDER_PARTITION_SWITCH_BURST_PROTECTION_INITIAL_DELAY, which defines the minimum delay (in seconds) between the first and second assignments of partition to each guider instance during a smart warm-up operation
- GUIDER_PARTITION_SWITCH_BURST_PROTECTION_DELAY, which defines the minimum delay (in seconds) between the assignment of partitions to each guider instance during a smart warm-up operation

As the Java Virtual Machine (JVM) is more solicited on startup due to additional internal operations, SAP SE recommends that you set an initial delay that is bigger than the delay used for the subsequent assignments.

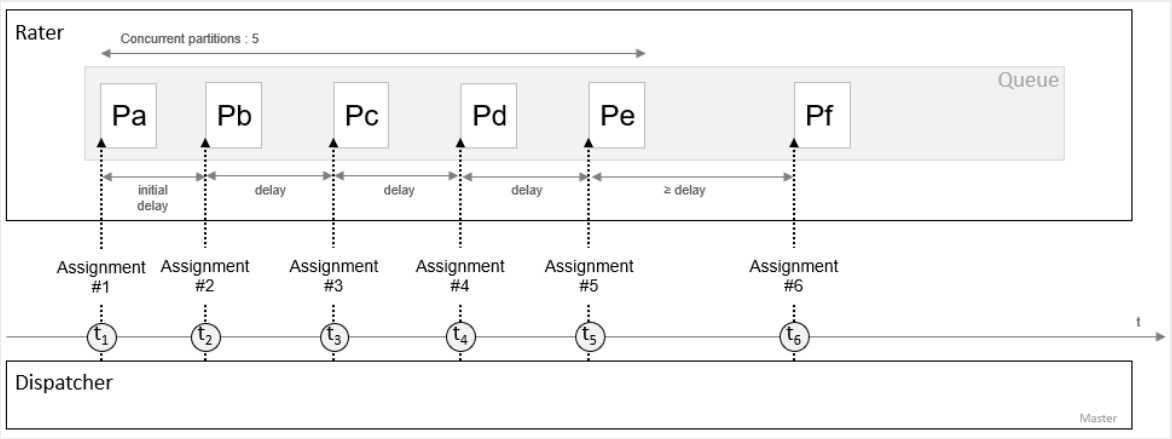
As the system administrator, you configure your SAP system (rater and guider instances).

Example

Considering a configuration where:

- 1 rater instance is starting
- 6 partitions (Pa to Pf) must be assigned by the primary dispatcher instance to this starting rater instance
- `RATER_PARTITION_SWITCH_CONCURRENT_PARTITIONS`=5, which means that a maximum of 5 partitions can be simultaneously assigned to this rater for warming purpose

The following schema shows how this list of partitions are assigned to the rater instance using delays for protection purpose:



- At **t1**, the primary dispatcher instance assigns the partition **Pa** to the rater instance, which represents the first partition assignment. The rater puts **Pa** in its dedicated queue and starts warming up this partition.
- At **t2**, the primary dispatcher assigns the partition **Pb** to the rater, which represents the second partition assignment, delayed using the value of the `RATER_PARTITION_SWITCH_BURST_PROTECTION_INITIAL_DELAY` system parameter. The rater puts **Pb** in its dedicated queue and starts warming up this partition.
- At **t3**, the primary dispatcher assigns the partition **Pc** to the rater, which represents the third partition assignment, delayed at least with the value of the `RATER_PARTITION_SWITCH_BURST_PROTECTION_DELAY` system parameter. The rater puts **Pc** in its dedicated queue and starts warming up this partition.

- At **t4**, the primary dispatcher assigns the partition **Pd** to the rater, which presents the fourth partition assignment, delayed at least with the value of the `RATER_PARTITION_SWITCH_BURST_PROTECTION_DELAY` system parameter. The rater puts **Pd** in its dedicated queue and starts warming up this partition.
- At **t5**, the primary dispatcher assigns the partition **Pe** to the rater, which represents the fifth partition assignment, delayed at least with the value of the `RATER_PARTITION_SWITCH_BURST_PROTECTION_DELAY` system parameter. The rater puts **Pe** in its dedicated queue and starts warming up this partition.
- At **t6**, which corresponds to an undefined time but that is greater than the value specified in the `RATER_PARTITION_SWITCH_BURST_PROTECTION_DELAY` system parameter, the rater successfully warmed up one of the previously assigned partitions (Pa to Pe). The primary dispatcher then assigns the partition **Pf** to the rater. The rater puts **Pf** in its dedicated queue and starts warming up this partition.

> ⓘ Note
>
> In case the smart execution mode has not been activated, all the 6 partitions (Pa to Pf) would be simultaneously assigned to the rater instance by the primary dispatcher instance, which could decrease the performance of the rater regarding the warm-up operations.

4.2 Modifying the Table Rollover Policy for Charging Sessions

Keywords [page 317]

Preliminary Notes [page 317]

Description [page 318]

Enabling the Table Rollover Mechanism [page 318]

Fine-Tuning the Table Rollover Policy [page 318]

4.2.1 Keywords

Session Database, session-based charging, rating session, charging session, rollover, fragmentation

4.2.2 Preliminary Notes

For further information, refer to the following SAP CC documentations:

- Table Rollover for Rating Sessions and Dual Database sections of the SAP CC 2021 Application Help documentation
- Database Management Settings group in the SAP CC 2021 System Parameter Reference documentation

4.2.3 Description

As described in the description of the Dual Database feature available in the SAP CC 2021 Application Help documentation, the Session Database is an optional database that contains the different data related to the business objects handled by SAP CC during the session-based charging operations.

When this database is used, each event that occurs on a given rating session leads to one or multiple operations that directly impact the content of the Session Database. These frequent modifications may lead to the fragmentation of the database tables, which decreases its performances and can consequently decrease the overall performance of your landscape.

You can use this procedure to configure a rollover mechanism that reduces the fragmentation of the storage related to the rating sessions within the Session Database.

4.2.4 Enabling the Table Rollover Mechanism

If the RATING_SESSION_TABLE_ROLLOVER_COUNT system parameter has been set to a value higher than 1, the table rollover mechanism is activated. In this case, the rater instances record the session data in the relevant table considered as destination for the current period.

> **Note**
>
> You can use the RATING_SESSION_TABLE_ROLLOVER_STATUS indicator to see the status of the mechanism, particularly the validity period of the currently used destination.

4.2.5 Fine-Tuning the Table Rollover Policy

If the Session Database is part of your SAP system landscape, fine-tune the table rollover mechanism.

As described in the SAP CC 2021 Application Help documentation, the configuration of the rollover mechanism is based on the following system parameters:

- RATING_SESSION_TABLE_ROLLOVER_COUNT, used to specify the number of tables that can be used as destinations by the rater instances to store the rating sessions
- RATING_SESSION_TABLE_ROLLOVER_PERIOD, that defines the validity period (in days) of the table that is currently used as destination for storing the rating sessions data

To fit your specific needs, SAP SE recommends that you take into consideration the maximum duration of your rating sessions. Indeed, the validity period must be large enough to avoid that a pending rating session that has been created within a given destination table locks the defragmentation of this table after the table switch.

Example

Considering 2 destination tables used to store rating sessions whose maximum observed duration is 1 day, you can set a validity period of 7 days. This validity period implies that:

- During the 7 first days, the rating sessions will be created in the current destination table
- The 8th day, the new rating sessions will be created in the newly switched destination table. The cleanup of the previous destination table begins and will lead to the deletion of the ended rating sessions, but will not impact the pending ones
- The 9th day, all rating sessions created in the previous destination table are theoretically ended and can be deleted
- The 10th day, the previous destination table is empty, and is thus no more fragmented

Once you determined a validity period, SAP SE recommends that you start using 2 destination tables. You can increase this number up to 4 destination tables, but you need to keep in mind that each destination table represents the same physical storage, that needs to be available in your landscape. Increasing the number of destination tables thus requires multiplying the physical storage space. Theoretically, when the validity is well-tuned, 2 destination tables should be enough.

Note

If you hot change the number of destination tables or the validity period, the “From” date available in the RATING_SESSION_TABLE_ROLLOVER_STATUS indicator will not reflect the real start date of the period.

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